
بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ
IN PC
PRESS CTRL+F
SO U CAN SRCH!
cs-
411
"Always Push Yourself
a little and watch your
Potential rise

1. **Which one of the following is not a routed event?**

   Direct

2. **To achieve DockPanel behavior in grid we can use _____.**

   Row span and column span

3. **_____ is the subclass of grid used to freeze some rows and columns of grid.**

   Selective scrolling Grid

4. **Multitouch events are available on _____.**

   Windows 7 or later

5. **Suppose we have two elements in grid one is button and other is rectangle, which property will be helpful to set position of button on top of the rectangle?**

   Z index…..confirm

   To set the position of a button on top of a rectangle in UWP, you can use the **Z index** property. The Z index property specifies the stack order of an element. An element with greater stack order is always in front of an

6. **_____ is the subclass of grid used to freeze some rows and columns of grid.**

   Selective Scrolling Grid…..confirm

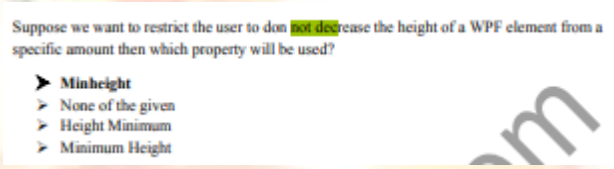
7. **To design a complex interface, which of the following technique is best to achieve the required results?**

   Panels composed within panels……confirm

The best technique to achieve the required results for designing a complex interface would depend on the specific requirements and complexity of the interface itself. However, from the options you provided, "Panel composed within panels" seems to be the most suitable option.

8. **Which of the following property will be used to restrict the user to not decrease the height of a WPF element form a specific value?**

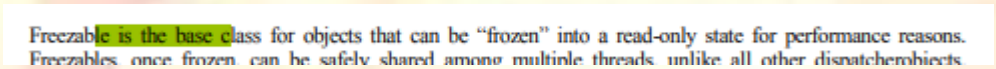
Minheight….confirm


Suppose we want to restrict the user to don not decrease the height of a WPF element from a specific amount then which property will be used?

- **Minheight**
- None of the given
- Height Minimum
- Minimum Height

9. **In WPF, _____ class defines an object which has read-only(frozen) state.**

Freezable…..confirm


Freezable is the base class for objects that can be "frozen" into a read-only state for performance reasons.

10. While working with grids in WPF, which attribute is used to see grid lines on screen?

Showgridlines….confirm

In WPF (Windows Presentation Foundation), the attribute used to display grid lines on the screen is "ShowGridLines." This attribute is

11. **When a user select element from listbox, after selecting compiler marked the selected element as _____.**

Selected….confirm

When a user selects an element from a list box, the compiler marks the selected element as **selected**. This is because the user has chosen that element from the list of options provided by the list box. The

12. Which of the followings are keyboard events?

Key down, key up

13. A _____ transform element after they are laid out by the parent panel (but before they are rendered).

Layout transform…..confirm

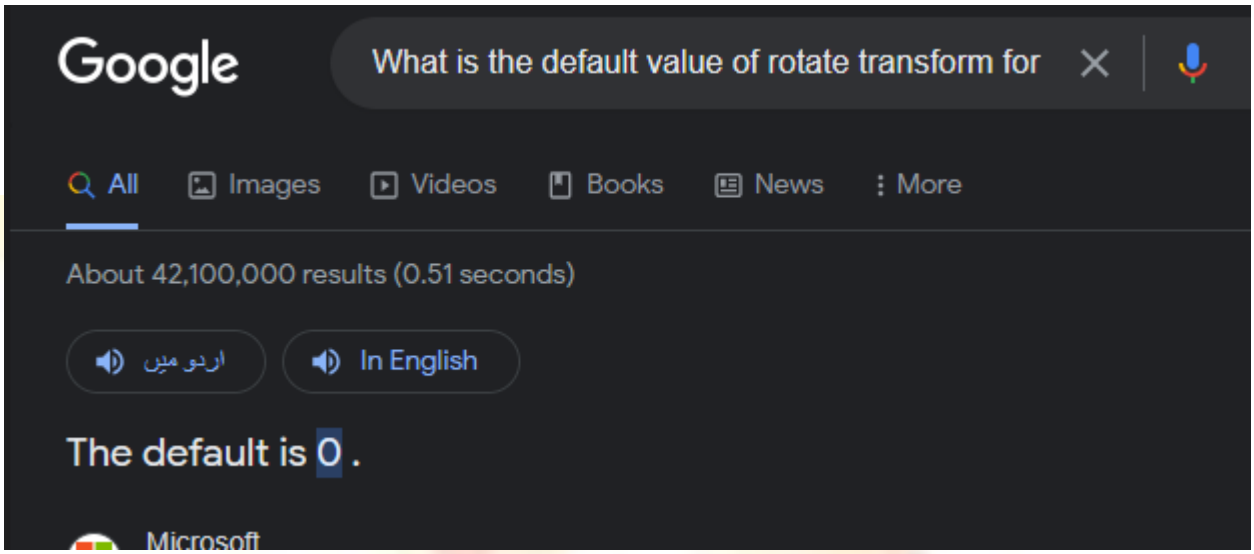
A _____ transforms elements before they are laid out by the parent panel.
Select correct option

- ☒ Skew transform
- ☒ Render transform
- ☒ Layout transform
- ☒ Matrix transform

The correct option is "Layout transform."

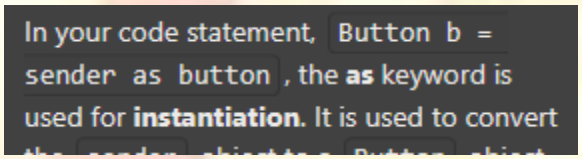
14. Rotate transform has default value _____for center-x, center-y, Angle

0….confirm



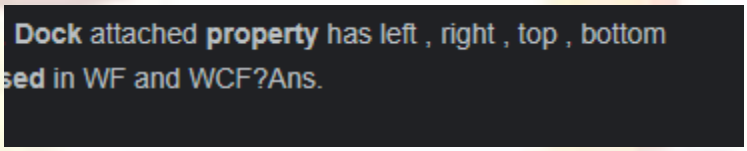
15. In this code statement
Button b = sender as button
As keyword is used for _____.

Instantiation……confirm



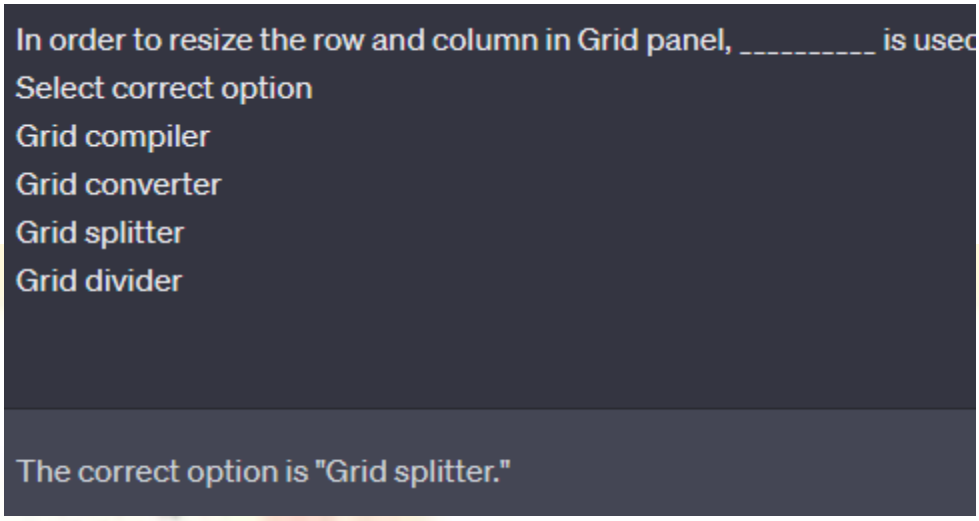
16. Dockpanel uses _____ property while docking elements in different orientations.

Attached…..confirm



17. In order to resize the row and column in Grid panel, _____ is used.

Grid splitter,,,,confirm

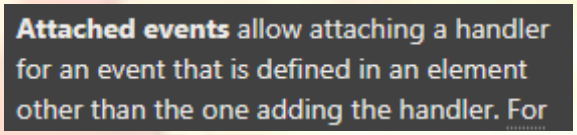


18. In dockpanel _____ attribute doesn't fill the remaining space at the end.

Lastchildfill=False…..confirm

19._____ allow attaching a handler for an event that is defined in an element other than the one adding the handler.

Attached event…..confirm



20. Which statement from the following statements is true about a java anonymous inner class?

All options…..confirm





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GRAND QUIZ

0304-1659294

JUNAID MALIK

LMS handling service are available

PAID

Assignment, QUIZ, GDBZ and GRAND QUIZ SERVICE are available

0304-1659294

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CS411 Grand Quiz

In WFP, if we try to add list box item in list box object which is already null, it will give us

- Runtime exception
- **Compile time error**
- Syntax error
- Logical error

XML is a stricter version of _____.

- Java
- CSS
- **HTML**
- C#

Which of these methods used to read strings from the console?

- Get ()
- Get line ()
- Read ()
- **Readline ()**

Which if the following statement is correct about WPF?

- WPF can use Python language foe designing
- WPF can developed mobile apps
- **WPF is resolution independence**
- WPF not support declarative programming

The code given bellow is equivalent to _____.

<SolidColorBrush>White</SolidColorBrush>

- <WhiteBrush>MyWhiteBrush</WhiteBrush>
- None of the given option
- **<SolidColorBrush Color = "White"/>**
- <Brush Style = "Solid White">

WPF4 was released in _____.

- 2012
- **2010**

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- 2005
- 2003

Which of the following event may or may not relate to an actual occurrence?

- **Probabilistic Events**
- Actual Events
- Real Events
- Expected Events

All elements of WFP have following property/properties.

- Width
- None of the given option
- **Height and Width**
- Height

_____ is used to declare a class into multiple files and all these files and all these files are combined into a single class file when the application is compiled.

- **Inheritance**
- Partial class
- Abstract Class
- Property Element

Which of the following is an example of tunnel key event?

- Key up
- Preview key entered
- **Preview key down**
- Key down

Is SharedSizeScope shares size information in _____grids(s)?

- **Multiple**
- Single
- Both 1 & 2
- Double

Wait for a single event is _____ operation.

- Waiting
- Waste
- Idle
- **Blocking**

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Which of the following components generates an input Event?

- Printer
- **Multi touch**
- Plotter
- Speaker

Arrays in C# are _____ objects.

- Logical
- Arithmetic
- **Reference**
- Value

A language supported by MS. Net platform.

- C
- C++
- Java
- **C#**

Extra space inside the element in WFP is known as _____.

- Margin
- Style
- Padding
- **Tunneling**

Which of the following statements is true about raw events?

- **It is introduced into an event processing system by an event procedure.**
- It is generated as a result of event processing.
- It is introduced into an event processing system by an event consumer.
- It is a subscription mechanism for events.

Dependency object .Clear Value method clears the _____ value in a dependency property.

- Default
- **Local**
- Setter
- Getter

All the members of interfaces are implicitly _____.

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- Private
- Static
- **Public**
- Abstract

Dock panal is used in _____ interface.

- **Top level**
- Bottom level
- Right level
- Left level

The data members of a class by default are?

- Protected ,Public
- Private ,Public
- **Public**
- Private

The main method returns zero value which indicates the _____.

- Source of programme
- Successful execution of a program
- **Termination of program**
- Exception in program

The _____, _____ and _____ of applications that use events, either directly or indirectly is called event-based programming.

- Model , Code , Operation
- Analyze , Design , Operation
- Code ,operation , Maintain
- **Design , Coding , Operation**

Silver light is a subset of _____.

- **NET**
- Python
- Window Forms
- C++

If we insert some object in a single cell in Grid panel, these objects are placed on _____.

- **One on top of the other**
- One after of the other

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- One bottom of the other
- One side of the other

A class can inherit from _____class/classes.

- Two
- **Single**
- Three
- Multiple

If we exclude the mechanism of calling the object through name then:

- Code will not compile
- Code will be difficult to debug
- Code will have an error
- Code will design dependent

Margin defines an extra space _____while considering edges of the element.

- Between
- Inside
- **Outside**
- Anywhere

Suppose we want to restrict the user to don not decrease the height of a WPF element from a specific amount then which property will be used?

- **Minheight**
- None of the given
- Height Minimum
- Minimum Height

Attribute parameter can be either positional parameters or _____ parameters.

- **Named**
- Static
- Global
- None of the given option

Suppose we have two elements in grid one is bottom and other is rectangle, which property will be helpful to set position of button on top of the rectangle?

- Z index
- **Sizing**
- Transform

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- Stack Panel

_____ Graphics work with shapes.

- Scalar
- Vector
- **Both 1 & 2**
- None of the given

UI element focusable value is _____ by default.

- **True**
- False
- Nil
- Zero

Tag of namespace in XML is just a_____.

- String
- Integer
- Float
- **Property**

In which type does the enumerators are stored by the compiler?

- String
- **Integer**
- Float
- None of the given

Visual Studio has a snippet called _____ that automatically expands into a definition of a dependency property, which makes defining one much faster than doing all the typing yourself.

- **Propdp**
- Property extension
- Dpprop
- Dependency property

Suppose we want to restrict the user to don't increase the height of a WPF element from a specific amount then which property will be used?

- MaximumHeight
- **MaxHeight**
- None of the given
- HeightMaximum

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Which of the following event may or may not relate to an actual occurrence?

- **Probabilistic Events**
- Actual Events
- Real Events
- Expected Events

Attribute parameter can be either positional parameters or _____ parameters.

- **Named**
- Static
- Global
- None of the given option

Which of the following keyword is used to monitor exception?

- Finally
- **Try**
- Throw
- Catch

Which keyword is used to declare the indexers?

- base
- **this**
- extract
- super

We want the width of button should cover the whole width of SnakPanel in WPF then what should be value of HorizantalAlignment property of button?

- parent
- whole
- fill
- **stretch #**

Which of these methods used to read strings from the console?

- Get ()
- Get line ()
- Read ()
- **Readline ()**

Which symbol is used to declare the proposer directives?

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- $
- *
- #
- ^

Which method is used to write single byte to the file?

- Write ()
- **Write Line()**
- Write Byte()
- Write Key()

Graphical user interfaces and the event-driven model are not applied on which of the following?

- Web Application
- **Console Application**
- Mobile Application
- Desktop Application

_____ Function will register a dependency property in WPF.

- Dependency Property .reg()
- Dependency Property. Registration ()
- Dependency Property. Reiterated()
- **Dependency Property. Register()**

Derived Event is:

- Introduced into an event processing system by an event consumer.
- **Generated as result of event that takes place inside an event processing system.**
- Introduced into an event processing system by an even producer.
- A set of associated events.

_____Key can be represented as text_ in XAML.

- Protected
- **Focus**
- Public
- None of the given

Silver light is a subset of _____.

- **NET**
- Python
- Window Forms

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- C++

Which type of sizing is used in WPF?

- Line
- Percentage
- **Proportional**
- Element

By convention _____ event names are prefixed with Preview.

- **Tunnelling**
- Bubbling
- Direct
- None of the given

We use "UIlement" property Clip to Bounds = "false" when _____.

- We want area child element don't cross the boundary of parent
- None of given
- **We want area of child element is allowed to cross the boundary of parent**
- Parent area also increase on increasing of area of child element

The _____ in the namespace indicate a hierarchy of nested namespaces.

- Arrows
- **Dots**
- Semicolon
- Brackets

If we want to place a component at top in "dock panel" the in is correct syntax?

- Panel Dock="Top"
- Dock = "Top"
- **Dock Panel .Dock = "Top"**
- Dock Panel = "Top"

If we insert some object in a single cell in Grid panel, thru these objects are placed on _____.

- **One on top of the other**
- One after of the other
- One bottom of the other
- One side of the other

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Which of the following is the object-oriented way to handle run time errors?

- On Error
- Error codes
- Interface
- **Exception**

The code given bellow is equivalent to _____.

<SolidColorBrush>White</SolidColorBrush>

- <WhiteBrush>MyWhiteBrush</WhiteBrush>
- None of the given option
- **<SolidColorBrush Color = "White"/>**
- <Brush Style = "Solid White">

_____ is an entity that receives events from the system.

- Event Producer
- **Event Consumer**
- Event Channel
- Event Generator

Wait for a single event is _____ operation.

- Waiting
- Waste
- Idle
- **Blocking**

_____ is used to declare a class into multiple files and all these files and all these files are combined into a single class file when the application is compiled.

- **Inheritance**
- Partial class
- Abstract Class
- Property Element

<Button. Content> tag in XAML is the syntax for _____.

- Namespace
- Property Element
- **Class**
- Form

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COOL stands for _____.

- Class Oriented Object Language
- Consumer Oriented Object Language
- **C-like Object Oriented Language**
- C++-like Oriented Language

Against _____ keyword a partial class is generated which we write in C# code.

- **X:class**
- Xmlns
- <button>
- X:Element

Suppose we have declared a button in XAML as shown in give n code. Choose the correct C# code which gets this button in C#.

<Button x: Name=”okbutton”>OK</Button>

- Button okbutton=(Button)window.(“okbutton”);
- Button okbutton=(Button)window.searchname(“okbutton”);
- Button okbutton=(Button)window.getbutton(“okbutton”);
- **Button okbutton=(Button)window.findname(“okbutton”);**

Additional XML namespaces (on the root or on children) must be _____to be used on any identifiers from that namespace.

- Given a common prefix
- Qualified with “Pre” keyword
- **Given a distinct prefix**
- Qualified with “Ext” keyword

Arrays in C# are _____ objects.

- Logical
- Arithmetic
- **Reference**
- Value

When your C++ program is compiled, it creates a file called _____

- Assembly
- Exe
- Object
- Class

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_____ Property value is used to set width and height of element in WPF according to its content.

- Name
- **X:class**
- Nan
- Tree

What is the return type of destruction?

- Int
- Float
- Void
- No return types

In context of XAML child element rules, if the parent supports a a#content property and the type of the child is compatible with that property, treat the child as its _____

- Super class
- Sub class
- **Value**
- Interface

Event –based programming is also known as:

- Service –oriented architecture(SOA)
- **Event – driven Architecture(EDA)**
- Service –driven Architecture(SDA)
- Event –based Architecture(EBA)

When a user select element from list box, after selecting compiler marked the selected element as

- Fix
- Selected
- Handled
- **Bubble**

Design of silver light mostly follows the design of _____

- C++
- **WPF**
- Console
- C#

Which of the following is/are not derived from "Control "class?

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- ListBox
- Status Bar
- **None of the given**
- Button

To design a complex interface, which of the following technique is best to achieve the required results?

- Panel composed in three panels
- Panel composed within panels
- Panel composed in two panels
- Panel composed in single panels

In tunnelling, event move from _____.

- Left to Right
- Right to Left
- **Top to Down**
- Diagonally

Attribute of a program can be retrieved from its assembly metadata through _____.

- Static member
- **Reflection**
- Garbage collector
- Constructor

Clean room design is also called _____.

- Clear Room design
- Wall Technique Design
- Chinese Wall Technique
- French Wall Technique

The code given bellow is equivalent to _____.

<SolidColorBrush>White</SolidColorBrush>

- <WhiteBrush>MyWhiteBrush</WhiteBrush>
- None of the given option
- **<SolidColorBrush Color = "White"/>**
- <Brush Style = "Solid White">

If we want to scale an object in the 2-D x-y coordinate system then which transform class will be used?

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- Rotate transform
- Matrix transform
- **Translate transform**
- Skew transform

Clean room design is useful as a defence against _____.

- Copyright
- Trade secret infringement
- **Copyright and trade secret infringement**
- None of the given

_____ Function will register a dependency property in WPF.

- Dependency Property .reg()
- Dependency Property. Registration ()
- Dependency Property. Reiterated()
- **Dependency Property. Register()**

The root object in XAML must specify a least _____XML namespace(s) that is/are used to qualify it and any child elements.

- Two
- None of the given
- Four
- **One**

Anonymous methods _____ the code size.

- Double
- Expand
- **Reduce**
- Expand

_____ works with Graphics Cards.

- OpenGL
- **GDI**
- DirectX
- XNA

System .Window .Control.Botton b = new System .Window .Controls. Button ();

b.Content = “OK”;

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For the above code (written in C#) the corresponding XAML code will be:

- **<Button Xmlns = http://schemes.microsoft.com/winfx/2006/xaml/presentation Content = "OK"/>**
- <Button Xmlns ="OK" Content = " http://schemes.microsoft.com/winfx/2006/xaml/presentation"/>
- <Button Xmlns = http://schemes.microsoft.com/winfx/2006/xaml/presentation Event = "OK"/>
- <Button Xmlns =" http://schemes.microsoft.com/winfx/2006/xaml/presentation" Call = "OK"/>

Which of these methods used to read strings from the console?

- Get ()
- Get line ()
- Read ()
- **Readline ()**

<Button. Content> tag in XAML is the syntax for _____.

- Namespace
- Property Element
- **Class**
- Form

We use "UlElement" property ClipToBounds = "false" when _____.

- We want area child element don't cross the boundary of parent
- None of given
- **We want area of child element is allowed to cross the boundary of parent**
- Parent area also increase on increasing of area of child element

To check that the user either pressed left Alt key or right Alt key, the _____ is used.

- isKeyDown
- is Down
- **keyboardDevice.IsKeyDown**
- KeyStates.ISKeyDown

Wait for a single event is _____ operation.

- Waiting
- Waste
- Idle
- **Blocking**

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UI element focusable value is _____ by default.

- **True**
- False
- Nil
- Zero

XAML specification defines rules that map.”NET” namespaces, types, properties, and events into _____.

I) XAML namespace

II) XML elements

III) XML Attributes

- (I) Only
- (II) Only
- **(I),(II)and (III) only**
- (III) Only

Which provider has the highest priority while providing a value to a dependency property?

- Style setters
- **Default Value**
- Local Value
- Theme Style Setters
- **Tunneling**

Which of the following statements is true about raw events?

- **It is introduced into an event processing system by an event procedure.**
- It is generated as a result of event processing.
- It is introduced into an event processing system by an event consumer.
- It is a subscription mechanism for events.

WPF Margin is defines by _____type.

- Thin
- Thinnest
- Thick
- **Thickness**

Which property is used to change the Z Order of elements in WPF?

- Z-Coordinate

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- ZPanal
- **ZIndex**
- ZValue

If we run the program from within Visual Studio, the console window disappears quickly so we use ____ to keep screen from going away.

- **Console .Readline ()**
- Console. Write Line()
- Console .Title
- Console .Clear

When parent element space is not enough for the content of child element, this type of situation arises due to _____.

- Short memory
- Content Overflow
- Indexing
- Grid

_____ is a base class similar to UlElement but for document-related pieces of content that don't have rendering behaviour on their own

- **Content Element**
- Visual Element
- System Element
- XAMLElement

Which of these of the class String is used to obtain length of string object?

- Get()
- Sizeof()
- **Length()**
- Length of()

Overloaded operators must be _____.

- Referenced
- **Static**
- Objective
- Delegates

Which symbol is used to declare the proposer directives?

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- $
- *
- #
- ^

Suppose we want to restrict the user to don not decrease the height of a WPF element from a specific amount then which property will be used?

- **Minheight**
- None of the given
- Height Minimum
- Minimum Height

A (n) _____ is a reference to a method.

- **Delegate**
- XAML
- Function
- Constructor

Default value of Grid. Row is _____

- **0**
- 1
- 2
- 3

In XAML 2006, x: AnsyncRecords works on _____thread(s).

- Single
- Multiple
- **Both 1 & 2**
- None of given

How can you specify null property in mark-up extension?

- (null)
- :null:
- (x:null)
- [x:null]

Dependency object .Clear Value method clears the _____ value in a dependency property.

- Default

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- **Local**
- Setter
- Getter

---

1. Attributes of a program can be retrieved from its assembly metadata through .
   Reflection
2. Attached event will be raised by the element while tunneling.
   Both 1 & 2
3. What is the key feature of Java?
   Both (a) and (b)
4. The , and of applications that use events, either directly or indirectly is called event-based programming.
   Design, Coding, Operation
5. All the members of interfaces are implicitly .
   Public
6. In WPF, if there is any name property then that property is also be taken as a substitute of .
   X: static
7. In tunneling, event move from .
   Top to Down
8. Suppose we want to restrict the user to don't decreases the width of a WPF element from a specific amount then which property will be used?
   Min Width
9. Overloaded operators must be .
   Static
10. Which symbols is used to declare the preprocessor directives?
    #
11. Suppose we have two elements in grid one is button and other is rectangle, which property will be helpful to set position of button on top of the rectangle?
    Sizing
12. To use kbhit() command, you need to include which header file in your program?
    Conio.h
13. Singleton method has instance(s).
    1
14. If we want to scale an object in 2-D x-y coordinate system then which transform class will be used?

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Translate Transform

15. Superset of canvas is .
    Grid
16. In event-driven architecture (EDA), sends event to .
    Event Producer, Event Consumer
17. The method is used to read the whole file at once.
    ReadAllText("text.txt")
18. In XAML 2006, X:Classmodifier by default access level is .
    Public
19. XML is stricter version of .
    HTML
20. XAML is good to concisely represent .
    User interface
21. allow attaching a handler for an event that is defined in an element other than the one adding the handler.
    Tunneling
22. The code given below is equivalent to .
    <SolidColorBrush>White</SolidColorBrush>
    <SolidColorBrush Color="White"/>
23. Wait for single event is operation.
    Blocking
24. How can you specify null property in markup extension?
    {x:null}
25. Button, listbox and status bar are derived from Control class in WPF.
    Base
26. All elements of WPF have following property/properties.
    Height and Width
27. Which type of sizing is used in Wpf?
    Proportional
28. In WPF, if we try to add listbox item in listbox object which is already null, it will give us
    Compile time error
29. Which of these methods used to read strings from the console?
    Read line()
30. Which of following statement is correct about WPF?
    WPF is resolution independence
31. WPF4 was released in .
    2010
32. Which of the following events may or may not relate to an actual occurrence?
    Probabilistic Events

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33. Which of following is an example of tunnel key event?
    Preview key down
34. IsSharedSizeScope shares size information in grid.
    Multiple
35. Which of the following components generates an input Event?
    Multi Touch
36. Arrays in C# are objects.
    Reference
37. A language supported by MS.Net platform.
    C#
38. Extra space inside the element in WPF is known as .
    Padding
39. DependencyObject.ClearValue method clears the value in a dependency property.
    Local
40. All the members of interfaces are implicitly .
    Public
41. Dock panel is used in interface.
    Top level
42. The data members of a class by default are?
    Public
43. The Main method returns zero value which indicates the .
    Termination of program
44. Silverlight is the subset of .
    .NET
45. If we insert some objects in a single cell in Grid panel, then these objects are placed on
    One on Top of the other
46. A class can inherit from class/classes.
    Single
47. If we exclude the mechanism of calling the objects through name then.
    Code will not compile
48. Margin defines an extra space while considering edges of the element.
    Outside
49. Attributes parameters can be either positional parameters or parameters.
    Named
50. UI elements focusable value is by default.
    True
51. Tag of namespace in XML is just a .
    Property

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52. In which type does the enumerators are stored by the compiler?
    Integer
53. Which of following events may or may not relate to an actual occurrence?
    Probabilistic Events
54. Which of the following keyword is used to monitor exceptions?
    Try
55. Which keywords is used to declare the indexers?
    This
56. Which method is used to write single byte to the file?
    WriteLine()
57. Derived Events is:
    Generated as a result of event that takes place inside an event processing system.
58. key can be represented as _text in XAML.
    Focus
59. Which type of sizing is used in Wpf?
    Proportional
60. We use "UIElement" property ClipToBounds= "false" when .
    We want area of child element is allowed to cross the boundary of parent
61. The in the namespace indicate a hierarchy of nested namespace.
    Dots
62. If we want to place a components at top in "dockpanel" then which one is correct syntax?
    DockPanel.Dock="Top"
63. Which of the following is the object-oriented way to handle run time errors?
    Exceptions
64. is an entity that receives events from the system.
    Events Consumer
65. <Button.Content> tag in XAML is the syntax for .
    Class
66. COOL stands for .
    C-like Object Oriented Language
67. Against keyword a partial class is generated which we written in C# code.
    X:class
68. When your C# program is compiled, it creates a file called .
    Assembly
69. Event –based programming is also known as:
    Event –driven Architecture (EDA)
70. Design of Silverlight mostly follow the design of .
    WPF

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71. Which of following is not derived from “Control” class?
    None of the given
72. To design a complex Interface, which of the following technique is best to achieve the required results?
    Panels composed within panels
73. In tunneling, event move from .
    Top to Down
74. Clean room design is useful as defense against .
    Copyright and trade secret infringement
75. Anonymous methods the code size.
    Reduce
76. works with Graphic Cards.
    GDI
77. To check that the user either pressed left Alt key or right Alt key, the is used.
    keyboardDevice.IskeyDown
78. Wait for a single event is operation.
    Blocking
79. WPF Margin is defined by type.
    Thickness
80. Which property is used to change the Z Order of elements in WPF?
    ZIndex
81. Which of these methods of the class String is used to obtain length of string object?
    Length()
82. Overloaded operators must be .
    Static
83. A is a reference to a method.
    Delegate
84. Default value of Grid.Row is .
    0
85. Which of the following is an example of tunnel key event?
    Preview key down
86. Which of the following components generates an input event?
    Multi Touch
87. Which method is used to write single byte to the file?
    WriteLine()
88. Additional XML namespaces must be to be used on any identifiers from that namespace.
    Given a distinct prefix
89. System. Windows. Controls. Button b = new System. Windows. Controls. Buttons();

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b. Content = “OK”;

for the above code (written in C#) the corresponding XAML code will be:

<Button xmlns=https://schemas. Microsoft. Com/winfx/2006/xaml/presentation content=”OK”/>

90. XAML specification defines rules that map”.NET” namespaces, types, properties and events into .

1) XML namespaces
2) XML elements
3) XML Attributes

(1), (2) and (3) only

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CS-411

Quiz 1

by Attiq Kundi

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CS-411

Up Dated on 24-May-2023

1. keyword is used to declare a base class method while performing overriding of base class method.
   A. Overload
   B. Virtual
   C. Extend
   D. Override
2. Choose the correct option among the following indexers which correctly allows to index in same way as an array?
   A. A Class
   B. A function
   C. An interface
   D. A property
3. A class can inherit from_____class/classes in C#.
   A. Three
   B. Single
   C. Two
   D. Multiple
4. Which one of the following is not the feature of C#?
   A. Multiple inheritance
   B. Interface supported
   C. Locals can not shadow global variables
   D. Operator Overloading is allowed
5. Which keyword is used to declare the indexers?
   A. super
   B. base
   C. this
   D. extract
6. The common type system of C# has value types and___types.
   A. Class
   B. Object

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C. Reference

D. Enum

7. Select the correct statement about the structure in C#?

A. Its syntax is similar to class

B. It can implement destructors

C. It is a reference type

D. It can implement inheritance

8. Reaction of events are either_____or_____.

A. Asynchronous, Blocking

B. Synchronous, Blocking

C. Blocking, Synchronous

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D. Synchronous, Asynchronous

9. is an example of request and response based application
   A. Web browser
   B. E –Commerce Site
   C. Pressing a button
   D. Swiping of credit card
10. Which of the following is built from request-response?
   A. Event-based Architecture (EBA)
   B. Service-driven Architecture (SDA)
   C. Service-oriented Architecture (SOA)
   D. Event-processing Architecture (EDA)
11. _____ and delegates work hand in hand. (Page-38)
   A. Static member
   B. Event
   C. Post-processor
   D. Pre-processor
12. In C#, when a value type is converted to object type, it is called__.
   A. Polymorphism
   B. Abstraction
   C. Boxing
   D. Data Conversion
13. Which keyword is used for correct implementation of an interface in C#?
   A. Intf
   B. Interface
   C. Int Interface
   D. Interfaces
14. The main method return a non-zero value which indicates the
   A. Exception in program
   B. Termination of program
   C. Error in program
   D. Success of program
15. is an occurrence within a particular system or domain.
   A. Program
   B. Class
   C. Event
   D. Object
16. Select wrong statement about destructor in C#?
   A. Destructor can have modifiers or parameter
   B. Destructor cannot be inherited or overloaded
   C. Destructor work opposite to constructor
   D. Class can have one destructor only
17. The main method returns zero value which indicates the
   A. Successful execution of program
   B. Source of program

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C. Termination of program
D. Exception in program

18. In C#, which of the following keyword is used for including the namespaces in the program?
A. using
B. extend
C. import
D. #

19. When your C# program is compiled, is creates a file called
A. Exe
B. Class
C. Object
D. Assembly

20. Which of the following is an insertion operator in C++?
A. <<
B. >
C. - -
D. &

21. C# is a class member that is activated whenever the event . it was designed for. Occurs
A. Attributes
B. Arguments
C. Method
D. Event

22. A fraud detection event on banking transaction is an example of___.
A. Linear event
B. Sequential event
C. Predictive event
D. Probabilistic event

23. In C++, which of the following library is include for 'cin' and 'cout' function?
A. #include<fstream.h>
B. #include<fstream>
C. #include<conio.h>
D. #include<iostream>

24. Which of the following doesn't have Graphical User Interface and the Event-Driven Model?
A. Console-Application
B. Web Application
C. Mobile Application
D. Desktop Application

25. A stream in which the events may be of different types is referred to as a **heterogeneous** event stream. (Page-10)
A. Continuous
B. Homogenous
C. Heterogeneous
D. Mutual

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26. Defining a method in subclass having same name and type signature as a method in its super class is known as
    A. Method hiding
    B. Method Overloading
    C. Method Binding
    D. Method Overriding
27. A single try block must be followed by which of these?
    A. Finally
    B. Catch
    C. Both Finally and catch
    D. Throw
28. By default the data member of a class are
    A. Protected, public
    B. Public
    C. Private
    D. Private, Public
29. Exception are derived from_____class. (Page-39)
    A. Windows error
    B. Exception error
    C. System exception
    D. Error Exception
30. Which of the following does NOT include in event processing operations?
    A. Terminating Event
    B. Deleting Event
    C. Transforming Event
    D. Reading Event
31. What is the return type of destruction?
    A. Float
    B. No return type
    C. Void
    D. Int
32. At the time an event is fired, the_____method (with that event) will be invoked.
    A. Public
    B. Protected
    C. Registered
    D. Private
33. C# is a ___and .NET is a__.
    A. Platform, Language
    B. Package, Language
    C. Language, Package
    D. Language, Platform
34. Which of the following is an entity that introduces event into the system?
    A. Event producer
    B. Event Stream

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C. Event Channel
D. Event Consumer

35. _____ is an entity that receive event from the system?
A. Event producer
B. Event Consumer
C. Raw event
D. Event stream

36. Which of the following is the object-oriented way to handle run time errors?
A. OnError () Method
B. Exception
C. Interface
D. Error Code

37. Why do we use cin.get() in C++ program?
A. It is used to print all values of array
B. It is used to get numeric data
C. It is used to access character array
D. It is used to take input from user

38. Types not defined in any namespace are said to reside in the _ _namespace.
A. Static
B. Virtual
C. Global
D. Local

Reason: Types not defined in any namespace are said to reside in the global namespace. The global namespace also includes top-level namespaces

39. Arrays in C# are_____object.
A. Logical
B. Arithmetic
C. Reference
D. Value

40. The initial name for C# was____.
A. COAL
B. COOP
C. COAP
D. COOL

41. Which of the following is declared like a class but have no implementation?
A. Encapsulation
B. Structure
C. Interface
D. Inheritance

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42. Which of the following keyword is used to change data and behavior of a base class by replacing a member of the base class with the new derived member.
    A. Base
    B. Overrides
    C. Overloads
    D. New
43. A stream in which all the events must be of same types is called **Homogenous** event stream. (Page-10)
    A. Continuous
    B. Homogenous
    C. Heterogeneous
    D. Mutual
44. Select the correct statement about an exception>
    A. It occurs during Just-in-time compilation
    B. It occurs at runtime
    C. It occurs during loading of program
    D. It occurs during data binding
45. _____ is not a defense against patents and clean room design?
    A. Independent Invention
    B. Event Invention
    C. Dependent Invention
    D. Clear room design
46. COOL stands for____.
    A. Class Oriented Object Language
    B. Consumer Oriented Object Language
    C. C-like Object Oriented Language
    D. C++-like Oriented Language
47. operator is used to signifies the destructor in C#?
    A. #
    B. =
    C. &
    D. ~
48. What is the return type of destruction?
    A. Int
    B. Float
    C. Void
    D. No return types
49. An Event that is generated as a result of event processing that take place inside an event processing system is called . (Page-10)

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A. Event Stream
B. Derived Event
C. Raw Event
D. Event Consumer

50. Choose the incorrect statement about delegates in C#?
A. Delegates are value type
B. Delegates are like function pointers in other language
C. Delegates are object oriented
D. Delegates are type safe

51. is the method of copying a design by reverse engineering and then recreating it without infringing any of the copyrights and trade secrets associated with the original design. (Page-12)
A. Clear Room Design
B. Event Room design
C. Clean Room Design
D. Wall Room Design

52. Which of the following statement is TRUE about raw event?
A. It is a subscription mechanism for event
B. It is introduced into an event processing system by an event consumer
C. It is generate as a result of event processing
D. It is introduced into an event processing system by an event producer

53. Which symbol is used to declare the preprocessor directives?
A. *
B. #
C. $
D. ^

54. Which of these keyword is used to manually throw an exception in C#?
A. Try
B. Finally
C. Catch
D. Throw

55. On successful compilation of a java source code _____ is generated.
A. Output
B. Bytecode
C. Hexadecimal code
D. Octal code

56. _____ is the necessary condition to implement delegates.
A. Class Declaration
B. Function overloading
C. Inheritance
D. Runtime Polymorphism

57. Select correct statement about properties of read and write in C#
A. A read only property will only have get accessor
B. A write only property will only have get accessor
C. A property cannot be either read only or write only
D. A property can simultaneously be read or write only

By Atiq Kundi

58. What will be the output of the following C# code? Static void Main{string[] args}
```
{
  int i;    for (i = 0; ;)
  {
    Console.WriteLine(&amp;quot;Virtual University&amp;quot;);
  }
 Console.ReadLine();
}
```

A. || will print "Virtual University" in continuous loop
**B. Syntax error **
C. Virtual University
D. =p= No output =/p=

59. Choose the wrong statement about structures in C#.Net.
A. Structures can have destructors
B. Structures can implement an interface but they cannot inherit from another structure
C. Structure members cannot be declared as protected
D. A structure cannot be empty

60. Which of the following keyword is used to monitor exceptions?
A. Catch
B. Finally
C. Try
D. Throw

61. A method performs an action in a series of statements, called a _____.
A. Code Block
B. Method Block
C. Routine Block
D. Statement Block

62. If we run the program from within Visual Studio, the console window disappears quickly so we can use _____ to keep screen from going away.
A. Console.ReadLine()
B. Console.WriteLine()
C. Console.Title
D. Console.Clear

63. _____ events may or may not relate to an actual occurrence.
A. Observed Events
B. Deducted Events
C. Probabilistic Events
D. None of the above

64. Clean room design is useful as a defense against _____.
A. Copyright
B. Trade secret infringement
C. Copyright and trade secret infringement
D. None of the given options

65. Which of these methods of the class String is used to obtain length of string object.
A. Lengthof()
B. Sizeof()

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C. Get()

D. Length()

66. If math class had added property with get and set accessors. Then which of the following statements will work correctly?

A. math.add = 40;

B. math m = new math();
m.add = 50;

C. Console.WriteLine(math.add);

D. System.out.Println("math add")

67. _____ language is supported by Microsoft.Net platform.

A. C#

B. C

C. Php

D. Java

68. The _____ in the namespace indicate a hierarchy of nested namespaces.

A. Semicolon

B. Brackets

C. Arrows

D. Dots

69. which keyword is used to define the macros in c++?

A. macro

B. #define

C. #macro

D. Define

70. In C#, when an object type is converted to a value type, it is called _____

A. Unboxing

B. Data Conversion

C. Abstraction

D. Polymorphism

71. The _____ pointer in methods refers to the object on which the method is called.

A. Int

B. String

C. Out

D. This

72. In C#, all binary operators are _____

A. Right-associative

B. Top-associative

C. Center-associative

D. Left-associative

73. Clean room design is also called_____.

A. Clear Room Design

B. Wall Technique Design

C. Chinese Wall Technique

D. French Wall Technique

74. To perform I/O operations, we must use _____ header file.

A. <conio>

B. <fstream>

C. <ifstream>

D. <ofstream>

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75. Which reference modifier is used to define reference variable?
    A. #
    B. def
    C. &
    D. ref
76. Waiting for a single event is _____operation.
    A. Unavoidable
    B. Waste
    C. Idle
    D. Blocking
77. In System.Console ,"Console" is a _____ which enables us to do console Input and Output.
    A. Namespace
    B. Structure
    C. Class
    D. Library
78. In Event-Driven Architecture (EDA). _____ sends event to _____
    A. Event Consumer, Event Producer
    B. Event Producer, Event Consumer
    C. Event Handler, Event Producer
    D. Event Generator, Event Producer
79. In C++ _____ function is used to check if this stream has raised any error or not
    A. eof()
    B. Peek()
    C. Fstream
    D. Good()
80. In C++ , _____ function is used to return the next character in the input sequence, without extracting it.
    A. Putback()
    B. Peek()
    C. Fstream
    D. Good()
81. which of these keywords is not a part of exception handling?
    A. Thrown
    B. Finally
    C. Try
    D. Catch

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CHAPTER 1 to 8

UpDated on 07 June 2022

3. _____ keyword is used to declare a base class method while performing overriding of base class method.
   A. Overload
   Virtual
   B. Extend
   C. Override
4. Choose the correct option among the following indexers which correctly allows to index in same way as an array? (Page-29)
   A Class
   A. A function
   B. An interface
   C. A property
5. A class can inherit from _____ class/classes in C#. (Page-22)
   A. Three
   Single
   B. Two
   C. Multiple
6. Which one of the following is not the feature of C#? (Page-12)
   Multiple inheritance
   A. Interface supported
   B. Locals can not shadow global variables
   C. Operator Overloading is allowed
7. Which keyword is used to declare the indexers? (Page-29)
   A. super
   B. base
   this
   C. extract
8. The common type system of C# has value types and _____ types. (Page-13)
   A. Class
   B. Object
   Reference
   C. Enum
9. Select the correct statement about the structure in C#? (Page-13)
   Its syntax is similar to class
   A. It can implement destructors
   B. It is a reference type
   C. It can implement inheritance
10. Reaction of events are either _____ or _____ (Page-07)
    A. Asynchronous, Blocking
    B. Synchronous, Blocking
    C. Blocking, Synchronous

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Synchronous, Asynchronous

6. _____ is an example of request and response based application (Page-09)

   Web browser

   A. E –Commerce Site
   B. Pressing a button
   C. Swiping of credit card

7. Which of the following is built from request-response? (Page-09)

   A. Event-based Architecture (EBA)
   B. Service-driven Architecture (SDA)

   Service-oriented Architecture (SOA)

   C. Event-processing Architecture (EDA)

8. _____ and delegates work hand in hand. (Page-38)

   A. Static member

   Event

   B. Post-processor
   C. Pre-processor

9. In C#, when a value type is converted to object type, it is called _____.

   A. Polymorphism
   B. Abstraction

   Boxing

   C. Data Conversion

10. Which keyword is used for correct implementation of an interface in C#?

    A. Intf

    Interface

    B. Int Interface
    C. Interfaces

11. The main method return a non-zero value which indicates the _____

    A. Exception in program
    B. Termination of program

    Error in program

    C. Success of program

12. _____ is an occurrence within a particular system or domain. (Page-06)

    A. Program
    B. Class

    Event

    C. Object

13. Select wrong statement about destructor in C#?

    Destructor can have modifiers or parameter

    A. Destructor cannot be inherited or overloaded
    B. Destructor work opposite to constructor
    C. Class can have one destructor only

14. The main method returns zero value which indicates the _____

    Successful execution of program

    A. Source of program

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E. Termination of program
F. Exception in program

8. In C#, which of the following keyword is used for including the namespaces in the program?
using
A. extend
B. import
C. #

9. When your C# program is compiled, is creates a file called _____
Exe
A. Class
B. Object
C. Assembly

10. Which of the following is an insertion operator in C++? (Google)
<<
A. >
B. - -
C. &

11. C# ___ is a class member that is activated whenever the event . it was designed for. Occurs (Page-37)
A. Attributes
B. Arguments
C. Method
Event

12. A fraud detection event on banking transaction is an example of _____. (Page-06)
A. Linear event
B. Sequential event
C. Predictive event
Probabilistic event

13. In C++, which of the following library is include for 'cin' and 'cout' function?
A. #include<fstream.h>
B. #include<fstream>
C. #include<conio.h>
#include<iostream>

14. Which of the following doesn't have Graphical User Interface and the Event-Driven Model?
Console-Application
A. Web Application
B. Mobile Application
C. Desktop Application

15. A stream in which the events may be of different types is referred to as a **heterogeneous** event stream. (Page-10)
A. Continuous
B. Homogenous
Heterogeneous
C. Mutual

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D. Defining a method in subclass having same name and type signature as a method in its super class is known as _____

Method hiding

Method Overloading

Method Binding

Method Overriding

E. A single try block must be followed by which of these?

Finally

Catch

Finally and catch

Throw

F. By default the data member of a class are _____

Protected, public

Public

Private

Private, Public

G. Exception are derived from_____ class. (Page-39)

Windows error

Exception error

System exception

Error Exception

H. Which of the following does NOT include in event processing operations? (Page-07)

Terminating Event

Deleting Event

Transforming Event

Reading Event

I. What is the return type of destruction?

Float

No return type

Void

Int

J. At the time an event is fired, the _____ method (with that event) will be invoked. (Page-38)

Public

Protected

Registered

Private

K. C# is a ___ and .NET is a ___. (Page-12)

Platform, Language

Package, Language

Language, Package

Language, Platform

L. Which of the following is an entity that introduces event into the system? (Page-10)

Event producer

Event Stream

by Atiq Kundi

E. Event Channel
F. Event Consumer

12. Which of the following is an entity that receive event from the system? (Page-10)
    A. Event producer

    Event Consumer

    B. Raw event
    C. Event stream
13. Which of the following is the object-oriented way to handle run time errors?
    A. OnError () Method

    Exception

    B. Interface
    C. Error Code
14. Why do we use cin get() in C++ program?
    A. It is used to print all values of array
    B. It is used to get numeric data
    C. It is used to access character array
    D. It is used to take input from user
15. Types not defined in any namespace are said to reside in the ___ namespace.

    Local

    Static

    Virtual

16. Select the correct statement about an exception. (Google)

2 It occurs during loading of program

3 It occurs during Just-In-Time compilation

It occurs at run time

4 All of the mentioned

2 Arrays in C# are _____ object.
    13. Logical
    14. Arithmetic

    Reference

    15. Value

3 The initial name for C# was _____. (Page-12)
    13. COAL
    14. COOP
    15. COAP

    COOL

4 Which of the following is declared like a class but have no implementation?(Page-34)
    13. Encapsulation
    14. Structure

    Interface

    15. Inheritance

by Atiq Kundi

17. Which of the following keyword is used to change data and behavior of a base class by replacing a member of the base class with the new derived member. (Google)
    16. Base
    17. Overrides
    18. Overloads

    New
18. A stream in which all the events must be of same types is called **Homogenous** event stream. (Page-10)
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20. Select the correct statement about the structure in C#?
    16. It can implement inheritance

    Its syntax similar to class
    17. It is a reference type
    18. It can implement destructors
21. ____ is not a defense against patents and clean room design?(Page-12)

    Independent Invention
    16. Event Invention
    17. Dependent Invention
    18. Clear room design
22. COOL stands for _____.
    16. Class Oriented Object Language
    17. Consumer Oriented Object Language

    C-like Object Oriented Language (Page-12)
    18. C++-like Oriented Language
23. _____ operator is used to signifies the destructor in C#?
    16. #
    17. =
    18. &

    ~
24. What is the return type of destruction?
    16. Int
    17. Float
    18. Void

    No return types
25. An Event that is generated as a result of event processing that take place inside an event processing system is called _____. (Page-10)

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E. Event Stream
Derived Event
F. Raw Event
G. Event Consumer

21. Choose the incorrect statement about delegates in C#?
2 Delegates are value type
3 Delegates are like function pointers in other language
4 Delegates are object oriented
Delegates are type safe

E. _____ is the method of copying a design by reverse engineering and then recreating it without infringing any of the copyrights and trade secrets associated with the original design. (Page-12)
Clear Room Design
Event Room design
Clean Room Design
Wall Room Design

F. Which of the following statement is TRUE about raw event? (Page-10)
It is a subscription mechanism for event
It is introduced into an event processing system by an event consumer
It is generate as a result of event processing
It is introduced into an event processing system by an event producer

G. Which symbol is used to declare the preprocessor directives?
*
#
$
^

H. Which of these keyword is used to manually throw an exception in C#?
Try
Finally
Catch
Throw

CS411 Quiz-1 BY Attiq Kundi

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by Attiq Kundi

Visual Programming

CS411



Virtual University of Pakistan

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Chapter 1

Lecture 1

This course is about graphical user interfaces and the event-driven model as applied to desktop, web, and mobile applications. It uses a ground up approach from what you already know.

The pre-requisites are C++ programming and data structures. I will introduce any new languages and concepts we are going to use.

This is a hands-on course. We will use in-class examples and you should focus on completing programming assignments to understand the material.

We will primarily use 2 books in this course. "Event processing in action" and "Windows resentation foundationunleashed". The first one is authored by Opher Etzion and Peter Niblett and is one of the very fewbooks on event processing. Event processing is often taught as a side-concept, but we will focus on it in this course. The other book is by Adam Nathan. It is about Windows presentation foundation or WPF in short. We use it as our gui example with C#. We will introduce C# concepts and WPF concepts. They are not a pre-requisite for this course.

Later in the course, we will touch event-driven programming in the browser, in particular made possible by techniques called AJAX. Again not a pre-requisite. I will introduce all the concepts in class. Towards the very end, we will discuss event driven programming on mobiles. Again no pre-requisite. I will introduce all the tools involved.

So what is Visual Programming and what you will learn in this course. Its primarily event driven concepts and application of these concepts on GUI programs on desktop, web, and mobile.

Here is our first example:

```
Using namespace std;

#include <iostream>

Int main()
{
  Char a;
  Do {
    A = cin.get();
    Cout << a;
  } while (a!='x');
  Return 0;
}
```

Let's compile and run using the commands

G++ -o example1 example1.cc ./example1

Our second example

```
#include <iostream>
#include <fstream>
```

```
Int main()
{
  Char a = '-';
  Do {
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good() && a != f.peek()) {
      A = f.get();
      Std::cout << a << std::endl;
    }
    F.close();
  } while (a!='x');
  Return 0;
}
```

Let's compile and run it. There are a few issues like busy wait etc. But we'll leave them on a side for now.

So why we saw these two very basic examples. Well, they were both very easy to write. With an OR.

```
#include <iostream>
#include <fstream>

Int main()
{
  Char a = '-', b;
  Do {
    If (iskeyboardpressed()) {
      B = std::cin.get();
      Std::cout << b << std::endl;
    }
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good() && a != f.peek()) {
      A = f.get();
      Std::cout << a << std::endl;
    }
    F.close();
  } while (a!='x' && b!='x');
  Return 0;
}
```

I.e. We react to either of the two actions. Here is how we can do it.

```
#include  <iostream>
#include<fstream>
 Int main()
 {
 Char a  =’-’, b;
 Do{
 If(iskeyboardpressed())  {
 B = std::cin.get();
 Std::cout  << b  << std::endl;
 }
 Std::ifstream f("test.txt", std::ifstream::in);
 If(f.good() && a  != f.peek())  {
 A = f.get();
 Std::cout  << a  << std::endl;
 }
```

```
F.close();
} while  (a!=’x’ && b!=’x’);
Return  0;
}
```

Or we could re-factor it into a cleaner version.

```
#include <iostream>
#include <fstream>

Int onkeypress()
{
  Char b = std::cin.get();
  Std::cout << b << std::endl;
  Return b == 'x';
}

Int onfilechanged()
{
  Char a = f.get();
  Std::cout << a << std::endl;
  Return a == 'x';
}

Int main()
{
  Char a = '-', b;
  Do {
    If (iskeyboardpressed()) {
      If (onkeypress())
Return 0;
    }
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good() && a != f.peek()) {
      If (onfilechanged())
Return 0;
    }
    F.close();
  } while (true);
  Return 0;
}
```

So what are the pros and cons of the re-factored approach. We do one thing at a time (more focused approach). Blocking is a huge issue. Nothing is parallel. The key idea is that once you understand the approach, it scales really well to a large number of things you want to respond to.

Next time we’ll define events, event loops, and different terms in the event driven model.

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Chapter 2

Lecture 2

Last time, we saw some examples and demonstrated the idea of event driven design. We can re-factor the last example even one step further like this:

```
#include <iostream>
#include <fstream>

Int onkeypress()
{
  Char b = std::cin.get();
  Std::cout << b << std::endl;
  Return b == 'x';
}

Int onfilechanged()
{
  Char a = f.get();
  Std::cout << a << std::endl;
  Return a == 'x';
}

Int main()
{
  Char a = '-', b;
  Do {
    If (iskeyboardpressed()) {
      If (onkeypress())
 Return 0;
    }
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good() && a != f.peek()) {
      If (onfilechanged())
 Return 0;
    }
    F.close();
  } while (true);
  Return 0;
}
```

So what is an event? It's an occurrence within a particular system or domain. There are two meanings: something that happened and the corresponding detection in computer world. An event captures "some" things from the actual occurrence and multiple events may capture "one" occurrence.

Probabilistic events may or may not relate to an actual occurrence e.g. A fraud detection event on a banking transaction. Every event is represented by an event-object. There are various types of events and information in the event describes details of the particular event type. E.g. Key press, file event etc.

Let's take a coffee shop analogy. Events are taking place all the time. Let's understand synchronous and asynchronous behavior with this example. So if an order is completed, coffee ready, pastery heated before the next order that's synchronous behavior. There are observed events and deduced events. Situation is

an event ocurrence that requires a reaction.

Let's take example of events in computing systems. Interrupts and exceptions on a computer e.g. Divide by zero. Patient monitored by sensors. Car sensors alerting to oil or pressure situations. Banking alerts. Road tolling. Etc. Event processing is computing that performs operations on events. Common event processing operations include reading, creating, transforming, and deleting events. The design, coding and operation of applications that use events, either directly or indirectly is called event-based programming or applications based on event-driven architecture. So what if we don't use event-driven programming. We will poll for events. Even if you make your application non-event-driven, you still wait for events. Wait for a single event is blocking operation.

Synchronous operations are completed before the next operation can be started. Asynchronous operations can be started and we can do something else before they are completed.

Why do we want our applications to be event based. They are easier to scale. Well suited to Visual programming where multiple GUI elements and many sources of events exist. It has a direct mapping to real world. Its deterministic-there is an exact mapping between a situation in the real world and its representation in the event processing system.

At the same time its approximate-the event processing system provides an approximation to real world events. So what kind of events are there in non-real time applications. Mouse and keyboard, secondary events of GUI elements, file events, message based parallel and local communication, network events etc.
Here is flower delivery example from the book.

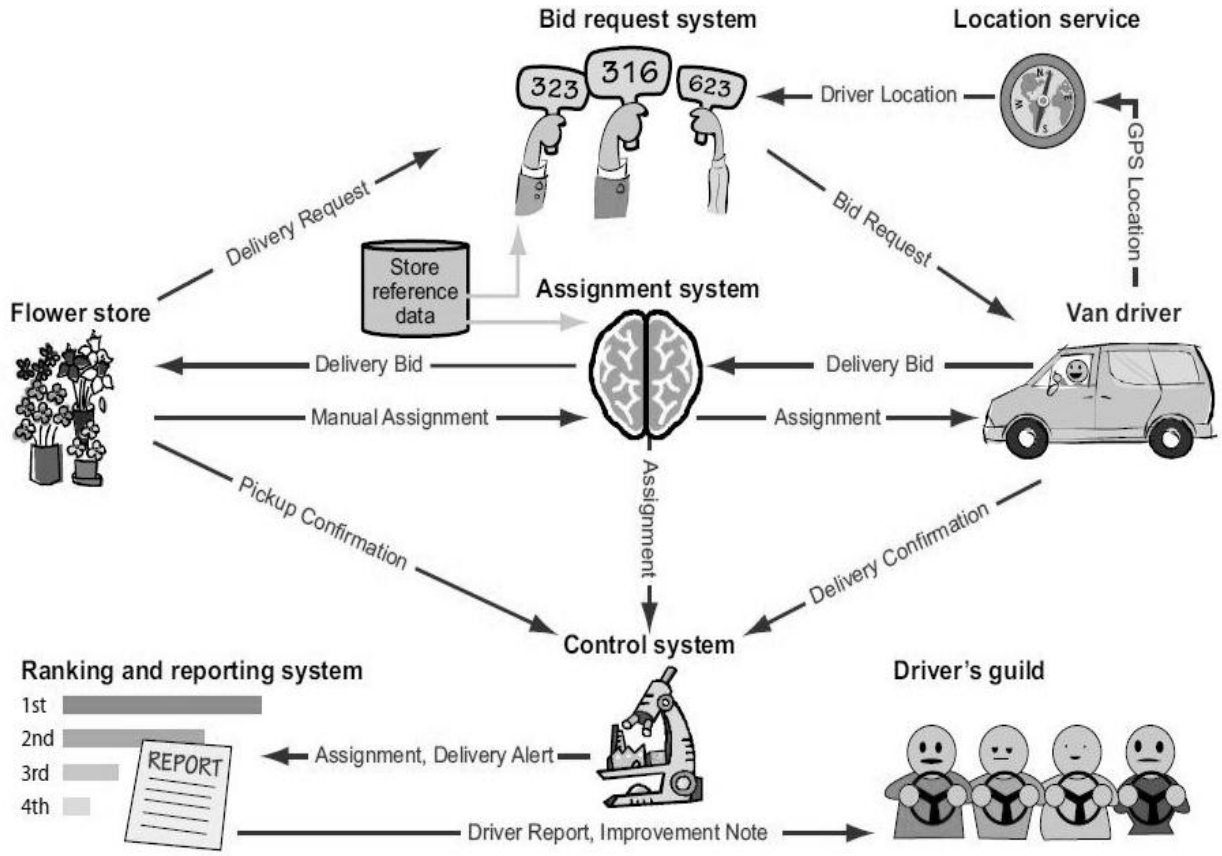


Figure 1.6 Parts of the Fast Flower Delivery application. The arrows represent the flow of events, and the pictures represent the various entities involved in the network.

Here are some excerpts of its description from the book. A consortium of flower stores in a large city has established an agreement with local independent van drivers to deliver flowers from the stores to their destinations. When a store gets a flower delivery order, it creates a request which is broadcast to relevant drivers within a certain distance from the store, with the time for pickup (typically now) and the required delivery time. A driver is then assigned and the customer is notified that a delivery has been scheduled. The driver makes the pickup and delivery, and the person receiving the flowers confirms the delivery time by signing for it on the drivers mobile device.

The system maintains a ranking of each individual driver based on his or her ability to deliver flowers on time. Each store has a profile that can include a constraint on the ranking of its drivers; for example, a store can require its drivers to have a ranking greater than 10. The profile also indicates whether the store wants the system to assign drivers automatically,or whether it wants to receive several applications and then make its own choice. A permanently weak driver is a driver with fewer than five assignments on all the days on which the driver has been active.

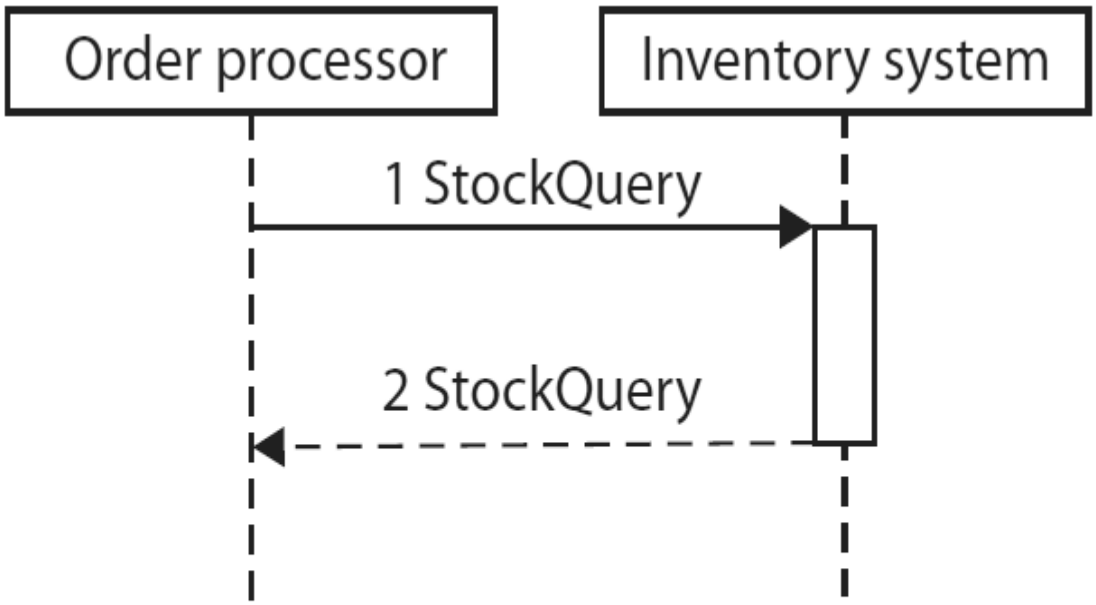
An idle driver is a driver with at least one day of activity that had no assignments. A consistently weak driver is a driver whose assignments, when active, are at least two standard deviations lower than the average assignments per driver on that day. A consistently strong driver is a driver whose daily assignments are at least two standard deviations higher than the average number of assignments per driver on each day in question. An improving driver is a driver whose assignments increase or stay the same day by day.

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Chapter 3

Lecture 3

Let's discuss request and response based applications. Example is a web browser which sends queries and updates. It performs synchronous interactions. It looks like this:



Events are based on the principle of decoupling. Let's compare events and request-response based architecture. Events have already happened whereas requests are asking something to happen. Service requester or client contacts the service provider or the server. In event-driven architecture event producer sends event to event consumer. A customer order can be represented as an event or a request: what are the benefits of each approach?

Similarly, should we send a request to query the state or use "change events". Event has meaning independent of its producers and consumers whereas the same cannot be said about requests. Events are often one way (push events). Events decouple producers and consumers. Events can be processed asynchronously. There can be more than one consumer, each processing it differently possibly processing a sequence of events that happened over time. Latency is reduced in comparison to the "pull" style.

Events can also be compared to messages. Messages may contain events. Events may live outside events like in event logs.

An event channel is a subscription mechanism for events. It further decouples consumers and producers. Can even be an intermediate XML file to store events.

Event-driven architecture and service-driven architecture can also be compared. Event-based programming, also called event-driven architecture (EDA) is an architectural style in which one or more components in a software system execute in response to receiving one or more event notifications. Service-oriented architecture (SOA) is built from request-response. It moves away from monolithic applications. There are

many similarities and often a component can provide both modes of contacting it.
Here are some definitions from the book.

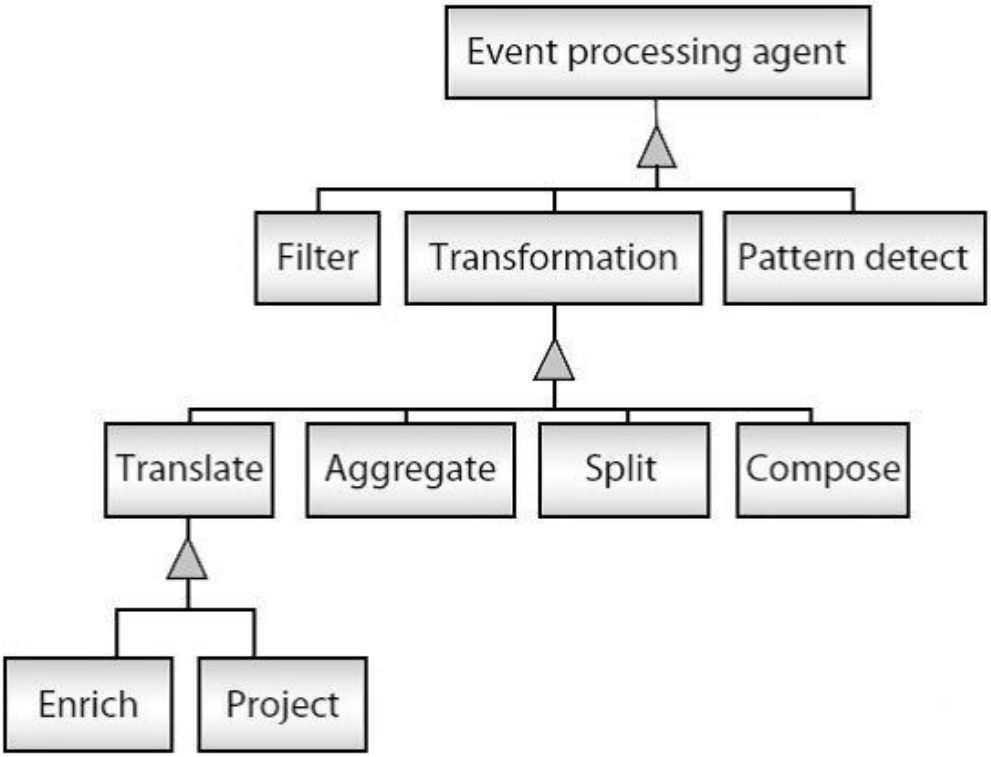
EVENT PRODUCER: An event producer is an entity at the edge of an event process- ing system that introduces events into the system.

EVENT CONSUMER: An event consumer is an entity at the edge of an event pro- cessing system that receives events from the system.

RAW EVENT: A raw event is an event that is introduced into an event processing system by an event producer. DERIVED EVENT: A derived event is an event that is generated as a result of event processing that takes place inside an event processing system.

STATELESS EVENT PROCESSING An event processing agent is said to be stateless if the way it processes one event does not influence the way it processes any subsequent events.

EVENT STREAM An event stream (or stream) is a set of associated events. It is often a temporally totally ordered set (that is to say, there is a well-defined timestamp-based order to the events in the stream). A stream in which all the events must be of the same type is called a homogeneous event stream; a stream in which the events may be of different types is referred to as a heterogeneous event stream.
Here are types of event agents.



Looking back at the program discussed before, can you map the event definitions to the code below:

```
#include <iostream>
#include <fstream>

Int filechanged = 0;
```

```
Char lastchar = '-';

Int onkeypress()
{
  Char b = std::cin.get();
  Std::cout << b << std::endl;
  Return b == 'x';
}

Int onfileread()
{
  Char a = f.get();
  If (a != lastchar) {
    Filechanged = 1;
    Lastchar = a;
  }
  Return 0;
}
Int onfilechanged(char a)
{
  Std::cout << a << std::endl;
  Return a == 'x';
}
Int main()
{
  Do {
    If (kbhit()) {
      If (onkeypress())
 Return 0;
    }
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good()) {
      If (onfileread(f.get()))
 Return 0;
    }
    F.close();
    If (filechanged) {
      If (onfilechanged())
 Return 0;
    }
  } while (true);
  Return 0;
}
```

What aspects of real occurrence do event objects capture. An event type is a specification for a set of event objects that have the same semantic intent and same structure; every event object is considered to be an instance of an event type. An event attribute is a component of the structure of an event. Each attribute has a name and a data type. It can also have occurrence/detection time, certainty, source, location. Events can be composed, generalized, and specialized. Let's experiment with flower delivery example. Can you find events and their attributes.

Event producers can produce events that are hardware generated, software generated, or from human interaction. Event consumers have similar types. Examples are locking or unlocking a door, raising or lowering a barrier, applying the brakes on a vehicle, opening or closing a valve, controlling a railroad switch, and turning a piece of equipment on or off. Try to find producers and consumers in flower delivery application.

Chapter 4

Lecture 4

We will get introduced to C# in this lecture. Before that, Let's start with some history of Java and J++. Microsoft wanted to extend Java to communicate it with COM. Sun did not want that as it would make Java platform dependent. Microsoft wanted a clean-room implementation of Java. What is a clean-room implementation?

Clean room design (also known as the Chinese wall technique) is the method of copying a design by reverse engineering and then recreating it without infringing any of the copyrights and trade secrets associated with the original design. Clean room design is useful as a defense against copyright and trade secret infringement because it relies on independent invention. However, because independent invention is not a defense against patents, clean room designs typically cannot be used to circumvent patent restrictions.

Where do the name for C# comes from. The initial name was "Cool", which stood for "C-like Object Oriented Language". Whats the difference between .NET and C#. C# design most directly reflects .NET (CLR) design but C# is a language and .NET is the platform.

As a first step, you should get Visual Studio from http://www.microsoft.com/visualstudio. Click Downloads and choose "Visual Studio Express 2012 for Windows Desktop". Choose "Download now" or "Install now". You can save iso file to disk (right click to mount in win8) otherwise there are free utilities to mount it or burn it on a cd.

Install and register online for continued use. To create a new project choose File, New Project, Visual C#, Console Application. Then click the Start key.
Here are some notable features of C#.

- No global variables or functions.
- Locals cannot shadow global variables.
- There is a strict boolean type.
- Memory address pointers can only be used in specifically marked "unsafe" blocks and require permissions
- No instruction to "free" memory. Only garbage collection.
- Try-finally block.
- No "multiple inheritance" but interfaces supported.
- Operator overloading allowed.
- More type-safe (only integer widening allowed).
- Enumeration members are scoped.
- Property syntax for getters and setters.
- No checked exceptions.
- some functional programming features like function objects and lambda expressions.

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The common type system of C# has value types and reference types. Instances of value types do not have referential identity nor referential comparison semantics. Value types are derived from System.valuetype and can always be created, copied and have a default value (int, float, char, System.datetime, enum, struct). In contrast, reference types have the notion of referential identity. Default equality and inequality comparisons test for referential rather than structural equality unless overloaded e.g System.String. Not "always" possible to create an instance of a reference type, copy an existing instance, perform a value comparison on two instances, though specific types "can" provide such services by exposing a public constructor or implementing a corresponding interface e.g. Object, System.String, System.Array.

Here is how to box and unbox variables in C#. Boxing stores a value type in a reference type.

```
Int foo = 42;// Value type.
Object bar = foo;// foo is boxed to bar.

Int foo2 = (int)bar;// Unboxed back to value type.
```

Here is an example of Generics in C# which are like templates in C++.

```
Public class genericlist<T>
{
 Void Add(T input) { }
}
Class  testgenericlist
{
 Private class exampleclass{ }
 Static void Main()
 {
 // Declare a list of type int.
 Genericlist<int> list1= new genericlist<int>();
 // Declare a list of type string.
 Genericlist<string> list2= new genericlist<string>();
 // Declare a list of type exampleclass.
 Genericlist<exampleclass> list3 = new genericlist<exampleclass>();
 }

   }
```

Pre-processor directives like #if, #else, #endif, #region, #endregion are supported. Comments are written using // and /* */.

It also provides an XML documentation system that is used like this.

```
Public class Foo{
/**<summary>A summary of the method.</summary>
* <param name="firstparam">A description of the parameter.</param>
* <remarks>Remarks about the method.</remarks> */
Public static void Bar(int firstparam){}

   }
```

The C# specification details a minimum set of types and class libraries that the compiler expects to have available. In practice, C# is most often used with some implementation of the Common Language Infrastructure (CLI), which is standardized as ECMA-335 Common Language Infrastructure (CLI).

Here is a hello world program in C#.

```
Using System;
Class Program {
```


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```
    Static void Main() {
        Console.writeline("Hello world!");
    }
}
```

Writing the using clause uses System as a candidate prefix for types used in the source code. When the compiler sees use of the Console type later in the source code. It allows the programmer to state all candidate prefixes to use during compilation instead of always using full type names. Console is a static class in the System namespace.

If you run the program from within Visual Studio and the console window disappears quickly you can use the following code.

```
1 // keep screen from going  away
2 // when run from VS. NET
3 Console.readline();
```

To test use csc.exe Welcome.cs or run from within Visual Studio. Now Let's write a GUI based hello world.

```
Using System.Windows.Forms;

Class Program
{
    Static void Main()
    {
        Messagebox.Show("Hello world!");
    }

  }
```

Here is a hello world with a call to Console.Write with an argument.

```
// Namespace Declaration
Using System;

// Program start class
Class interactivewelcome
{
 // Main begins program execution.
 Public static void Main()
 {
 // Write to console/get input
Console.Write("What is your name?: ");
Console.Write("Hello, {0}! ", Console.readline());
Console.writeline("Welcome to the C\# Station Tutorial!");
 }
}
```


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Chapter 5

Lecture 5

Here is an example with boolean types.

```
Using System;

Class Booleans
{
Public static void Main()
 {
Bool content = true;
Bool nocontent = false;

Console.writeline("It is {0} that C\# Station provides C\# programming language
content.", content);
Console.writeline("The statement above is not {0}.", nocontent);
 }
}
```

Here are the integer types in C#.

Type	Size (in bits)	Range
sbyte	8	-128 to 127
byte	8	0 to 255
short	16	-32768 to 32767
ushort	16	0 to 65535
int	32	-2147483648 to 2147483647
uint	32	0 to 4294967295
long	64	-9223372036854775808 to 9223372036854775807
ulong	64	0 to 18446744073709551615
char	16	0 to 65535

And here are the floating point and decimal types in C#.

Type	Size (in bits)	precision	Range
float	32	7 digits	1.5×10^{-45} to 3.4×10^{38}
double	64	15-16 digits	5.0×10^{-324} to 1.7×10^{308}
decimal	128	28-29 decimal places	1.0×10^{-28} to 7.9×10^{28}

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The System.String type supports the following sequences.

Char	Meaning	Value
\'	Single quote	0x0027
\"	Double quote	0x0022
\\	Backslash	0x005C
\0	Null	0x0000
\a	Alert	0x0007
\b	Backspace	0x0008
\f	Form feed	0x000C
\n	New line	0x000A
\r	Carriage return	0x000D
\t	Horizontal tab	0x0009
\v	Vertical tab	0x000B

The escape character is while @ is the verbatim character. The following operators are supported by C#.

Category (by precedence)	Operator(s)	Associativity
Primary	x.y f(x) a[x] x++ x-- new typeof default checked unchecked delegate	left
Unary	+ - ! ~ ++x --x (T)x	right
Multiplicative	* / %	left
Additive	+ -	left
Shift	<< >>	left
Relational	< > <= >= is as	left
Equality	== !=	right
Logical AND	&	left
Logical XOR	^	left
Logical OR	|	left
Conditional AND	&&	left
Conditional OR	||	left
Null Coalescing	??	left
Ternary	?:	right
Assignment	= *= /= %= += -= <<= >>= &= ^= |= =>	right

Here is an example of using the Array type in C#.

```
Using System;
Class Array{
 Public static void Main(){
    Int[] myints = { 5, 10, 15 };
    Bool[][] mybools = new bool[2][];
    Mybools[0] = new bool[2];
    Mybools[1] = new bool[1];
    Double[,] mydoubles = new double[2, 2];
    String[] mystrings = new string[3];
    Console.writeline("myints[0]: {0}, myints[1]: {1}, myints[2]: {2}", myints[0],
myints[1], myints[2]);
    Mybools[0][0] = true;
    Mybools[0][1] = false;
    Mybools[1][0] = true;
    Console.writeline("mybools[0][0]: {0}, mybools[1][0]: {1}", mybools[0][0],
mybools[1][0]);
    Mydoubles[0, 0] = 3.147;
    Mydoubles[0, 1] = 7.157;
    Mydoubles[1, 1] = 2.117;
    Mydoubles[1, 0] = 56.00138917;
    Console.writeline("mydoubles[0, 0]: {0}, mydoubles[1, 0]: {1}", mydoubles[0, 0],
mydoubles[1, 0]);
    Mystrings[0] = "Joe";
    Mystrings[1] = "Matt";
    Mystrings[2] = "Robert";
    Console.writeline("mystrings[0]: {0}, mystrings[1]: {1}, mystrings[2]: {2}",
mystrings[0], mystrings[1], mystrings[2]);
 }
}
```

You can make jagged arrays or multi-dimensional arrays. Jagged is basically array of arrays. Array size is any integer type value. It uses a zero-based index.

Control statements like if and else work much like C++.

```
If (myint < 0 || myint == 0) {

Console.writeline("Your number {0} is less than or equal to zero.", myint);

}  else  if (myint > 0 && myint <= 10){

Console.writeline("Your number {0} is in the range from 1 to 10.", myint);

}  else  if (myint > 10 && myint <= 20){

Console.writeline("Your number {0} is in the range from 11 to 20.", myint);

} else if (myint > 20 && myint <= 30){

Console.writeline("Your number {0} is in the range from 21 to 30.", myint);

} else{

Console.writeline("Your number {0} is greater than 30.", myint); }
```

Switch statement can work for booleans, enums, integral types, and strings. You break the switch statement

Branching statement	Description
break	Leaves the switch block
continue	Leaves the switch block, skips remaining logic in enclosing loop, and goes back to loop condition to determine if loop should be executed again from the beginning. Works only if switch statement is in a loop as described in Lesson 04: Control Statements - Loops.
goto	Leaves the switch block and jumps directly to a label of the form "<labelname>:"
return	Leaves the current method. Methods are described in more detail in Lesson 05: Methods.
throw	Throws an exception, as discussed in Lesson 15: Introduction to Exception Handling.

with one of the following branching statements:
The following four kinds of loops are available.

While (<boolean expression>) { <statements> }
Do { <statements> } while (<boolean expression>);
For (<initializer list>; <boolean expression>; <iterator list>) { <statements> }
foreach (<type> <iteration variable> in <list>) { <statements> }

Here is an example of a while loop:

```
Using System;
Class whileloop
{
    Public static void Main()
    {
        Int myint = 0;

        While (myint < 10)
        {
            Console.Write("{0} ", myint);
            Myint++;
        }
        Console.writeline();
    }
}
```

Another example with a for loop.

```
Using System;
Class forloop
{
    Public static void Main()
    {
        For (int i=0; i < 20; i++)
        {
            If (i == 10)
                Break;

            If (i % 2 == 0)
```


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```
                Continue;
            Console.Write("{0} ", i);
        }
        Console.writeline();
    }

}
```

And one of foreach loop.

```
Using System;
Class foreachloop
{
    Public static void Main()
    {
        String[] names = {"Cheryl", "Joe", "Matt", "Robert"};

        Foreach (string person in names)
        {
            Console.writeline("{0} ", person);
        }
    }

}
```

Methods are of the form.

Attributes modifiers return-type method-name(parameters) { statements }

Elements are accessed using the dot operator. Object variables are references to the original object not the object themselves. Here is an example of a method.

```
Using System;
Class onemethod
{
    Public static void Main()
    {
        String mychoice;
        Onemethod om = new onemethod();
            Mychoice = om.getchoice();
    }
    String getchoice()
    {
        Return “example”;
    }

}
```

The “this” pointer in methods refers to the object on which the method is called. Parameters can be “ref” parameter which are passed by reference, “out” parameters which are used for return values or the “params” argument for variable arguments.

Here is an example of the last:

```
// show  addresses
Void viewaddresses(params string[] names)
{
 Foreach (string name in names){
 Console.writeline("Name: {0}", name);
 }
}
```


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Chapter 6

Lecture 6

Namespaces allow name reuse. For example, a Console class can reside in multiple libraries. Here is an example of namespace declaration:

```
// Namespace Declaration

Using System;

// The C\# Station Namespace

Namespace csharp_station
{
// Program start class
Class namespacecss  {
 // Main begins program execution.

Public static void Main()
 {

// Write to console

  Console.writeline("This is the new C\# Station Namespace.");

}

 }


}
```

Nested namespaces can be used like this:

Namespace testnamespace {

Namespace tutorial { OR

Namespace testnamespace.tutorial {

Calling this namespace from within testnamespace we use "tutorial.". The using directive can be used to rename a long namespace in the current file like:

Using theexample = testnamespace.tutorial.myexample

Class use the word class followed by name, curly braces and the class body. It has constructor (ctor) to create objects. Ctor does not return any values and it initializes class members.

```
// Namespace Declaration
Using System;
// helper class
Class outputclass
{
String mystring;

// Constructor
```


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```
Public outputclass(string inputstring)
    {

        Mystring = inputstring;
    }

// Instance Method

Public void printstring()
    {
        Console.writeline("{0}", mystring);

    }

// Destructor

~outputclass()
    {

        // Some resource cleanup routines

    }

}

// Program start class

Class exampleclass
{
    // Main begins program execution.

Public static void Main()
    {

    // Instance of outputclass
    Outputclass outcl = new
    Outputclass("This is printed by the output class.");
    // Call Output class' method
    Outcl.printstring();
    }

}
```

Multiple classes are conventionally stored in multiple files. To create a new class in IDE, use the new class wizard. Multiple .cs files can be compiled as csc a.cs b.cs c.cs. Default ctors are written with no arguments when no ctor is written. An initializer list can be used to use an alternate constructor.

```
Public outputclass() : this("Default Constructor String") { }
```

Multiple ctors can be written. Types of class members in C# are instance and static. For every new occurrence there are new instance vars.

Outputclass oc1 = new outputclass("outputclass1");

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Outputclass oc2 = new outputclass("outputclass2");

Now oc1.printstring or oc2.printstring will access different instance variables. On the other hand static members have only one copy.

```
Public static void staticprinter(){
 Console.writeline("There is only one of me.");

}
```

Then you could call that function from Main() like this: outputclass.staticprinter();

Static ctor exists to initialize class static members. Its called only once. It has no parameters. Destructors (dtors) are called by garbage collector. "public" methods accessed outside class. "public" classes accessed outside assembly. A class can contain Constructors, Destructors, Fields, Methods, Properties, Indexers, Delegates, Events, and Nested Classes.

Inheritance introduces the concept of base classes and derived classes. Declaration of inheritance is done as Derived : Base. Only single inheritance is supported. Multiple interface inheritance discussed later.

```
Using System;

Public class parentclass
{

  Public parentclass()
 {

    Console.writeline("Parent Constructor.");

  }

  Public void print() {

    Console.writeline("I'm a Parent Class.");

  }

}

Public class childclass : parentclass
{

  Public childclass()
 {

    Console.writeline("Child Constructor.");

  }

  Public static void Main()
 {

    Childclass child = new childclass();

    Child.print();
```


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```
  }

}
```

The output is:

Parent Constructor. Child Constructor. I'm a Parent Class

.

Derived class is exactly the same as base. Child class "is a" parent class except that derived is more special. Base are automatically initialized before derived. How can derived communicate with the base i.e. The specialized part communicate with the generic part. By using ": base()" at ctor time or "base.x()" later. "new" keyword used to override methods. Cast back to base type to call an overridden method of base.

Here is another example:

```
Using System;
Public class Parent {
  String parentstring;
  Public Parent() {
    Console.writeline("Parent Constructor.");
  }
  Public Parent(string mystring) {
    Parentstring = mystring;
    Console.writeline(parentstring);
  }

  Public void print() {
    Console.writeline("I'm a Parent Class.");
  }
}
  Public class Child : Parent {
    Public Child() : base("From Derived") {
      Console.writeline("Child Constructor.");
    }

    Public new void print() {
      Base.print(); Console.writeline("I'm a Child Class.");
    }

    Public static void Main() {
      Child child = new Child();
      Child.print(); ((Parent)child).print();
    }
  }
```

Output is:

From Derived
Child Constructor. I'm a Parent Class. I'm a Child Class. I'm a Parent Class.

Polymorphism means to invoke derived class methods through base class reference during run-time. Normally (as we saw) base class casting means base class method is called. Its handy when a group of objects is stored in array/list and we invoke some method on all of them. They dont have to be the same type but if related by inheritance, they can be cast.

```
Using System;
```


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```
Public class drawingobject
{
 Public virtual void Draw()
 {
     Console.writeline("I'm just a generic drawing object.");
   }

}

Public class Line : drawingobject
{
 Public override void Draw()
 {
     Console.writeline("I'm a Line.");
 }
}

Public class Circle : drawingobject
{
 Public override void Draw()
 {
     Console.writeline("I'm a Circle.");
  }
}

Public class Square : drawingobject
{
 Public override void Draw()
 {
    Console.writeline("I'm a Square.");
 }
}

Public class drawdemo
{
  Public static int Main( )
    {

    Drawingobject[] dobj = new drawingobject[4];
    Dobj[0] = new Line();
    Dobj[1] = new Circle();
    Dobj[2] = new Square();
    Dobj[3] = new drawingobject();
    Foreach (drawingobject drawobj in dobj)
    {
      Drawobj.Draw();
    }
    Return 0;
  }
}
```

Polymorphic methods are declared with “virtual” keyword and the “override” keyword is used to provide an implementation. Polymorphism needs the signatures to be the same. Because of inheritance, the three derived classes can be treated as the base class. Because of polymorphism, the methods from drived can still be called.

Output is:

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I'm a Line.
I'm a Circle. I'm a Square.
I'm just a generic drawing object.

Properties allow protected reads and writes to a field of a class. Other languages require custom getters and setters.

C# allows accessing properties like fields. Properties allows changing representation, public fields dont. Other languages use custom encapsulation to protect data.

```
Using System;

Public class Customer
{
    Private int m_id = -1;

    Public int getid()
    {
        Return m_id;
    }

    Public void setid(int id)
    {
        M_id = id;
    }

    Private string m_name = string.Empty;

    Public string getname()
    {
        Return m_name;
    }

    Public void setname(string name)
    {
        M_name = name;
    }
}

Public class customermanagerwithaccessormethods
{
    Public static void Main()
    {
        Customer cust = new Customer();

        Cust.setid(1);
        Cust.setname("Amelio Rosales");

        Console.writeline(
            "ID: {0}, Name: {1}",
            Cust.getid(),
            Cust.getname());

        Console.readkey();
    }
}
```


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Its such a common pattern that C# has a dedicated language feature for it.

```
Using System;
Public class Customer
{
    Private int m_id = -1;

    Public int ID
    {
        Get
        {
            Return m_id;
        }
        Set
        {
            M_id = value;
        }
    }

    Private string m_name = string.Empty;

    Public string Name
    {
        Get
        {
            Return m_name;
        }
        Set
        {
            M_name = value;
        }
    }
}

Public class customermanagerwithproperties
{
    Public static void Main()
    {
        Customer cust = new Customer();

        Cust.ID = 1;
        Cust.Name = "Amelio Rosales";

 Console.writeline(
            "ID: {0}, Name: {1}",
            Cust.ID,
            Cust.Name);

        Console.readkey();
    }
}
```

Getters “return”, setters use “value”. Next we discuss read-only peroperties. Here is an example that uses read-only properties.

```
Using System;

Public class Customer
{
```


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```
    Private int m_id = -1;
    Private string m_name = string.Empty;

    Public Customer(int id, string name)
    {
        M_id = id;
        M_name = name;
    }

    Public int ID
    {
        Get
        {
            Return m_id;
        }
    }

    Public string Name
    {
        Get
        {
            Return m_name;
        }
    }
}

Public class readonlycustomermanager
{
    Public static void Main()
    {
        Customer cust = new Customer(1, "Amelio Rosales");

        Console.writeline(
            "ID: {0}, Name: {1}",
            Cust.ID,
            Cust.Name);

        Console.readkey();
    }
}
```

This example has only get and no implementation of set. So what about write-only properties. Yes, they exist and have an implementation of set but not get.

```
  Using System;

Public class Customer
{
    Private int m_id = -1;

    Public int ID
    {
        Set
        {
            M_id = value;
        }
    }
```


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```
    Private string m_name = string.Empty;

    Public string Name
    {
        Set
        {
            M_name = value;
        }
    }

    Public void displaycustomerdata()
    {
        Console.writeline("ID: {0}, Name:
            {1}", m_id, m_name);
    }
}

Public class writeonlycustomermanager
{
    Public static void Main()
    {
        Customer cust = new Customer();

        Cust.ID = 1;
        Cust.Name = "Amelio Rosales";

        Cust.displaycustomerdata();

        Console.readkey();
    }
}
```


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Chapter 7

Lecture 7

Auto-implemented properties improve the common-case we saw. Here is the same example using auot-implemented properties. Properties have the same idea as getters and setters. They just allow simplified syntax. Same idea of backing store and a simplified syntax for the general case is enabled by auto-implemented properties.

```
  Using System;


Public class Customer
{
    Public int ID { get; set; }
    Public string Name { get; set; }
}

Public class autoimplementedcustomermanager
{
    Static void Main()
    {
        Customer cust = new Customer();

        Cust.ID = 1;
        Cust.Name = "Amelio Rosales";

        Console.writeline(
            "ID: {0}, Name: {1}",
            Cust.ID,
            Cust.Name);

        Console.readkey();
    }
}
```

An indexer enables your class to be treated like an array. However you have your internal data representation not an actual array. Its implemented using "this" keyword and square brackets syntax. It looks like a property implementation. Here is an example.

```
Class intindexer
{
    Private string[] mydata;

    Public intindexer(int size)
    {
        Mydata = new string[size];

        For (int i=0; i < size; i++)
        {
            Mydata[i] = "empty";
        }
```


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```
    }

    Public string this[int pos]
    {
        Get
        {
            Return mydata[pos];
        }
        Set
        {
            Mydata[pos] = value;
        }
    }

    Static void Main(string[] args)
    {
        Int size = 10;

        Intindexer myind = new intindexer(size);

        Myind[9] = "Some Value";
        Myind[3] = "Another Value";
        Myind[5] = "Any Value";

        Console.writeline("\nindexer Output\n");

        For (int i=0; i < size; i++)
        {
            Console.writeline("myind[{0}]: {1}", i, myind[i]);
        }
    }
}
```

The output of this program looks like this:

Indexer Output

Myind[0]: empty
Myind[1]: empty
Myind[2]: empty
Myind[3]: Another Value myind[4]: empty
Myind[5]: Any Value
Myind[6]: empty
Myind[7]: empty
Myind[8]: empty
Myind[9]: Some Value

Indexers can take any number of parameters. The parameters can be integers, strings, or enums. Additionally they can be overloaded with different type and number of parameters. The following example has overloaded indexers.

```
Using System;

/// <summary>
///     Implements overloaded indexers.
/// </summary>
Class ovrindexer
```


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```
{
    Private string[] mydata;
    Private int         arrsize;

    Public ovrindexer(int size)
    {
        Arrsize = size;
        Mydata = new string[size];

        For (int i=0; i < size; i++)
        {
            Mydata[i] = "empty";
        }
    }

    Public string this[int pos]
    {
        Get
       {
            Return mydata[pos];
        }
        Set
       {
            Mydata[pos] = value;
        }
    }

     Public string this[string data]
    {
        Get
       {
            Int count = 0;

            For (int i=0; i < arrsize; i++)
            {
                If (mydata[i] == data)
                    Count++;
            }
            Return count.tostring();
        }
        Set
       {
            For (int i=0; i < arrsize; i++)
            {
                If (mydata[i] == data)
                    Mydata[i] = value;
            }
        }
    }
    Static void Main(string[] args)
    {
        Int size = 10;
        Ovrindexer myind = new ovrindexer(size);

        Myind[9] = "Some Value";
        Myind[3] = "Another Value";
        Myind[5] = "Any Value";
```


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```
        Myind["empty"] = "no value";

        Console.writeline("\nindexer Output\n");

        For (int i=0; i < size; i++)
        {
            Console.writeline("myind[{0}]: {1}", i, myind[i]);
        }

        Console.writeline("\nnumber of \"no value\" entries: {0}", myind["no value"]);
    }
}
```

Indexers with multiple parameters look like this.

```
Public object this[int param1, ..., int paramn] {
Get {
 // process and return some class data
 }
Set {
 // process and assign some class data
 }

   }
```

Our next topic is a "struct". Its a value-type, whereas class is a reference-type. Value types hold their value in memory where they are declared, but reference types hold a reference to an object in memory. If you copy a class, C# creates a new copy of the reference to the object and assigns the copy of the reference to the separate class instance. Structs cant have destructors, cant have implementation inheritance (still can have interface inheritance). Many built-in types are structs like System.Int32 is a C# int. C# built-in types are aliases for .NET Framework types. Syntax of struct and class are very similar. Here is an example of using a struct.

```
/// <summary>
/// Custom struct type, representing
    A rectangular shape
/// </summary>
Struct Rectangle
{
    /// <summary>
    /// Backing Store for Width
    /// </summary>
    Private int m_width;

    /// <summary>
    /// Width of rectangle
    /// </summary>
    Public int Width
    {
        Get
        {
            Return m_width;
        }
        Set
        {
            M_width = value;
        }
    }
```


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```
    /// <summary>
    /// Backing store for Height
    /// </summary>
    Private int m_height;

    /// <summary>
    /// Height of rectangle
    /// </summary>
    Public int Height
    {
        Get
        {
            Return m_height;
        }
        Set
        {
            M_height = value;
        }
    }
}
```

Its instantiated and used just like a class.

```
Using System;

/// <summary>
/// Example of declaring and using
///    a struct
/// </summary>
Class structexample
{
    /// <summary>
    /// Entry point: execution starts
///        here
    /// </summary>
    Static void Main()
    {
        // instantiate a new Rectangle struct
        // where Width is set to 1 and Height is set to 3
       Rectangle rect1 = new Rectangle();
        Rect1.Width = 1;
        Rect1.Height = 3;

        // show the value of Width and Height for rect1
        Console.writeline("rect1: {0}:{1}", rect1.Width, rect1.Height);

        Console.readkey();
    }
}
```

The output of this program is: Output:

Rect1: 1:3

We can add constructors to a struct but cant overlaod the default ctor which initializes everything to default values. We can add methods like classes and they are called in exactly the same way.

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An object initializer is a syntax for initializing a struct without using a ctor. It looks like this:

```
// you can also use object initialization syntax
        Rectangle rect11 = new Rectangle
        {
            Width = 1,
            Height = 3
        };
```

Next we discuss "Interfaces". They are declared like a class but have no implementation. There is only declarations of events, indexers, methods and/or properties. They are inherited by classes which provide the real implementation. So, what are interfaces good for if they don't implement functionality? They are great for putting together plug-n-play like architectures where components can be interchanged at will. The interface forces each component to expose specific public members that will be used in a certain way.

Interfaces define a contract. For instance, if class foo implements the idisposable interface, it is making a statement that it guarantees it has the Dispose() method, which is the only member of the idisposable interface. Any code that wishes to use class foo may check to see if class foo implements idisposable. When the answer is true, then the code knows that it can call foo.Dispose()

To define an interface, we use the following syntax.

```
Interface imyinterface
{
Void methodtoimplement();
}
```

The "I" prefix is a convention. There is no implementation, only semi-colon. There are only method signatures.
Let's now see how its used.

```
Class  interfaceimplementer : imyinterface
{
Static void Main()
{
 Interfaceimplementer iimp = new interfaceimplementer();
 Iimp.methodtoimplement();
}
Public void methodtoimplement()
{
 Console.writeline("methodtoimplement() called.");
}
}
```

The syntax used is same as for inheritance. Any signature difference or not implementing some method in the interface is an error. Interfaces can also be inherited. Here is an example.

```
Using System;

Interface iparentinterface
{
    Void parentinterfacemethod();
}

Interface imyinterface : iparentinterface
{
    Void methodtoimplement();
}
```


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```
Class interfaceimplementer : imyinterface
{
    Static void Main()
    {
        Interfaceimplementer iimp = new interfaceimplementer();
        Iimp.methodtoimplement();
        Iimp.parentinterfacemethod();
    }

    Public void methodtoimplement()
    {
        Console.writeline("methodtoimplement() called.");
    }

    Public void parentinterfacemethod()
    {
        Console.writeline("parentinterfacemethod() called.");
    }
}
```

We can also do polymorphism using interfaces.

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Chapter 8

Lecture 8

A delegate is a reference to a method. Its like function pointers in some other languages. Methods are algorithms that operate on data. Sometimes the data needs a special operation e.g. A comparator in a sorting routine. We can use a bad solution which would an if then else for all types we want our sorting routine to work for. There are two good solutions. One is to implement a comparator interface in all types we want and then pass an instance of a type that provides that interface. The other solution is using delegates i.e. References to functions. Now we can pass a comparator delegate to the sorting algorithm.

```
Using System;

// this is the delegate declaration
Public delegate int Comparer(object obj1, object obj2);

Public class Name
{
    Public string firstname = null;
    Public string lastname = null;

    Public Name(string first, string last)
    {
        Firstname = first;
        Lastname = last;
    }

    // this is the delegate method handler
    Public static int comparefirstnames(object name1, object name2)
    {
        String n1 = ((Name)name1).firstname;
        String n2 = ((Name)name2).firstname;

        If (String.Compare(n1, n2) > 0)
        {
            Return 1;
        }
        Else if (String.Compare(n1, n2) < 0)
        {
            Return -1;
        }
        Else
        {
            Return 0;
        }
    }

    Public override string tostring()
    {
        Return firstname + " " + lastname;
    }
}
```


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```
Class simpledelegate
{
    Name[] names = new Name[5];

    Public simpledelegate()
    {
      Names[0] = new Name("Joe", "Mayo");
        Names[1] = new Name("John", "Hancock");
        Names[2] = new Name("Jane", "Doe");
        Names[3] = new Name("John", "Doe");
        Names[4] = new Name("Jack", "Smith");
    }
    Static void Main(string[] args)
    {
        Simpledelegate sd = new simpledelegate();

        // this is the delegate instantiation
        Comparer cmp = new Comparer(Name.comparefirstnames);
        Console.writeline("\nbefore Sort: \n");
        Sd.printnames();
        // observe the delegate argument
        Sd.Sort(cmp);
        Console.writeline("\nafter Sort: \n");
        Sd.printnames();
    }
    // observe  the delegate parameter
    Public void Sort(Comparer compare)
    {
        Object temp;

        For (int i=0; i < names.Length; i++)
        {
            For (int j=i; j < names.Length; j++)
            {
                // using delegate "compare" just like
                // a normal method
                If ( compare(names[i], names[j]) > 0 )
                {
                    Temp = names[i];
                    Names[i] = names[j];
                    Names[j] = (Name)temp;
                }
            }
        }
    }
    Public void printnames()
    {
        Console.writeline("Names: \n");

        Foreach (Name name in names)
        {
            Console.writeline(name.tostring());
        }
    }
}
```

A C# event is a class member that is activated whenever the event it was designed for occurs (fires).

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Anyone interested in the event can register and be notified as soon as the event fires. At the time an event fires, methods registered with the event will be invoked.

Events and delegates work hand in hand. Any class, including the same class that the event is declared in, may register one of its methods with the event. This occurs through a delegate, which specifies the signature of the method that is registered for the event. The delegate may be one of the pre-defined .NET delegates or one that you declare yourself. Then, you assign the delegate to the event, which effectively registers the method that will be called when the event fires.

Here is an example:

```
Using System;
Using System.Drawing;
Using System.Windows.Forms;

// custom delegate
Public delegate void Startdelegate();

Class Eventdemo : Form
{
    // custom event
    Public event Startdelegate startevent;
    Public Eventdemo()
    {
        Button clickme = new Button();
        Clickme.Parent = this;
        Clickme.Text = "Click Me";
        Clickme.Location = new Point(
            (clientsize.Width - clickme.Width) /2,
            (clientsize.Height - clickme.Height)/2);
        // an eventhandler delegate is assigned
        // to the button's Click event
        Clickme.Click += new eventhandler(onclickmeclicked);
        // our custom "Startdelegate" delegate is assigned
        // to our custom "startevent" event.
        Startevent += new Startdelegate(onstartevent);
        // fire our custom event
        Startevent();
    }
    // this method is called when the "clickme" button is pressed
    Public void onclickmeclicked(object sender, eventargs ea)
    {
        Messagebox.Show("You Clicked My Button!");
    }
    // this method is called when the "startevent" Event is fired
    Public void onstartevent()
    {
        Messagebox.Show("I Just Started!");
    }
    Static void Main(string[] args)
    {
        Application.Run(new Eventdemo());
    }
}
```

To run this example in Visual Studio you will either have to add a reference to System.Drawing.dll and

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Sys-tem.Windows.Forms.dll or you can make a new Windows Forms Project. This class inherits from Form, which essen-tially makes it a Windows Form. This automatically gives you all the functionality of a Windows Form, including Title.Bar, Minimize/Maximize/Close buttons, System Menu, and Borders. It is started by calling the Run() method of the static Application object with a reference to the form object as its parameter. The += syntax registers a delegate with the event. Its fired like a method call. The Click event is pre-defined, eventhandler delegate is also already in .NET. All you do is define your callback method (delegate handler method) that is invoked when someone presses the clickme button. The onclickmeclicked() method, conforms to the signature of the eventhandler delegate.

Next, we discuss exception handling. There can be unforeseen errors in the program. There are some normal errors and some exceptional errors e.g. File i/o error, system out of memory, null pointer exception. Exceptions are "thrown". They are derived from System.Exception class. They have a message property, contain a stacktrace, a tostring method. Identifying the exceptions you'll need to handle depends on the routine you're writing e.g. System.IO.File.openread() could throw securityexception, argumentexception, argumentnullexception, pathtoolongexception, directorynotfoundexception, unauthorizedaccessexception, filenotfoundexception, or notsupportedexception.

Here is an example of exception handling:

```
Using System;
Using System.IO;

Class trycatchdemo
{
    Static void Main(string[] args)
    {
        Try
        {
            File.openread("nonexistentfile");
        }
        Catch(Exception ex)
        {
            Console.writeline(ex.tostring());
        }
    }
}
```

A single exception can be handled differently as well:

```
1           catch (filenotfoundexception fnfex)
2           {
3               Console.writeline(fnfex.tostring());
4           }
5           catch (Exception ex)

6           {
7               Console.writeline(ex.tostring());
8           }
```

First handler that handles an exception catches it. Otherwise the exception bubbles up the caller stack until someone handles it.

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Chapter 9

Lecture 9

Exception can leave your program in an inconsistent state by not releasing resources or doing some other type of cleanup. Sometimes you need to perform clean up actions whether or not your program succeeds. These are good candidates for using a finally block e.g. A filestream must be closed.

```
Using System;
Using System.IO;
Class finallydemo
{
    Static void Main(string[] args)
    {
        Filestream outstream = null;
        Filestream instream = null;
        Try
        {
            Outstream = File.openwrite("destinationfile.txt");
            Instream = File.openread("bogusinputfile.txt");
        }
        Catch (Exception ex)
        {
            Console.writeline(ex.tostring());
        }
        Finally
        {
            If (outstream != null)
            {
                Outstream.Close();
                Console.writeline("outstream closed.");
            }
            If (instream != null)
            {
                Instream.Close();
                Console.writeline("instream closed.");
            }
        }
    }
}
```

The alternate would be to duplicate code after catch and in catch. But finally is a neat cleanup solution.

Next, we discuss attributes. Attributes add declarative information to your programs. They are used for various purposes during runtime. They can be used at design time by application development tools. For example, dllimpor-tattribute allows a program to communicate with Win32 libraries. Obsoleteattribute causes a compile-time warning to appear. Such things would be difficult to accomplish with normal code. Attributes add metadata to your programs.

When your C# program is compiled, it creates a file called an assembly, which is normally an executable or DLL library. Assemblies are self-describing because they have metadata. Via reflection, a program's

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attributes can be retrieved from its assembly metadata. Attributes are classes that can be written in C# and are used to decorate your code with declarative information. Its a powerful concept. It Let's you extend your language by creating customized declarative syntax with attributes.

Attributes are generally applied physically in front of type and type member declarations. They are declared with square brackets, "[" and "]" surrounding the attribute such as " [obsoleteattribute]". The "Attribute" part of the attribute name is optional i.e. "[Obsolete]" is correct as well. Parameter lists are also possible.

```
Using System;
Class basicattributedemo
{
    [Obsolete]
    Public void myfirstdeprecatedmethod()
    {
        Console.writeline("Called myfirstdeprecatedmethod().");
    }
    [obsoleteattribute]
    Public void myseconddeprecatedmethod()
    {
        Console.writeline("Called myseconddeprecatedmethod().");
    }
    [Obsolete("You shouldn't use this method anymore.")]
    Public void mythirddeprecatedmethod()
    {
        Console.writeline("Called mythirddeprecatedmethod().");
    }
    // make the program thread safe for COM
    [stathread]
    Static void Main(string[] args)
    {
        Basicattributedemo attrdemo = new basicattributedemo();
        Attrdemo.myfirstdeprecatedmethod();
        Attrdemo.myseconddeprecatedmethod();
        Attrdemo.mythirddeprecatedmethod();
    }
}
```

When you compile this program you'll see.

```
>csc basicattributedemo.cs
Microsoft (R) Visual C\# .NET Compiler version 7.10.2292.4 for Microsoft (R) .NET Framework version 1.1.4322
Copyright (C) Microsoft Corporation 2001-2002. All rights reserved.

Basicattributedemo.cs(29,3): warning CS0612:
'Basicattributedemo.myfirstdeprecatedmethod()'is obsolete basicattributedemo.cs(30,3): warning CS0612:
'Basicattributedemo.myseconddeprecatedmethod()' is obsolete basicattributedemo.cs(31,3):
warning CS0618:
'Basicattributedemo.mythirddeprecatedmethod()' is obsolete: 'You shouldn't use this method anymore.'
```

Stathread is a common attribute you will see later. It stands for Single Threaded Apartment model which is used for communicating with unmanaged COM.

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Attribute parameters can be either positional parameters or named parameters. Usually named parameters with optional stuff, but positional can be optional as well e.g. "[Obsolete("You shouldn't use this method anymore.", true)] public void mythirddeprecatedmethod()" will give an error instead of warning. The attribute dllimportattribute has

Both positional and named parameters "[dllimport("User32.dll", entrypoint="messagebox")]". Positional parameters come before named parameters. There is no order requirement on named parameters.

```
Using System;
[assembly: clscompliant(true)]
Public class attributetargetdemo
{
    Public void nonclscompliantmethod(uint nclsparam)
    {
        Console.writeline("Called nonclscompliantmethod().");
    }
    [stathread]
    Static void Main(string[] args)
    {
        Uint myuint = 0;
        Attributetargetdemo tgtdemo = new attributetargetdemo();
        Tgtdemo.nonclscompliantmethod(myuint);
    }
}
```

We rarely write new attributes but we use them extensively.

Enums (or enumerations) are strongly typed constants. They are unique types that allow you to assign symbolic names to integral values. Enum of one type may not be implicitly assigned to an enum of another type (even though the underlying value of their members is the same). All assignments between different enum types and integral types require an explicit cast. It allows you to work with integral values, but using a meaningful name. North, South, East, and West or the zet of integers 0, 1, 2, and 3. C# type, enum, inherits the Base Class Library (BCL) type, Enum.

```
Using System;
// declares the enum
Public enum Volume
{
    Low,_
    Medium,
    High
}
// demonstrates how to use the enum
Class enumswitch
{
    Static void Main()
    {
        // create and initialize
        // instance of enum type
        Volume myvolume = Volume.Medium;
        // make decision based
        // on enum value
        Switch (myvolume)
        {
            Case Volume.Low:
                Console.writeline("The volume has been turned Down.");
                Break;
```


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```
            Case Volume.Medium:
                Console.writeline("The volume is in the middle.");
                Break;
            Case Volume.High:
                Console.writeline("The volume has been turned up.");
                Break;
        }
        Console.readline();
    }
}
```

Default underlying type of an enum is "int". It can be changed by specifying a "base". Valid base types include byte, sbyte, short, ushort, int, uint, long, and ulong. Default value of first member is 0. You can assign any member any value. If it is unassigned it gets +1 the value of its predecessor.

```
Public enum Volume  : byte
{
 Low= 1,
 Medium,
 High
}
```

To convert user input to enum:

```
// get value user provided
String volstring  = Console.readline();
Int volint  = Int32.Parse(volstring);
// perform  explicit cast from
// int to Volume enum type

Volume myvolume  =(Volume)volint
```

Enum inherits from System.Enum in base class library.

```
// get a list of member names from Volume enum,
// figure out the numeric value, and display
Foreach  (string volume in Enum.getnames(typeof(Volume)))
{
Console.writeline("Volume Member:  {0}\n Value:  {1}",
Volume,  (byte)Enum.Parse(typeof(Volume), volume));
}
```

To enumerate values.

```
// get all values  (numeric values) from the Volume
// enum type, figure out member name, and display
Foreach  (byte val in Enum.getvalues(typeof(Volume)))
{
Console.writeline("Volume Value:  {0}\n Member:  {1}",
Val, Enum.getname(typeof(Volume), val));
}
```

Next topic is operator overloading. You can add operators to your own types e.g. A matrix type can have an add and a dot product operator etc.

```
Matrix result  = mat1.Add(mat2); // instance
Matrix result  = Matrix.Add(mat1, mat2);
Matrix result  = mat1.dotproduct(mat2).dotproduct(mat3);
Matrix result  = mat1  + mat2;
Matrix result  = mat1  * mat2; 6  Matrix result = mat1 * mat2 * mat3 * mat4;
```

It should not be used when its not natural to be used. Its implementation syntax is:

```
Public static Matrix operator *(Matrix mat1, Matrix mat2)
```


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```
{
 // dot product  implementation
}
```

Here is an example.

```
Using System;

Class Matrix
{
    Public const int dimsize = 3;
    Private double[,] m_matrix = new double[dimsize, dimsize];

    // allow callers to initialize
    Public double this[int x, int y]
    {
        Get { return m_matrix[x, y]; }
        Set { m_matrix[x, y] = value; }
    }

    // let user add matrices
    Public static Matrix operator +(Matrix mat1, Matrix mat2)
    {
        Matrix newmatrix = new Matrix();

        For (int x = 0; x < dimsize; x++)
            For (int y = 0; y < dimsize; y++)
                Newmatrix[x, y] = mat1[x, y] + mat2[x, y];

        Return newmatrix;
    }
}
```

Overloaded operators must be static. They must be declared in the class for which the operator is defined. It is required that matching operators are both defined, e.g. == and != so that the behavior is consistent. Access modifiers on types and assemblies provide encapsulation. They are: private, protected, internal, protected internal, and public.

Next, we see how generic collections are used: There is a very useful List collection. It is better than arraylist of objects in earlier .NET versions. Another useful collection is Dictionary¡tkey,tvalue¿ vs. A non-generic hashtable of objects. We write "using System.Collections.Generic" to include these collections.

```
List<int> myints  = new List<int>();

Myints.Add(1);
Myints.Add(2);
Myints.Add(3);

For(int i= 0; i < myints.Count; i++)
{
Console.writeline("myints:{0}", myints[i]);
}

Public class Customer
{
Public Customer(int id, string name)
{
ID= id;
Name = name;
```


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```
}
Public int ID
{
Get;
Set;
}
Public string Name
{
Get;
Set;
}
}
Dictionary<int, Customer> customers = new Dictionary<int, Customer>();

Customer cust1 = new Customer(1, "Cust 1");
Customer cust2 = new Customer(2, "Cust 2");
Customer cust3 = new Customer(3, "Cust 3");

Customers.Add(cust1.ID, cust1);
Customers.Add(cust2.ID, cust2);
Customers.Add(cust3.ID, cust3);

Foreach (keyvaluepair<int, Customer> custkeyval in customers)
{
Console.writeline(
"Customer ID:{0}, Name: {1}",
Custkeyval.Key,
Custkeyval.Value.Name);
}
```


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Chapter 10

Lecture 10

Let's discuss Anonymous methods. It is used with delegates and handlers and events. Anonymous methods result in much less code. Anonymous method is a method without a name. You don't declare anonymous methods, Instead they get hooked up directly to events. With delegates 1) you declare the delegate, 2) write a method with a signature defined by the delegate interface, 3) declare the event based on that delegate, and 4) write code to hook the handler method up to the delegate. To delcare an anonymous method, you just use keyword "delegate" followed by method body.

```
Public partial class Form1: Form
{
 Public Form1()
 {
  Button btnhello= new Button();
  Btnhello.Text= "Hello";
  Btnhello.Click +=
  Delegate
  {
   Messagebox.Show("Hello");
  };
  Controls.Add(btnhello);
 }
}
```

We can skip parameters when not using them. Here is an example with parameters.

```
Using System;
Using System.Windows.Forms;

Public partial class Form1 : Form
{
    Public Form1()
    {
        Button btnhello = new Button();
        Btnhello.Text = "Hello";

        Btnhello.Click +=
            Delegate
            {
                Messagebox.Show("Hello");
            };

        Button btngoodbye = new Button();
        Btngoodbye.Text = "Goodbye";
        Btngoodbye.Left = btnhello.Width + 5;
        Btngoodbye.Click +=
            Delegate(object sender, eventargs e)
            {
                String message = (sender as Button).Text;
                Messagebox.Show(message);
            };
```


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```
        Controls.Add(btnhello);
        Controls.Add(btngoodbye);
    }
}
```

We can create nullable value types by appending a question mark to a type name.

```
Int? Unitsinstock = 5;
Datetime? Startdate;
Startdate= datetime.Now;
Startdate= null;
Int availableunits;

If(unitsinstock== null)
{
Availableunits = 0;
}
Else
{
Availableunits = (int)unitsinstock;
}
// OR just
Int availableunits = unitsinstock ?? 0;
```

We will now discuss debugging in Visual Studio.By printing output frequently we can debug. We can use Console.Write to print. Breakpoints allow stopping the program during execution. Press F5 and execution will stop at breakpoint. You can hover over variabeles for their current values. Locals, watch, call stack, immediate window give you different information about the program. Your changes will have effect on the current execution. Conditional breakpoints have a hit count and when hit condition.

We now discuss XML. XML or extensible Markup Language is widely used for exchanging data. Its readable for both humans and machines. Its a stricter version of HTML. Its made of tags, attributes, and values.

```
<users>
<user name="John Doe" age="42" />
<user name="Jane Doe" age="39" />
</users >
```

And here is a larger XML file.

```
<gesmes:Envelope xmlns:gesmes="http://www.gesmes.org/xml/2002-08-01"
xmlns="http://www.ecb.int/vocabulary/2002-08-01/eurofxref">
 <gesmes:subject>Reference rates</gesmes:subject>
 <gesmes:Sender>
  <gesmes:name>European Central Bank</gesmes:name>
 </gesmes:Sender>
 <Cube>
  <Cube time="2012-12-18">
   <Cube currency="USD" rate="1.3178"/>
   <Cube currency="JPY" rate="110.53"/>
   <Cube currency="BGN" rate="1.9558"/>
   <Cube currency="CZK" rate="25.200"/>
   <Cube currency="DKK" rate="7.4603"/>
   <Cube currency="GBP" rate="0.81280"/>
   <Cube currency="HUF" rate="288.40"/>
   <Cube currency="LTL" rate="3.4528"/>
   <Cube currency="LVL" rate="0.6961"/>
```


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```
   <Cube currency="PLN" rate="4.0928"/>
   <Cube currency="RON" rate="4.4700"/>
   <Cube currency="SEK" rate="8.7378"/>
   <Cube currency="CHF" rate="1.2080"/>
   <Cube currency="NOK" rate="7.3850"/>
   <Cube currency="HRK" rate="7.5380"/>
   <Cube currency="RUB" rate="40.6850"/>
   <Cube currency="TRY" rate="2.3476"/>
   <Cube currency="AUD" rate="1.2512"/>
   <Cube currency="BRL" rate="2.7595"/>
   <Cube currency="CAD" rate="1.2972"/>
   <Cube currency="CNY" rate="8.2079"/>
   <Cube currency="HKD" rate="10.2131"/>
   <Cube currency="IDR" rate="12707.71"/>
   <Cube currency="ILS" rate="4.9615"/>
   <Cube currency="INR" rate="72.2880"/>
   <Cube currency="KRW" rate="1413.30"/>
   <Cube currency="MXN" rate="16.7795"/>
   <Cube currency="MYR" rate="4.0236"/>
   <Cube currency="NZD" rate="1.5660"/>
   <Cube currency="PHP" rate="54.068"/>
   <Cube currency="SGD" rate="1.6053"/>
   <Cube currency="THB" rate="40.285"/>
   <Cube currency="ZAR" rate="11.2733"/>
  </Cube>
 </Cube>
</gesmes:Envelope>
```

We will now learn to read XML with the xmlreader class. There are two methods to read XML document. Using xmldocument and xmlreader. Xmldocuments reads entire document in memory, Let's you go forward, backward, even apply xpath searches on it. Xmlreader is fast, uses less memory and provides one element at a time.

```
Using System;
Using System.Text;
Using System.Xml;

Namespace parsingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmlreader xmlreader =
xmlreader.Create("http://www.ecb.int/stats/eurofxref/eurofxref-daily.xml");
            While (xmlreader.Read())
            {
                If ((xmlreader.nodetype == xmlnodetype.Element) && (xmlreader.Name ==
"Cube"))
                {
                    If (xmlreader.hasattributes)
                        Console.writeline(xmlreader.getattribute("currency") + ": " +
xmlreader.getattribute("rate"));
                }
            }
            Console.readkey();
        }
```


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```
    }
}
```

Using xmldocument is easier. No order is required. Documentelement is the root element and childnodes is the set of children of any node.

```
Namespace  parsingxml
{
Class Program
{
Static void Main(string[] args)
{
Xmldocument  xmldoc = new xmldocument();
Xmldoc.Load("http://www.ecb.int/stats/eurofxref/eurofxref-daily.xml");
Foreach (xmlnode xmlnode in xmldoc.documentelement.childnodes[2].childnodes[0].Child
Console.writeline(xmlnode.Attributes["currency"].Value + ": "+ xmlnode.Attribut
Console.readkey();
}
}
}
```

An xmlnode is derived from xmlelement and contains Name, innertext, innerxml, outerxml, and Attributes. Xpath is a cross-platform Xml Query language. We will look at basic examples only. We will see xmldocument methods that take xpath queries and return xmlnode(s) in particular selectsinglenode and selectnodes.

We will use the RSS feed from CNN, located at http://rss.cnn.com/rss/edition world.rss. It has a hierarchy of rss containing channel containing item.

```
Using System;
Using System.Text;
Using System.Xml;

Namespace parsingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmldocument xmldoc = new xmldocument();
            Xmldoc.Load("http://rss.cnn.com/rss/edition_world.rss");
            Xmlnode titlenode = xmldoc.selectsinglenode("//rss/channel/title");
            If (titlenode != null)
                Console.writeline(titlenode.innertext);
            Console.readkey();
        }
    }
}
```

Another example with selectnodes instead of selectsinglenode.

```
Using System;
Using System.Text;
Using System.Xml;

Namespace parsingxml
{
    Class Program
    {
```


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```
        Static void Main(string[] args)
        {
            Xmldocument xmldoc = new xmldocument();
            Xmldoc.Load("http://rss.cnn.com/rss/edition_world.rss");
            Xmlnodelist itemnodes = xmldoc.selectnodes("//rss/channel/item");
            Foreach (xmlnode itemnode in itemnodes)
            {
                Xmlnode titlenode = itemnode.selectsinglenode("title");
                Xmlnode datenode = itemnode.selectsinglenode("pubdate");
                If ((titlenode != null) && (datenode != null))
                    Console.writeline(datenode.innertext + ": " + titlenode.innertext);
            }
            Console.readkey();
        }
    }

}
```

If we only need title:

```
Using System;
Using System.Text;
Using System.Xml; 4
Namespace  parsingxml
{
Class Program
{
Static void Main(string[] args)
{
Xmldocument  xmldoc = new xmldocument();
Xmldoc.Load("http://rss.cnn.com/rss/edition_world.rss");
Xmlnodelist  itemnodes= xmldoc.selectnodes("//rss/channel/item");
Foreach (xmlnode  itemnode in itemnodes)
{
Xmlnode  titlenode= itemnode.selectsinglenode("title");
Xmlnode  datenode = itemnode.selectsinglenode("pubdate");
If((titlenode != null) && (datenode != null))
Console.writeline(datenode.innertext + ": "+ titlenode.innertext);
}
Console.readkey();
}
}
}
```

Let's write some XML using xmlwriter.

```
Using System;
Using System.Text;
Using System.Xml;

Namespace writingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmlwriter xmlwriter = xmlwriter.Create("test.xml");

            Xmlwriter.writestartdocument();
```


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```
            Xmlwriter.writestartelement("users");

            Xmlwriter.writestartelement("user");
            Xmlwriter.writeattributestring("age", "42");
            Xmlwriter.writestring("John Doe");
            Xmlwriter.writeendelement();

            Xmlwriter.writestartelement("user");
            Xmlwriter.writeattributestring("age", "39");
            Xmlwriter.writestring("Jane Doe");

            Xmlwriter.writeenddocument();
            Xmlwriter.Close();
        }
    }
}
```

The above code will create the following XML:

```
<users>
  <user age="42">John Doe</user>
  <user age="39">Jane Doe</user>
</users>
```

Let's now see writing with xmldocument (especially for updates).

```
Using System;
Using System.Text;
Using System.Xml;
Using System.Xml.Serialization;

Namespace writingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmldocument xmldoc = new xmldocument();
            Xmlnode rootnode = xmldoc.createelement("users");
            Xmldoc.appendchild(rootnode);

            Xmlnode usernode = xmldoc.createelement("user");
            Xmlattribute attribute = xmldoc.createattribute("age");
            Attribute.Value = "42";
            Usernode.Attributes.Append(attribute);
            Usernode.innertext = "John Doe";
            Rootnode.appendchild(usernode);

            Usernode = xmldoc.createelement("user");
            Attribute = xmldoc.createattribute("age");
            Attribute.Value = "39";
            Usernode.Attributes.Append(attribute);
            Usernode.innertext = "Jane Doe";
            Rootnode.appendchild(usernode);

            Xmldoc.Save("test-doc.xml");
        }
    }
}
```


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Updating is hard with a combination of xmlreader and xmlwriter.

```
Using System;
Using System.Text;
Using System.Xml;
Using System.Xml.Serialization;

Namespace writingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmldocument xmldoc = new xmldocument();
            Xmldoc.Load("test-doc.xml");
            Xmlnodelist usernodes = xmldoc.selectnodes("//users/user");
            Foreach (xmlnode usernode in usernodes)
            {
                Int age = int.Parse(usernode.Attributes["age"].Value);
                Usernode.Attributes["age"].Value = (age + 1).tostring();
            }
            Xmldoc.Save("test-doc.xml");
        }
    }
}
```


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Chapter 11

Lecture 11

Let's start discussing extension methods.

```
Public class myutils
{
Public static bool isnumeric(string s)
{
Float output;
Return float.tryparse(s, out output);
}
}
String test = "4";
If(myutils.isnumeric(test))
Console.writeline("Yes");
Else

Console.writeline("No");
```

We can simplify the above example with an extension method.

```
Public class myutils
{
Public static bool isnumeric(string s)
{
Float output;
Return float.tryparse(s, out output);
}
}

String test = "4";
If(myutils.isnumeric(test))
Console.writeline("Yes");
Else

Console.writeline("No");
```

Let's now see how we can handle files.

```
Using System;
Using System.IO;

Namespace filehandlingarticleapp
{
    Class Program
    {
        Static void Main(string[] args)
        {
            If(File.Exists("test.txt"))
            {
                String content = File.readalltext("test.txt");
                Console.writeline("Current content of file:");
                Console.writeline(content);
            }
            Console.writeline("Please enter new content for the file:");
```


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```
            String newcontent = Console.readline();
            File.writealltext("test.txt", newcontent);
        }
    }
}
```

And another example.

```
Using System;
Using System.IO;

Namespace filehandlingarticleapp
{
    Class Program
    {
        Static void Main(string[] args)
        {
            If(File.Exists("test.txt"))
            {
                String content = File.readalltext("test.txt");
                Console.writeline("Current content of file:");
                Console.writeline(content);
            }
            Console.writeline("Please enter new content for the file - type exit and
press enter to finish editing:");
            String newcontent = Console.readline();
            While (newcontent != "exit")
            {
                File.appendalltext("test.txt", newcontent + Environment.newline);
                Newcontent = Console.readline();
            }
        }
    }
}
```

If using keyword is not used, manually call the Close method of the stream.

```
Console.writeline("Please enter new content for the file - type exit and press enter to
finish”);
Using (streamwriter sw= new streamwriter("test.txt"))
{
String  newcontent = Console.readline();
While (newcontent != "exit")
{
Sw.Write(newcontent+ Environment.newline);
Newcontent = Console.readline();
}
}
```

We can delete a file like this.

```
Console.writeline("Please enter new content for the file  - type exit and press enter to
finish”);
Using (streamwriter sw= new streamwriter("test.txt"))
{
String  newcontent = Console.readline();
While (newcontent != "exit")
{
Sw.Write(newcontent+ Environment.newline);
Newcontent = Console.readline();
}
```


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```
}
```

We can delete a directory like this.

```
Console.writeline("Please enter new content for the file -type exit and press enter to
finish”);
Using (streamwriter sw= new streamwriter("test.txt"))
{
String  newcontent = Console.readline();
While (newcontent != "exit")
{
Sw.Write(newcontent+ Environment.newline);
Newcontent = Console.readline();
}
}
```

We can rename a file like this.

```
If (File.Exists("test.txt"))
{
Console.writeline("Please enter a new name for this file:");
String  newfilename= Console.readline();
If (newfilename != String.Empty)
{
File.Move("test.txt", newfilename);
If (File.Exists(newfilename))
{
Console.writeline("The file was renamed to " + newfilename);
Console.readkey();
}
}
}
```

We can rename a directory like this.

```
If (Directory.Exists("testdir"))
{
Console.writeline("Please enter a new name for this directory:");
String  newdirname = Console.readline();
If (newdirname != String.Empty)
{
Directory.Move("testdir", newdirname);
If (Directory.Exists(newdirname))
{
Console.writeline("The directory was renamed to "+ newdirname);
Console.readkey();
}
}
}
```

We can create a new directory like this.

```
Console.writeline("Please enter a name for the new directory:");
String  newdirname = Console.readline();
If (newdirname != String.Empty)
{
Directory.createdirectory(newdirname);
If (Directory.Exists(newdirname))
{
Console.writeline("The directory was created!");
```


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```
Console.readkey();
}
}
```

A fileinfo object gives you information about a file.

```
Static void Main(string[] args)
{
Fileinfo fi = new fileinfo(System.Reflection.Assembly.getexecutingassembly().Location);
If(fi!= null)
Console.writeline(String.Format("Information about file: {0}, {1} bytes, last modified o
Console.readkey();
}
```

Directoryinfo gives you information about a directory.

```
Directoryinfo di= new
directoryinfo(Path.getdirectoryname(System.Reflection.Assembly.getexecuti
If(di!= null)
{
Fileinfo[] subfiles= di.getfiles();
If(subfiles.Length> 0)
{
Console.writeline("Files:");
Foreach (fileinfo  subfile in subfiles)
{
Console.writeline("" + subfile.Name + " (" + subfile.Length + " bytes)");
}
}
Console.readkey();
}
```

Finding directories as well is as easy.

```
Directoryinfo[] subdirs = di.getdirectories();
If(subdirs.Length > 0)
{
Console.writeline("Directories:");
Foreach (directoryinfo  subdir in subdirs)
{
Console.writeline("" + subdir.Name);
}
}
```

We can use reflection to get information about types in our program.

```
Using System;
Using System.Collections.Generic;
Using System.Text;
Using System.Reflection;

Namespace  reflectiontest
{
Class Program
{
Static void Main(string[] args)
{
String test = "test";
Console.writeline(test.gettype().fullname);
Console.writeline(typeof(Int32).fullname);
Console.readkey();
}
}
```


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```
}
```

Here is another example.

```
Using System;
Using System.Collections.Generic;
Using System.Text;
Using System.Reflection;

Namespace reflectiontest
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Assembly assembly = Assembly.getexecutingassembly();
            Type[] assemblytypes = assembly.gettypes();
            Foreach(Type t in assemblytypes)
                Console.writeline(t.Name);
            Console.readkey();
        }
    }

    Class dummyclass
    {
        //Just here to make the output a tad less boring :)
    }
}
```

Let's see how methods are shown in reflection.

```
Using System;
Using System.Collections.Generic;
Using System.Text;
Using System.Reflection;

Namespace reflectiontest
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Type testtype = typeof(testclass);
            Constructorinfo ctor = testtype.getconstructor(System.Type.emptytypes);
            If (ctor != null)
            {
                Object instance = ctor.Invoke(null);
                Methodinfo methodinfo = testtype.getmethod("testmethod");
                Console.writeline(methodinfo.Invoke(instance, new object[] { 10 }));
            }
            Console.readkey();
        }
    }

    Public class testclass
    {
        Private int testvalue = 42;

        Public int testmethod(int numbertoadd)
        {
```


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```
            Return this.testvalue + numbertoadd;
        }
    }
}
```

We now have a more detailed example. We will now build a complete profile save load system with reflection.

```
Using System;
Using System.IO;
Using System.Reflection;


Class Program
{
    Public class Person
    {
        Public void Load()
        {
            If (File.Exists("settings.dat"))
            {
                Type type = this.gettype();

                String propertyname, value;
                String[] temp;
                Char[] splitchars = new char[] { '|' };
                Propertyinfo propertyinfo;

                String[] settings = File.readalllines("settings.dat");
                Foreach (string s in settings)
                {
                    Temp = s.Split(splitchars);
                    If (temp.Length == 2)
                    {
                        Propertyname = temp[0];
                        Value = temp[1];
                        Propertyinfo = type.getproperty(propertyname);
                        If (propertyinfo != null)
                            This.setproperty(propertyinfo, value);
                    }
                }
            }
        }

        Public void Save()
        {
            Type type = this.gettype();
            Propertyinfo[] properties = type.getproperties();
            Textwriter tw = new streamwriter("settings.dat");
            Foreach (propertyinfo propertyinfo in properties)
                Tw.writeline(propertyinfo.Name + "|" + propertyinfo.getvalue(this,
null));
            Tw.Close();
        }

        Public void setproperty(propertyinfo propertyinfo, object value)
        {
```


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```
            Switch (propertyinfo.propertytype.Name)
            {
                Case "Int32":
                    Propertyinfo.setvalue(this, Convert.toint32(value), null);
                    Break;
                Case "String":
                    Propertyinfo.setvalue(this, value.tostring(), null);
                    Break;
            }
        }

        Public int Age { get; set; }
        Public string Name { get; set; }
    }

    Static void Main(string[] args)
    {
        Person person = new Person();
        Person.Load();
        If ((person.Age > 0) && (person.Name != String.Empty))
        {
            Console.writeline("Hi " + person.Name + " - you are " + person.Age + " years
old!");
        }
        Else
        {
            Console.writeline("I don't seem to know much about you. Please enter the
following information:");
            Type type = typeof(Person);
            Propertyinfo[] properties = type.getproperties();
            Foreach (propertyinfo propertyinfo in properties)
            {
                Console.writeline(propertyinfo.Name + ":");
                Person.setproperty(propertyinfo, Console.readline());
            }
            Person.Save();
            Console.writeline("Thank you! I have saved your information for next time.");
        }
        Console.readkey();
    }
}
```


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Chapter 12

Lecture 12

We will start discussing WPF (Windows Presentation Foundations) now. It was publicly announced in 2003 (codenamed Avalon). WPF 4 was released in April 2010. It has a steep learning curve. Code has to be written in many places. There are multiple ways to do a particular task.

WPF enables polished user interfaces which are getting a lot of attention. It enables rapid iterations and major interface changes throughout the development process. It allows to keep user interface description and implementation separate. Developers can create an "ugly" application which designers can re-theme. Win32 style of programming makes such re-theming difficult. The code to re-paint the user interface is mixed up with program logic. GDI was an earlier user interface library introduced in windows 1.0 in 1985. Opengl was a leap ahead introduced in the 90's with directx coming in 95 and directx 2 in 96. GDI+ is a newer user interface library based on directx. It is also used behind Xbox graphics. Next was Windows Forms which is the primary way of user interface design in C#. XNA comes with managed library for directx and is great for game development (.net/com interoperability not required). A simple example is drawing bitmaps on buttons which can be efficiently done using GDI.

The highlights of WPF are 1) broad integration (2D, 3D, video, speech libraries etc.) 2) Resolution Independence with WPF giving emphasis on vector graphics 3) Hardware accelration as it is based on Direct3D but it can work using software pipeline also if Direct3D hardware is not available. 4) Declrative progrmaming using extensible Application Markup Language (XAML; pronounced "Zammel"). Custom attribute and configuration files were always there but XAML is very rich. 5) Rich composition and customization e.g. You can create a combobox filled with animated

Buttons or a Menu filled with live video clips! And it is quite easy to skin applications.

In short, WPF aims to combine the best attributes of systems such as directx (3D and hardware acceleration), Windows Forms (developer productivity), Adobe Flash (powerful animation support) and HTML (declarative markup). The first release in November 2006 was WPF 3.0 because it shipped as part of the .NET Framework 3.0. WPF 3.5 came an year later. Next version as part of .NET 3.5 SP1 came in August 2008. WPF Toolkit released in Aug 2008 was experimental. The toolkit has quick releases. Regarding tool support, WPF extensions for Visual Studio 2005 came a few months after the first WPF release and a public release of expression blend. Now, Visual Studio 2012 is a first class WPF devevelopment environment. Its mostly re-written using WPF and expression blend is 100% WPF and is great for designing and prototyping WPF apps.

New things that came in WPF 3.5/3.5SP1 include Interactive3D with 2d elements in 3d scenes, first class interoperability with directx, Better data binding using XLINQ and better validation and debugging which reduces code, Better special effects, High performance custom drawing, Text improvements, enhancements to Partial-trust apps, improved deployment, and improved performance.

Things that came with WPF 4.0 include multi-touch support — compatible with surface API v2, win7 support like jump lists, new common dialogs etc., new controls like datagrid, calendar etc, easing animation functions (bounce, elastic), enhanced styling with visual state manager, improved layout on pixel boundaries, non-blurry text but some limitations so must opt-in, deployment improvements, and performance improvements.

Silverlight in comparison is a light-weight version for web. It chose to follow WPF appraoch. First released in 2007 and in April 2010 4th version was released near WPF 4. There is often confusion when to use one or

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the other. Both can run in and outside the web. Silverlight is ostly a subset of WPF but there are some incompatibilities. Should I use full .net or partial .net and should I have the ability to run on other devices e.g. Macs. Ideally common codebase should work for both but today best case is using #ifdefs to handle incompatibilities.

XAML is primarily used to describe interfaces in WPF and Silverlight. It is also used to express activities and configurations in workflow foundation (WF) and windows communication foundation (WCF). Its a common language for programmers and other experts e.g. UI design experts. Field specific development tools can be made. Field experts are graphic designers. They can use a design tool such as expression blend. Other than co-ordinating with designers, XAML is good for a concise way to represent UI or hierarchies of objects, encourages separation of front-end and back-end, tool support with copy and paste, used by all WPF tools.

XAML is xml with a set of rules. Its a declrative programming language for creating and initializing objects. Itsa way to use .net apis.

Comparisons with SVG, HTML etc are misguided. It has few keywords, not much elemetns. It doesn't make sense without .net, like C# doesnt make sense without .net. Microsoft formalized XAML vocabluries, e.g. WPF XAML vocabluary.

Online specifications for XAML object specification language, and WPF and Silverlight vocabluaries are available. XAML is used by other technologies as well although originally it was designed for WPF. Also using XAML in WPF projects is optional. Everything can be done in procedural code as well (althought its rare to find it done that way).

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Chapter 13

Lecture 13

XAML specification defines rules that map .NET namespaces, types, properties, and events into XML namespaces, elements, and attributes. Let's see XAML and equivalent C#.

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK"/>
```

And the corresponding C# code is

```
System.Windows.Controls.Button b= new System.Windows.Controls.Button();

B.Content= "OK";
```

Declaring an XML element in XAML (known as an object element) is equivalent to instantiating the corresponding .NET object via a default constructor. Setting an attribute on the object element is equivalent to setting a property of the same name (called a property attribute) or hooking up an event handler of the same name (called an event attribute).

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK"
Click="button_Click"/ and the corresponding C# code is
System.Windows.Controls.Button b=new System.Windows.Controls.Button();
B.Click += new System.Windows.routedeventhandler(button_Click);

   B.Content = "OK";
```

XAML can no longer run standalone in the browser becuase of the butto click method — event handlers are attached before properties are set.

Let's discuss namespaces. Mapping to the namespace used above and other WPF namespaces is hard-coded in-side the WPF assemblies. Its just an arbitrary string like any namespace. The root object element in XAML must specify at least one XML namespace that is used to qualify itself and any child lements. Additional XML namespaces (on the root or on children) must be given a distinct prefix to be used on any identifiers from that namespace. E.g, typically a second namespace with the prefix x (denoted by using xmlns:x instead of just xmlns): xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"

This is the XAML language namespace, which maps to types in the System.Windows.Markup namespace but also defines some special directives for the XAML compiler or parser.

All of the following are mapped with http://schemas.microsoft.com/winfx/2006/xaml/presentation.

```
System.Windows
System.Windows.Automation
System.Windows.Controls
System.Windows.Controls.Primitives.System.Windows.Data
System.Windows.Documents
System.Windows.Forms.Integration
System.Windows.Ink
System.Windows.Input
System.Windows.Media
System.Windows.Media.Animation
System.Windows.Media.Effects
System.Windows.Media.Imaging
```


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```
System.Windows.Media.Media3D
System.Windows.Media.textformatting.System.Windows.Navigation
System.Windows.Shapes
System.Windows.Shell
```

Property elements in XAML enable rich composition.

```
System.Windows.Controls.Button b= new System.Windows.Controls.Button();
System.Windows.Shapes.Rectangle r = new System.Windows.Shapes.Rectangle();
R.Width = 40;
R.Height = 40;
R.Fill= System.Windows.Media.Brushes.Black;
B.Content= r; // Make the square the content of the Button
```

And in XAML

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
<Button.Content>
<Rectangle  Height="40" Width="40" Fill="Black"/>
</Button.Content ></Button >
```

The period distinguishes property elements from object elements. It doesn't have attributes except x:uid for localization. The syntax can be used for simple properties as well. The following two are equivalent.

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK"
Backgroun
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
<Button.Content>OK</Button.Content >
<Button.Background>White</Button.Background >
</Button >
```

Let's discuss type converters.

```
System.Windows.Controls.Button b= new System.Windows.Controls.Button();
B.Content= "OK";
B.Background =System.Windows.Media.Brushes.White;
```

WPF provides type converters for many common data types: Brush, Color, fontweight, Point, and so on. Classes deriving from typeconverter (brushconverter, colorconverter, and so on) convert from string to the corresponding types. You can also write your own type converters for custom data types.

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK">
<Button.Background>
<solidcolorbrush  Color="White"/>
</Button.Background >
</Button>
```

Without the type converter for color

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK">
<Button.Background>
<solidcolorbrush>
<solidcolorbrush.Color>
<Color A="255" R="255" G="255" B="255"/>
</solidcolorbrush.Color>
</solidcolorbrush>
</Button.Background>
</Button>
```

Even that requires a type converter for Byte.

```
System.Windows.Controls. Button b = new System.Windows.Controls. Button ();
B.Content = "OK";
```


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```
B.Background = (Brush) System.componentmodel. Typedescriptor. Getconverter (
Typeof (Brush)). Converterfrominvariantstring ("White");
```

Constants in strings cause problems with run time exception not caught at compile time. Althought for xaml Visual Studio will check it at compile time. Let's see how to find the type converter for a type.

Let's discuss markup extensions now. Markup extensions, like type converters, enable you to extend the expres-siveness of XAML. Both can evaluate a string attribute value at runtime (except for a few built-in markup extensions for performance reasons) and produce an appropriate object based on the string. As with type converters, WPF ships with several markup extensions built in.

Unlike type converters, however, markup extensions are invoked from XAML with explicit and consistent syntax. For this reason, using markup extensions is a preferred approach for extending XAML.

For example, if you want to set a control's background to a fancy gradient brush with a simple string value, you can write a custom markup extension that supports it even though the built-in brushconverter does not. Whenever an attribute value is enclosed in curly braces ({}), the XAML compiler/parser treats it as a markup extension value rather than a literal string (or something that needs to be type-converted). Here are few examples:

```
<Button xmlns=''http://schemas.microsoft.com/winfx/2006/xaml/presentation''
Content=''{}{This is not a markup extension!}''/>
<Button xmlns=''http://schemas.microsoft.com/winfx/2006/xaml/presentation''>
<Button.Content>{This is not a markup extension!}</Button.Content> </Button>
```

Because markup extensions are just classes with default constructors, they can be used with property element syntax.

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
            Xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
            Background = "{x:Null}"
            Height = "{x:Static systemparameters.iconheight}"
            Content = "{Binding Path=Height, relativesource={relativesource
Mode=Self}}"/>
```

It works because positional parameters have corresponding property value. Markup extension has the real code to be executed.

An object element can have three types of children: a value for a content property, collection items, or a value that can be type-converted to the object element. Designated property is the content property.

```
<Button xmlns=''http://schemas.microsoft.com/winfx/2006/xaml/presentation''
Content=''OK''/>
```

Could be rewritten as follows:

```
<Button xmlns=''http://schemas.microsoft.com/winfx/2006/xaml/presentation''> OK
</Button>
```

Also

```
<Button xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”>
<Button.Content>
<Rectangle Height=”40” Width=”40” Fill=”Black”/> </Button.Content>
</Button>
```

Could be rewritten as follows:

```
<Button xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”> <Rectangle
Height=”40” Width=”40” Fill=”Black”/>
```


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```
</Button>
```

There is no requirement that the content property must actually be called Content; classes such as combobox, listbox, and tabcontrol (also in the System.Windows.Controls namespace) use their Items property as the content property. Its designated with a custom attribute.

XAML enables you to add items to the two main types of collections that support indexing: lists and dictionaries.

```
<listbox xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”>
<listbox.Items>
<listboxitem Content=”Item 1”/>
<listboxitem Content=”Item 2”/> </listbox.Items>
</listbox>
```

Its equivalent to:

```
System.Windows.Controls.listbox listbox= new System.Windows.Controls.listbox();
System.Windows.new System.Windows.Controls.listboxitem();
System.Windows.Controls.listboxitem item2= new System.Windows.Controls.listboxitem();
Item1.Content=Item 1 ;
Item2.Content= Item 2;
```

OR

```
<listbox xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”> <listboxitem
Content=”Item 1”/>
<listboxitem Content=”Item 2”/>
</listbox>
```

Let’s see how to use a dictionary.

```
<resourcedictionary xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”
xmlns:x=”http://schemas.microsoft.com/winfx/2006/xaml”>
<Color x:Key=”1” A=”255” R=”255” G=”255” B=”255”/>
<Color x:Key=”2” A=”0” R=”0” G=”0” B=”0”/>
</resourcedictionary>
```

We used the XAML Key keyword. The equivalent code is:

```
System.Windows. Resourcedictionary d = new System.Windows. Resourcedictionary ();
System.Windows.Media. Color color1 = new System.Windows.Media. Color ();
System.Windows.Media. Color color2 = new System.Windows.Media. Color ();
Color1.A = 255; color1.R = 255; color1.G = 255; color1.B = 255;
Color2.A = 0; color1.R = 0; color1.G = 0; color1.B = 0;
D.Add ("1", color1);
D.Add ("2", color2);
```


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Chapter 14

Lecture 14

<solidcolorbrush>White</solidcolorbrush>

As discussed in the last lecture, this is equivalent to the following:

<solidcolorbrush Color="White"/>

It works because a typeconverter exists that converts string to solidcolorbrush, although no designated content property exists.
<Brush>White</Brush>

In fact, the above also works even although Brush is abstract. It works because of a type converter can convert a string to a solidcolorbrush.

XAML works with classes designated with marked attributes. Usually these classes have default ctors and useful properties. But what about other classes not designed for XAML. E.g. Hashtable.

```
System.Collections.Hashtable h
H.Add("key1",  7);
H.Add("key2",  23);
```

Heres how it can be represented in XAML:

```
<Collections: Hashtable
       Xmlns: collections = "clr-namespace: System.Collections; assembly = mscorlib"
       Xmlns: sys = "clr-namespace: System; assembly = mscorlib"
       Xmlns: x = "
<Sys: Int32 x: Key = "key1"> 7 </ sys: Int32>
<Sys: Int32 x: Key = "key2"> 23 </ sys: Int32>
</ Collections: Hashtable>
```

Let's study XAML child element rules. If the type implements ilist, call ilist.Add for each child. Otherwise, if the type implements idictionary, call idictionary.Add for each child, using the x:Key attribute value for the key and the element for the value. (Although XAML2009 checks idictionary before ilist and supports other collection interfaces, as described earlier.)

Otherwise, if the parent supports a content property (indicated by System.Windows.Markup.content property attribute) and the type of the child is compatible with that property, treat the child as its value. Otherwise, if the child is plain text and a type converter exists to transform the child into the parent type (and no properties are set on the parent element), treat the child as the input to the type converter and use the output as the parent object instance. Otherwise, treat it as unknown content and potentially raise an error

We can mix XAML and procedural code. Let's try this with xamlreader and xamlwriter.

```
Window window= null;
Using (filestream fs = new filestream("mywindow.xaml", filemode.Open, fileaccess.Read))
{
//Get the root element, which we know is a Window
Window= (Window)xamlreader.Load(fs);
}
```

Let's see how to grab the OK button by walking the children (with hard-coded knowledge!)

```
Stackpanel panel= (stackpanel)window.Content;
```


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```
Button okbutton = (Button)panel.Children[4];
```

For easier fetching, we can name XAML elements.

```
 <Button x:Name="okbutton">OK</Button>
```

Let's grab the OK button, knowing only its name

```
 Button okbutton=(Button)window.findname("okbutton");
```

Classes that already have a name property designate it as the special name using a custom attribute.

Let's see how to use named properties. Let's first use a Binding markup extension.

```
<stackpanel
  Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
  <Label Target="{Binding elementname=box}" Content="Enter _text:"/>
  <textbox Name="box"/>
</stackpanel>
```

It gives it focus when the Label's access key is pressed.) WPF 4 includes a new, simpler markup extension which works at parse time.

```
<stackpanel
  Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
  Xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"> <Label
  Target="{x:Reference box}" Content="Enter _text:"/> <textbox
  Name="box"/>
  </stackpanel>
```

If a relevant property is marked with the System.Windows.Markup.namereferenceconverter type converter:

```
  <stackpanel
  Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
  <Label Target="box" Content="Enter _text:"/> <textbox Name="box"/>
  </stackpanel>
```

Compiling xaml involves 3 steps — converting a XAML file into a special binaryformat, embedding the converted content asa binary resource in the assembly being built,and performing the plumbing that connects XAML with procedural code automatically. C# and VB have best support in typical case, the first step is specifying a subclass for the root element in a XAML file. Use the Class keyword for that.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
Xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
X:Class="mynamespace.mywindow"></Window >
Namespace  mynamespace
{
Partial class mywindow: Window
{
Public mywindow()
{
// Necessary to call in order to load XAML-defined content!
Initializecomponent();

}
Any other members can go here...
}
}
```

This is often refered as the code-behind file. We define event handlers here. The partial keyword is important. When we add new items to certain WPF projects, Visual Studio does all this automatically for us.

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Chapter 15

Lecture 15

BAML is binary application markup language. It just a compressed representation of XAML. There is even a BAML reader available. Earlier there was CAML which stands for compiled application markup language but its not used now.

Some glue code is generated when we use x:Class. Its kind of same as loading and parsing the XAML file. We must call initializecomponent and we can refer named elements like class members.

Procedural code can even be written inside the XAML file.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml" x:Class="mynamespace.mywindow">
    <Button Click="button_Click">OK</Button>
    <x:Code>
        <![CDATA[
            Void button_Click(object sender, routedeventargs e)
            {
                This.Close();
            }
        ]]>
    </x:Code>
</Window>
```

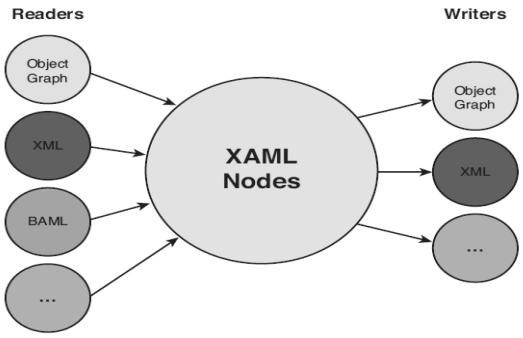
We must avoid]]. If we have to use them, we need &lt &amp etc. There is no good reason to embed code. Internally its build into .cs file by build system. BAML can be converted back to XAML.

```
System.Uri uri = new System.Uri("/wpfapplication1;component/mywindow.xaml",
System.urikind.Relative);
Window window= (Window)Application.loadcomponent(uri);
String xaml = xamlwriter.Save(window);
```

There are different loading mechanisms. We can load from resource, identified by original xaml file, or actually integrated baml loaded.

Key features of XAML 2009 include full generics support using typearguments, dictionary keys of any type, builtin system data types, creating objects with non-default ctors, getting instances via factory methods, event handlers can be markup extensions returning delegates, define additional members and properties.

System.Xaml.xamlxmlreader etc. Can be extended with readers and writers for a lot of formats. It abstracts differences in Xaml formats like accessing a content property, property element, or a property attribute.

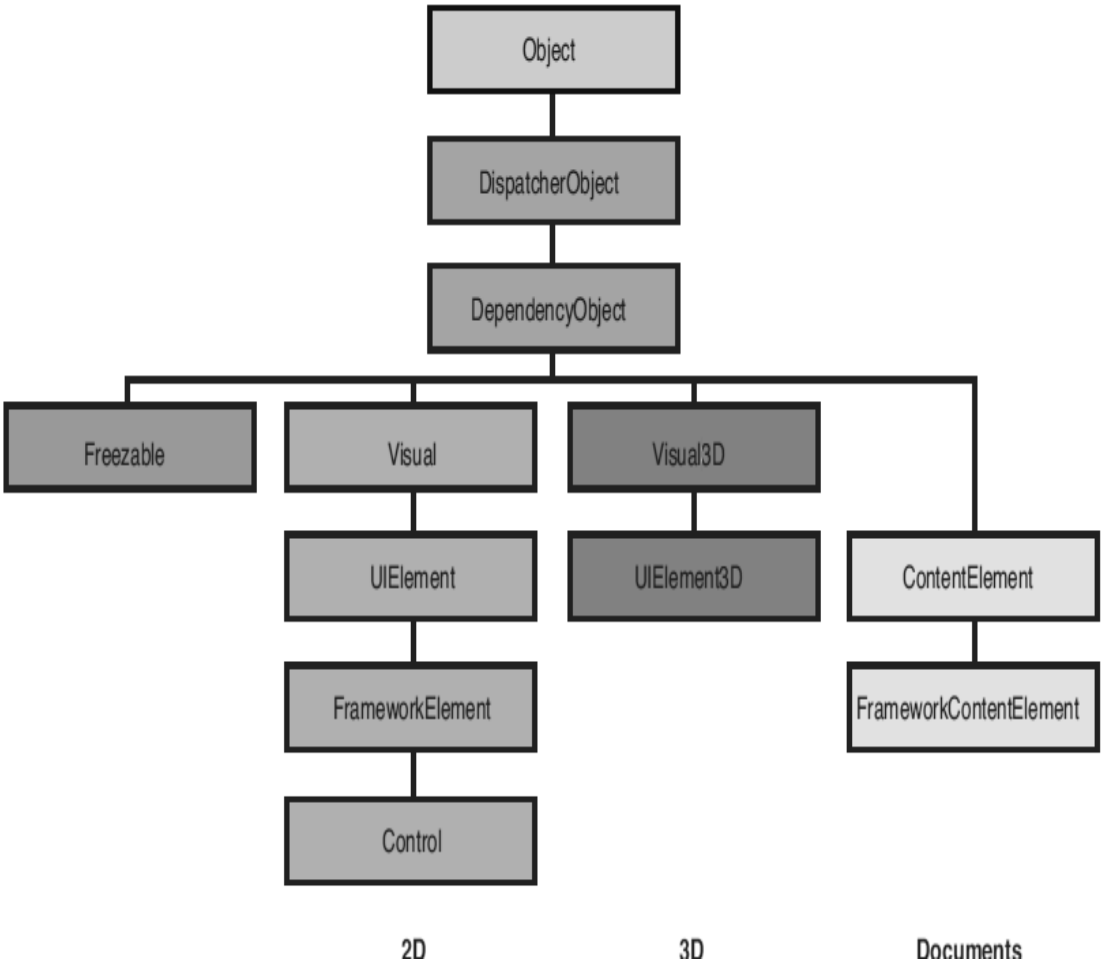


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XAML 2006 keywords include x:asyncrecords - Controls the size of asynchronous XAML-loading chunks, x:Class, x:classmodifier - visibility, public by default, x:Code, x:connectionid - not for public use, x:fieldmodifier - field visibility (internal def), x:Key, x:Name, x:Shared - =false means same resource instance not shared, x:subclass - only needed when partial not supported, x:Synchronous mode - xaml loaded in async mode, x:typearguments - used only with root with x:Class in xaml2006, x:Uid - represents system.uri, and x:xdata - data opaque for xaml parser.

Markup extensions often confused as keywords are x:Array - use with x:Type to define type of array, x:Null, x:Reference - to named element, x:Static - static property field, and x:Type - just like typeof operator. In summary, we can convert xaml into C#, often apis are optimized for xaml and can look clunky in C#, wpf applications have deep hierarchy, small building blocks. Two complaints are that xml too verbose to type and slow to parse (tools and baml fix it partially).

Let's discuss wpf fundamentals now. WPF concepts are above and beyond .net concepts. Its the cause of the steep learning curve of wpf. It has a deep inheritance hierarchy. There are a handful of fundamental classes.



Object is the base class for all .NET classes and the only class in the figure that isn't WPF specific.

Dispatcherobject is the base class meant for any object that wishes to be accessed only on the thread that created it. Most WPF classes derive from dispatcherobject and are therefore inherently thread-unsafe. The Dispatcher part of the name refers to wpfs version of a Win32-like message loop, discussed further in

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Chapter 7, “Structuring and Deploying an Application” of the book.

Dependencyobject is the base class for any object that can support dependency properties, one of the main topics in this chapter.

Freezable is the base class for objects that can be “frozen” into a read-only state for performance reasons. Freezables, once frozen, can be safely shared among multiple threads, unlike all other dispatcherobjects. Frozen objects can never be unfrozen, but you can clone them to create unfrozen copies. Most Freezables are graphics primitives such as brushes, pens, and geometries or animation classes.

Visual is the base class for all objects that have their own 2D visual representation. Visuals are discussed in depth in Chapter 15, “2D Graphics” of the book.

Uielement is the base class for all 2D visual objects with support for routed events, command binding, layout, and focus. These features are discussed in Chapter 5, “Layout with Panels,” and Chapter 6, “Input Events: Keyboard, Mouse, Stylus, and Multi-Touc.” Of the book.

Visual3D is the base class for all objects that have their own 3D visual representation. Visual3Ds are discussed in depth in Chapter 16, “3D Graphics” of the book.

Uielement3d is the base class for all 3D visual objects with support for routed events, command binding, and focus, also discussed in Chapter 16.

Contentelement is a base class similar to uielement but for document-related pieces of content that don’t have rendering behavior on their own. Instead, contentelements are hosted in a Visual-derived class to be rendered onthe screen. Each contentelement often requires multiple Visuals to render correctly (spanning lines, columns, and pages).

Frameworkelement is the base class that adds support for styles, data binding, resources, and a few common mechanisms for Windows-based controls, such as tooltips and context menus.

Frameworkcontentelement is the analog to frameworkelement for content. Chapter 11, “Images, Text, and Other Controls,” of the book examines the frameworkcontentelements in WPF.

Control is the base class for familiar controls such as Button, listbox, and statusbar. Control adds many properties to its frameworkelement base class, such as Foreground, Background, and fontsize, as well as the ability to be completely restyled. Part III, “Controls,” examines wpfs controls in depth.

Let’s discuss logical and visual trees.

Xaml good for UI because of hierarchical nature. Logical tree exists even if there is no xaml. Properties, events, resources are tied to logical trees. Properties propagated down and events can be routed up or down the tree. Its a simplification of whats actually going on when rendered. Visual tree can be thought of as an extension of the logical tree though some things can be dropped as well. Visual tree exposes visual implementation details e.g. A listbox is a border, two scrollbars and more. Only things from visual or visual3d appear in a visual tree. Avoid depending on visual tree in your code.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Title="About
WPF 4 Unleashed" sizetocontent="widthandheight" Background="orangered">

    <stackpanel>
        <Label fontweight="Bold" fontsize="20" Foreground="White">WPF 4 Unleashed
        </Label>
        <Label>© 2010 SAMS Publishing
        </Label>
```


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```
        <Label>Installed Chapters:
        </Label>
        <listbox>
            <listboxitem>Chapter 1</listboxitem>
            <listboxitem>Chapter 2</listboxitem>
        </listbox>
        <stackpanel Orientation="Horizontal" horizontalalignment="Center">
            <Button minwidth="75" Margin="10">Help</Button>
            <Button minwidth="75" Margin="10">OK</Button>
        </stackpanel>
        <statusbar>You have successfully registered this product.</statusbar>
    </stackpanel>
</Window>
```

We can traverse using System.Windows.logicaltreehelper and System.Windows.Media.visualtreehelper. Here is a code-behind file that will print these trees, assuming the gluing code is there i.e. X:Class and xmlns:x.

```
Using System;
Using System.Diagnostics;
Using System.Windows;
Using System.Windows.Media; 5

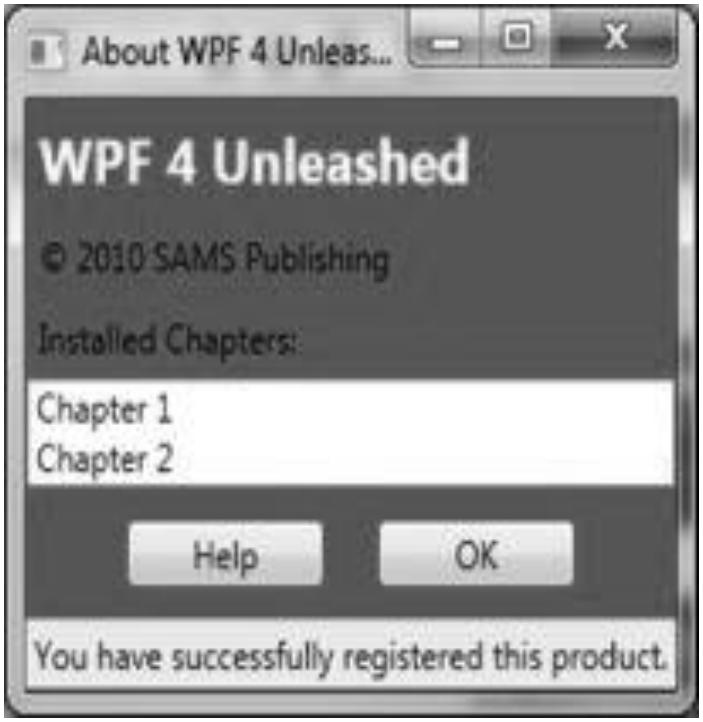
Public partial class aboutdialog  : Window
{
Public  aboutdialog()
{
Initializecomponent();
Printlogicaltree(0, this);
}
Protected  override void oncontentrendered(eventargs e)
{
Base.oncontentrendered(e);
Printvisualtree(0, this);
}
Void printlogicaltree(int depth, object obj) 19 {
// Print the object with preceding  spaces that represent its depth
Debug.writeline(new string(’ ’, depth)+ obj);
// Sometimes leaf nodes  aren  t  dependencyobjects (e.g. Strings)
If (!(obj is dependencyobject))
Return;

// Recursive call for each logical child
Foreach (object child in logicaltreehelper.getchildren( obj as dependencyobject))
Printlogicaltree(depth+ 1, child);
}
Void printvisualtree(int depth, dependencyobject obj)
{
// Print the object with preceding  spaces that represent its depth
Debug.writeline(new string(’ ’, depth)+ obj);
// Recursive call for each visual child
For(int i= 0; i < visualtreehelper.getchildrencount(obj); i++)
Printvisualtree(depth + 1, visualtreehelper.getchild(obj, i));

}

}
```

Visual tree is empty until the dialog box is rendered. Navigating either can be done in instance methods of the elements themselves e.g. Visual class has protected members visualparent, visualchildrencount, and getvisualchild. Frameworkelement and frameworkcontentelement have Parent property and logicalchildren property.



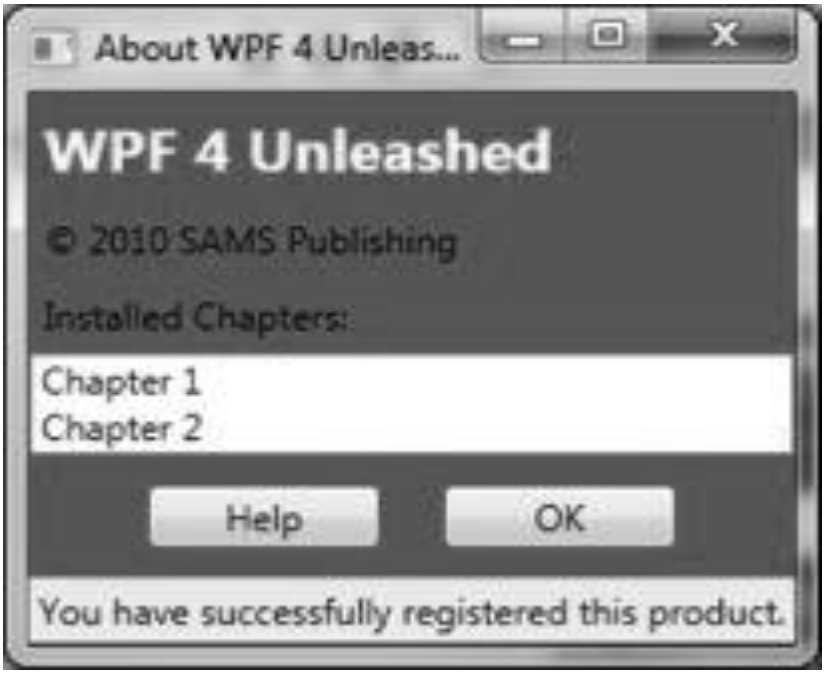
Chapter 16

Lecture 16

Let's discuss dependency properties. They complicates .net types but once you realize the problem it solves, you realize its importance. Dependency properties depends on multiple providers for determining its value at any point in time. These providers can be an animation continuously changing its values. A parent whose property propagates down. Arguably biggest feature is change notification. Motivation is to add rich functionality from declarative markup. Key to declarative-friendly design is heavy use of properties. Button e.g. Has 111 public properties (98 inherited). Properties can be set using xaml, directly or in a design tool, without procedural code. Without the extra work in dependency properties, it would be hard. We looked at implementation of a dependency property and then see its benefits on top of .net properties. Change notification, property value inheritance, and support for multiple providers are key features.

Understanding most nuances is important only for custom control authors. However what they are and how they work important for everyone. Can only style and animate dependency properties. After using wpf for a while, one wishes all properties were dependency properties. In practice, dependency properties are normal .net properties with extra wpf infrastructure. No .net language other than xaml has an intrinsic understanding of dependency properties.

Let's look at a standard dependency property implementation.




```
Public class Button: buttonbase
{
// The dependency  property
Public static readonly  dependencyproperty  isdefaultproperty;
Static Button()
{
```


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```
// Register the property
Button.isdefaultproperty = dependencyproperty.Register("isdefault", typeof(bool), typeof

}
// A .NET property  wrapper(optional)
Public bool isdefault
{
Get
{
Return(bool)getvalue(Button.isdefaultproperty);
}
Set
{
Setvalue(Button.isdefaultproperty, value);
}
}
// A property  changed  callback(optional)
Private static void onisdefaultchanged(_dependencyobject o,
dependencypropertychangedeventar
{
//...
}
//...
}
```

Dependency properties are represented by System.Windows.dependencyproperty. By convention, public static and Property suffix are used. Its required by several infrastructure pieces e.g. Localization tools, xaml loading. Optionally (via different overloads of Register), you can pass metadata that customizes how the property is treated by WPF, as well as callbacks for handling property value changes, coercing values, and validating values. .net prop wrapper optional. It helps setting from xaml and isnatural. Otherwise getvalue and setvalue are to be used. Getvalue and setvalue methods inherited from System.Windows.dependencyobject. Getvalue setvalue does not support generic. Dependency properties were introduced before generic support was added in C#.

Visual Studio has a snippet called propdp that automatically expands into a definition of a dependency property, which makes defining one much faster than doing all the typing yourself! .NET property wrappers are bypassed at runtime when setting dependency prop- erties in XAML! Although the XAML compiler depends on the property wrapper at compile time, WPF calls the underlying getvalue and setvalue methods directly at runtime! Therefore, to maintain parity between setting a property in XAML and procedural code, its crucial that property wrappers not contain any logic in addition to the getvalue/setvalue calls. If you want to add custom logic, thats what the registered callbacks are for. All of wpfs built-in property wrappers abide by this rule, so this warning is for anyone writing a custom class with its own dependency properties.

On the surface, it too much code but saves per-instance cost. Only static field and an efficient sparse storage system. If all were .net props with backing store, would consume lot more space e.g. 111 fields for button, 104 for label but 89 and 82 are dependency properties. Also code to check thread access, prompt containing element to be re-rendered.

Dependency properties support change notification. Its based on metadata at register time. Actions can be re-rendering the appropriate elements, updating the current layout, refreshing data bindings, and much property triggers - imagine you want color change on hovering.

```
<Button  mouseenter="Button_mouseenter" mouseleave="Button_mouseleave" minwidth="75"
Margin="10">
<Button  mouseenter="Button_mouseenter" mouseleave="Button_mouseleave" minwidth="75"
```


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```
Margin="10">

// Change the foreground to blue when the mouse enters the button
Void Button_mouseenter(object sender, mouseeventargs e)
{
Button b = sender as Button;
If(b != null)
B.Foreground = Brushes.Blue;
}
// Restore the foreground to black when the mouse exits the button
Void Button_mouseleave(object sender, mouseeventargs e)
{
Button b = sender as Button;
If(b != null)
B.Foreground = Brushes.Black;
}
```

And with Xaml

```
<Trigger  Property="ismouseover" Value="True">
<Setter Property="Foreground" Value="Blue"/>
</Trigger >
Eed to worry about reverting but cannot be applied directly to button elements.

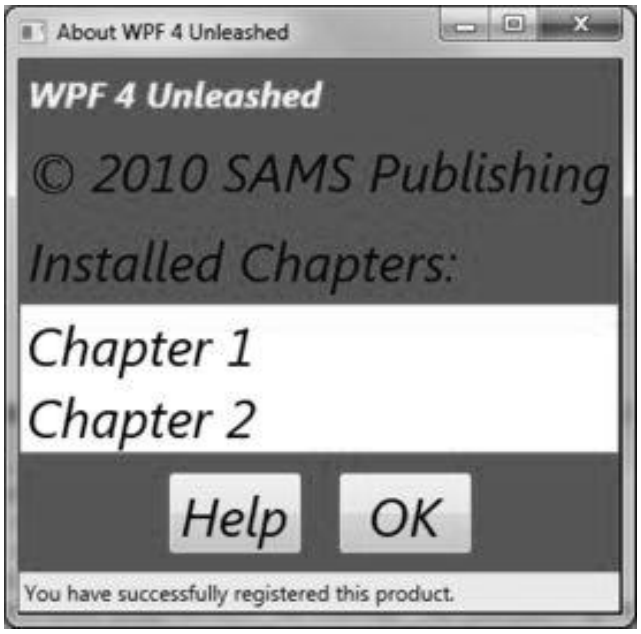
<Button minwidth="75" Margin="10">
<Button.Style>
<Style targettype="{x:Type Button}">
<Style.Triggers>
<Trigger  Property="ismouseover" Value="True">
<Setter Property="Foreground" Value="Blue"/>
</Trigger >
</Style.Triggers >
</Style >
</Button.Style >  OK
</Button >
```

Property value inheritance is flowing of prop values down an element tree.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Title="About WPF
 4 Unl
<stackpanel>
<Label fontweight="Bold" fontsize="20" Foreground="White">
WPF4 Unleashed
</Label >
<Label> 2010 SAMS Publishing</Label>
<Label>Installed  Chapters:</Label>
<listbox>
<listboxitem>Chapter 1</listboxitem>
<listboxitem>Chapter 2</listboxitem>
</listbox >
<stackpanel  Orientation="Horizontal" horizontalalignment="Center">
<Button minwidth="75" Margin="10">Help</Button>
<Button minwidth="75" Margin="10">OK</Button>
</stackpanel >
<statusbar>You have successfully  registered this product.</statusbar>
</stackpanel >
```


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```
</Window >
```



It does not effect status bar. Not every dependency property participates in inheritance (Actually they can opt-in). May be other higher priority sources setting property value. Some controls like status bar internally set their values to system defaults. Property value inheritance in other places like to triggers inside a definition.

WPF contains many powerful mechanisms that independently attempt to set the value of dependency properties. Dependency properties were designed to depend on these providers in a consistent and orderly manner. Its a 5 step process

Step 1: determine base value. Most providers factor into base value determination. Highest to lowest precedence. Ten providers that can set value for most dep. Props. 1. Local valuedependencyobject.setvalue, prop assignment in xaml or2. Parent template trigger3. Parent template4. Style triggers5. Template triggers6. Style setters7. Theme style triggers8. Theme style setters9. Property value inheritancealready seen10. Default valueinitial value registered with the property. Thats why status bar did not get font propagated. Use dependencypropertyhelper.getvaluesource to find which was the source used.

Step 2: evaluate. Expressions need evaluation. It is used in data binding.

Step 3: apply animation. Animations can alter value from step 2 or replace it.

Step 4: coerce. The almost final value passed to coercevaluecallback delegate if one is registered.

Step 5: validate. Passed to validatevaluecallback delegate. If one was registered. Returning false causes exception canceling the entire process. Wpf 4 adds new value in dependencyobject called setcurrentvalue. It updates current value without changing value source.

Let's discuss attached properties now. They are special dependency properties that can be attached to arbitrary objects. Sounds strange but there are many applications for it. There is new xaml syntax for attached props.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
```


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```
Title="About WPF4 Unleashed" sizetocontent="widthandheight"
Background="orangered">
<stackpanel>
<Label fontweight="Bold" fontsize="20" Foreground="White">
WPF4 Unleashed
</Label >
<Label> 2010 SAMS Publishing</Label>
<Label>Installed  Chapters:</Label>
<listbox>
<listboxitem>Chapter 1</listboxitem>
<listboxitem>Chapter 2</listboxitem>
</listbox >
<stackpanel  textelement.fontsize="30" textelement.fontstyle="Italic"
Orientation="Horizontal”>
<Button minwidth="75" Margin="10">Help</Button>
<Button minwidth="75" Margin="10">OK</Button>
</stackpanel >
<statusbar>You have successfully  registered this product.</statusbar>
</stackpanel >
</Window >
```


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Chapter 17

Lecture 17

We discussed this example in the last lecture.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
Title="About WPF4 Unleashed" sizetocontent="widthandheight"
Background="orangered">
<stackpanel>
<Label fontweight="Bold" fontsize="20" Foreground="White">
WPF4 Unleashed
</Label >
<Label> 2010 SAMS Publishing</Label>
<Label>Installed  Chapters:</Label>
<listbox>
<listboxitem>Chapter 1</listboxitem>
<listboxitem>Chapter 2</listboxitem>
</listbox >
<stackpanel  textelement.fontsize="30" textelement.fontstyle="Italic" Orientation="Horizo
<Button minwidth="75" Margin="10">Help</Button>
<Button minwidth="75" Margin="10">OK</Button>
</stackpanel >
<statusbar>You have successfully  registered this product.</statusbar>
</stackpanel >
</Window >
```

Xaml parser requires setfontsize, setfontstyle on textelement and therefore equivalent code in C# is.

```
Stackpanel panel= new stackpanel();
Textelement.setfontsize(panel, 30);
Textelement.setfontstyle(panel, fontstyles.Italic);
Panel.Orientation = Orientation.Horizontal;
Panel.horizontalalignment= horizontalalignment.Center;
Button  helpbutton= new Button();
Helpbutton.minwidth= 75;
Helpbutton.Margin = new Thickness(10);
Helpbutton.Content = "Help";
Button okbutton= new Button();
Okbutton.minwidth = 75;
Okbutton.Margin= new Thickness(10);
Okbutton.Content= "OK";
Panel.Children.Add(helpbutton);
Panel.Children.Add(okbutton);
```

Enumeration values such as fontstyles.Italic, Orientation.Horizontal, and horizontalalignment.Center were pre-viously specified in XAML simply as Italic, Horizontal, and Center, respectively. This is possible thanks to the enumconverter type converter C#. C# code shows no real magic, no .net property involved internally. There are just calls to dependencyobject.setvalue and getvalue like normal property wrappers, must not go anything else.

```
Public static void setfontsize(dependencyobject element, double value)
{
Element.setvalue(textelement.fontsizeproperty, value);
}

Public static double  getfontsize(dependencyobject  element)
{
```


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```
Return(double)element.getvalue(textelement.fontsizeproperty);
}
```

We are setting the fontsize property by unrelated class textelement. We could also have used textblock but textelement.fontsizeproperty is a separate dependencyproperty field from Control.fontsizeproperty.

```
Textelement.fontsizeproperty= dependencyproperty.registerattached(
 "fontsize", typeof(double), typeof(textelement), new frameworkpropertymetadata(
  Systemfonts.messagefontsize, frameworkpropertymetadataoptions.Inherits |
  Frameworkpropertymetadataoptions.affectsrender |
  Frameworkpropertymetadataoptions.affectsmeasure),
 New validatevaluecallback(textelement.isvalidfontsize));
```

Using registerattached us optimized for attached property metadata. Control on the other hand just calls addowner.

```
Control.fontsizeproperty = textelement.fontsizeproperty.addowner(
Typeof(Control), new frameworkpropertymetadata(systemfonts.messagefontsize,
Frameworkpropertymetadataoptions.Inherits));
```

These font related properties all controls inherit are from textelement in most cases, the class exposing the attached property is the same that defines the normal dependency propery. Many dependency properties have a Tag property for storing arbitrary custom data. A great mechanism for even extending "sealed" classes. In procedural code, you can actually use "any" property by calling setvalue.

```
// Attach an unrelated  property to a Button and set its value to true:
Okbutton.setvalue(itemscontrol.istextsearchenabledproperty, true);
```

We can use this property any way we want but its better way to use Tag.

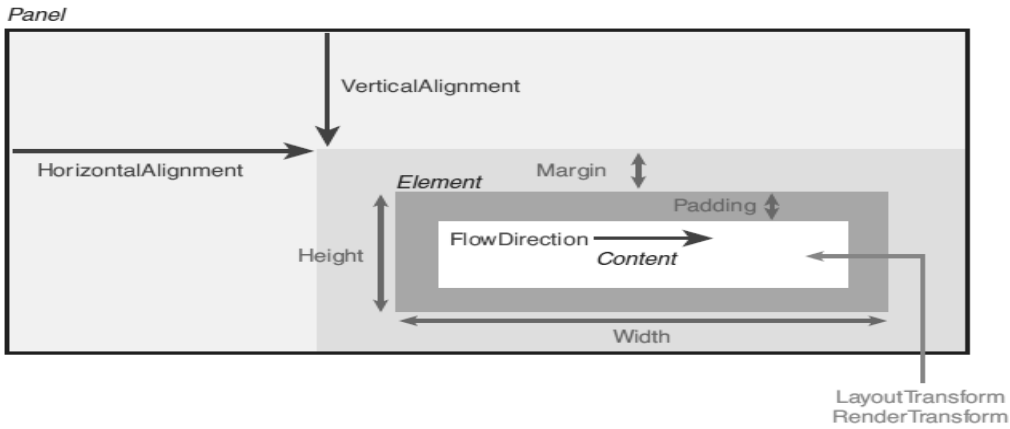
```
Geometrymodel3d  model= new geometrymodel3d();
Model.setvalue(frameworkelement.tagproperty, "my custom data");
```

Its most commonly used for layout of UI elements. Various panel derived classes define attached properties designed to be attached to their children. This way each panel can define custom behavior without burdening all possible children with their own set of properties. It allows extensibility.

In summary, wpf could have exposed its features by api's but instead they focused on adding features to the core. Multiple types of properties, multiple tree, multiple ways to achieve the same thing. Hopefully you can appreciate the value in coming lectures. These concepts will fade in the background as we accomplish different tasks in coming lectures.

Let's discuss sizing, positioning, and transforming elements. Sizing and positioning called "layout".

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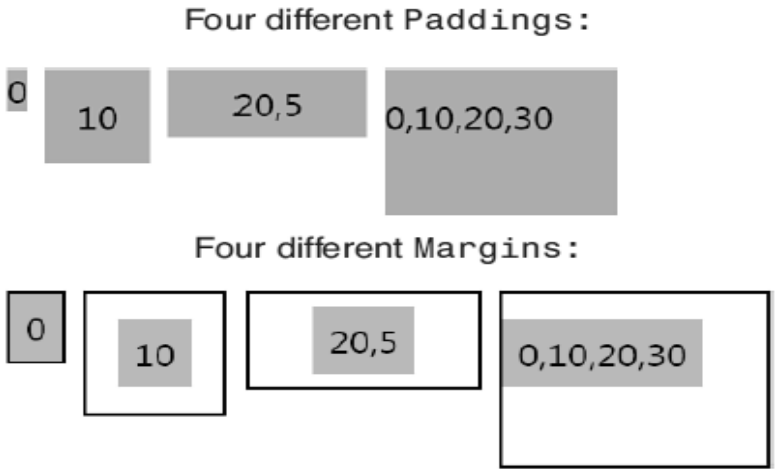


It boils down to parent child relationships. They work together to determine their final sizes and positions. Parents tell where to render and how much space they get. But more like collaborators they also ask children how much they really need. Panels are parents supporting layout of multiple children. They derive from the abstract System.Windows.Controls.Panel class. All the elements involved in the layout process derive from System.Windows.uielement. Several properties control various aspects of layout.

Size related properties are shown in blue, position related in red, green transforms can effect both. We will discuss various panels that arrange their children in specific ways.

Let's first discuss size. Children tell their size to parents. Wpf elements tend to size to content and no larger. Even the whole window when you set sizetocontent property. All frameworkelements have Height and Width properties of type double. Also have minheight, maxheight, minwidth, maxwidth. If explicit height width they takeprecedence when in range (avoid explicit). Min=0 and max=Infinity by default. Nan, Auto, Double.isnan mean size to content. Read only properties desiredsize, rendersize, actualheight, actualwidth. They are useful to act programatically on final size.

Next is margin and padding. All framework elements have margin, controls (and Border) have padding. Margin is extra space "outside" and padding is "inside" edges of the element. They are of type Thickness that can represent 1, 2, or 4 values. Negative margin is possible. Label's default padding=5. Xaml specification is helped by typeconverter.



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```
Mylabel.Margin = new   Thickness (10); //(10); // Same as Margin="10" in XAML
Mylabel.Margin = new Thickness (20,5,20,5); // Same as Margin="20,5" in XAML
Mylabel.Margin = new Thickness (0,10,20,30);// Same as Margin="0,10,20,30" in XAML
```

The lengthconverter type converter associated with the various length properties supports specifying explicit units of cm, pt, in, or px (the default). Default are device independent pixels. These logical pixels are 1/96 in regardless of screen DPI setting. They are always stored as "double". A typical display is 96 DPI. Important to note that all measurements are DPI independent.

Next is Visibility. It can have three values: Visible - The element is rendered and participates in layout, Collapsed The element is invisible and does not participate in layout, Hidden - The element is invisible yet still participates in layout.

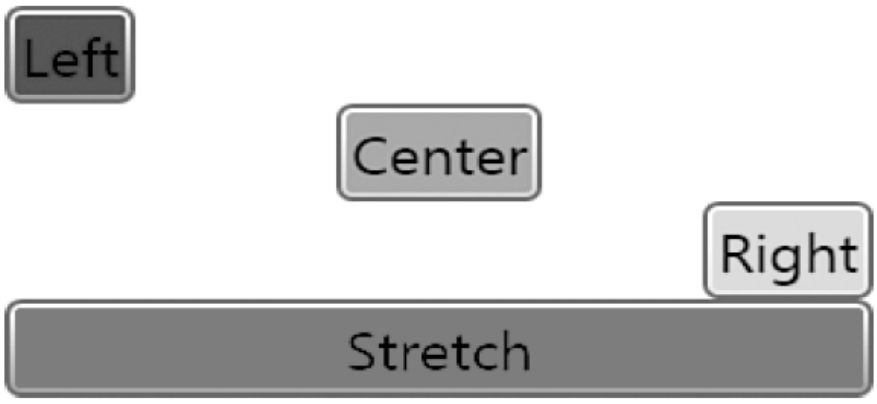
Compare the following two stackpanels.

```
<stackpanel  Height="100"  Background="Aqua">
<Button  Visibility="Collapsed">Collapsed Button</Button>
<Button>Below a Collapsed Button</Button>
</stackpanel >

<stackpanel  Height="100"  Background="Aqua">
<Button  Visibility="Hidden">Hidden Button</Button>
<Button>Below a Hidden Button</Button>
</stackpanel >
```

Next is Position and alignmnet. Position is not given as (x,y), instead its parent controlled. Rather alignment and flow direction is given vs. Explicit height. Horizontalalignment can taken values Left, Center, Right, and Stretch and verticalalignment can take values Top, Center, Bottom, and Stretch.

```
<stackpanel>
<Button  horizontalalignment="Left" Background="Red">Left</Button>
<Button  horizontalalignment="Center" Background="Orange">Center</Button>
<Button  horizontalalignment="Right" Background="Yellow">Right</Button>
<Button  horizontalalignment="Stretch" Background="Lime">Stretch</Button>
</stackpanel >
```

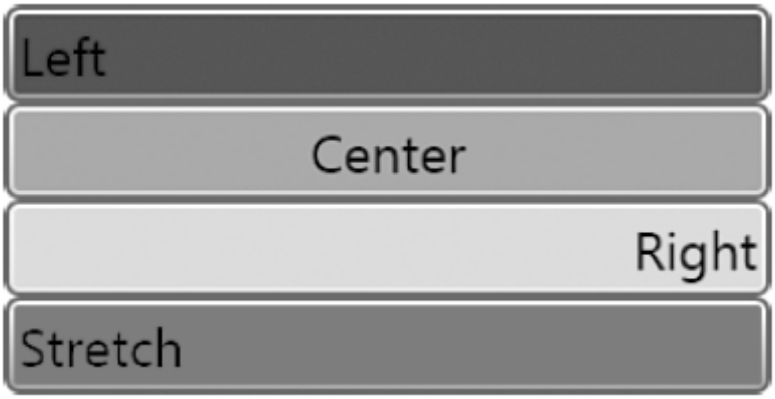


Its only useful when parent gives more space than needed.

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Control class also has horizontalcontentalignment and verticalcontentalignment. It sets how a control content fills the space inside, more like the relationship between margin n padding. Default is left and top except button which overrides it. Textblock doesnt strech like a control.

```
<stackpanel>
<Button  horizontalcontentalignment="Left" Background="Red">Left</Button>
<Button  horizontalcontentalignment="Center" Background="Orange">Center</Button>
<Button  horizontalcontentalignment="Right" Background="Yellow">Right</Button>
<Button  horizontalcontentalignment="Stretch" Background="Lime">Stretch</Button>
</stackpanel >
```



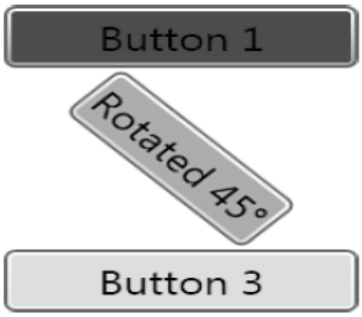
Frameworkelement.flowdirection can reverse the direction. It can be used for languages that are read right to left.

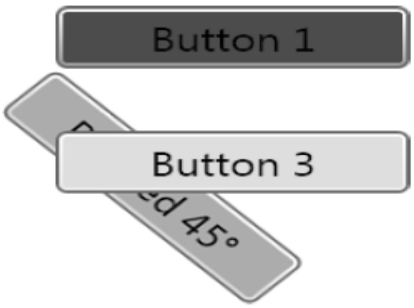
```
<stackpanel>
<Button  flowdirection="lefttoright" horizontalcontentalignment="Left"
verticalcontentalignme
<Button  flowdirection="righttoleft" horizontalcontentalignment="Left"
verticalcontentalignme
</stackpanel >
```

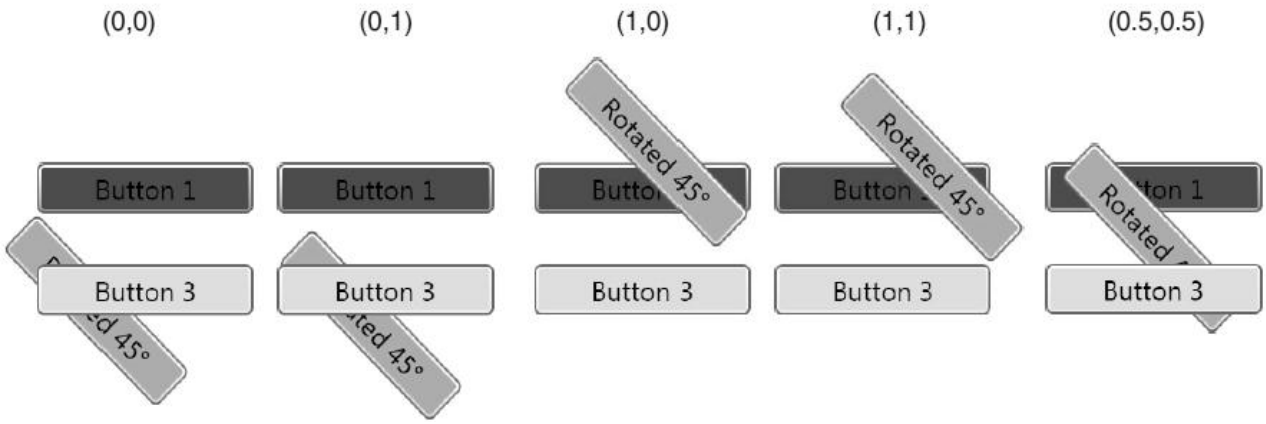


Transforms are from the base class System.Windows.Media.Transform. All framework elements have layouttransform and rendertransform. One applied before layout and one after. Uielements have rendertransformorigin.

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Button 1
Rotated 45°
Button 3


Button 1
Button 3
45°


(0,0)
(0,1)
(1,0)
(1,1)
(0.5,0.5)
Button 1
Button 3
Rotated 45°

Chapter 18

Lecture 18

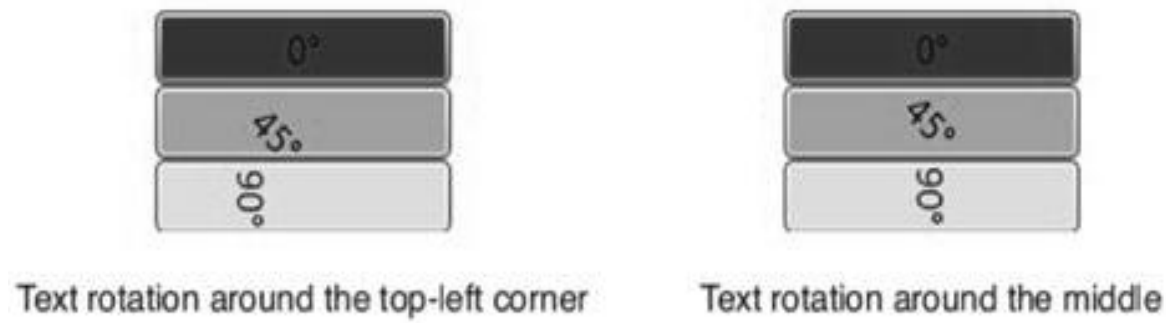
Let's revise size and position properties.
Layout transform vs render transform. One is applied before rendering and one after.

Pointconverter is used to specify the origin.

```
<Button  rendertransformorigin="0.5,0.5" Background="Orange">
<Button.rendertransform>
<rotatetransform  Angle="45"/>
</Button.rendertransform >
Rotated 45
</Button >
```

Builtin transforms are Rotate Transform, Scale Transform, Skew Transform, Translate Transform, and Matrix Transform. Rotatetransform has Angle, centerx, centery with defaults 0. Center useless for layout transform. Center is also useful when grouping rendertransforms.
Transform can be on inner element.

```
<Button  Background="Orange">
<textblock  rendertransformorigin="0.5,0.5">
<textblock.rendertransform>
<rotatetransform  Angle="45"/>
</textblock.rendertransform >
45
</textblock >
</Button >
```

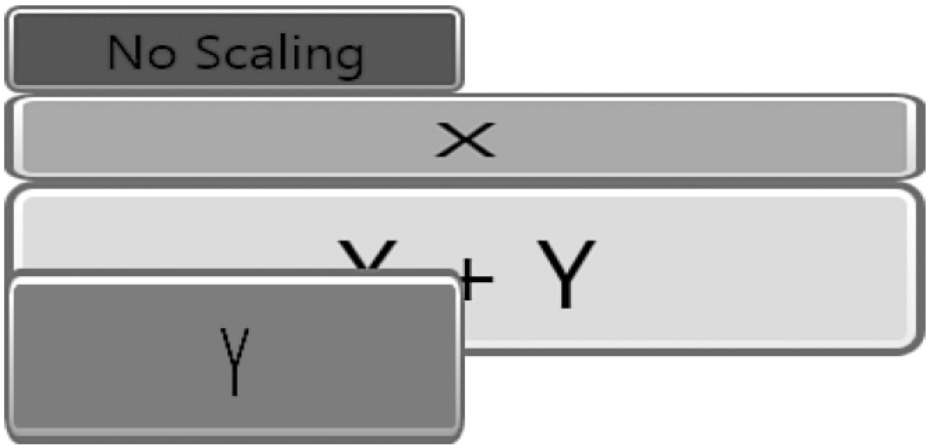


Scale Transform has scalex, scaley, centerx, centery.

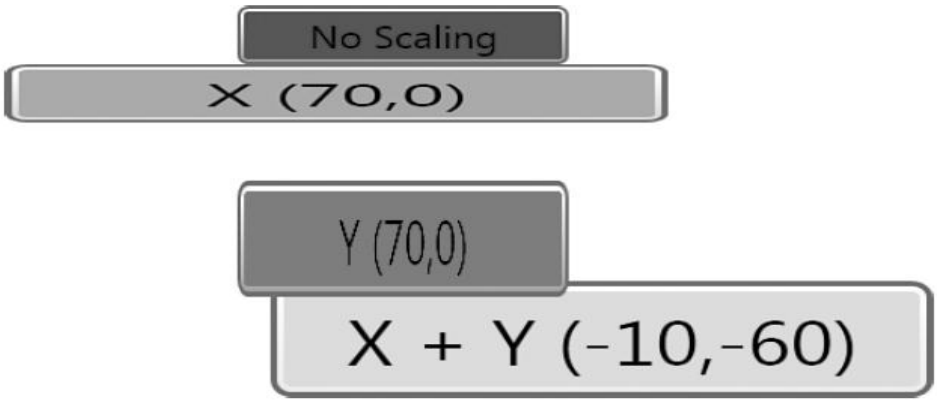
```
<stackpanel Width="100">
  <Button  Background="Red">No Scaling</Button>
  <Button  Background="Orange">
    <Button.rendertransform>
      <scaletransform  scalex="2"/>
    </Button.rendertransform >
    X
  </Button >
  <Button  Background="Yellow">
    <Button.rendertransform>
      <scaletransform  scalex="2" scaley="2"/>
    </Button.rendertransform >
```


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```
    X + Y
  </Button >
  <Button  Background="Lime">
    <Button.rendertransform>
      <scaletransform  scaley="2"/>
    </Button.rendertransform >
    Y
  </Button> 21
</stackpanel >
```

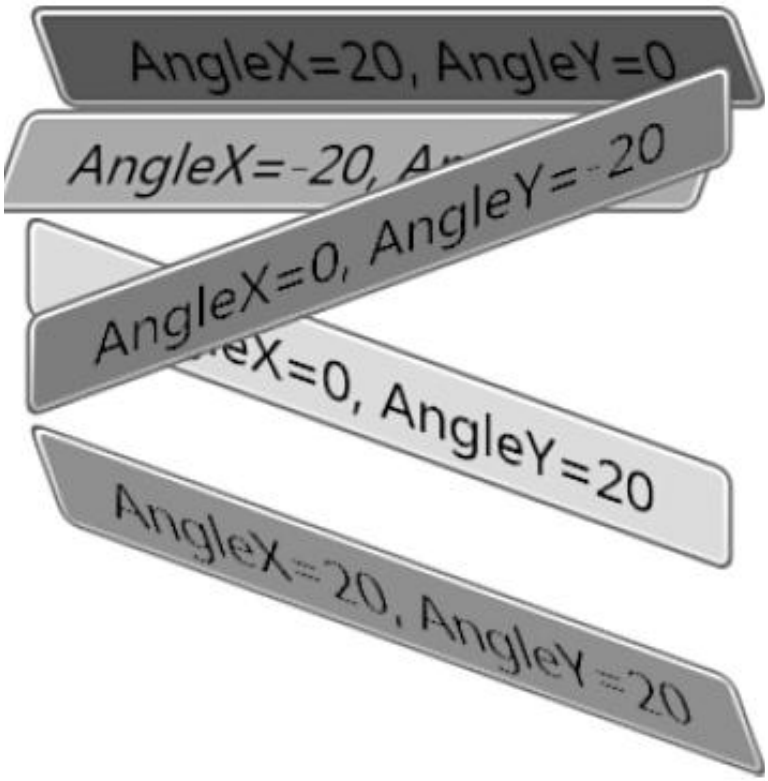


Stretch and scaletransform only effect if more than stretch. Padding is scaled but margin is not. Does not effect actualhieght actualwidth rendersize etc.
Skew Transform has anglex — Amount of horizontal skew (default value = 0), angley — Amount of vertical skew



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(default value = 0), centerx — Origin for horizontal skew (default value = 0), centery — Origin for vertical skew (default value = 0). It is applied as a render transform here.



Translate Transform has X — Amount to move horizontally (default value = 0), Y — Amount to move vertically (default value = 0). It has no effect as a layout transform.

Matrix Transform has a single Matrix property (of type System.Windows.Media.Matrix) representing a 3x3 affine transformation matrix.

```
<Button  rendertransform=1,0,0,1,10,20/>
```

M11	M12	0
M21	M22	0
OffsetX	OffsetY	1

Its the only transform with a type converter to convert a string.

Combining transforms can be done. We can apply both layout and render transforms. Can figure out a

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combined matrix and apply matrix transform or use transformgroup a Transform derived class and let it be calculated for you.

```
<Button>
  <Button.rendertransform>
    <transformgroup>
      <rotatetransform  Angle="45"/>
      <scaletransform  scalex="5" scaley="1"/>
      <skewtransform  anglex="30"/>
    </transformgroup >
  </Button.rendertransform >
  OK
</Button >
```



Not all elements can be transformed.

In Summary, we learned about layout of individual elements. The layout is finally decided by parent. We have used stackpanel. Let's see this and other panels for layout.

Static pixel based placement and sizes results in limited but different screen sizes. Builtin panels that make it easy.

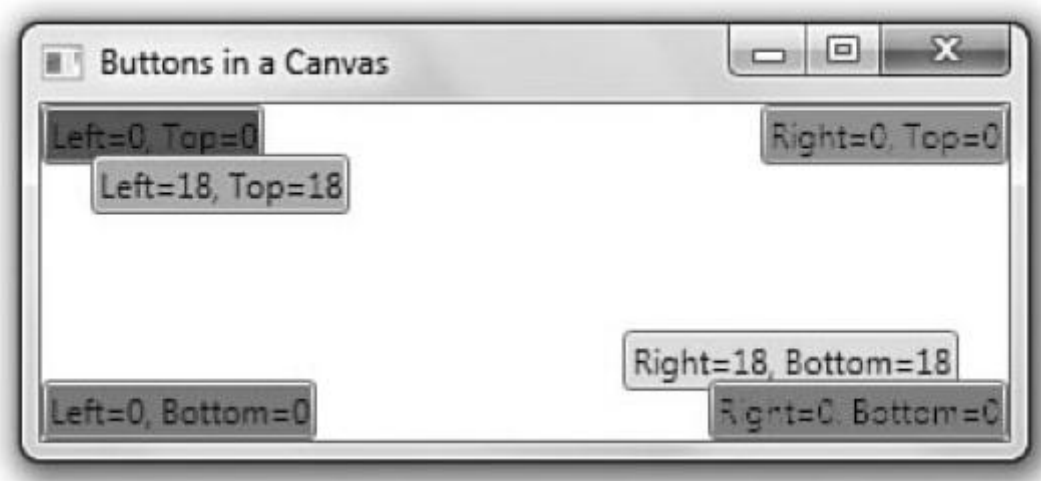
5 main builtin panels in System.Windows.Controls: Canvas, stackpanel, wrappanel, dockpanel, and Grid. Content overflow is when parents and children cant agree on the use of available space.

Canvas is the most basic. Probably never used. "classic" notion of explicit coordinates but device independent pixels and relative to any corner of the window. We can position elements in a Canvas by using its attached properties: Left, Top, Right, and Bottom. Default offsets are from top left.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
Title="Buttons in a Canvas">
  <Canvas>
    <Button  Background="Red">Left=0, Top=0</Button>
    <Button Canvas.Left="18" Canvas.Top="18"
    Background="Orange"> Left=18, Top=18</Button>
    <Button Canvas.Right="18" Canvas.Bottom="18"
    Background="Yellow">Right=18, Bottom=18</Button>
```

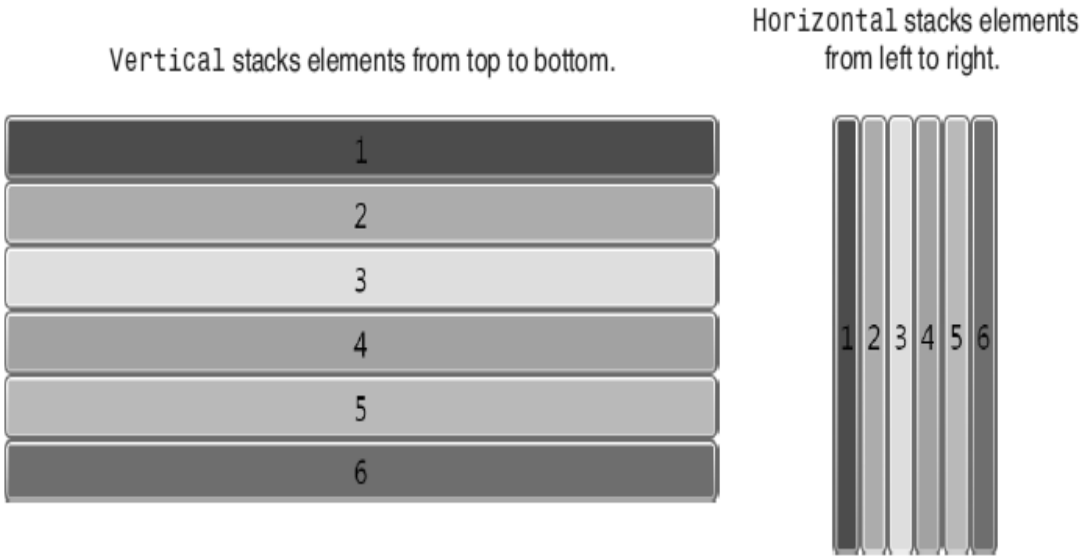

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```
    <Button Canvas.Right="0" Canvas.Bottom="0"
    Background="Lime">Right=0, Bottom=0</Button>
    <Button Canvas.Right="0" Canvas.Top="0"
    Background="Aqua">Right=0, Top=0</Button>
    <Button Canvas.Left="0" Canvas.Bottom="0"
    Background="Magenta">Left=0, Bottom=0</Button>
  </Canvas ></Window >
```



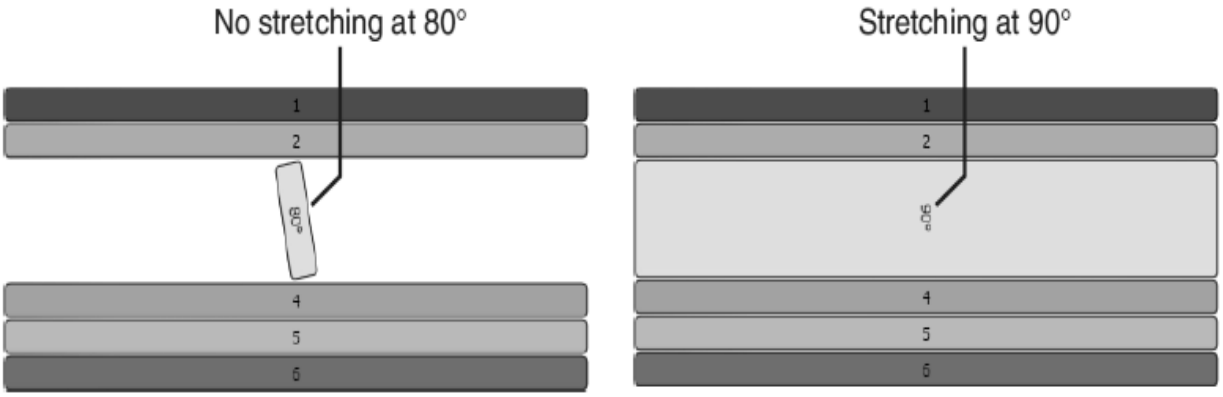
If left and right, right is ignored, top and bottom, bottom is ignored. Whats z order. Z-order decides which element to show in overlapping elements.

```
<Canvas>
  <Button Canvas.zindex="1" Background="Red">On Top!</Button>
  <Button  Background="Orange">On Bottom with a Default zindex=0</Button>
</Canvas >
```



Stack Panel is popular, simple, useful, stacks sequentially. It is one of the few panels that dont even define an attached property. Orientation can be horizontal or vertical (which is default). Default horizontal direction is basedon flowdirection.

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Virtualizing panels e.g. Virtualizingstackpanel save space for offscreen content when data binding e.g. Listbox uses it Wrap panel wraps to additional rows or columns when not enough space. It has no attached properties. Orientation is by default horizontal. Itemheight, itemwidth are uniform for all children and not set by default.



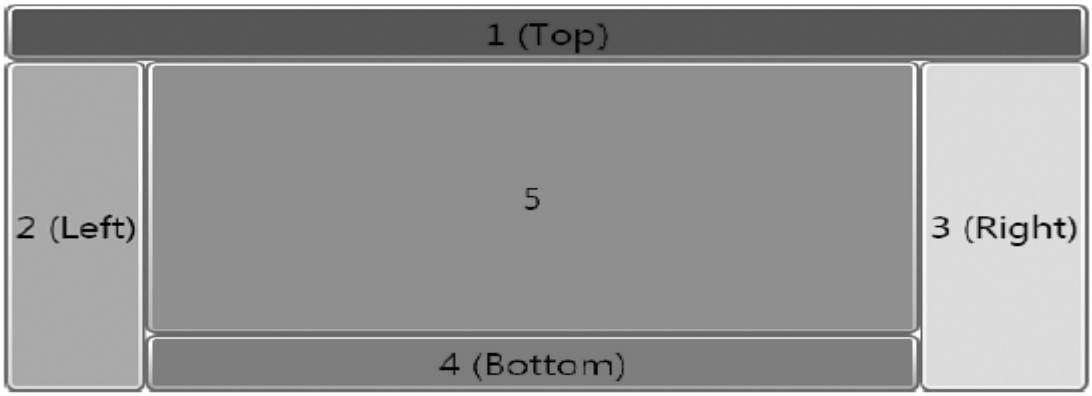
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Chapter 19

Lecture 19

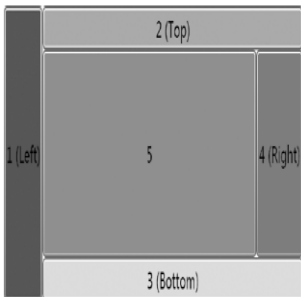
Dock panel allows easy docking of elements to an entire side. Dock attached property has left, right, top, bottom values. Last child fills space unless lastchildfill=false.

```
<dockpanel>
  <Button dockpanel.Dock="Top" Background="Red">1(Top)</Button>
  <Button dockpanel.Dock="Left" Background="Orange">2 (Left)</Button>
  <Button dockpanel.Dock="Right" Background="Yellow">3 (Right)</Button>
  <Button dockpanel.Dock="Bottom" Background="Lime">4 (Bottom)</Button>
  <Button  Background="Aqua">5</Button>
</dockpanel >
```



Why are elements streching. Let's change alignment and see.

```
<dockpanel>
  <Button dockpanel.Dock="Top" horizontalalignment=”Right” Background=”Red”>1 (Top,
Align=Right) </Button>
  <Button dockpanel.Dock=”Left” verticalalignment=”Bottom” Background=”Orange”>2 (Left,
Align=Bottom)</Button>
  <Button dockpanel.Dock=”Right” verticalalignment=”Bottom” Background=”Yellow”>3 (Right,
Align=Bottom)</Button>
  <Button dockpanel.Dock=”Bottom” horizontalalignment=”Right” Background=”Lime”>4
(Bottom, Align=Right)</Button>
  <Button Background=”Aqua”>5</Button></dockpanel>
```



Its useful in a toplevel interface. Order of adding matters for the corners.

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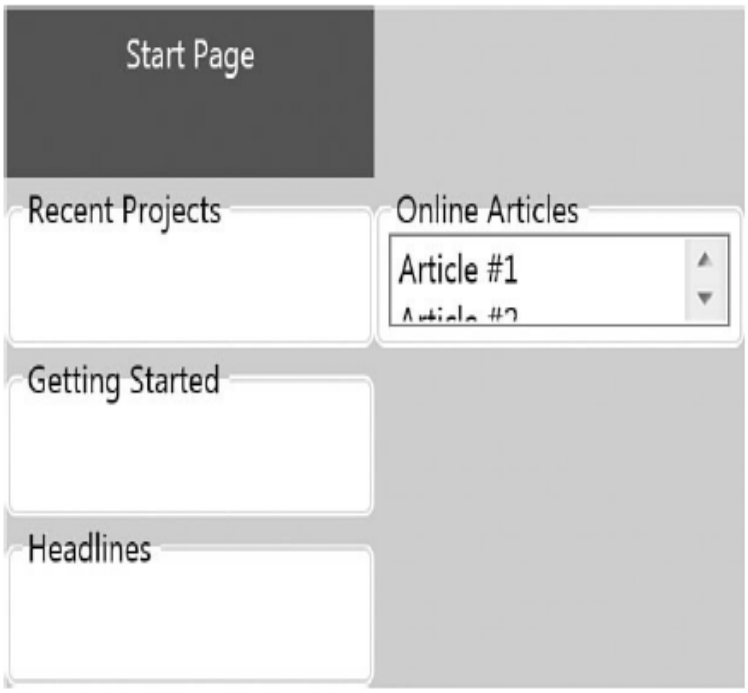


More elements can be added to any side. Its a superset of stackpanel.

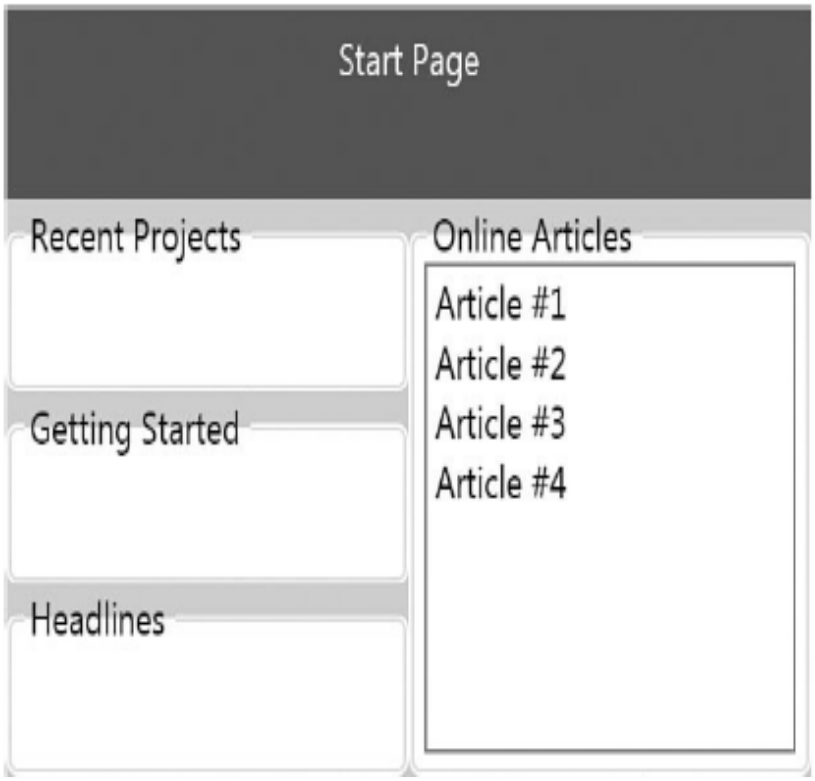
Grid is the most versatile. Its the default panel in VS and expression blend. You arrange things in multirow multicolumn, like a table in html. Also a Table class which is a frameworkcontentelement not a uielement.

```
<Grid Background="lightblue">
  <!--  Define  four  rows: -->
  <Grid.rowdefinitions>
    <rowdefinition/>
    <rowdefinition/>
    <rowdefinition/>
    <rowdefinition/>
  </Grid.rowdefinitions >
  <!--  Define  two  columns:-->
  <Grid.columndefinitions>
    <columndefinition/>
    <columndefinition/>
  </Grid.columndefinitions >
  <!--  Arrange  the  children: -->
  <Label   Grid.Row="0" Grid.Column="0" Background="Blue" Foreground="White"
horizontalcontenta=""
<groupbox Grid.Row="1" Grid.Column="0" Background="White" Header="Recent Projects">
 </groupbox>
  <groupbox Grid.Row="2" Grid.Column="0" Background="White" Header="Getting
Started"></groupbox>
  <groupbox Grid.Row="3" Grid.Column="0" Background="White"
Header="Headlines"></groupbox>
  <groupbox Grid.Row="1" Grid.Column="1" Background="White" Header="Online
Articles"></groupbox>
  20 <listbox>
    <listboxitem>Article #1</listboxitem>
    <listboxitem>Article #2</listboxitem>
    <listboxitem>Article #3</listboxitem>
    <listboxitem>Article #4</listboxitem>
  </listbox >
  </groupbox >
</Grid >
```

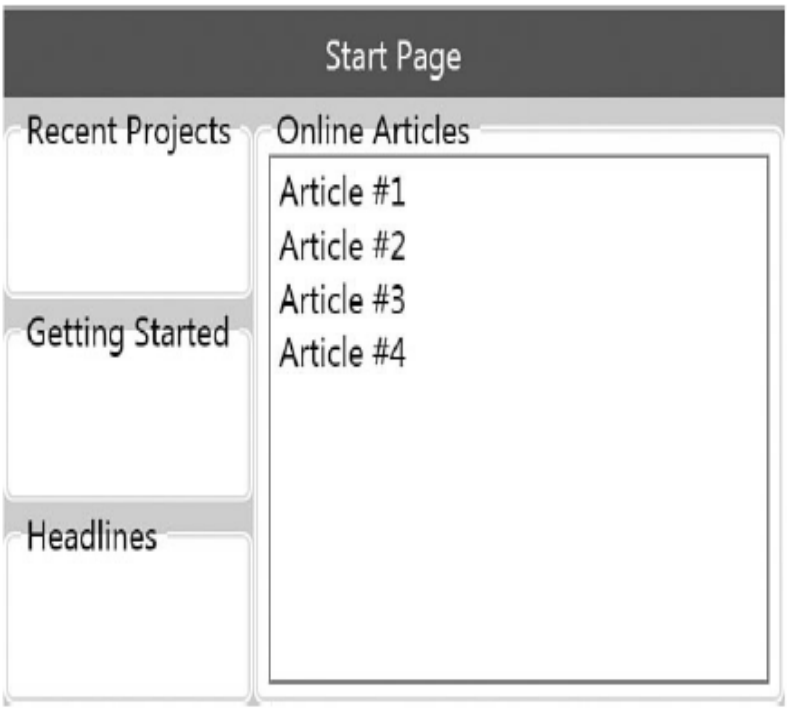

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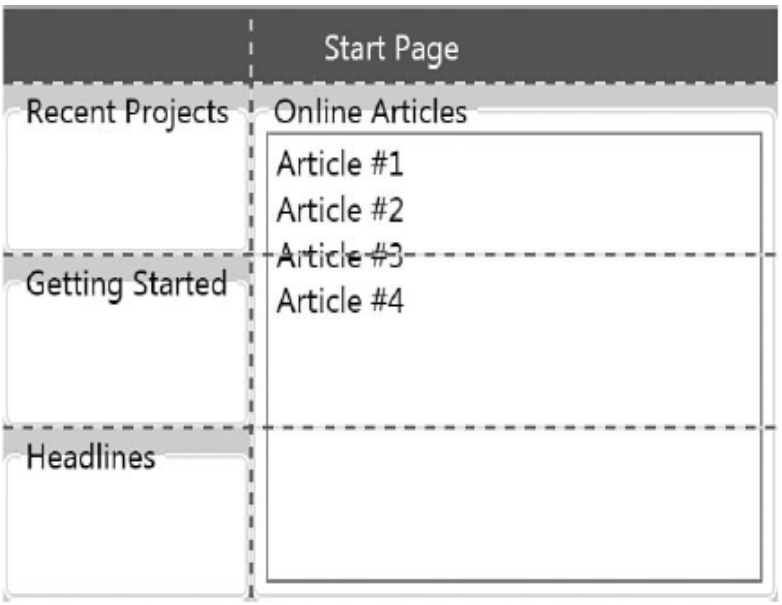


We specify individual rows and cols. Its verbose but useful to specify details. Default is 1 cell. We position elements using Row and Column attached properties (zero-based) default is 0,0. Multiple elements in same cell simply overlap by z-order. Cells can be empty.



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Online article list too small and and start page label not full width. We can make spanning rows and cols. Rowspan and columnspan 1 by default. By default height n width are same. Height and Width = "Auto" for sizing to content. Showgridlines = "True" to show grid lines.

For sizing rows and columns, use rowdefinition and Col. Height and Width not double but gridlength and not default to Auto or nan. Three types of sizing. Absoulte sizing: device independent pixels means no grow shrink, Autosizing: size to content, or Proportional or star sizing: grows or shrinks. When 1 row col * all remaining space is taken by it. When more row col * they divides remaining space. It can an be 2* or

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5.5*. 2* is twice as * (1*). 5.5* is twice as 2.75*. Remaining space is after absolute and autosized rows or cols. Default width and height are *.

Here is how gridlength is used in procedural code.




```
Gridlength  length = new gridlength(100);
Gridlength  length = new gridlength(0, gridunittype.Auto);
Gridlength  length = new gridlength(100, gridunittype.Pixel);
Gridlength  length = new gridlength(2, gridunittype.Star);
```

Grid splitter is used for interactive sizing. Any number of gridsplitters can be added to a Grid. Can be used to move entire row or column. Atleast one cell resizes and other cells behavior depends on weather absolute sizing or proportional is applied on them. It fits in one cell, but behavior affects entire row or col so better to use span. Which cells are affected depends on gridsplitters alignment values. Default horizontal alignment=right and vertical alignment=stretch. Reasonable use requires stretch in one dimension.

HorizontalAlignment

VerticalAlignment	Left	Right	Center	Stretch
Top	Current cell and cell to the left	Current cell and cell to the right	Cells to the left and right	Current cell and cell above
Bottom	Current cell and cell to the left	Current cell and cell to the right	Cells to the left and right	Current cell and cell below
Center	Current cell and cell to the left	Current cell and cell to the right	Cells to the left and right	Cells above and below
Stretch	Current cell and cell to the left	Current cell and cell to the right	Cells to the left and right	Cells to the left and right if `GridSplitter` is taller than it is wide, *or* cells to the top and bottom if `GridSplitter` is wider than it is tall

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When all proportionally sized, changes co-efficients accordingly. When absolute sizing, changes only top or left of the cells. Remaining cells pushed down or right. Also has resize direction (default Auto, or can be set to Rows or Cols) and resize behavior for explicit control. Resize direction effect only when stretching in both directions. Resize behavior defaults to based on alignment. It can be set to previous and current, current and next, or previous and next to control which 2 rows or columns should be directly affected by resizing. Best way is to place it in its own auto sized row or column so it doesn't overlap existing content.

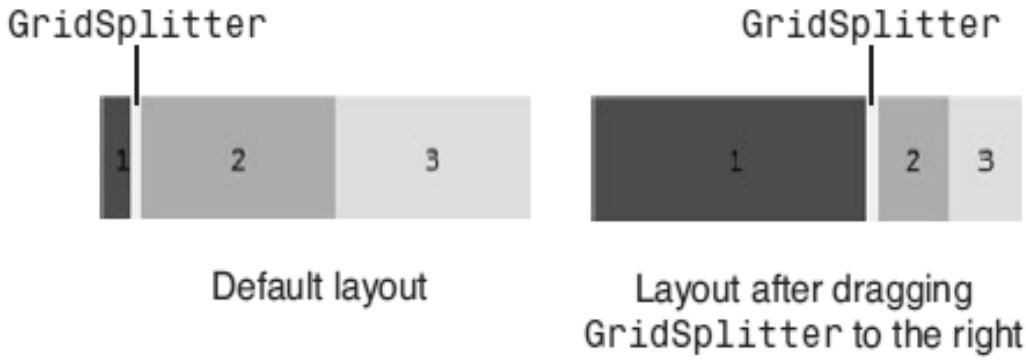
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Chapter 20

Lecture 20

Let's revise grid.

```
<Grid>
  <Grid.columndefinitions>
    <columndefinition  Width="Auto"/>
    <columndefinition/>
    <columndefinition/>
  </Grid.columndefinitions >
  <Label Grid.Column="0" Background="Red" horizontalcontentalignment="Center"
verticalcontenta=""
  </Label >
  <gridsplitter Grid.Column="0" Width="5"/>
  <Label Grid.Column="1" Background="Orange" horizontalcontentalignment="Center"
verticalconte=""
  </Label >
  <Label Grid.Column="2" Background="Yellow" horizontalcontentalignment="Center"
verticalconte=""
  </Label >
</Grid >
```

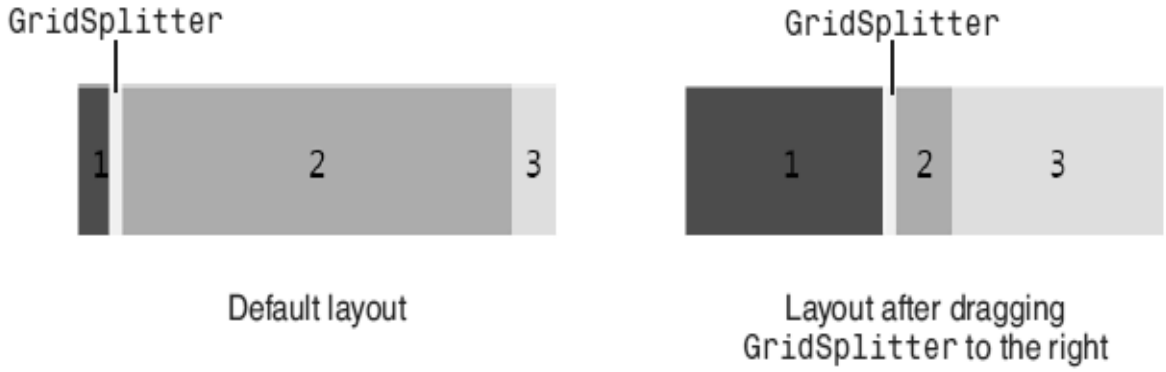


Sharedsizegroup enables multiple row cols to remain the same width height when length changed via gridsplitter.

```
<Grid issharedsizescope="True">
  <Grid.columndefinitions>
    <columndefinition  Width="Auto" sharedsizegroup="mygroup"/>
    <columndefinition/>
    <columndefinition  sharedsizegroup="mygroup"/>
  </Grid.columndefinitions >
  <Label Grid.Column="0" Background="Red" horizontalcontentalignment="Center"
verticalcontenta=""
  </Label >
  <gridsplitter Grid.Column="0" Width="5"/>
  <Label Grid.Column="1" Background="Orange" horizontalcontentalignment="Center"
verticalconte=""
  </Label >
  <Label Grid.Column="2" Background="Yellow" horizontalcontentalignment="Center"
verticalconte=""
```


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```
  </Label >
</Grid >
```



Issharedscope because sizegroups can be shared across multiple grids. All uses must be under a common parent with issharedscope set to true. It is an attached property of grid as well. Grid is usually the best choice. Except wraping, it can do what most panels do.

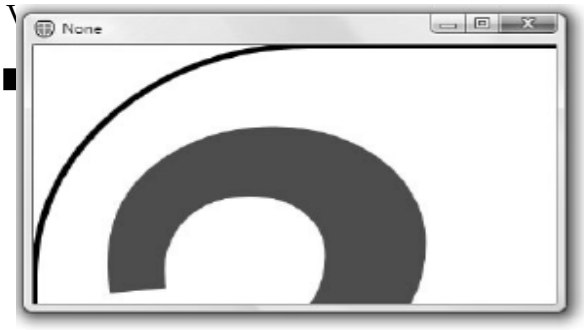


Content overflow can be dealt with Clipping, Scrolling, Scaling, Wrapping and Trimming. Wrapping already seen. Only way for non-text to wrap is using wrappanel. Trimming is intelligent form of clipping. Supported for text by textblock and accesstext. They have texttrimming=None (default), or characterellipsis, orwordellipsis.

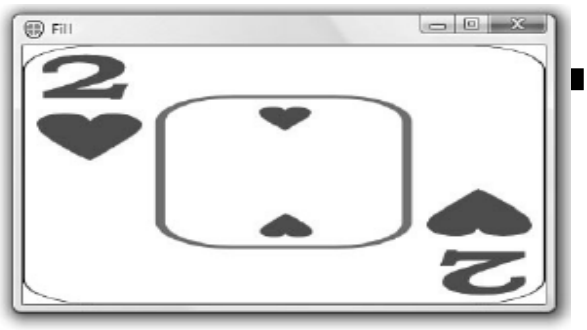
Clipping is default behavior. Edges of panel or cell area of dock area. All uielements have cliptobounds. Controls if it can draw outside bounds. Still cannot draw outside the window or page. Most panels clip regardless of this property. Canvas and uniformgrid dont unless this property is set. Button has this property false by default. Place canvas inside a grid cell to avert clipping. Clipping is done before rendertransform. Cant shrink back.



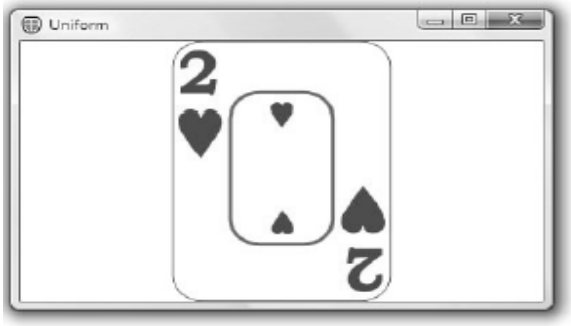
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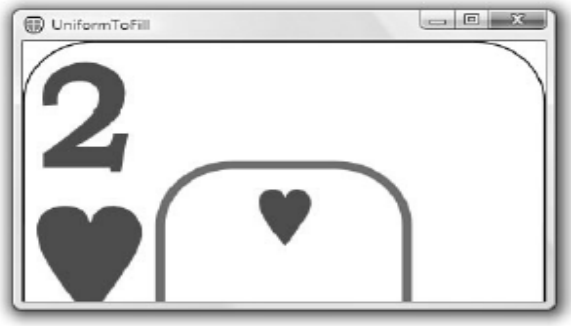
Stretch="None"



Stretch="Fill"



Stretch="Uniform"



Stretch="UniformToFill"

System.Windows.Controls.scrollviewer control can be used for scrolling. Its content property is set and verticalscrollbarvisibility and horizontalscrollbarvisibility properties determine the display.

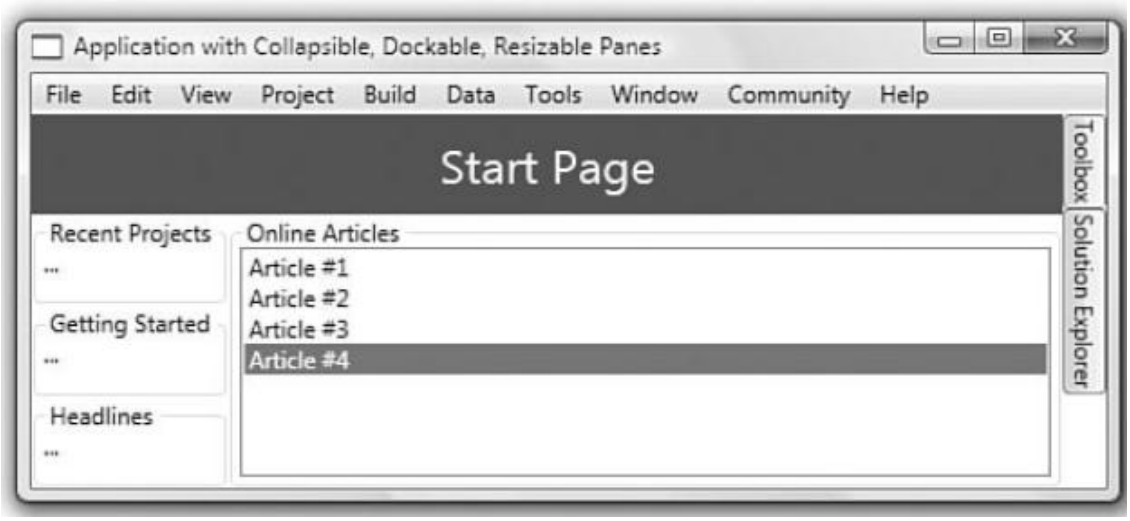
```
<Window Title="Using scrollviewer"
Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
    <scrollviewer>
        <stackpanel>
            …
        </stackpanel>
    </scrollviewer>
</Window>
```

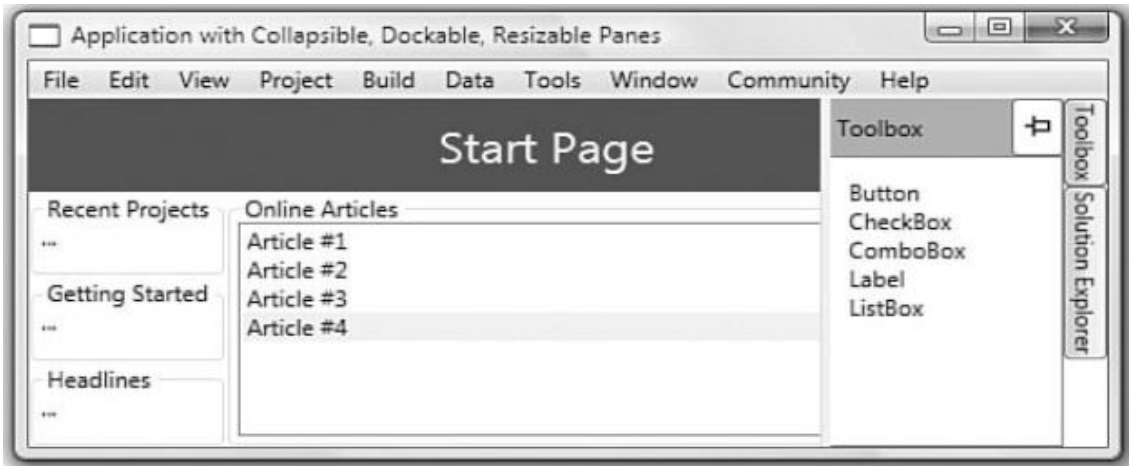

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Scrolling more popular but e.g. Card game needs scaling. Scaletransform works relative to their own size not the available size. System.Windows.Controls.Viewbox, a type of class called Decorator (also Border). A panel-like thing but has only one child. Stretches to fill available space by default but also Stretch=None (like not using it at all), Fill, Uniform (aspect ratio, default), uniformtofill (cropped) strechdirection=uponly, downonly, Both (default).

```
<Window Title="Using Viewbox"
Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
    <Viewbox>
        <stackpanel>
            …
        </stackpanel>
    </Viewbox>
</Window>
```

Let's make a visual studio like interface.





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```
Using System.Windows;
Using System.Windows.Input;
Using System.Windows.Media;
Using System.Windows.Controls;

Public partial class aboutdialog : Window
{
    Public aboutdialog()
    {
        Initializecomponent();
    }
    Void aboutdialog_mouserightbuttondown(object sender, mousebuttoneventargs e)
    {
        // Display information about this event
        This.Title = "Source = " + e.Source.gettype().Name + ", originalsource = " +
            E.originalsource.gettype().Name + " @ " + e.Timestamp;
        // In this example, all possible sources derive from Control
        Control source = e.Source as Control;
        // Toggle the border on the source control
        If (source.borderthickness != new Thickness(5))
        {
            Source.borderthickness = new Thickness(5);
            Source.borderbrush = Brushes.Black;
        }
        Else
            Source.borderthickness = new Thickness(0);
    }
}
```


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GDB 95% Results

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1. Every ____ is represented by an event-object.

- Information
- Entity
- Object
- Event page10

2. Which statement is True about interfaces?

- An interface can contain only methods, variables, events, and indexers.
- An interface can contain only variables, properties, events, and indexers.
- An interface can contain only methods, properties, variables, and indexers.
- An interface can contain only methods, properties, events, and indexers. Page 40

3. Which one is the correct syntax for declaring attribute?

- [Obsolete] page 48
- {Obsolete}
- (Obsolete)
- *Obsolete

4. An "object" element can have____type/s of children.

- Four
- Three Page 73
- Two
- One

5_____allows to keep the user interface description and implementation separate.

- GDI

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- WPF (Page#61)
- Avalon
- Altia

6. There is/are _____type(s) of padding in WPF element.

- One
- Two
- Three page 91
- Four

7. Which of the following is not a feature of "C#"?

- Operator overloading is not allowed. Page 17
- Multiple Inheritances is not supported but interfaces are supported.
- Enumeration members are scoped.
- Global variables or functions are not allowed.

8. Stylus can behave like a_____ but has_____resolution.

- Keyboard, lower
- Mouse, lower
- Mouse, higher page 122
- Keyboard, higher

9. Attributes are declared_____the class/function.

- Within
- After
- Above
- Outside

10. If we want that "Stretch" property of child element takes the available "height or

width of parent" and shape of child element doesn't change, then we should write

- "Stretch =_____".

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- Fill page 109
- UniformToFill
- Uniform

11.The code given below is equivalent to ___.

- <SolidColorBrush>White</SolidColorBrush>
- <WhiteBrush >MyWhiteBrush </ WhiteBrush>
- <SolidColorBrush Color="White"/> (Page#65)
- <Brush Style= “SolidWhite”>

12.Visual Studio has a snippet called _____that automatically expands into a definition of a

dependency property, which makes defining one much faster than doing all the typing yourself!

- Propdp (Page#77)
- Property extension
- Dpprop
- Dependency property

13.Multiple_____are conventionally stored in multiple files.

- Functions
- Values
- Variables
- Classes (Page#22)

14. "Situation" is an event occurrence that requires a (n)_____.

- Reaction page 10
- Class
- Object
- Action

15. We have two buttons "b1" and "b2". If we want to place "b1" on "b2" then value of

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- "ZIndex" of "b1" should be___the value of "ZIndex" of "b2".
- Less than
- Greater than
- Equal to
- No need to give value of ZIndex of b1 page99

16. Events can be_____.

- Specialized
- Composed
- Generalized
- All of the given options page 15

17. Wpf 3.5 released in

- 2005
- 2006
- 2008 Page No 61 In AUGUST
- 2009

18.The corresponding type converter class for "Brush" class will be ___.

- BrushConverter
- CastBrush
- BrushCast
- None of the given options

19.Extension methods_____access the private data members of the class in which they are extended.

- may
- **cannot**
- may not
- can

20. Which "transform" property can help us to flip the element from its center?

- ScaleY

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- None of given options
- ScaleX
- RenderTransformOrigin page 126

21. CLI stands for_____.

- Compact Language Infrastructure
- Common Language Infrastructure page18
- Console Language Infrastructure
- Control Language Infrastructure

22. The elements in a/an___ are always stored in a contiguous block of memory.

- Tree
- Array
- Variable
- Struct

23. The _____statement iterates over each element in an "enumerable" object.

- Foreach page 191
- Dowhile
- While
- For

24. We are checking that whether an element is eligible for focus or not, which property

will help us in this testing?

- Focus
- IsFocus
- IsFocusable
- Focusable page 121

25. Directory is the class of_____.

- System.Collections.Generic

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- System.Reflection
- System.Text
- System.IO MPOV and GOOGLE

25. WCF stands for
- Windows Communication Foundation
- Windows Class Foundation
- Windows Classic Foundation
- None

26. C# is a ___and .NET is a____.
- Platform, Language
- Package, Language
- Language, Package
- Language, Platform

27. The ___,____ and ___of applications that use events, either directly or indirectly is called event-based programming.
- Model, Code, Operation
- Analyze, Design, Operation
- Code, Operation, Maintain
- Design, Coding, Operation

28. Suppose we want to restrict the user to don't decrease the height of a WPF element from a specific amount, then which property will be used?
- MinHeight
- None
- HeightMinimum
- MinimumHeight

29. Types not defined in any namespace are said to reside in the ___ namespace.
- Static
- Virtual
- Local
- Global

30. Overloaded operators must be ____
- Referenced
- Static

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- Objective
- Delegates

31. If we want to scale an object in the 2-D x-y coordinate system then which transform class will be used?

- Rotate Transform
- Matrix Transform
- Translate Transform
- Skew Transform

32. Polymorphism works on the basis that child class should have _____ feature/s of its parent class.

- Only one
- No
- Few
- All

33. The root object eement in XAML must specify at least_____XML namespace that is usedto qualify itself and any child elements.

- **one(Page#63)**
- two
- three
- five

34. In event-driven architecture (EDA),_____sends event to _____.

- Event Consumer, Event Producer
- **Event Producer, Event Consumer(Page#10)**
- Event Handler, Event Producer
- Event Generator, Event Producer

35. Auto-implemented properties do not allow simplified syntax.

- True
- **False(Page#30)**

36. Clean room design is useful as a defense against_____.

- Copyright
- Trade secret infringement
- **Copyright and trade secret infringement(Page#13)**
- None of the given options

37._____is an occurrence within a particular system or domain.

- Function
- **Event (Page#7)**
- Information
- Transaction

38.BAML stands for:

- binary application map language
- binary application markup linkage
- **binary application markup language(Page#69)**
- binary application map linkage

39. If we want to place a component at top in "dockpanel" then which one is correct syntax?

- PanelDock="Top"
- Dock="Top"
- **DockPanel.Dock="Top"(Page#93)**
- DockPanel="Top

40. Dock panel allows easy docking of elements to an entire side.

- **True(Page#91)**
- False

41. _____processing operations include reading, creating, transforming, and deleting events.

- Observed Events
- **Common event(Page#8)**
- Deducted Events
- Probabilistic Events

42. The _____in the namespace indicate a hierarchy of nested namespaces.

- Arrows
- Dots GOOGLE
- Semicolon
- **Brackets HO (Page#22)**

43. The interface forces each component to expose specific _____ members that will be used ina certain way.

- **Public(Page#35)**
- Private
- Both of above

- None of the given

44. Static ctor exists to initialize class static members and called only____.
- **Once(Page#23)**
- Twice
- Anytime
- All of the given

45. Interfaces define a contract.
- **True(Page#35)**
- False

46. Additional XML namespaces (on the root or on children) must be _____ to be used on anyidentifiers from that namespace.
- Given a common prefix
- Qualified with "Pre" keyword
- **Given a distinct prefix(Page#63)**
- Qualified with "Ext" keyword

47. We want the width of button should cover the whole width of StackPanel in WPF then whatshould be value of HorizontalAlignment property of button?
- Parent
- Whole
- Fill
- **Stretch(Page#97)**

48. _____decouple producers and consumers.
- Requests
- **Events(Page#10)**
- Functions
- All of these

49. To declare an anonymous method, you just use keyword "__".
- handler
- event
- **delegate (Page#47)**

- none of the given

50. A (n)_____is a subscription mechanism for events.

- Raw Event
- Event Stream
- **Event Channel (Page#10)**
- Event Producer

51. "Situation" is an event occurrence that requires a (n)_____.

- Class
- Object
- **Reaction**
- Action

52. XML is a stricter version of_____.

- CSS
- C#
- Java
- **HTML**

53. _____contacts the service provider or the server.

- Event-driven
- **Service Requester (Page#10)**
- Both of the above
- None of the given options

54. Any class, including the same class that the event is declared in, may register one of its methods with the_____.

- **Event (Page 39)**
- Delegate
- Function
- Class

55. There are_____methods to read XML document.

- **Two (1.xmldocument 2.xmlreader) Page#47**

- Three
- Four
- Five

56. releasing resources or doing some other type of cleanup.

- Functions
- **Exception (Page#41)**
- Abstraction
- All of these

57. _____are called by garbage collector.

- Constructors
- **Destructors (Page#23)**
- Both of above
- None of the given

58. Attributes are generally applied_____in front of type and type member declarations.

- Logically
- **Physically (Page#42)**
- Both of above
- None of given

59. Multiple_____are conventionally stored in multiple files.

- **Classes (Page#22)**
- Values
- Variables
- Functions

60. The_____,_____and_____of applications that use events, either directly
or indirectly is called event-based programming.

- Model, Code, Operation

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- Analyze, Design, Operation
- **Design, Coding, Operation (Page#8)**
- Code, Operation, Maintain

61. C# is a _____ and .NET is a _____.

- Platform, Language
- **Language, platform (Page#13)**
- Package, Language
- Language, Package

62. Windows Presentation Foundations (WPF) was publicly announced in _____.

- 2000
- 2001
- 2002
- **2003 (Page#61)**

63. WPF stands for:

- Windows Powerful Foundations
- Windows Presentation Formations
- Windows Presentation Functions
- **Windows Presentation Foundations (Page#61)**

64. Value-types hold their value in_____where they are declared.

- Variable
- **Memory (Page#33)**
- Object
- Reference

65. Indexers can take any number of _____.

- Arrays
- Variables
- **Parameters (Page#31)**

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- All of these

66. Attributes add declarative_____to your programs.

- Clearance
- **information (Page#41)**
- Meta Data
- Ambiguity

67. Cast back to base type to call a/an _____method of base.

- Instance
- Base Class
- **Overridden (Page#24)**
- Derived Class

68. _____is said to be stateless if the way it processes one event does not influence the way it processes any subsequent events.

- Event Producer
- Raw Event
- **Event Processing (Page#11)**
- Event Stream

69. _____is built from request-response and it moves away from monolithic applications.

- Event-driven architecture (EDA)
- **Service-oriented architecture (SOA)(Page#10)**
- Both of above
- None of the given option

70. Object variables are references to the_____object not the object themselves.

- **Original (Page#20)**
- Alternative
- Primary

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- None of the given

71. An indexer enables your class to be treated like a/an_____.

- Class
- Function
- **Array (Page#30)**
- Pointer

72. Polymorphism needs the signatures to be the _____.

- Different
- **Same (Page#25)**
- Alternative
- None of the given

73. Types of class members in C# are _____and_____.

- static, instance
- **instance, static (Page#22)**
- public, private
- static, private

74. Events may live _____events like in event logs.

- Inside
- **Outside (Page#10)**
- Both of above
- None of the given

75. Additional XML namespaces (on the root or on children) must be _____to be used on any identifiers from that namespace.

- Given a common prefix
- Qualified with "Pre" keyword
- **Given a distinct prefix (Page#63)**
- Qualified with "Ext" keyword

76. **The interface forces each component to expose specific _____ members that will be used in a certain way.**

- **Public (Page#35)**
- Private
- Both of above
- None of the given

77. _____allows to keep the user interface description and implementation separate.

- GDI
- **WPF (Page#61)**
- Avalon
- Altia

79. Dependency properties are represented by_____.

- **System.Windows.DependencyProperty (Page#75)**
- System.Windows.Dependency.Property
- System.Windows.Dependency
- System.Windows.Property.Dependency

80. Suppose we want to restrict the user to don‟t decrease the height of a WPF element from a specific amount then which property will be used?

- HeightMinimum
- MinimumHeight
- **MinHeight**
- None of the given

81. If we want to scale an object in the 2-D x-y coordinate system then which transform class will be used?

- Rotate Transform
- Matrix Transform
- **Translate Transform**
- Skew Transform

82. Which of the following is/are not derived from “Control” class?

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- ListBox
- StatusBar
- Button
- **None of the given**

83. Wait for a single event is_____operation.

- Waiting
- Waste
- **Blocking (Page#8)**
- Idle

84. _____method reads the whole document in memory.

- XPath
- **XmlDocument (Page#49)**
- XmlReader
- XmlLine

85. Which of the following events may or may not relate to an actual occurrence?

- **Probabilistic Events (Page#7)**
- Actual Events
- Real Events
- Expected Events

86. The System.Console is a_____ which enables us to do Console Input and Output.

- Structure
- **Class**
- Namespace
- Library

88. Clean room design is useful as a defense against _____.

- Copyright
- Trade secret infringement
- **Copyright and trade secret infringement (Page#13)**

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- None of the given options

89. Clean room design is also called_____.

- Clear Room Design
- Wall Technique Design
- **Chinese Wall Technique (Page#13)**
- French Wall Technique

90. Which of the following is an occurrence within a particular system or domain?

- Object
- **Event (Page#7)**
- Result
- Message

91. During the program execution, program can be stopped with the help of _____.

- Attributes
- Compiler
- **Breakpoints (Page#48)**
- Exception Handling

92. Which of the following does NOT include in event processing operations?

- Deleting Events
- Reading Events
- **Terminating Events (Page#8)**
- Transforming Events

93. The result of Exception can bring_____in the program.

- Reliability
- Stability
- **Inconsistency**
- Consistency

95. Graphical user interfaces and the event-driven model is not applied on which of the following?

- Web Application

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- **Console Application (Page#4)**
- Mobile Application
- Desktop Application

96. Events are based on the principle of_____.

- Request
- Response
- Coupling
- **Decoupling (Page#10)**

97. _____methods do not have names.

- Attribute
- Non-Anonymous
- **Anonymous (Page#47)**
- None of give

98. Which of the following is an entity that introduces event into the system?

- Event Channel
- **Event Producer (Page#11)**
- Event Stream
- Event Consumer

99. Anonymous methods_____the code size.

- Double
- Expand
- **Reduce (Page#47)**
- Extend

100. _____events may or may not relate to an actual occurrence

- Layer
- Properties
- Event properties
- **Probabilistic (Page#7)**

101. Task Parallel_____exploit multicore for real parallel tasks.

- Structure
- **Library (Page#175)**

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- Class
- Namespace

102. The Main method returns a non-zero value which indicates the_.

- Source of Program
- **Error in Program**
- Termination of Program
- Exception in Program

103. WPF4 was released in_____.

- 2012
- **2010 (Page#61)**
- 2005
- 2003

104. Extension methods can only be defined in_____class.

- **Static class**
- Dynamic class
- Normal class
- String class

105. BAML is compressed representation of_____.

- CAML:NS
- **XAML (Page#69)**
- XAML:NS
- CAML

106. _____processing operations include reading, creating, transforming,

- **and deleting events.**
- Observed Events
- **Common event (Page#8)**
- Deducted Events
- Probabilistic Events

107. Static ctor exists to initialize class static members and called only _____.

- **Once (Page#23)**
- Twice
- Anytime
- All of the given

108. _____Interact with web page layout.

- Java script
- **HTML**
- XML
- DOM

109. _____supports the creation of applications that run directly in a web browser.

- C#
- **WPF**
- HTML
- C/C++

110. Which one of the following panels is most powerful, versatile and customizable?

- Doc panel
- Stack panel
- **Grid Panel (Page#92)**
- Canvas panel

111. With the help of WPF, an ugly looking application can be _____ by the designers.

- Rearranged
- Re-created
- Rejected

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- **Re-themed (Page#61)**

112. Graphical user interfaces and the event-driven model are not applied on which of the following?

- Web Applications
- Mobile applications
- Desktop Applications
- **Console Applications (Page#4)**

113. C# is most often used with some implementation of the _____.

- Compact Language Interface
- **Common Language Infrastructure (Page#14)**
- Control Language Infrastructure
- Control Language Interface

114. In C#, comments are written using_____.

- # and */ /*
- **// and /* */**
- # and /*
- // and */ /*

115. XML document is readable by_____.

- Human only
- Machine only
- No one
- **Machine and Human both (Page#48)**

116. "Console" is a/an_____class in the "System" namespace.

- Input
- Dynamic
- **Static (Page#15)**
- Virtual

117. The only _____.Net language has an intrinsic understanding of dependency Properties.

- Vb.net

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- **Xaml (Page#74)**
- C#
- J#

118. Which of the following grabs the "OK" button by only knowing its name?

- Button okButton = (Button)window.GetName("okButton")
- Button okButton = (Button)window.getElementByName("okButton")
- Button okButton = (Button)window.FindName("okButton")
- **Button okButton = (Button)window.GetButton("okButton")**

120. Unlike type converters, however, markup extensions are invoked from XAML with _____.

- Bridge assemblies
- **Explicit and consistent syntax (Page#65)**
- Wrapper code
- None of the given options

121. Which statement is True about interfaces?

- An interface can contain only methods, variables, events, and indexers.
- An interface can contain only variables, properties, events, and indexers.
- An interface can contain only methods, properties, variables, and indexers.
- **An interface can contain only methods, properties, events, indexers.**

122. Events can be_____.

- Specialized
- Composed
- Generalized
- **All of the given options**

123. g++ is a command of_____.

- Dev C++

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- **Unix based system**
- Windows based system
- None of the given options

124. _____reads entire document in memory.

- **Xmldocuments (Page#49)**
- Xmlreader
- Both of above
- None of the given

125. _____may contain events.

- Requests
- Service
- **Messages (Page#10)**
- Applications

126. A customer order can be represented as_____.

- Event
- Request
- **Event or Request (Page#10)**
- All of the given

127. Markup extensions are invoked from_____with explicit and consistent syntax.

- C#
- HTML
- **XAML (Page#65)**
- None of the given

128. All the members of interfaces are implicitly_____.

- Private
- Static
- **Public**
- Abstract

129. A compiled C# file is called_____.

- Complied File

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- Source File
- **Assembly File**
- EXE File

130. _____add declarative information to your programs.

- **Attributes (Page#41)**
- Functions
- Exception
- Objects

131. Anonymous method is a method without a name.

- **True (Page#47)**
- False

132. "out" parameters which are not used for return values or the "params" argument for variable arguments.

- True
- **False (Page#20)**

133. _____types have the notion of referential identity.

- Value
- **Reference (Page#14)**
- Both of above
- None of the given

134. _____let any classes behave like an array.

- Properties
- Fields
- **Indexers**
- Finalizers

135. _____are algorithms that operate on data.

- Variables
- Functions
- **Methods (Page#37)**
- Classes

136. _____is an event that is introduced into an event processing system by an event producer.

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- Event Producer
- Event Consumer
- **Raw Event (Page#11)**
- Event Stream

137. Events and_____work hand in hand.

- Parameters
- **Delegates (Page#39)**
- Functions
- Data Members

138. C# event is a class member that is _____whenever the event it was designed for occurs (fires).

- **Activated (Page#38)**
- Deactivated
- Transformed
- Associated

139. _____is a static class in the System namespace.

- **Console (Page#15)**
- Pointer
- Variable
- Function

140. Default value of first member is _____.

- **(Page#44)**
- 1
- 2
- 3

141. _____are inherited by classes which provide the real implementation.

- FunctionsVariables
- **Interfaces (Page#35)**

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- Objects

142. _____is basically array of arrays.

- Stack
- Pointer
- **Jagged (Page#18)**
- Function

143. The initial name for C# was_____.

- COAL
- COOP
- COAP
- **COOL (Page#13)**

144. BAML is compressed representation of_____.

- CAML:NS
- **XAML (Page#69)**
- XAML:NS
- CAML

145. Extension methods_____access the private data members of the class in which they are extended.

- May
- **Cannot**
- May not
- Can

146. Which of the following is/are not derived from "Control" class?

- ListBox
- StatusBar
- Both of above
- **None of the given**

147. _____events may or may not relate to an actual occurrence.

- Observed Events
- Deducted Events
- **Probabilistic Events (Page#7)**
- None of the above

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148. _____is an entity that receives events from the system.

- Event Producer
- **Event Consumer (Page#11)**
- Event Channel
- Event Generator

149. The easiest way to declare an event is to put the event keyword in front of a_____member.

- Interface
- **Delegate**
- Class
- Struct

150. keyword is used to declare a base class method while performing overriding of baseclass method.

- Overload
- **Virtual**
- Extend
- Override

151. Choose the correct option among the following indexers which correctly allows to indexin sameway as an array?

- **A Class**
- A function
- An interface
- A property

152. A class can inherit from_____class/classes in C#.

- Three
- **Single**
- Two
- Multiple

153. Which one of the following is not the feature of C#?

- **Multiple inheritance**
- Interface supported
- Locals can not shadow global variables
- Operator Overloading is allowed

154. Which keyword is used to declare the indexers?

- super
- base
- **this**

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- extract

155. The common type system of C# has value types and_____types.

- Class
- Object
- **Reference**
- Enum

156. Select the correct statement about the structure in C#?

- **Its syntax is similar to class**
- It can implement destructors
- It is a reference type
- It can implement inheritance

157. Reaction of events are either_or_____.

- Asynchronous, Blocking
- Synchronous, Blocking
- Blocking, Synchronous
- **Synchronous, Asynchronous**

158. is an example of request and response based application

- **Web browser**
- E –Commerce Site
- Pressing a button
- Swiping of credit card

159. Which of the following is built from request-response?

- Event-based Architecture (EBA)
- Service-driven Architecture (SDA)
- **Service-oriented Architecture (SOA)**
- Event-processing Architecture (EDA)

160. and delegates work hand in hand. (Page-38)

- Static member
- **Event**
- Post-processor
- Pre-processor

161. In C#, when a value type is converted to object type, it is called_.

- Polymorphism
- Abstraction
- **Boxing**
- Data Conversion

162. Which keyword is used for correct implementation of an interface in C#?

- Intf
- **Interface**
- Int Interface

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- Interfaces

163. The main method return a non-zero value which indicates the

- Exception in program
- Termination of program
- **Error in program**
- Success of program

164. is an occurrence within a particular system or domain.

- Program
- Class
- **Event**
- Object

165. Select wrong statement about destructor inC#?

- **Destructor can have modifiers or parameter**
- Destructor cannot be inherited or overloaded
- Destructor work opposite to constructor
- Class can have one destructor only

166. The main method returns zero value which indicates the

- **Successful execution of program**
- Source of program
- Termination of program
- Exception in program

167. In C#, which of the following keyword is used for including the namespaces in theprogram?

- **using**
- extend
- import
- #

168. When your C# program is compiled, it creates a file called

- Class
- EXE
- Object
- Assembly According to the Instructor

169. Which of the following is an insertion operator in C++?

- <<
- >
- -
- &

170. C#_is a class member that is activated whenever the event . it was designed for.Occurs

- Attributes
- Arguments
- Method

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- **Event**

171. A fraud detection event on banking transaction is an example of.

- Linear event
- Sequential event
  - Predictive event
- **Probabilistic event**

172. In C++, which of the following library is include for 'cin' and 'cout' function?

- #include<fstream.h>
- #include<fstream>
- #include<conio.h>
- **#include<iostream>**

173. Which of the following doesn't have Graphical User Interface and the Event-DrivenModel?

- **Console-Application**
- Web Application
- Mobile Application
- Desktop Application

174. A stream in which the events may be of different types is referred to as a **heterogeneous** eventstream.

- Continuous
- Homogenous
- **Heterogeneous**
- Mutual

175. Defining a method in subclass having same name and type signature as a method in itssuperclass is known as

- Method hiding
- Method Overloading
- Method Binding
- **Method Overriding**

176. A single try block must be followed by which of these?

- Finally
- Catch
- **Both Finally and catch**
- Throw

177. By default the data member of a class are

- Protected, public
- Public
- **Private**
- Private, Public

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178. Exception are derived from___class. (Page-39)

- Windows error
- Exception error
- **System exception**
- Error Exception

179. Which of the following does NOT include in event processing operations?

- **Terminating Event**
- Deleting Event
- Transforming Event
- Reading Event

180. What is the return type of destruction?

- Float
- **No return type**
- Void
- Int

181. At the time an event is fired, the_method (with that event) will be invoked.

- **Public**
- Protected
- Registered
- Private

182. C# is a____and .NET is a_.

- Platform, Language
- Package, Language
- Language, Package
- **Language, Platform**

183. Which of the following is an entity that introduces event into the system?

- **Event producer**
- Event Stream
- Event Channel
- Event Consumer

184. Which of the following is an entity that receive event from the system?

- Event producer
- **Event Consumer**
- Raw event
- Event stream

185. Which of the following is the object-oriented way to handle run time errors?

- OnError () Method
- **Exception**
- Interface
- Error Code

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186. Why do we use cin get() in C++ program?

- ❖ It is used to print all values of array
- ❖ It is used to get numeric data
- ❖ **It is used to access character array**
- ❖ It is used to take input from user

187. Types not defined in any namespace are said to reside in the _namespace.

- Static
- Virtual
- C. Global

❖ D. Local

188. Arrays in C# are_object.

- ❖ Logical
- ❖ Arithmetic
- ❖ **Reference**
- ❖ Value

189. The initial name for C# was_.

- ❖ COAL
- ❖ COOP
- ❖ COAP
- ❖ **COOL**

190. Which of the following is declared like a class but have no implementation?

- ❖ Encapsulation
- ❖ Structure
- ❖ **Interface**
- ❖ Inheritance

191. Which of the following keyword is used to change data and behavior of a base class byreplacinga member of the base class with the new derived member.

- ❖ Base
- ❖ Overrides
- ❖ Overloads
- ❖ **New**

192. A stream in which all the events must be of same types is called **Homogenous** eventstream

- ❖ Continuous
- ❖ **Homogenous**
- ❖ Heterogeneous
- ❖ Mutual

193. Select the correct statement about an exception>

- ❖ It occurs during Just-in-time compilation
- ❖ **It occurs at runtime**
- ❖ It occurs during loading of program

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- It occurs during data binding

194. is not a defense against patents and clean room design?

- **Independent Invention**
- Event Invention
- Dependent Invention
- Clear room design

195. COOL stands for_____.

- Class Oriented Object Language
- Consumer Oriented Object Language
- **C-like Object Oriented Language**
  - C++-like Oriented Language

196. operator is used to signifies the destructor in C#?

  - #

- B. =
- C. &
- D. ~

197. What is the return type of destruction?

- Int
- Float
- Void
- **No return types**

198. An Event that is generated as a result of event processing that take place inside an eventprocessing system is called . _

- Event Stream
- **Derived Event**
- Raw Event
- Event Consumer

199. Choose the incorrect statement about delegates in C#?

- Delegates are value type
- Delegates are like function pointers in other language
- Delegates are object oriented
- **Delegates are type safe**

200. is the method of copying a design by reverse engineering and then recreating it without infringing any of the copyrights and trade secrets associated with the originaldesign.(Page-12)

- Clear Room Design
- Event Room design
- **Clean Room Design**
- Wall Room Design

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201. Which of the following statement is TRUE about raw event?

- It is a subscription mechanism for event
- It is introduced into an event processing system by an event consumer
- It is generate as a result of event processing
- **It is introduced into an event processing system by an event producer**

202. Which symbol is used to declare the preprocessor directives?

- *
- **#**
- $
- ^

203. Which of these keyword is used to manually throw an exception in C#?

- Try
- Finally
- Catch
- **Throw**

204. On successful compilation of a java source code_____is generated.

- Output
- **Bytecode**
- Hexadecimal code
- Octal code

205. _____ is the necessary condition to implement delegates.

- **Class Declaration**
- Function overloading
- Inheritance
- Runtime Polymorphism

206. Select correct statement about properties of read and write in C#

- **A read only property will only have get accessor**
- A write only property will only have get accessor
- A property cannot be either read only or write only
- A property can simultaneously be read or write only

207. <p> What will be the output of the following C# code?Static void Main{string[] args}

```
{
  int i;   for  (i = 0;  ;)
  {
    Console.WriteLine(&amp;quot;Virtual University&amp;quot;);
  }
  Console.ReadLine();
}
```

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- || will print "Virtual University" in continuous loop
- ** Syntax error **
- Virtual University
  - D. =p= No output =/p=

208. Choose the wrong statement about structures in C#.Net.

- **Structures can have destructors**
- Structures can implement an interface but they cannot inherit from another structure
- Structure members cannot be declared as protected
- A structure cannot be empty

209. Which of the following keyword is used to monitor exceptions?

- Catch
- Finally
- **Try**
- Throw

210. _____ library is used for dictionary in C#.

- System.Windows.Dictionary
- System.Windows.DictionaryResource
- System.Windows.ResourceDictionary
- System.Windows.Resource

211. In WPF,_____class define an object which has read-only(forzen)state.

- Read-Only
- Freeze
- Frozen
- Freezable

212. Which one of the following is the default panel in Visual Studio and expression blend?

- Stack Panel
- Wrap Panel
- Grid
- Canvas

213. Which one of the following is the first step in compiling XAML?

Performing the plumbing that connects XAML with procedural code automatically

- Converting XAML into.exe file
- Converting XAML into a special binaryformat
- Converting XAML into bytecode

214. Which of the following property will be used to restrict the user to not increase the width of a WPF element from a specific value.

- MinimumWidth
- MinWidth

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- WidthMinumum
- DecreaseWidth

215. ____transform requires an angle in transformation.

- Button transform
- Rotate transform Not SUre
- Element transform
- Scale transform

216. "Viewbox" element uses different types of stretching with the help of stretch property. There is/are _____possible value(s) of stretch property.

- 4
- 3
- 1
- 2

217. Markup extensions are surrounded by____

- Curly braces
- Square Brackets
- Parentheses
- Angle Brackets

218. ____is a timer based change of value.

- Themes
- Template
- Animation
- Styles Not Sure

219. _____Element is used in XAML to embed the C# code in XAML file.

- X:Code
- X:Content
- X:Name
- X:Button

220. The reason behind why XAML is good for user Interface is because of its_____nature.

- Class
- Hierarchical OR UI
- Base

221. _____ is used to declare a class into multiple files and all these files are combined into a single class file when the application is compiled.

- Property Element
- Abstract Class
- Inheritance
- Partial Class

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222. Margin defines an extra space_____while considering edges of the element.

- Outside
- Inside
- Between
- Anywhere

223. Which provider has the highest priority while providing a value to a dependency property?

- Style Setters
- Theme Style Setters
- Default Value
- Local Value

224. 'FrameworkElement' is the base class that adds support for _____

- Polymorphism
- Inheritance
- Abstract Class
- Data binding

225. Type Converter exists for_____transform.

- Render
- Matrix
- Scale
- Translate

226. _____classes are used to restrict the inheritance in C#.

- Sealed
- Property
- Object

227. _____class provides type conversion to convert a string name into an object reference to the object with that name.

- NameReferenceConverter
- NameConverter
- NameObjectConverter

228. Dockpanel.Dock is the___ in which child element describe its dock position in dock panel.

- Button Property
- Grid Property
- Attach Property

229. Type converters in WPF are similar to_____

- Multimedia Integration
- Markup extensions Not Sure
- Namespaces

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- Rich compositions

230. Which of the following device is sensitive to pressure?

- Mouse
- Printer
- Stylus
- Keyboard

231. Which will take precedence in sizing if explicit given width and height value of WPF application are not in range?

- Width=-1,height =-1
- Min,max(default values)
- Width=0,height=0

232. Which one is considered to be a keyword with reference to XAML 2006?

- X Name
- X Static
- X Null
- X Array

233. In WPF Grids have _____base indexing for Rows& Columns.

- One
- -1
- Zero
- Two

234. By default there will be ____ number of columns if we do not mention any column definition in grids.

- 2
- 3
- 1
- None

235. Translate Transform work on____ axes.

- X,Y
- X,Y,Z
- Y

236. In WPF, if Trigger is _____then its Setter's value will be Set.

- Hidden
- Inactive
- Active

237. Default value if Grid Row is_____

- 0
- 2
- **1**

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- 3

238. Superset of Canvas is ____

- Stack Panel
- Dock Panel
- Wrap Panel
- Grid

239. _____function will be register a Dependency property in WPF.

- DependencyProperty.Registration()
- DependencyProperty.Registered()
- DependencyProperty.Reg()
- DependencyProperty.Register()

240. Equivalent norms on a vector space 'X'defines _____topology for X.

- Coarser
- None of these
- Same Not Sure
- Finer

241. Dock panel is a ___of stack panel.

- Intersection
- Superset
- Union
- Subset

242. All classes of Windows Presentation foundation are derived form___class.

- Object
- Element
- XAML
- Abstract

243. Dock Panel is used in____interface.

- Right level
- Bottom level
- Top level
- Left level

244. In XAML namespace other than the default one is denoted by_____

- Xmlns(x)
- Xlmns/x
- Xlmns.x
- Xlmns:x

245. In WPF, if there is any name property than that property is also be taken as a substitute of_____

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- X:static
- X:reference
- X:binding
- X:name

246. In XAML 2006,X:Classmodifier by default access level is_____

- Public
- Internal
- Private
- Protected

247. _____sequence is not allowed within a CDATA section.

- {}
-]]
- <>
- ()

248. If we want to place a component at top in "dockpanel" then which one is correct syntax?

- DockPanel.Dock="top"
- Dock="Top"
- PanelDock="Top"
- DockPanel="Top"

249. In WPF, the default value of Horizontal/Alignment properties of child Element is_____

- Right
- Center
- Stretch
- Left

250. Which property is used to change the Z order of elements in WPF?

- ZPanel
- Zvalue
- ZIndex
- Z-coordinates

251. 1/96 pixel are equal to ___inch(es) in DPI display setting.

- 3
- 2
- 4
- 1

252. In WPF,_____control will allow to you visually group a set of control together.

- GroupBox
- Group

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- Box_Group
- Boxed_group

253. In WPF,_____function registers an attached property with the property system.

- Register_Attached(String,Type,Type,PropertyMetadata,ValidateValueCallback)
- RegisterAttached(String,Type,Type,PropertyMetadata,ValidateValueCallback)
- Reg_Attached(String,Type,Type,PropertyMetadata,ValidateValueCallback)
- None

254. In grid, elements are arranged in multicolumn and multirow like a____in HTML.

- Column
- Paragraph
- List
- Table

255. In XAML, tag declaration is equivalent to C#'s_____

- Object declaration
- Object Validation
- Variable declaration
- Object instantiation

256. Mapping to the WPF namespaces is _____inside the WPF assemblies.

- Dynamically specified
- Hard-coded
- Statically declared
- Specified at compile time

257. _____convert from a string in XAML markup into particular objects/data types in an object graph.

- Rich composition
- Resolution Independence
- Type converters
- Data binding

258. In WPF, if we try to add listbox item in listbox object which is already null, it will give us____

- Runtime exception
- Syntax Error
- Compile time error
- Logical error

259. In WPF, Subtags don't add as a(n)_____

- List
- Array
- Designated Attribute

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- Dictionary

260. Tag of namespace in XAML is just a_____

- Float
- Property
- Integer
- String

261. Is SharedSizeScope share size information in _____grid(s).

- Multiple
- Single
- Both 1&2
- Double

262. How can you specify null property in markup extension?

- :null:
- {x:null}
- (null)
- [x:null]

263. _____property value is used to set width and height of element in WPF according to its content.

- Tree
- Nan
- X:class
- Name

264. In WPF,____library is used for scrolling.

- Systems.Windows.Controls.ScrollViewer
- Systems.Windows.Controls.Scroll
- Systems.Windows.Controls.Viewer
- Systems.Media.ScrollViewer

265. When element in XAML overlaps in that scenario which technique is used to decide which element should come at the top of other element?

- Scaling
- Alignment
- Rotate
- Z Order

266. Which one of the following is the built-in transform function in WPF?

- Left Move
- Query transform
- Top transform
- Matrix transform

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267. In WPF, _____use(s) the method to add additional data with the sealed class object.

- Class
- Tree Structure
- Xmlns
- Dependency Properties

268. Grid act as a stack panel if it has____

- 1 row and autosize column
- 1 row and autosize rows
- Multiple rows and single column
- Single row & multiple column

269. In context of XML child element rules, if the child is plain text and a type converter existsto transform the child into the parent type (and no properties are set on the parent element), treat the child as input to the type converter and use the output as the _____.

- Child Object instance
- **Parent object instance**
- Interface object instance
- None of these

270. The root object in XAML must specify a least___XML namespace(s) that is/are used toqualify itself and any child elements (Page-62)

- **One**
- Two
- Three
- None

271. _____library is used to import the dependency poperties in WPF project (Page-74)

- System.Windows.Property.Dependency
- System.Windows.Dependancy
- System.Windows.Dependancy.Property
- **System.Windows.DependancyProperty**

272. A_transforms elements before they are laid out by the parent panel.

- Skew transform
- Render transform
- **Layout transform**
- Matrix transform

273. Defining any type of controls as a content of another control is __

- Vector based rendering
- **Rich Composition**
- Data Bining
- Resolution dependency

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274. The goal of WPF and XAML is used to keep separate the user interface design and ___

- Tools
- Content
- Element tree
- **Implementation**

275. Wgen elements in XAML overlaps, in that scenario which technique is used to decidewhich element should come at the top of other element?

- **Z order**
- Rotate
- Scaling
- Alignment

276. Visual Studio has a snippet called _ that automatically expands into a definition of a dependencyproperty, which makes defining one much faster than doing all the typing yourself.(Page-74)

- Property Extension
- **Propdp**
- Dependency property
- Dpprop

277. CAML stands for_.

- **Compiled Application Markup language**
- Composite Application Markup language
- Compiled Application Matrix language
- Centralized Application Markup language

278. _____is the callback which can be registered with property that can change the value ofproperty.

- Bubbling
- Element Property
- Tunneling
- **Coerce**

279. _library is used for Transforms in C#? (Page-82)

- System.Control.Transform
- System.Windows.Transform
- **System.Windows.Media.Transform**
- System.Windows.Controls.Transform

280. In WPF, Translate transform has_default value for X and Y? (Page-84)

- **0**
- 3
- 2
- 1

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281. DependancyProperty.Register is the__type of method of dependency property.

- Abstract
- Element
- **Static**
- Class

282. In XML, CDATA Section looks like

- <[CDATA[…]]>
- <![CDATA[…]]
- **<![CDATA[…]]>**
- ![CDATA[…]]>

283. Default Value of Grid.Row is

- 1
- 3
- 2
- **0**

284. ____is a base class similar to UIElement but for document-related pieces ofcontent that don't have rendering behavior on their own.

- VisualElement
- SystemElement
- **ContentElement**
- XAMLElement

285. In WPF ,_____function adds another type as an owner of a dependency property that hasalready been registered to a type.

- Owner(Type.propertyMetadata)
- Add_Owner (Type.Propert Metadata)
- **AddOwner(Type.PropertyMetadata)**
- Owner_Add(Type.Property Metadata)

286. xmlns namespace must be declare once in a XAML file at_element. (Page-67)

- Form
- Middle
- **Root**
- Child

287. Which type of panel is used when we have to customize the builtin controls in wpf?

- Wrap
- Grid
- Primitive
- **Stack**

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288. If we want to rotate an object clockwise about a specified point in a 2-D x-y coordinate systemthen which transform class will be used? (Page 87)

- **Rotate Transform**
- Translate Transform
- Skew Transform
- Scale Transform

289. A___enables to extend XAML and set any property that can be set in XAML using attributessyntax.

- **Markup extension**
- Reduction independence
- Data binding
- Rich composition

290. In WPF, Subtags don't add as a(n)___

- **Designated attribute**
- List
- Dictionary
- Array

291. ____key will provide a label access in WPF

- **Alt**
- CTRL
- Ctrl+alt
- Shift

292. Which one of the following is the first step in compiling XAML?

- Converting XAML to byte code
- Performing the plumbing that connects XAML with procedural code automatically
- **Converting a XAML file into a special binary format**
- Converting XAML into .exefile

293. "Viewbox" element uses different types of stretching with the help of stretch property .There is/are _____possible value(s) of stretch property

- 2
- 1
- 3
- **4**

294. Singleton method has__instance

- 4
- 3
- 2
- **1**

295. Suppose the property of x coordinate at the button timely changes in such a way thatafter every 100 second value of x coordinate increments by 1, then this case refers to

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- Transformation
- Property element
- Tunneling
- **Animation**

296. __refers to the space between an element and the content inside it.

- Tunneling
- Margin
- Pading
- **Style**

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Visual Programming CS411

Grand Quiz 2020 (COVID-19)

Solved by Sonu Ilyas Mughal

Sonu Mughal

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CS411:Midterm Quiz

Question # 1 of 30 (Start time: 04:28:38 PM, 11 July 2020)

In WPF, If we try to add listbox item in listbox object which is already null, it will give us

Select the correct option

- ○ Runtime exception
- ● Compile time error
- ○ Syntax error
- ○ Logical error

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Question # 2 of 30 (Start time: 04:29:57 PM, 11 July 2020)

XML is a stricter version of _____.

Select the correct option

- ○ Java
- ○ CSS
- ● HTML
- ○ C#

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Question # 3 of 30 (Start time: 04:30:26 PM, 11 July 2020)

Which of these methods used to read strings from the console?

Select the correct option

- Get()
- Getline()
- Read()
- ReadLine()

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CS411:Midterm Quiz

Question # 4 of 30 (Start time: 04:31:22 PM, 11 July 2020)

Which of the following statement is correct about WPF?

Select the correct option

- ○ WPF can use Python language for designing
- ○ WFP can developed mobile apps
- ● WPF is resolution independence
- ○ WPF not support declarative programming

Original

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Question # 5 of 30 (Start time: 04:31:47 PM, 11 July 2020)

The code given below is equivalent to _____ .
<SolidColorBrush>White</SolidColorBrush>

Select the correct option

- <WhiteBrush >MyWhiteBrush </ WhiteBrush>
- None of the given options
- <SolidColorBrush Color="White"/>
- <Brush Style= “SolidWhite” >

Mughal

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Question # 6 of 30 (Start time: 04:32:13 PM, 11 July 2020)

WPF4 was released in _____.

Select the correct option

- 2012
- 2010
- 2005
- 2003

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Question # 7 of 30 (Start time: 04:32:45 PM, 11 July 2020)

Which of the following events may or may not relate to an actual occurrence?

Select the correct option

- Probabilistic Events
- Actual Events
- Real Events
- Expected Events

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Question # 8 of 30 (Start time: 04:33:30 PM, 11 July 2020)

All elements of WPF have following property/properties.

Select the correct option

- ○ Width
- ○ None of given options
- ● Height and Width
- ○ Height

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Question # 9 of 30 (Start time: 04:34:13 PM, 11 July 2020)

_____ is used to declare a class into multiple files and all these files are combined into a single class file when the application is compiled.

Select the correct option

- (●) Inheritance
- ( ) Partial Class
- ( ) Abstract Class
- ( ) Property Element

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Question # 10 of 30 (Start time: 04:35:18 PM, 11 July 2020)

Which of the following is an example of tunnel key event?

Select the correct option

- Key up
- Preview key entered
- Preview key down
- Key down

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Question # 11 of 30 (Start time: 04:35:54 PM, 11 July 2020)

IsSharedSizeScope shares size information in _____ grid(s).

Select the correct option

- (●) Multiple
- ( ) Single
- ( ) Both 1 & 2
- ( ) Double

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MC180400061: SIDRA FATIMA

CS411:Midterm Quiz

Question # 12 of 30 (Start time: 04:37:24 PM, 11 July 2020)

Wait for a single event is _____ operation.

Select the correct option

- Waiting
- Waste
- Idle
- Blocking

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Question # 13 of 30 (Start time: 04:37:58 PM, 11 July 2020)

Which of the following components generates an input Event?

Select the correct option

- ○ Printer
- ● Multi Touch
- ○ Plotter
- ○ Speaker

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Question # 14 of 30 (Start time: 04:38:52 PM, 11 July 2020)

Arrays in C# are _____ objects.

Select the correct option

- Logical.
- Arithmetic.
- Reference.
- Value.

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A language supported by MS.Net platform.

Select the correct option

- C
- C++
- Java
- C#

Original

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Question # 16 of 30 (Start time: 04:40:46 PM, 11 July 2020)

Extra space inside the element in WPF is known as _____

Select the correct option

- Margin
- Style
- Padding
- Tunneling

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Question # 17 of 30 (Start time: 04:41:16 PM, 11 July 2020)

Which of the following statements is TRUE about raw event?

Select the correct option

- (●) It is introduced into an event processing system by an event producer.
- ( ) It is generated as a result of event processing.
- ( ) It is introduced into an event processing system by an event consumer.
- ( ) It is a subscription mechanism for events.

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Question # 18 of 30 (Start time: 04:42:10 PM, 11 July 2020)

DependencyObject.ClearValue method clears the _____ value in a dependency property.

Select the correct option

- Default
- Local
- Setter
- Getter

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Question # 19 of 30 (Start time: 04:43:23 PM, 11 July 2020)

All the members of interfaces are implicitly _____.

Select the correct option

- Private
- Static
- Public
- Abstract

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Question # 20 of 30 (Start time: 04:44:05 PM, 11 July 2020)

Dock panel is used in _____ interface.

Select the correct option

- (●) Top level
- ( ) Bottom level
- ( ) Left level
- ( ) Right level

The data members of a class by default are?

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Question # 21 of 30 (Start time: 04:45:23 PM, 11 July 2020)

The data members of a class by default are?

Select the correct option

- Protected, Public
- Private, Public
- Public
- Private

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Question # 22 of 30 (Start time: 04:46:28 PM, 11 July 2020)

The Main method returns zero value which indicates the_____.

Select the correct option

- Source of Program
- Successful execution of Program
- Termination of Program
- Exception in Program

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Question # 23 of 30 (Start time: 04:47:38 PM, 11 July 2020)

The _____, _____ and _____of applications that use events, either directly or indirectly is called event-based programming.

Select the correct option

- Model, Code, Operation
- Analyze, Design, Operation
- Code, Oeration, Maintain
- Design, Coding , Operation

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Question # 24 of 30 (Start time: 04:48:18 PM, 11 July 2020)

Silverlight is the subset of _____

Select the correct option

- (●) .NET
- ( ) Python
- ( ) Window Forms
- ( ) C++

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Question # 25 of 30 (Start time: 04:49:16 PM, 11 July 2020)

If we insert some objects in a single cell in Grid panel, then these objects are placed on _____ .

Select the correct option

- (●) One on Top of the other
- ( ) One after the other
- ( ) One on Bottom of the other
- ( ) One on Side of the other

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Question # 26 of 30 (Start time: 04:50:17 PM, 11 July 2020)

A class can inherit from _____class/classes.

Select the correct option

- ○ Two
- ● Single
- ○ Three
- ○ Multiple

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Question # 27 of 30 (Start time: 04:50:59 PM, 11 July 2020)

If we exclude the mechanism of calling the objects through name then:

Select the correct option

- Code will not compile
- Code will be difficult to debug
- Code will have errors
- Code will become design dependent

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Margin defines an extra space_____ while considering edges of the element

Select the correct option

- Between
- Inside
- Outside
- Anywhere

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Question # 29 of 30 (Start time: 04:52:33 PM, 11 July 2020)

Suppose we want to restrict the user to don't decrease the height of a WPF element from a specific amount then which property will be used?

Select the correct option

- (●) MinHeight
- ( ) None of given
- ( ) HeightMinimum
- ( ) MinimumHeight

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Attribute parameters can be either positional parameters or _____ parameters.

Select the correct option

- (●) Named
- ( ) Static
- ( ) Global
- ( ) None of the given options

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Question # 1 of 30 (Start time: 04:56:51 PM, 11 July 2020)

Suppose we have two elements in grid one is button and other is rectangle,which property will be helpful to set position of button on top of the rectangle?

Select the correct option

- ○ Z index
- ● Sizing
- ○ Transform
- ○ Stack Panel

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Question # 2 of 30 (Start time: 04:58:03 PM, 11 July 2020)

_____ graphics work with shapes.

Select the correct option

- Scalar
- Vector
- Both 1 & 2
- None of the above

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CS411:Midterm Quiz

Question # 3 of 30 (Start time: 04:59:20 PM, 11 July 2020)

UI element focusable value is _____ by default.

Select the correct option

- (●) True
- ( ) False
- ( ) Nil
- ( ) Zero

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Question # 4 of 30 (Start time: 05:00:23 PM, 11 July 2020)

Tag of namespace in XML is just a_____

Select the correct option

- String
- Integer
- Float
- Property

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Question # 3 of 30 (Start time: 05:01:44 PM, 11 July 2020)

In which type does the enumerators are stored by the compiler?

Select the correct option

- string
- integer
- float
- none of these

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Question # 6 of 30 (Start time: 05:03:00 PM, 11 July 2020) **Total Marks**

Visual Studio has a snippet called _____ that automatically expands into a definition of a dependency property, which makes defining one much faster than doing all the typing yourself!

Select the correct option

- (●) Propdp
- ( ) Property extension
- ( ) Dpprop
- ( ) Dependency property

Saving

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Question # 7 of 30 (Start time: 05:03:30 PM, 11 July 2020)

Suppose we want to restrict the user to don't increase the height of a WPF element from a specific amount then which property will be used?

Select the correct option

- MaximumHeight
- MaxHeight
- None of given
- HeightMaximum

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Which of the following events may or may not relate to an actual occurrence?

Select the correct option

- (●) Probabilistic Events
- ( ) Actual Events
- ( ) Real Events
- ( ) Expected Events

SOL

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Question # 9 of 30 (Start time: 05:05:18 PM, 11 July 2020)

Attribute parameters can be either positional parameters or _____ parameters.

Select the correct option

- (●) Named
- ( ) Static
- ( ) Global
- ( ) None of the given options

03063548328 For Solving Quiz



Question # 10 of 30 (Start time: 05:06:01 PM, 11 July 2020)

Which of the following keyword is used to monitor exceptions?

Select the correct option

- finally
- try
- throw
- catch

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Question # 11 of 30 (Start time: 05:07:04 PM, 11 July 2020)

Which keyword is used to declare the indexers?

Select the correct option

- base
- this
- extract
- super

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We want the width of button should cover the whole width of StackPanel in WPF then what should be value of HorizontalAlignment property of button?

Select the correct option

- Parent
- Whole
- Fill
- Stretch

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Question # 13 of 30 (Start time: 05:09:15 PM, 11 July 2020)

Which of these methods used to read strings from the console?

Select the correct option

- Get()
- Getline()
- Read()
- ReadLine()

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Question # 14 of 30 (Start time: 05:09:38 PM, 11 July 2020)

Which symbol is used to declare the preprocessor directives?

Select the correct option

- $
- *
- #
- ^

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Question # 15 of 30 (Start time: 05:10:16 PM, 11 July 2020)

Which method is used to write single byte to the file?

Select the correct option

- Write( )
- WriteLine( )
- WriteByte ( )
- WriteKey( )

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Question # 10 of 50 (Start time: 08:11:44 PM, 11 July 2020)

Graphical user interfaces and the event-driven model is not applied on which of the following?

Select the correct option

- Web Applications
- Console Applications
- Mobile applications
- Desktop Applications

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Question # 18 of 30 (Start time: 05:13:37 PM, 11 July 2020)

_____ Function will register a dependency property in WPF.

Select the correct option

- DependencyProperty.Reg()
- DependencyProperty.Registeration()
- DependencyProperty.Registered()
- DependencyProperty.Register()

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Question # 19 of 30 (Start time: 05:14:43 PM, 11 July 2020)

Derived Event is:

Select the correct option

○	Introduced into an event processing system by an event consumer.
●	Generated as a result of event that takes place inside an event processing system.
○	Introduced into an event processing system by an event producer.
○	A set of associated events.

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_____ key can be represented as _text in XAML.

Select the correct option

- Protected
- Focus
- Public
- None of the above

Solved

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Question # 20 of 50 (Start time: 05:15:09 PM, 11 July 2020)

_____ key can be represented as _text in XAML.

Select the correct option

- Protected
- Focus
- Public
- None of the above

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Question # 21 of 50 (Start time: 08:10:07 PM, 11 July 2020)

Silverlight is the subset of _____

Select the correct option

- [x] .NET
- [ ] Python
- [ ] Window Forms
- [ ] C++

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Question # 22 of 30 (Start time: 05:17:31 PM, 11 July 2020)

Which type of sizing is used in Wpf?

Select the correct option

- ○ Line
- ○ Percentage
- ● Proportional
- ○ Element

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Question # 23 of 30 (Start time: 05:18:34 PM, 11 July 2020)

By convention _____ event names are prefixed with Preview.

Select the correct option

- (●) Tunneling
- ( ) Bubbling
- ( ) Direct
- ( ) None of the given

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Question # 24 of 30 (Start time: 05:19:31 PM, 11 July 2020)

We use "UIElement" property ClipToBounds= “false” when _____.

Select the correct option

- We want area child element don’t cross the boundary of parent
- None of given
- We want area of child element is allowed to cross the boundary of parent
- Parent area also increases on increasing of area of child element

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Question # 25 of 30 (Start time: 05:20:44 PM, 11 July 2020)

The _____ in the namespace indicate a hierarchy of nested namespaces.

Select the correct option

- Arrows
- Dots
- Semicolon
- Brackets

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Question # 26 of 30 (Start time: 05:21:39 PM, 11 July 2020)

If we want to place a component at top in "dockpanel" then which one is correct syntax?

Select the correct option

○	PanelDock="Top"
○	Dock="Top"
●	DockPanel.Dock="Top"
○	DockPanel="Top"

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Question # 27 of 30 (Start time: 05:22:45 PM, 11 July 2020)

If we insert some objects in a single cell in Grid panel, then these objects are placed on _____ .

Select the correct option

- (●) One on Top of the other
- ( ) One after the other
- ( ) One on Bottom of the other
- ( ) One on Side of the other

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Question # 28 of 30 (Start time: 05:23:51 PM, 11 July 2020)

Which of the following is the object-oriented way to handle run time errors?

Select the correct option

- OnError
- Error codes
- Interface
- Exceptions

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Question # 29 of 30 (Start time: 05:24:54 PM, 11 July 2020)

The code given below is equivalent to _____ .
<SolidColorBrush>White</SolidColorBrush>

Select the correct option

- <WhiteBrush >MyWhiteBrush </ WhiteBrush>
- None of the given options
- <SolidColorBrush Color="White"/>
- <Brush Style= “SolidWhite” >

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_____ is an entity that receives events from the system.

Select the correct option

- ○ Event Producer
- ■ Event Consumer
- ○ Event Channel
- ○ Event Generator

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Question # 1 of 30 (Start time: 05:33:42 PM, 11 July 2020)

Wait for a single event is _____ operation.

Select the correct option

- Waiting
- Waste
- Idle
- Blocking

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Question # 2 of 50 (Start time: 08:04:07 PM, 11 July 2020)

_____ is used to declare a class into multiple files and all these files are combined into a single class file when the application is compiled.

Select the correct option

- Inheritance
- Partial Class
- Abstract Class
- Property Element

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Question # 3 of 30 (Start time: 05:35:05 PM, 11 July 2020)

<Button.Content> tag in XAML is the syntax for _____

Select the correct option

- Namespace
- Property Element
- Class
- Form

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Question # 4 of 30 (Start time: 05:36:21 PM, 11 July 2020)

COOL stands for _____.

Select the correct option

- Class Oriented Object Language
- Consumer Oriented Object Language
- C-like Object Oriented Language
- C++-like Object Oriented Language

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Against _____ keyword a partial class is generated which we write in C# code.

Select the correct option

- (●) X:class
- ( ) Xmlns
- ( ) <button>
- ( ) X:Element

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Question # 8 of 30 (Start time: 05:39:08 PM, 11 July 2020)

Suppose we have declared a button in XAML as shown in given code.Choose the correct C# code which get this button in C#.
<Button x:Name="okbutton">OK</Button>

Select the correct option

○	Button okbutton=(Button)window.("okbutton");
○	Button okbutton=(Button)window.searchname("okbutton");
○	Button okbutton=(Button)window.getbutton("okbutton");
●	Button okbutton=(Button)window.findname("okbutton");

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Question # 9 of 50 (Start time: 08:40:10 PM, 11 July 2020)

Additional XML namespaces (on the root or on children) must be _____ to be used on any identifiers from that namespace.

Select the correct option

- Given a common prefix
- Qualified with “Pre” keyword
- Given a distinct prefix
- Qualified with “Ext” keyword

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Question # 10 of 30 (Start time: 05:41:23 PM, 11 July 2020)

Arrays in C# are _____ objects.

Select the correct option

- Logical.
- Arithmetic.
- Reference.
- Value.

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Question # 11 of 30 (Start time: 05:41:51 PM, 11 July 2020)

When your C# program is compiled, it creates a file called _____

Select the correct option

- (●) Assembly
- ( ) Exe
- ( ) Object
- ( ) Class

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Question # 12 of 30 (Start time: 05:42:50 PM, 11 July 2020)

_____ property value is used to set width and height of element in WPF according to its content.

Select the correct option

- Name
- X:class
- Nan
- Tree

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Question # 13 of 30 (Start time: 05:43:34 PM, 11 July 2020)

What is the return type of destructors?

Select the correct option

- int
- float
- void
- no return types

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Question # 14 of 30 (Start time: 05:44:34 PM, 11 July 2020)

In context of XAML child element rules, if the parent supports a content property and the type of the child is compatible with that property, treat the child as its _____

Select the correct option

- Super class
- Sub class
- Value
- Interface

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Question # 20 of 30 (Start time: 05:49:44 PM, 11 July 2020)

Event-based programming is also known as:

Select the correct option

- Service-oriented architecture (SOA)
- Event-driven Architecture (EDA)
- Service-driven Architecture (SDA)
- Event-based Architecture (EBA)

“FrameworkElement” is the base class that adds support for _

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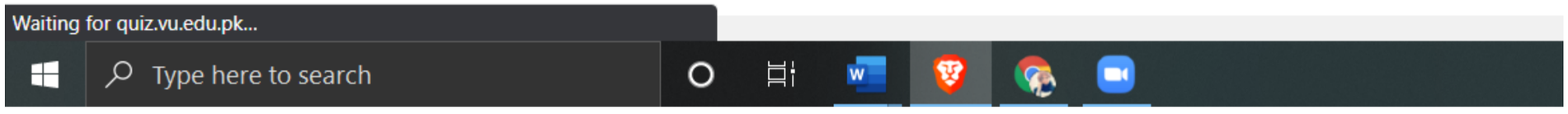
CS411:Midterm Quiz

Question # 21 of 30 (Start time: 05:51:13 PM, 11 July 2020)

When a user select element from listbox, after selecting compiler marked the selected element as____

Select the correct option

- Fix
- Selected
- Handled
- Bubble



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Design of Silverlight mostly follow the design of _____

Select the correct option

- C++
- WPF
- Console
- C#

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Question # 23 of 30 (Start time: 05:54:00 PM, 11 July 2020)

Which of the following is/are not derived from “Control” class?

Select the correct option

- ListBox
- StatusBar
- None of the given
- Button

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Question # 24 of 30 (Start time: 05:55:08 PM, 11 July 2020)

To design a complex Interface, which of the following technique is best to achieve the required results?

Select the correct option

- ○ Panels composed in three panels
- ● Panels composed within panels
- ○ Panels composed in two panels
- ○ Panels composed in single panel

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Question # 25 of 30 (Start time: 05:56:09 PM, 11 July 2020)

In tunneling, event move from _____.

Select the correct option

- ○ Left to Right
- ○ Right to Left
- ● Top to Down
- ○ Diagonally

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Question # 26 of 30 (Start time: 05:57:17 PM, 11 July 2020)

Attributes of a program can be retrieved from its assembly metadata through _____.

Select the correct option

- Static member
- Reflection
- Garbage collector
- Constructor

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Question # 27 of 30 (Start time: 05:58:20 PM, 11 July 2020)

Clean room design is also called _____.

Select the correct option

- Clear Room Design
- Wall Technique Design
- Chinese Wall Technique
- French Wall Technique

Sohail Mughal

03063548328 For Solving Quiz



Question # 28 of 30 (Start time: 05:58:46 PM, 11 July 2020)

The code given below is equivalent to _____ .
<SolidColorBrush>White</SolidColorBrush>

Select the correct option

- <WhiteBrush >MyWhiteBrush </ WhiteBrush>
- None of the given options
- <SolidColorBrush Color="White"/>
- <Brush Style= “SolidWhite” >

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CS411:Midterm Quiz

Question # 29 of 30 (Start time: 05:59:27 PM, 11 July 2020)

If we want to scale an object in the 2-D x-y coordinate system then which transform class will be used?

Select the correct option

- ○ Rotate Transform
- ○ Matrix Transform
- ● Translate Transform
- ○ Skew Transform

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Question # 30 of 30 (Start time: 06:00:43 PM, 11 July 2020)

Clean room design is useful as a defense against _____.

Select the correct option

○	Copyright
○	Trade secret infringement
●	Copyright and trade secret infringement
○	None of the given options

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Question # 1 of 30 (Start time: 06:44:55 PM, 11 July 2020)

_____ Function will register a dependency property in WPF.

Select the correct option

- DependencyProperty.Reg()
- DependencyProperty.Registeration()
- DependencyProperty.Registered()
- DependencyProperty.Register()

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Question # 1 of 30 (Start time: 06:44:55 PM, 11 July 2020)

_____ Function will register a dependency property in WPF.

Select the correct option

- DependencyProperty.Reg()
- DependencyProperty.Registeration()
- DependencyProperty.Registered()
- DependencyProperty.Register()

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Question # 2 of 30 (Start time: 06:46:10 PM, 11 July 2020)

The root object element in XAML must specify at least _____ XML namespace(s) that is/are used to qualify itself and any child elements.

Select the correct option

- Two
- None of the given options
- Four
- One

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CS411:Midterm Quiz

Question # 3 of 30 (Start time: 06:47:19 PM, 11 July 2020)

Anonymous methods _____ the code size.

Select the correct option

- Double
- Expand
- Reduce
- Extend

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Question # 4 of 30 (Start time: 06:48:15 PM, 11 July 2020)

_____ works with Graphic Cards.

Select the correct option

- OpenGL
- GDI
- DirectX
- XNA

Sughal

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Question # 5 of 30 (Start time: 06:49:37 PM, 11 July 2020)

System.Windows.Controls.Button b = new System.Windows.Controls.Button();
b.Content = "OK";

For the above code (written in C#) the corresponding XAML code will be:

Select the correct option

●	<Button xmlns=http://schemas.microsoft.com/winfx/2006/xaml/presentation Content="OK"/>
○	<Button xmlns= “OK” Content="http://schemas.microsoft.com/winfx/2006/xaml/presentation "/>
○	<Button xmlns=http://schemas.microsoft.com/winfx/2006/xaml/presentation Event="OK"/>
○	<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Call="OK"/>

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Which of these methods used to read strings from the console?

Select the correct option

- Get()
- Getline()
- Read()
- ReadLine()

Sumaira Mughal

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Question # 7 of 30 (Start time: 06:50:57 PM, 11 July 2020)

<Button.Content> tag in XAML is the syntax for _____

Select the correct option

- Namespace
- Property Element
- Class
- Form

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Question # 8 of 30 (Start time: 06:51:50 PM, 11 July 2020)

We use "UIElement" property ClipToBounds= “false” when _____.

Select the correct option

- We want area child element don’t cross the boundary of parent
- None of given
- We want area of child element is allowed to cross the boundary of parent
- Parent area also increases on increasing of area of child element

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Question # 9 of 30 (Start time: 06:53:14 PM, 11 July 2020)

To check that the user either pressed left Alt key or right Alt key, the _____ is used.

Select the correct option

- IsKeyDown
- IsDown
- keyboardDevice.IsKeyDown
- KeyStates.IsKeyDown

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CS411:Midterm Quiz

Question # 10 of 30 (Start time: 06:54:02 PM, 11 July 2020)

Wait for a single event is _____ operation.

Select the correct option

- Waiting
- Waste
- Idle
- Blocking

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UI element focusable value is _____ by default.

Select the correct option

●	True
○	False
○	Nil
	Zero

Sonia Mughal

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Question # 12 of 30 (Start time: 06:54:59 PM, 11 July 2020)

XAML specification defines rules that map ".NET" namespaces, types, properties, and events into _____.

I) XML namespaces
II) XML elements
III) XML Attributes

Select the correct option

○	(I) Only
○	(II) Only
●	(I), (II) and (III) only
○	(III) Only

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Question # 13 of 30 (Start time: 06:55:17 PM, 11 July 2020)

Which provider has the highest priority while providing a value to a dependency property?

Select the correct option

- Style Setters
- Default Value
- Local Value
- Theme Style Setters

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Question # 14 of 30 (Start time: 06:56:53 PM, 11 July 2020)

Which of the following statements is TRUE about raw event?

Select the correct option

- [x] It is introduced into an event processing system by an event producer.
- [ ] It is generated as a result of event processing.
- [ ] It is introduced into an event processing system by an event consumer.
- [ ] It is a subscription mechanism for events.

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CS411:Midterm Quiz

Question # 15 of 30 (Start time: 06:57:40 PM, 11 July 2020)

WPF Margin is defined by _____ type.

Select the correct option

- Thin
- Thinnest
- Thick
- Thickness

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Question # 16 of 30 (Start time: 06:59:08 PM, 11 July 2020)

Which property is used to change the Z Order of elements in WPF?

Select the correct option

- Z-Coordinate
- ZPanel
- ZIndex
- ZValue

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Question # 18 of 30 (Start time: 07:01:45 PM, 11 July 2020)

If we run the program from within Visual Studio, the console window disappears quickly so we can use _____ to keep screen from going away.

Select the correct option

- (●) Console.ReadLine()
- ( ) Console.WriteLine()
- ( ) Console.Title
- ( ) Console.Clear

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Question # 19 of 30 (Start time: 07:02:41 PM, 11 July 2020)

When parent element space is not enough for the content of child element,this type of situation arises due to _____

Select the correct option

- Short memory
- Content Overflow
- Indexing
- Grid

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Question # 20 of 30 (Start time: 07:04:06 PM, 11 July 2020)

_____ is a base class similar to UIElement but for document-related pieces of content that don't have rendering behavior on their own.

Select the correct option

- ● ContentElement
- ○ VisualElement
- ○ SystemElement
- ○ XAMLElement

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Question # 21 of 30 (Start time: 07:04:41 PM, 11 July 2020)

Which of these methods of the class String is used to obtain length of string object?

Select the correct option

- get()
- Sizeof()
- length()
- lengthof()

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Question # 22 of 30 (Start time: 07:05:37 PM, 11 July 2020)

Overloaded operators must be _____.

Select the correct option

- Referenced
- Static
- Objective
- Delegates

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Question # 23 of 30 (Start time: 07:06:01 PM, 11 July 2020)

Which symbol is used to declare the preprocessor directives?

Select the correct option

- $
- *
- #
- ^

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Question # 24 of 30 (Start time: 07:06:39 PM, 11 July 2020)

Suppose we want to restrict the user to don't increase the height of a WPF element from a specific amount then which property will be used?

Select the correct option

- MaximumHeight
- MaxHeight
- None of given
- HeightMaximum

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Question # 25 of 30 (Start time: 07:07:00 PM, 11 July 2020)

A(n) _____ is a reference to a method.

Select the correct option

- (●) Delegate
- ( ) XAML
- ( ) Function
- ( ) Constructor

ughal
SO

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CS411:Midterm Quiz

Question # 26 of 30 (Start time: 07:07:47 PM, 11 July 2020)

Default value of Grid.Row is _____

Select the correct option

- (●) 0
- ( ) 1
- ( ) 2
- ( ) 3

Vughal

03063548328 For Solving Quiz



Question # 28 of 30 (Start time: 07:09:53 PM, 11 July 2020)

In XAML 2006, x:AsyncRecords works on _____ thread(s).

Select the correct option

- Single
- Multiple
- Both 1 & 2
- None of the given

03063548328 For Solving Quiz



Question # 29 of 30 (Start time: 07:10:51 PM, 11 July 2020)

How can you specify null property in markup extension?

Select the correct option

- (null)
- :null:
- {x:null}
- [x:null]

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Question # 30 of 30 (Start time: 07:11:25 PM, 11 July 2020)

DependencyObject.ClearValue method clears the _____ value in a dependency property.

Select the correct option

- Default
- Local
- Setter
- Getter

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https://web.facebook.com/groups/vu1234/

Question No: 1 (Marks: 1) - Please choose one
Event-based programming is also known as:

Service-oriented architecture (SOA)
Event-driven Architecture (EDA) (Page 11)
Service-driven Architecture (SDA)
Event-based Architecture (EBA)

Question No: 2 (Marks: 1) - Please choose one
Polymorphism works on the basis that child class should have_____feature/s of its parent class.

1. Only one
2. No
3. Few
4. All

Question No: 3 (Marks: 1) - Please choose one
The base class will have the _____ functionality of all derived classes

1. Common
2. Specialized
3. Unique
4. Different

Question No: 4 (Marks: 1) - Please choose one
Clean room design is also called_____

1. Clear Room Design
2. Wall Technique Design (Page 13)
3. Chinese Wall Technique
4. French Wall Technique

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Campus:- Institute of E-Learning & Moderen Studies (IEMS) Samundari

Question No: 5 (Marks: 1) - Please choose one
Indexers are similar to properties, but are accessed via a/an_____rather than a property name.

1. Index Argument Click here 4 detail
2. Reference
3. Pointer
4. Integer Argument

Question No: 6 (Marks: 1) - Please choose one
Types not defined in any namespace are said reside in the_____namespace.

1. Local
2. Static
3. Virtual ✓
4. Global Click here 4 detail ✗

Question No: 7 (Marks: 1) - Please choose one
A compiled C# file is called

1. Complied File
2. Source File
3. Assembly File (Page 41)
4. EXE File

Question No: 8 (Marks: 1) - Please choose one
Cool stands for_____.

1. Class Oriented Object Language
2. Consumer Oriented Object Language
3. C-like Object Oriented Language ✓
4. C++-like Object Oriented Language (Page 13) ✗

Question No: 9 (Marks: 1) - Please choose one
The initial name for C# was_____.

1. COAL
2. COOP
3. COAP
4. COOL (Page 13)

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Question No: 10 (Marks: 1) - Please choose one
The Main method returns a non-zero value which indicates the_____.

1. Source of Program
2. Error in Program Click here 4 more detail
3. Termination of Program
4. Exception in Program

Question No: 11 (Marks: 1) - Please choose one
Attributes add _____to year program.

1. Clearance
2. Errors
3. Meta Data (Page 40)
4. Ambiguity

Question No: 12 (Marks: 1) - Please choose one
Wait for a single event is _____operation.

1. Waiting
2. Waste
3. Idle
4. Blocking (Page 8)

Question No: 13 (Marks: 1) - Please choose one
To use kbhit() command, you need to include which header file in your program?

1. fstream.h
2. stdio.h
3. iostream.h
4. conio.h Click here 4 more detail

Question No: 14 (Marks: 1) - Please choose one
Which of the following is built from request-response?

1. Event-processing Architecture (EDA)
2. Service-driven Architecture (SDA)
3. Service-oriented Architecture (SOA) (Page 11)
4. Event-bas Architecture (EBA)

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Question No: 15 (Marks: 1) - Please choose one
_____let any classes behave like an array.

1. Properties
2. Fields
3. Indexers (Page 29)
4. Finalizers

Question No: 16 (Marks: 1) - Please choose one
All the members of interfaces are implicitly_____.

1. Private
2. Static
3. Public (Page 34)
4. Abstract

Question No: 17 (Marks: 1) - Please choose one
_____methods do not have names.

1. Attribute
2. Anonymous (Page 46)
3. None of given
4. Non-Anonymous

Question No: 18 (Marks: 1) - Please choose one
Derived Event is:

Introduced into an event processing system by an event consumer.
Generated as a result of event that takes place inside an event processing system. (Page 11)
Introduced into an event processing system by an event producer.
A set of associated events.

Question No: 19 (Marks: 1) - Please choose one
Which of the following is an occurrence within a particular system or domain?

Object
Event (Page 7)
Result
Message

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Question No: 20 (Marks: 1) - Please choose one
_____ is an entity that receives events from the system.

Event Producer
Event Consumer (Page 11)
Event Channel
Event Generator

Question No: 21 (Marks: 1) - Please choose one
Writing higher-level functions that call upon lower-level functions _____ a program.

Decode
Harder
Refactor
Simplify

Question No: 22 (Marks: 1) - Please choose one
Which statement is True?

A "catch" block always executes whether or not an exception is thrown and whether or not the "try" block runs to completion.
A "finally" block executes only, when an exception is thrown and when the "try" block runs to completion.
A "finally" block may or may not executes whether or not an exception is thrown and whether or not the "try" block runs to completion.
A "finally" block always executes whether or not an exception is thrown and whether or not the "try" block runs to completion. Click here 4 detail

Question No: 23 (Marks: 1) - Please choose one
Attributes of a program can be queried at run time through _____.

Reflection (Page 41, for more detail**)**
Value
Reference
Extension

Question No: 24 (Marks: 1) - Please choose one
During the program execution, program can be stopped with the help of _____.

Attributes
Compiler
Exception Handling
Breakpoints (Page 47)

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Question No: 25 (Marks: 1) - Please choose one
The _____, _____ and _____of applications that use events, either directly or indirectly is called event-based programming.

Model, Code, Operation
Analyze, Design, Operation
Code, Operation, Maintain
Design, Coding, Operation (Page 8)

Question No: 26 (Marks: 1) - Please choose one
Clean room design is useful as a defense against _____.

Copyright
Trade secret infringement
Copyright and trade secret infringement (Page 13)
None of the given options

Question No: 27 (Marks: 1) - Please choose one
The result of Exception can bring _____ in the program.

Reliability
Stability
Inconsistency (Page 40)
Consistency

Question No: 28 (Marks: 1) - Please choose one
_____ method reads the whole document in memory.

XmlDocument (Page 48)
XmlLine
XPath
XmlReader

Question No: 29 (Marks: 1) - Please choose one
The _____ in the namespace indicate a hierarchy of nested namespaces.

Arrows
Dots
Semicolon
Brackets (Page 22)

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Question No: 30 (Marks: 1) - Please choose one
If we run the program from within Visual Studio, the console window disappears quickly so we can use _____ to keep screen from going away.

Console.ReadLine() (Page 15)
Console.WriteLine()
Console.Title
Console.Clear

Question No: 31 (Marks: 1) - Please choose one
Which of the following is an entity that introduces event into the system?

Event Channel
Event Stream
Event Consumer
Event Producer (Page 11)

Question No: 32 (Marks: 1) - Please choose one
At the time an event is fired, the _____ methods will be invoked.

Registered Click here 4 detail
Public
Static
Unregistered

Question No: 33 (Marks: 1) - Please choose one
The easiest way to declare an event is to put the event keyword in front of a _____ member.

Interface
Delegate Click here 4 detail
Class
Struct

Question No: 34 (Marks: 1) - Please choose one
Which of the following does NOT include in event processing operations?

Deleting Events
Reading Events
Terminating Events (Page 8)
Transforming Events

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Question No: 35 (Marks: 1) - Please choose one
An event is a list of_____

constants
variable
pointers
delgates

Question No: 36 (Marks: 1) - Please choose one
C# is a _____ and .NET is a _____.

Platform, Language
Package, Language
Language, Package
Language platform (Page 13)

Question No: 37 (Marks: 1) - Please choose one
Anonymous methods _____ the code size.

Double
Expand
Reduce (Page 46)
Extend

Question No: 38 (Marks: 1) - Please choose one
g++ is a command of _____.

Dev C++
Unix based system
Windows based system
None of the given options

Question No: 39 (Marks: 1) - Please choose one
A class can inherit from _____class/classes.

Two
Single Click here 4 detail
Three
Multiple

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Question No: 40 (Marks: 1) - Please choose one
Which of the following events may or may not relate to an actual occurrence?

Probabilistic Events (Page 7)
Actual Events
Real Events
Expected Events

Question No: 41 (Marks: 1) - Please choose one
Which of the following statements is TRUE about raw event?

It is introduced into an event processing system by an event producer. (Page 11)
It is generated as a result of event processing.
It is introduced into an event processing system by an event consumer.
It is a subscription mechanism for events.

Question No: 42 (Marks: 1) - Please choose one
A method performs an action in a series of statements, called a _____.

Code Block
Method Block
Routine Block
Statement Block Click here for detail

Question No: 43 (Marks: 1) - Please choose one
Graphical user interfaces and the event-driven model are not applied on which of the following?

Web Applications
Console Applications
Mobile applications
Desktop Applications

Question No: 44 (Marks: 1) - Please choose one
The System.Console is a _____ which enables us to do Console Input and Output.

Structure
Class (Page 14)
Namespace
Library

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Question No: 45 (Marks: 1) - Please choose one
Events are based on the principle of _____.

Request
Response
Coupling
Decoupling (Page 10)

Question No: 46 (Marks: 1) - Please choose one
In event-driven architecture (EDA), _____ sends event to _____.

Event Consumer, Event Producer
Event Producer, Event Consumer (Page 10)
Event Handler, Event Producer
Event Generator, Event Producer

Question No: 47 (Marks: 1) - Please choose one
XML is a stricter version of _____.

Java
CSS
HTML (Page 47)
C#

Question No: 48 (Marks: 1) - Please choose one
Extension methods _____ access the private data members of the class in which they are extended.

May
Cannot click here 4 more detail
May not
Can

Question No: 49 (Marks: 1) - Please choose one
BAML is compressed representation of _____.

CAML:NS
XAML (Page 71)
XAML:NS
CAML

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Question No: 50 (Marks: 1) - Please choose one
In context of XAML child element rules, if the child is plain text and a type converter exists to transform the child into the parent type (and no properties are set on the parent element), treat the child as the input to the type converter and use the output as the _____.

Parent object instance (Page 68)
Interface object instance
none
Child object instance

Question No: 51 (Marks: 1) - Please choose one
IsNumeric(string s) is a function of _____ class.

Math class
Program class
String class
MyUtills class (Page 53)

Question No: 52 (Marks: 1) - Please choose one
XAML specification defines rules that map ".NET" namespaces, types, properties, and events into _____.
I) XML namespaces II) XML elements III) XML Attributes

(I) Only
(II) Only
(I), (II) and (III) only (Page 63)
(III) Only

Question No: 53 (Marks: 1) - Please choose one
Visual Studio has a snippet called _____ that automatically expands into a definition of a dependency property, which makes defining one much faster than doing all the typing yourself!

Propdp (Page 77)
Property extension
Dpprop
Dependency property

Question No: 54 (Marks: 1) - Please choose one
Suppose we want to restrict the user to don't decrease the height of a WPF element from a specific amount then which property will be used?

MinHeight (Page 83)
none
HeightMinimum
MinimumHeight

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Question No: 55 (Marks: 1) - Please choose one
WPF element in "DockPanel" has property "dock". There is/are _____ possible value(s) of "dock" property.

two
three
four (Page 93)
one

Question No: 56 (Marks: 1) - Please choose one
WPF4 was released in

2012
2010 Click here for detail
2005
2003

Question No: 57 (Marks: 1) - Please choose one
_____ allows to keep the user interface description and implementation separate.

GDI
WPF (Page 61)
Avalon
Altia

Question No: 58 (Marks: 1) - Please choose one
When the data is passed from one place to another in chunks, this data is called _____.

Bits
Line
Byte
Stream

Question No: 59 (Marks: 1) - Please choose one
We want the width of button should cover the whole width of StackPanel in WPF then what should be value of HorizontalAlignment property of button?

Parent
Whole
Fill
Stretch (Page 97)

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Question No: 60 (Marks: 1) - Please choose one
Which of the following is/are not derived from “Control” class?

ListBox
StatusBar
None of the given (Page 73)

Question No: 61 (Marks: 1) - Please choose one
If we want to move an object in the 2-D x-y coordinate system then which transform class will be used?

Translate Transform (Page 89)
Skew Transform
Skew Transform
Rotate Transform

Question No: 62 (Marks: 1) - Please choose one
Dependency properties are represented by _____ .

System.Windows.DependencyProperty (Page 77)
System.Windows.Dependency.Property
System.Windows.Dependency
System.Windows.Property.Dependency

Question No: 63 (Marks: 1) - Please choose one
System.Windows.Controls.Button b = new System.Windows.Controls.Button(); b.Content = "OK"; For the above code (written in C#) the corresponding XAML code will be:

<Button xmlns=http://schemas.microsoft.com/winfx/2006/xaml/presentation Content="OK"/> (Page 63)
<Button xmlns= “OK” Content="http://schemas.microsoft.com/winfx/2006/xaml/presentation "/>
<Button xmlns=http://schemas.microsoft.com/winfx/2006/xaml/presentation Event="OK"/>
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Call="OK"/>

Question No: 64 (Marks: 1) - Please choose one
_____ is a base class similar to UIElement but for document-related pieces of content that don’t have rendering behavior on their own.

ContentElement (Page 73)
VisualElement
SystemElement
XAMLElement

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Question No: 65 (Marks: 1) - Please choose one
In context of XAML child element rules, if the parent supports a content property and the type of the child is compatible with that property, treat the child as its _____.

Super class
Sub class
Value (Page 68)
Interface

Question No: 66 (Marks: 1) - Please choose one
Mapping to the WPF namespaces is _____ in-side the WPF assemblies.

Specified at compile time
Hard-coded (Page 63)
None of the given
Dynamically specified

Question No: 67 (Marks: 1) - Please choose one
The _____ user interfaces, enabled by WPF, is getting a lot of attention.

Loutish
Polished (Page 61)
Coarse
Distracted

Question No: 68 (Marks: 1) - Please choose one
The _____ method is used to read the whole file at once.

ReadAllText("test.txt"); Click here for detail
ReadAll();
Read("test.txt");
ReadLine();

Question No: 69 (Marks: 1) - Please choose one
With the help of WPF, an ugly looking application can be _____ by the designers.

Rearranged
Re-created
Re-themed(Page 61)
Rejected

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Question No: 70 (Marks: 1) - Please choose one
Additional XML namespaces (on the root or on children) must be _____ to be used on any identifiers from that namespace.

Given a common prefix
Qualified with “Pre” keyword
Given a distinct prefix (Page 63)
Qualified with “Ext” keyword

Question No: 71 (Marks: 1) - Please choose one
Extension methods can only be defined in _____ class.

Dynamic class
Normal class
String class
Static class Click here for detail

Question No: 72 (Marks: 1) - Please choose one
If we want to rotate an object clockwise about a specified point in a 2-D x-y coordinate system then which transform class will be used?

None of given (Page 87)
Translate Transform
Skew Transform
Scale Transform

Question No: 73 (Marks: 1) - Please choose one
The code given below is equivalent to _____ . <SolidColorBrush>White</SolidColorBrush>

<WhiteBrush >MyWhiteBrush </ WhiteBrush>
None of the given options
<SolidColorBrush Color="White"/> (Page 65)
<Brush Style= “SolidWhite” >

Question No: 74 (Marks: 1) - Please choose one
“FrameworkElement” is the base class that adds support for _____.

Some common mechanisms for Windows-based controls
Data binding
Styles
All of the given options (Page 73)

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Question No: 75 (Marks: 1) - Please choose one
All elements of WPF have following property/properties.

Width
None of given options
Height and Width
Height

Question No: 76 (Marks: 1) - Please choose one
If we exclude the mechanism of calling the objects through name then:

Code will not compile
Code will be difficult to debug
Code will have errors
Code will become design dependent

Question No: 77 (Marks: 1) - Please choose one
If we want to place a component at top in "dockpanel" then which one is correct syntax?

PanelDock="Top"
Dock="Top"
DockPanel.Dock="Top" (Page 93)
DockPanel="Top"

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CS-411

Quiz 1

by Attiq Kundi

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CS-411

Up Dated on 24-May-2023

1. keyword is used to declare a base class method while performing overriding of base class method.
   A. Overload
   B. Virtual
   C. Extend
   D. Override
2. Choose the correct option among the following indexers which correctly allows to index in same way as an array?
   A. A Class
   B. A function
   C. An interface
   D. A property
3. A class can inherit from_____class/classes in C#.
   A. Three
   B. Single
   C. Two
   D. Multiple
4. Which one of the following is not the feature of C#?
   A. Multiple inheritance
   B. Interface supported
   C. Locals can not shadow global variables
   D. Operator Overloading is allowed
5. Which keyword is used to declare the indexers?
   A. super
   B. base
   C. this
   D. extract
6. The common type system of C# has value types and___types.
   A. Class
   B. Object

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C. Reference

D. Enum

7. Select the correct statement about the structure in C#?

A. Its syntax is similar to class

B. It can implement destructors

C. It is a reference type

D. It can implement inheritance

8. Reaction of events are either_____or_____.

A. Asynchronous, Blocking

B. Synchronous, Blocking

C. Blocking, Synchronous

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D. Synchronous, Asynchronous

9. is an example of request and response based application
   A. Web browser
   B. E –Commerce Site
   C. Pressing a button
   D. Swiping of credit card
10. Which of the following is built from request-response?
   A. Event-based Architecture (EBA)
   B. Service-driven Architecture (SDA)
   C. Service-oriented Architecture (SOA)
   D. Event-processing Architecture (EDA)
11. _____ and delegates work hand in hand. (Page-38)
   A. Static member
   B. Event
   C. Post-processor
   D. Pre-processor
12. In C#, when a value type is converted to object type, it is called__.
   A. Polymorphism
   B. Abstraction
   C. Boxing
   D. Data Conversion
13. Which keyword is used for correct implementation of an interface in C#?
   A. Intf
   B. Interface
   C. Int Interface
   D. Interfaces
14. The main method return a non-zero value which indicates the
   A. Exception in program
   B. Termination of program
   C. Error in program
   D. Success of program
15. is an occurrence within a particular system or domain.
   A. Program
   B. Class
   C. Event
   D. Object
16. Select wrong statement about destructor in C#?
   A. Destructor can have modifiers or parameter
   B. Destructor cannot be inherited or overloaded
   C. Destructor work opposite to constructor
   D. Class can have one destructor only
17. The main method returns zero value which indicates the
   A. Successful execution of program
   B. Source of program

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C. Termination of program
D. Exception in program

18. In C#, which of the following keyword is used for including the namespaces in the program?
A. using
B. extend
C. import
D. #

19. When your C# program is compiled, is creates a file called
A. Exe
B. Class
C. Object
D. Assembly

20. Which of the following is an insertion operator in C++?
A. <<
B. >
C. - -
D. &

21. C# is a class member that is activated whenever the event . it was designed for. Occurs
A. Attributes
B. Arguments
C. Method
D. Event

22. A fraud detection event on banking transaction is an example of___.
A. Linear event
B. Sequential event
C. Predictive event
D. Probabilistic event

23. In C++, which of the following library is include for 'cin' and 'cout' function?
A. #include<fstream.h>
B. #include<fstream>
C. #include<conio.h>
D. #include<iostream>

24. Which of the following doesn't have Graphical User Interface and the Event-Driven Model?
A. Console-Application
B. Web Application
C. Mobile Application
D. Desktop Application

25. A stream in which the events may be of different types is referred to as a **heterogeneous** event stream. (Page-10)
A. Continuous
B. Homogenous
C. Heterogeneous
D. Mutual

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26. Defining a method in subclass having same name and type signature as a method in its super class is known as
    A. Method hiding
    B. Method Overloading
    C. Method Binding
    D. Method Overriding
27. A single try block must be followed by which of these?
    A. Finally
    B. Catch
    C. Both Finally and catch
    D. Throw
28. By default the data member of a class are
    A. Protected, public
    B. Public
    C. Private
    D. Private, Public
29. Exception are derived from_____class. (Page-39)
    A. Windows error
    B. Exception error
    C. System exception
    D. Error Exception
30. Which of the following does NOT include in event processing operations?
    A. Terminating Event
    B. Deleting Event
    C. Transforming Event
    D. Reading Event
31. What is the return type of destruction?
    A. Float
    B. No return type
    C. Void
    D. Int
32. At the time an event is fired, the_____method (with that event) will be invoked.
    A. Public
    B. Protected
    C. Registered
    D. Private
33. C# is a ___and .NET is a__.
    A. Platform, Language
    B. Package, Language
    C. Language, Package
    D. Language, Platform
34. Which of the following is an entity that introduces event into the system?
    A. Event producer
    B. Event Stream

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C. Event Channel
D. Event Consumer

35. _____ is an entity that receive event from the system?
A. Event producer
B. Event Consumer
C. Raw event
D. Event stream

36. Which of the following is the object-oriented way to handle run time errors?
A. OnError () Method
B. Exception
C. Interface
D. Error Code

37. Why do we use cin.get() in C++ program?
A. It is used to print all values of array
B. It is used to get numeric data
C. It is used to access character array
D. It is used to take input from user

38. Types not defined in any namespace are said to reside in the _ _namespace.
A. Static
B. Virtual
C. Global
D. Local

Reason: Types not defined in any namespace are said to reside in the global namespace. The global namespace also includes top-level namespaces

39. Arrays in C# are_____object.
A. Logical
B. Arithmetic
C. Reference
D. Value

40. The initial name for C# was____.
A. COAL
B. COOP
C. COAP
D. COOL

41. Which of the following is declared like a class but have no implementation?
A. Encapsulation
B. Structure
C. Interface
D. Inheritance

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42. Which of the following keyword is used to change data and behavior of a base class by replacing a member of the base class with the new derived member.
    A. Base
    B. Overrides
    C. Overloads
    D. New
43. A stream in which all the events must be of same types is called **Homogenous** event stream. (Page-10)
    A. Continuous
    B. Homogenous
    C. Heterogeneous
    D. Mutual
44. Select the correct statement about an exception>
    A. It occurs during Just-in-time compilation
    B. It occurs at runtime
    C. It occurs during loading of program
    D. It occurs during data binding
45. _____ is not a defense against patents and clean room design?
    A. Independent Invention
    B. Event Invention
    C. Dependent Invention
    D. Clear room design
46. COOL stands for____.
    A. Class Oriented Object Language
    B. Consumer Oriented Object Language
    C. C-like Object Oriented Language
    D. C++-like Oriented Language
47. operator is used to signifies the destructor in C#?
    A. #
    B. =
    C. &
    D. ~
48. What is the return type of destruction?
    A. Int
    B. Float
    C. Void
    D. No return types
49. An Event that is generated as a result of event processing that take place inside an event processing system is called . (Page-10)

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A. Event Stream
B. Derived Event
C. Raw Event
D. Event Consumer

50. Choose the incorrect statement about delegates in C#?
A. Delegates are value type
B. Delegates are like function pointers in other language
C. Delegates are object oriented
D. Delegates are type safe

51. is the method of copying a design by reverse engineering and then recreating it without infringing any of the copyrights and trade secrets associated with the original design. (Page-12)
A. Clear Room Design
B. Event Room design
C. Clean Room Design
D. Wall Room Design

52. Which of the following statement is TRUE about raw event?
A. It is a subscription mechanism for event
B. It is introduced into an event processing system by an event consumer
C. It is generate as a result of event processing
D. It is introduced into an event processing system by an event producer

53. Which symbol is used to declare the preprocessor directives?
A. *
B. #
C. $
D. ^

54. Which of these keyword is used to manually throw an exception in C#?
A. Try
B. Finally
C. Catch
D. Throw

55. On successful compilation of a java source code _____ is generated.
A. Output
B. Bytecode
C. Hexadecimal code
D. Octal code

56. _____ is the necessary condition to implement delegates.
A. Class Declaration
B. Function overloading
C. Inheritance
D. Runtime Polymorphism

57. Select correct statement about properties of read and write in C#
A. A read only property will only have get accessor
B. A write only property will only have get accessor
C. A property cannot be either read only or write only
D. A property can simultaneously be read or write only

By Atiq Kundi

58. What will be the output of the following C# code? Static void Main{string[] args}
```
{
  int i;    for (i = 0; ;)
  {
    Console.WriteLine(&amp;quot;Virtual University&amp;quot;);
  }
 Console.ReadLine();
}
```

A. || will print "Virtual University" in continuous loop
**B. Syntax error **
C. Virtual University
D. =p= No output =/p=

59. Choose the wrong statement about structures in C#.Net.
A. Structures can have destructors
B. Structures can implement an interface but they cannot inherit from another structure
C. Structure members cannot be declared as protected
D. A structure cannot be empty

60. Which of the following keyword is used to monitor exceptions?
A. Catch
B. Finally
C. Try
D. Throw

61. A method performs an action in a series of statements, called a _____.
A. Code Block
B. Method Block
C. Routine Block
D. Statement Block

62. If we run the program from within Visual Studio, the console window disappears quickly so we can use _____ to keep screen from going away.
A. Console.ReadLine()
B. Console.WriteLine()
C. Console.Title
D. Console.Clear

63. _____ events may or may not relate to an actual occurrence.
A. Observed Events
B. Deducted Events
C. Probabilistic Events
D. None of the above

64. Clean room design is useful as a defense against _____.
A. Copyright
B. Trade secret infringement
C. Copyright and trade secret infringement
D. None of the given options

65. Which of these methods of the class String is used to obtain length of string object.
A. Lengthof()
B. Sizeof()

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C. Get()

D. Length()

66. If math class had added property with get and set accessors. Then which of the following statements will work correctly?

A. math.add = 40;

B. math m = new math();
m.add = 50;

C. Console.WriteLine(math.add);

D. System.out.Println("math add")

67. _____ language is supported by Microsoft.Net platform.

A. C#

B. C

C. Php

D. Java

68. The _____ in the namespace indicate a hierarchy of nested namespaces.

A. Semicolon

B. Brackets

C. Arrows

D. Dots

69. which keyword is used to define the macros in c++?

A. macro

B. #define

C. #macro

D. Define

70. In C#, when an object type is converted to a value type, it is called _____

A. Unboxing

B. Data Conversion

C. Abstraction

D. Polymorphism

71. The _____ pointer in methods refers to the object on which the method is called.

A. Int

B. String

C. Out

D. This

72. In C#, all binary operators are _____

A. Right-associative

B. Top-associative

C. Center-associative

D. Left-associative

73. Clean room design is also called_____.

A. Clear Room Design

B. Wall Technique Design

C. Chinese Wall Technique

D. French Wall Technique

74. To perform I/O operations, we must use _____ header file.

A. <conio>

B. <fstream>

C. <ifstream>

D. <ofstream>

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75. Which reference modifier is used to define reference variable?
    A. #
    B. def
    C. &
    D. ref
76. Waiting for a single event is _____operation.
    A. Unavoidable
    B. Waste
    C. Idle
    D. Blocking
77. In System.Console ,"Console" is a _____ which enables us to do console Input and Output.
    A. Namespace
    B. Structure
    C. Class
    D. Library
78. In Event-Driven Architecture (EDA). _____ sends event to _____
    A. Event Consumer, Event Producer
    B. Event Producer, Event Consumer
    C. Event Handler, Event Producer
    D. Event Generator, Event Producer
79. In C++ _____ function is used to check if this stream has raised any error or not
    A. eof()
    B. Peek()
    C. Fstream
    D. Good()
80. In C++ , _____ function is used to return the next character in the input sequence, without extracting it.
    A. Putback()
    B. Peek()
    C. Fstream
    D. Good()
81. which of these keywords is not a part of exception handling?
    A. Thrown
    B. Finally
    C. Try
    D. Catch

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CHAPTER 1 to 8

UpDated on 07 June 2022

3. _____ keyword is used to declare a base class method while performing overriding of base class method.
   A. Overload
   Virtual
   B. Extend
   C. Override
4. Choose the correct option among the following indexers which correctly allows to index in same way as an array? (Page-29)
   A Class
   A. A function
   B. An interface
   C. A property
5. A class can inherit from _____ class/classes in C#. (Page-22)
   A. Three
   Single
   B. Two
   C. Multiple
6. Which one of the following is not the feature of C#? (Page-12)
   Multiple inheritance
   A. Interface supported
   B. Locals can not shadow global variables
   C. Operator Overloading is allowed
7. Which keyword is used to declare the indexers? (Page-29)
   A. super
   B. base
   this
   C. extract
8. The common type system of C# has value types and _____ types. (Page-13)
   A. Class
   B. Object
   Reference
   C. Enum
9. Select the correct statement about the structure in C#? (Page-13)
   Its syntax is similar to class
   A. It can implement destructors
   B. It is a reference type
   C. It can implement inheritance
10. Reaction of events are either _____ or _____ (Page-07)
    A. Asynchronous, Blocking
    B. Synchronous, Blocking
    C. Blocking, Synchronous

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Synchronous, Asynchronous

6. _____ is an example of request and response based application (Page-09)

   Web browser

   A. E –Commerce Site
   B. Pressing a button
   C. Swiping of credit card

7. Which of the following is built from request-response? (Page-09)

   A. Event-based Architecture (EBA)
   B. Service-driven Architecture (SDA)

   Service-oriented Architecture (SOA)

   C. Event-processing Architecture (EDA)

8. _____ and delegates work hand in hand. (Page-38)

   A. Static member

   Event

   B. Post-processor
   C. Pre-processor

9. In C#, when a value type is converted to object type, it is called _____.

   A. Polymorphism
   B. Abstraction

   Boxing

   C. Data Conversion

10. Which keyword is used for correct implementation of an interface in C#?

    A. Intf

    Interface

    B. Int Interface
    C. Interfaces

11. The main method return a non-zero value which indicates the _____

    A. Exception in program
    B. Termination of program

    Error in program

    C. Success of program

12. _____ is an occurrence within a particular system or domain. (Page-06)

    A. Program
    B. Class

    Event

    C. Object

13. Select wrong statement about destructor in C#?

    Destructor can have modifiers or parameter

    A. Destructor cannot be inherited or overloaded
    B. Destructor work opposite to constructor
    C. Class can have one destructor only

14. The main method returns zero value which indicates the _____

    Successful execution of program

    A. Source of program

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E. Termination of program
F. Exception in program

8. In C#, which of the following keyword is used for including the namespaces in the program?
using
A. extend
B. import
C. #

9. When your C# program is compiled, is creates a file called _____
Exe
A. Class
B. Object
C. Assembly

10. Which of the following is an insertion operator in C++? (Google)
<<
A. >
B. - -
C. &

11. C# ___ is a class member that is activated whenever the event . it was designed for. Occurs (Page-37)
A. Attributes
B. Arguments
C. Method
Event

12. A fraud detection event on banking transaction is an example of _____. (Page-06)
A. Linear event
B. Sequential event
C. Predictive event
Probabilistic event

13. In C++, which of the following library is include for 'cin' and 'cout' function?
A. #include<fstream.h>
B. #include<fstream>
C. #include<conio.h>
#include<iostream>

14. Which of the following doesn't have Graphical User Interface and the Event-Driven Model?
Console-Application
A. Web Application
B. Mobile Application
C. Desktop Application

15. A stream in which the events may be of different types is referred to as a **heterogeneous** event stream. (Page-10)
A. Continuous
B. Homogenous
Heterogeneous
C. Mutual

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D. Defining a method in subclass having same name and type signature as a method in its super class is known as _____

Method hiding

Method Overloading

Method Binding

Method Overriding

E. A single try block must be followed by which of these?

Finally

Catch

Finally and catch

Throw

F. By default the data member of a class are _____

Protected, public

Public

Private

Private, Public

G. Exception are derived from_____ class. (Page-39)

Windows error

Exception error

System exception

Error Exception

H. Which of the following does NOT include in event processing operations? (Page-07)

Terminating Event

Deleting Event

Transforming Event

Reading Event

I. What is the return type of destruction?

Float

No return type

Void

Int

J. At the time an event is fired, the _____ method (with that event) will be invoked. (Page-38)

Public

Protected

Registered

Private

K. C# is a ___ and .NET is a ___. (Page-12)

Platform, Language

Package, Language

Language, Package

Language, Platform

L. Which of the following is an entity that introduces event into the system? (Page-10)

Event producer

Event Stream

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E. Event Channel
F. Event Consumer

12. Which of the following is an entity that receive event from the system? (Page-10)
    A. Event producer

    Event Consumer

    B. Raw event
    C. Event stream
13. Which of the following is the object-oriented way to handle run time errors?
    A. OnError () Method

    Exception

    B. Interface
    C. Error Code
14. Why do we use cin get() in C++ program?
    A. It is used to print all values of array
    B. It is used to get numeric data
    C. It is used to access character array
    D. It is used to take input from user
15. Types not defined in any namespace are said to reside in the ___ namespace.

    Local

    Static

    Virtual

16. Select the correct statement about an exception. (Google)

2 It occurs during loading of program

3 It occurs during Just-In-Time compilation

It occurs at run time

4 All of the mentioned

2 Arrays in C# are _____ object.
    13. Logical
    14. Arithmetic

    Reference

    15. Value

3 The initial name for C# was _____. (Page-12)
    13. COAL
    14. COOP
    15. COAP

    COOL

4 Which of the following is declared like a class but have no implementation?(Page-34)
    13. Encapsulation
    14. Structure

    Interface

    15. Inheritance

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17. Which of the following keyword is used to change data and behavior of a base class by replacing a member of the base class with the new derived member. (Google)
    16. Base
    17. Overrides
    18. Overloads

    New
18. A stream in which all the events must be of same types is called **Homogenous** event stream. (Page-10)
    A. Continuous
    B. Homogenous

    Heterogeneous
    C. Mutual
19. Select the correct statement about an exception>
    16. It occurs during Just-in-time compilation

    It occurs at runtime
    17. It occurs during loading of program
    18. It occurs during data binding
20. Select the correct statement about the structure in C#?
    16. It can implement inheritance

    Its syntax similar to class
    17. It is a reference type
    18. It can implement destructors
21. ____ is not a defense against patents and clean room design?(Page-12)

    Independent Invention
    16. Event Invention
    17. Dependent Invention
    18. Clear room design
22. COOL stands for _____.
    16. Class Oriented Object Language
    17. Consumer Oriented Object Language

    C-like Object Oriented Language (Page-12)
    18. C++-like Oriented Language
23. _____ operator is used to signifies the destructor in C#?
    16. #
    17. =
    18. &

    ~
24. What is the return type of destruction?
    16. Int
    17. Float
    18. Void

    No return types
25. An Event that is generated as a result of event processing that take place inside an event processing system is called _____. (Page-10)

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E. Event Stream
Derived Event
F. Raw Event
G. Event Consumer

21. Choose the incorrect statement about delegates in C#?
2 Delegates are value type
3 Delegates are like function pointers in other language
4 Delegates are object oriented
Delegates are type safe

E. _____ is the method of copying a design by reverse engineering and then recreating it without infringing any of the copyrights and trade secrets associated with the original design. (Page-12)
Clear Room Design
Event Room design
Clean Room Design
Wall Room Design

F. Which of the following statement is TRUE about raw event? (Page-10)
It is a subscription mechanism for event
It is introduced into an event processing system by an event consumer
It is generate as a result of event processing
It is introduced into an event processing system by an event producer

G. Which symbol is used to declare the preprocessor directives?
*
#
$
^

H. Which of these keyword is used to manually throw an exception in C#?
Try
Finally
Catch
Throw

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Visual Programming

CS411



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Chapter 1

Lecture 1

This course is about graphical user interfaces and the event-driven model as applied to desktop, web, and mobile applications. It uses a ground up approach from what you already know.

The pre-requisites are C++ programming and data structures. I will introduce any new languages and concepts we are going to use.

This is a hands-on course. We will use in-class examples and you should focus on completing programming assignments to understand the material.

We will primarily use 2 books in this course. "Event processing in action" and "Windows resentation foundationunleashed". The first one is authored by Opher Etzion and Peter Niblett and is one of the very fewbooks on event processing. Event processing is often taught as a side-concept, but we will focus on it in this course. The other book is by Adam Nathan. It is about Windows presentation foundation or WPF in short. We use it as our gui example with C#. We will introduce C# concepts and WPF concepts. They are not a pre-requisite for this course.

Later in the course, we will touch event-driven programming in the browser, in particular made possible by techniques called AJAX. Again not a pre-requisite. I will introduce all the concepts in class. Towards the very end, we will discuss event driven programming on mobiles. Again no pre-requisite. I will introduce all the tools involved.

So what is Visual Programming and what you will learn in this course. Its primarily event driven concepts and application of these concepts on GUI programs on desktop, web, and mobile.

Here is our first example:

```
Using namespace std;

#include <iostream>

Int main()
{
  Char a;
  Do {
    A = cin.get();
    Cout << a;
  } while (a!='x');
  Return 0;
}
```

Let's compile and run using the commands

G++ -o example1 example1.cc ./example1

Our second example

```
#include <iostream>
#include <fstream>
```

```
Int main()
{
  Char a = '-';
  Do {
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good() && a != f.peek()) {
      A = f.get();
      Std::cout << a << std::endl;
    }
    F.close();
  } while (a!='x');
  Return 0;
}
```

Let's compile and run it. There are a few issues like busy wait etc. But we'll leave them on a side for now.

So why we saw these two very basic examples. Well, they were both very easy to write. With an OR.

```
#include <iostream>
#include <fstream>

Int main()
{
  Char a = '-', b;
  Do {
    If (iskeyboardpressed()) {
      B = std::cin.get();
      Std::cout << b << std::endl;
    }
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good() && a != f.peek()) {
      A = f.get();
      Std::cout << a << std::endl;
    }
    F.close();
  } while (a!='x' && b!='x');
  Return 0;
}
```

I.e. We react to either of the two actions. Here is how we can do it.

```
#include  <iostream>
#include<fstream>
 Int main()
 {
 Char a  =’-’, b;
 Do{
 If(iskeyboardpressed())  {
 B = std::cin.get();
 Std::cout  << b  << std::endl;
 }
 Std::ifstream f("test.txt", std::ifstream::in);
 If(f.good() && a  != f.peek())  {
 A = f.get();
 Std::cout  << a  << std::endl;
 }
```

```
F.close();
} while  (a!=’x’ && b!=’x’);
Return  0;
}
```

Or we could re-factor it into a cleaner version.

```
#include <iostream>
#include <fstream>

Int onkeypress()
{
  Char b = std::cin.get();
  Std::cout << b << std::endl;
  Return b == 'x';
}

Int onfilechanged()
{
  Char a = f.get();
  Std::cout << a << std::endl;
  Return a == 'x';
}

Int main()
{
  Char a = '-', b;
  Do {
    If (iskeyboardpressed()) {
      If (onkeypress())
Return 0;
    }
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good() && a != f.peek()) {
      If (onfilechanged())
Return 0;
    }
    F.close();
  } while (true);
  Return 0;
}
```

So what are the pros and cons of the re-factored approach. We do one thing at a time (more focused approach). Blocking is a huge issue. Nothing is parallel. The key idea is that once you understand the approach, it scales really well to a large number of things you want to respond to.

Next time we’ll define events, event loops, and different terms in the event driven model.

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Chapter 2

Lecture 2

Last time, we saw some examples and demonstrated the idea of event driven design. We can re-factor the last example even one step further like this:

```
#include <iostream>
#include <fstream>

Int onkeypress()
{
  Char b = std::cin.get();
  Std::cout << b << std::endl;
  Return b == 'x';
}

Int onfilechanged()
{
  Char a = f.get();
  Std::cout << a << std::endl;
  Return a == 'x';
}

Int main()
{
  Char a = '-', b;
  Do {
    If (iskeyboardpressed()) {
      If (onkeypress())
 Return 0;
    }
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good() && a != f.peek()) {
      If (onfilechanged())
 Return 0;
    }
    F.close();
  } while (true);
  Return 0;
}
```

So what is an event? It's an occurrence within a particular system or domain. There are two meanings: something that happened and the corresponding detection in computer world. An event captures "some" things from the actual occurrence and multiple events may capture "one" occurrence.

Probabilistic events may or may not relate to an actual occurrence e.g. A fraud detection event on a banking transaction. Every event is represented by an event-object. There are various types of events and information in the event describes details of the particular event type. E.g. Key press, file event etc.

Let's take a coffee shop analogy. Events are taking place all the time. Let's understand synchronous and asynchronous behavior with this example. So if an order is completed, coffee ready, pastery heated before the next order that's synchronous behavior. There are observed events and deduced events. Situation is

an event ocurrence that requires a reaction.

Let's take example of events in computing systems. Interrupts and exceptions on a computer e.g. Divide by zero. Patient monitored by sensors. Car sensors alerting to oil or pressure situations. Banking alerts. Road tolling. Etc. Event processing is computing that performs operations on events. Common event processing operations include reading, creating, transforming, and deleting events. The design, coding and operation of applications that use events, either directly or indirectly is called event-based programming or applications based on event-driven architecture. So what if we don't use event-driven programming. We will poll for events. Even if you make your application non-event-driven, you still wait for events. Wait for a single event is blocking operation.

Synchronous operations are completed before the next operation can be started. Asynchronous operations can be started and we can do something else before they are completed.

Why do we want our applications to be event based. They are easier to scale. Well suited to Visual programming where multiple GUI elements and many sources of events exist. It has a direct mapping to real world. Its deterministic-there is an exact mapping between a situation in the real world and its representation in the event processing system.

At the same time its approximate-the event processing system provides an approximation to real world events. So what kind of events are there in non-real time applications. Mouse and keyboard, secondary events of GUI elements, file events, message based parallel and local communication, network events etc.
Here is flower delivery example from the book.

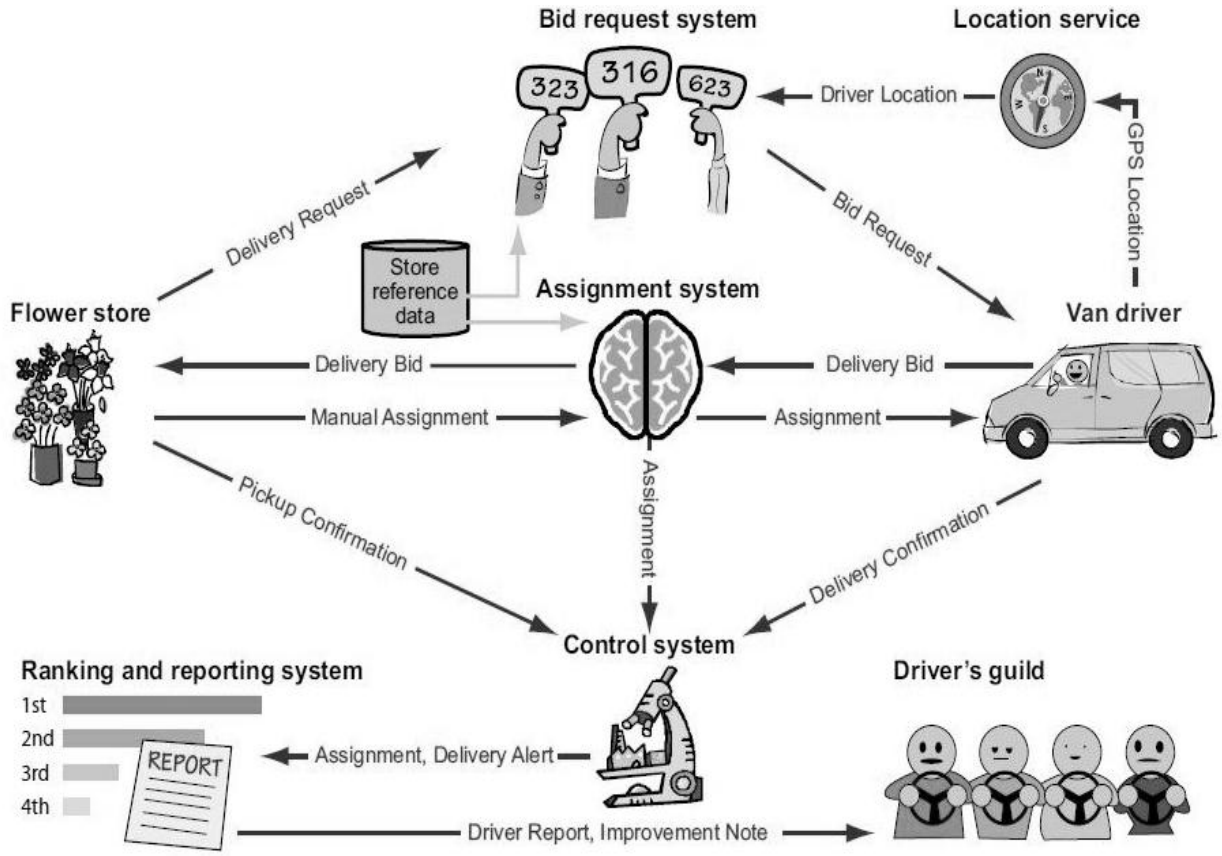


Figure 1.6 Parts of the Fast Flower Delivery application. The arrows represent the flow of events, and the pictures represent the various entities involved in the network.

Here are some excerpts of its description from the book. A consortium of flower stores in a large city has established an agreement with local independent van drivers to deliver flowers from the stores to their destinations. When a store gets a flower delivery order, it creates a request which is broadcast to relevant drivers within a certain distance from the store, with the time for pickup (typically now) and the required delivery time. A driver is then assigned and the customer is notified that a delivery has been scheduled. The driver makes the pickup and delivery, and the person receiving the flowers confirms the delivery time by signing for it on the drivers mobile device.

The system maintains a ranking of each individual driver based on his or her ability to deliver flowers on time. Each store has a profile that can include a constraint on the ranking of its drivers; for example, a store can require its drivers to have a ranking greater than 10. The profile also indicates whether the store wants the system to assign drivers automatically,or whether it wants to receive several applications and then make its own choice. A permanently weak driver is a driver with fewer than five assignments on all the days on which the driver has been active.

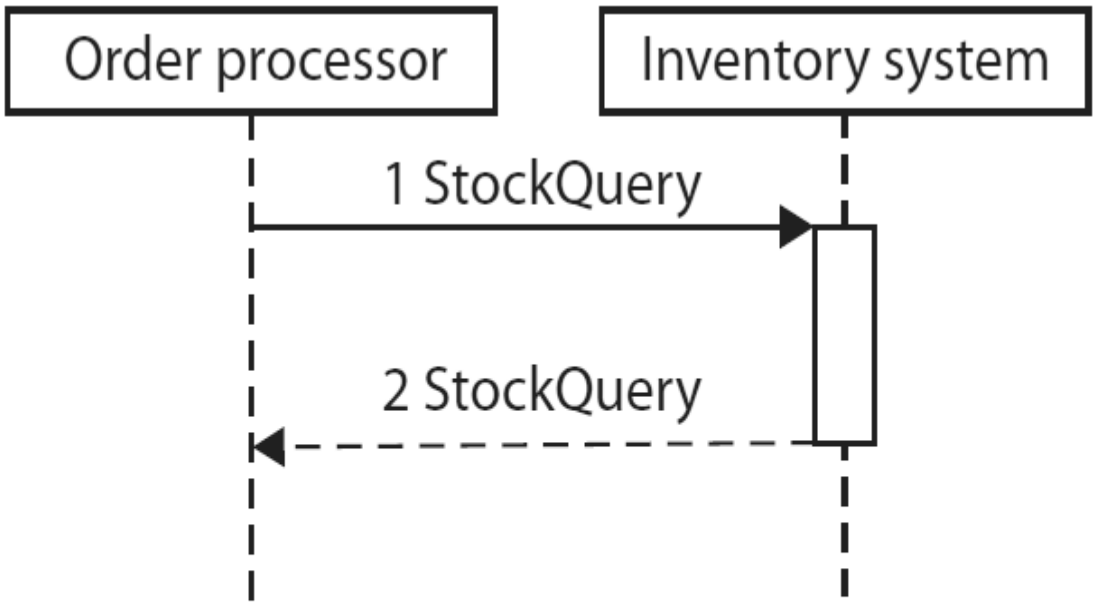
An idle driver is a driver with at least one day of activity that had no assignments. A consistently weak driver is a driver whose assignments, when active, are at least two standard deviations lower than the average assignments per driver on that day. A consistently strong driver is a driver whose daily assignments are at least two standard deviations higher than the average number of assignments per driver on each day in question. An improving driver is a driver whose assignments increase or stay the same day by day.

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Chapter 3

Lecture 3

Let's discuss request and response based applications. Example is a web browser which sends queries and updates. It performs synchronous interactions. It looks like this:



Events are based on the principle of decoupling. Let's compare events and request-response based architecture. Events have already happened whereas requests are asking something to happen. Service requester or client contacts the service provider or the server. In event-driven architecture event producer sends event to event consumer. A customer order can be represented as an event or a request: what are the benefits of each approach?

Similarly, should we send a request to query the state or use "change events". Event has meaning independent of its producers and consumers whereas the same cannot be said about requests. Events are often one way (push events). Events decouple producers and consumers. Events can be processed asynchronously. There can be more than one consumer, each processing it differently possibly processing a sequence of events that happened over time. Latency is reduced in comparison to the "pull" style.

Events can also be compared to messages. Messages may contain events. Events may live outside events like in event logs.

An event channel is a subscription mechanism for events. It further decouples consumers and producers. Can even be an intermediate XML file to store events.

Event-driven architecture and service-driven architecture can also be compared. Event-based programming, also called event-driven architecture (EDA) is an architectural style in which one or more components in a software system execute in response to receiving one or more event notifications. Service-oriented architecture (SOA) is built from request-response. It moves away from monolithic applications. There are

many similarities and often a component can provide both modes of contacting it.
Here are some definitions from the book.

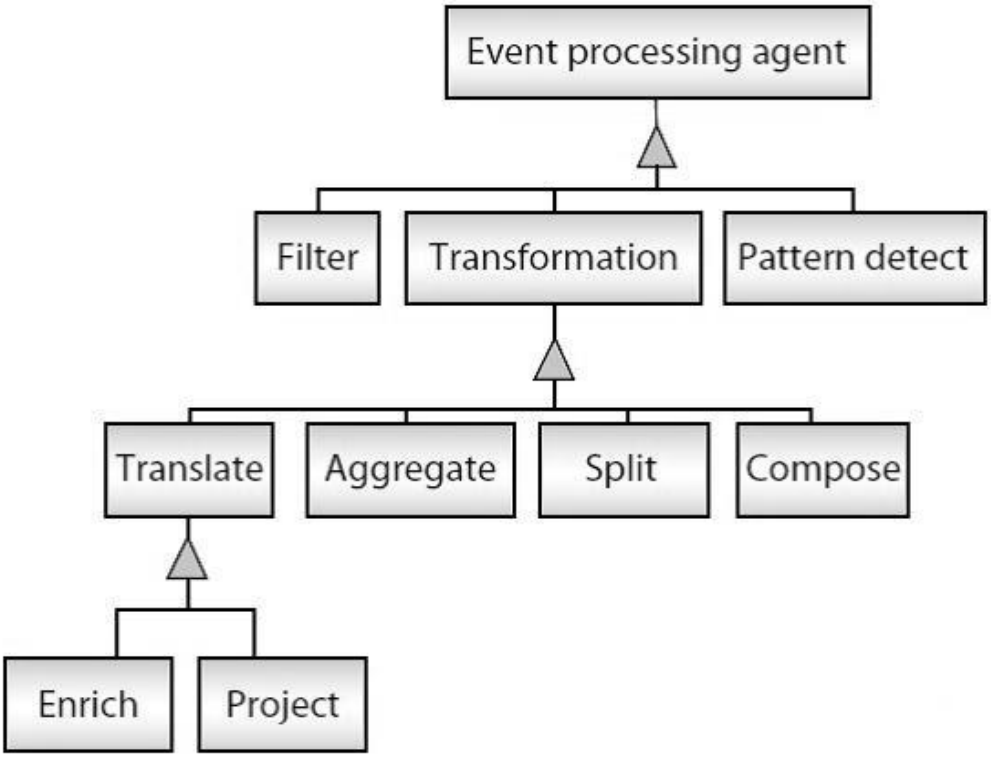
EVENT PRODUCER: An event producer is an entity at the edge of an event process- ing system that introduces events into the system.

EVENT CONSUMER: An event consumer is an entity at the edge of an event pro- cessing system that receives events from the system.

RAW EVENT: A raw event is an event that is introduced into an event processing system by an event producer. DERIVED EVENT: A derived event is an event that is generated as a result of event processing that takes place inside an event processing system.

STATELESS EVENT PROCESSING An event processing agent is said to be stateless if the way it processes one event does not influence the way it processes any subsequent events.

EVENT STREAM An event stream (or stream) is a set of associated events. It is often a temporally totally ordered set (that is to say, there is a well-defined timestamp-based order to the events in the stream). A stream in which all the events must be of the same type is called a homogeneous event stream; a stream in which the events may be of different types is referred to as a heterogeneous event stream.
Here are types of event agents.



Looking back at the program discussed before, can you map the event definitions to the code below:

```
#include <iostream>
#include <fstream>

Int filechanged = 0;
```

```
Char lastchar = '-';

Int onkeypress()
{
  Char b = std::cin.get();
  Std::cout << b << std::endl;
  Return b == 'x';
}

Int onfileread()
{
  Char a = f.get();
  If (a != lastchar) {
    Filechanged = 1;
    Lastchar = a;
  }
  Return 0;
}
Int onfilechanged(char a)
{
  Std::cout << a << std::endl;
  Return a == 'x';
}
Int main()
{
  Do {
    If (kbhit()) {
      If (onkeypress())
 Return 0;
    }
    Std::ifstream f("test.txt", std::ifstream::in);
    If (f.good()) {
      If (onfileread(f.get()))
 Return 0;
    }
    F.close();
    If (filechanged) {
      If (onfilechanged())
 Return 0;
    }
  } while (true);
  Return 0;
}
```

What aspects of real occurrence do event objects capture. An event type is a specification for a set of event objects that have the same semantic intent and same structure; every event object is considered to be an instance of an event type. An event attribute is a component of the structure of an event. Each attribute has a name and a data type. It can also have occurrence/detection time, certainty, source, location. Events can be composed, generalized, and specialized. Let's experiment with flower delivery example. Can you find events and their attributes.

Event producers can produce events that are hardware generated, software generated, or from human interaction. Event consumers have similar types. Examples are locking or unlocking a door, raising or lowering a barrier, applying the brakes on a vehicle, opening or closing a valve, controlling a railroad switch, and turning a piece of equipment on or off. Try to find producers and consumers in flower delivery application.

Chapter 4

Lecture 4

We will get introduced to C# in this lecture. Before that, Let's start with some history of Java and J++. Microsoft wanted to extend Java to communicate it with COM. Sun did not want that as it would make Java platform dependent. Microsoft wanted a clean-room implementation of Java. What is a clean-room implementation?

Clean room design (also known as the Chinese wall technique) is the method of copying a design by reverse engineering and then recreating it without infringing any of the copyrights and trade secrets associated with the original design. Clean room design is useful as a defense against copyright and trade secret infringement because it relies on independent invention. However, because independent invention is not a defense against patents, clean room designs typically cannot be used to circumvent patent restrictions.

Where do the name for C# comes from. The initial name was "Cool", which stood for "C-like Object Oriented Language". Whats the difference between .NET and C#. C# design most directly reflects .NET (CLR) design but C# is a language and .NET is the platform.

As a first step, you should get Visual Studio from http://www.microsoft.com/visualstudio. Click Downloads and choose "Visual Studio Express 2012 for Windows Desktop". Choose "Download now" or "Install now". You can save iso file to disk (right click to mount in win8) otherwise there are free utilities to mount it or burn it on a cd.

Install and register online for continued use. To create a new project choose File, New Project, Visual C#, Console Application. Then click the Start key.
Here are some notable features of C#.

- No global variables or functions.
- Locals cannot shadow global variables.
- There is a strict boolean type.
- Memory address pointers can only be used in specifically marked "unsafe" blocks and require permissions
- No instruction to "free" memory. Only garbage collection.
- Try-finally block.
- No "multiple inheritance" but interfaces supported.
- Operator overloading allowed.
- More type-safe (only integer widening allowed).
- Enumeration members are scoped.
- Property syntax for getters and setters.
- No checked exceptions.
- some functional programming features like function objects and lambda expressions.

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The common type system of C# has value types and reference types. Instances of value types do not have referential identity nor referential comparison semantics. Value types are derived from System.valuetype and can always be created, copied and have a default value (int, float, char, System.datetime, enum, struct). In contrast, reference types have the notion of referential identity. Default equality and inequality comparisons test for referential rather than structural equality unless overloaded e.g System.String. Not "always" possible to create an instance of a reference type, copy an existing instance, perform a value comparison on two instances, though specific types "can" provide such services by exposing a public constructor or implementing a corresponding interface e.g. Object, System.String, System.Array.

Here is how to box and unbox variables in C#. Boxing stores a value type in a reference type.

```
Int foo = 42;// Value type.
Object bar = foo;// foo is boxed to bar.

Int foo2 = (int)bar;// Unboxed back to value type.
```

Here is an example of Generics in C# which are like templates in C++.

```
Public class genericlist<T>
{
 Void Add(T input) { }
}
Class  testgenericlist
{
 Private class exampleclass{ }
 Static void Main()
 {
 // Declare a list of type int.
 Genericlist<int> list1= new genericlist<int>();
 // Declare a list of type string.
 Genericlist<string> list2= new genericlist<string>();
 // Declare a list of type exampleclass.
 Genericlist<exampleclass> list3 = new genericlist<exampleclass>();
 }

   }
```

Pre-processor directives like #if, #else, #endif, #region, #endregion are supported. Comments are written using // and /* */.

It also provides an XML documentation system that is used like this.

```
Public class Foo{
/**<summary>A summary of the method.</summary>
* <param name="firstparam">A description of the parameter.</param>
* <remarks>Remarks about the method.</remarks> */
Public static void Bar(int firstparam){}

   }
```

The C# specification details a minimum set of types and class libraries that the compiler expects to have available. In practice, C# is most often used with some implementation of the Common Language Infrastructure (CLI), which is standardized as ECMA-335 Common Language Infrastructure (CLI).

Here is a hello world program in C#.

```
Using System;
Class Program {
```


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```
    Static void Main() {
        Console.writeline("Hello world!");
    }
}
```

Writing the using clause uses System as a candidate prefix for types used in the source code. When the compiler sees use of the Console type later in the source code. It allows the programmer to state all candidate prefixes to use during compilation instead of always using full type names. Console is a static class in the System namespace.

If you run the program from within Visual Studio and the console window disappears quickly you can use the following code.

```
1 // keep screen from going  away
2 // when run from VS. NET
3 Console.readline();
```

To test use csc.exe Welcome.cs or run from within Visual Studio. Now Let's write a GUI based hello world.

```
Using System.Windows.Forms;

Class Program
{
    Static void Main()
    {
        Messagebox.Show("Hello world!");
    }

  }
```

Here is a hello world with a call to Console.Write with an argument.

```
// Namespace Declaration
Using System;

// Program start class
Class interactivewelcome
{
 // Main begins program execution.
 Public static void Main()
 {
 // Write to console/get input
Console.Write("What is your name?: ");
Console.Write("Hello, {0}! ", Console.readline());
Console.writeline("Welcome to the C\# Station Tutorial!");
 }
}
```


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Chapter 5

Lecture 5

Here is an example with boolean types.

```
Using System;

Class Booleans
{
Public static void Main()
 {
Bool content = true;
Bool nocontent = false;

Console.writeline("It is {0} that C\# Station provides C\# programming language
content.", content);
Console.writeline("The statement above is not {0}.", nocontent);
 }
}
```

Here are the integer types in C#.

Type	Size (in bits)	Range
sbyte	8	-128 to 127
byte	8	0 to 255
short	16	-32768 to 32767
ushort	16	0 to 65535
int	32	-2147483648 to 2147483647
uint	32	0 to 4294967295
long	64	-9223372036854775808 to 9223372036854775807
ulong	64	0 to 18446744073709551615
char	16	0 to 65535

And here are the floating point and decimal types in C#.

Type	Size (in bits)	precision	Range
float	32	7 digits	1.5×10^{-45} to 3.4×10^{38}
double	64	15-16 digits	5.0×10^{-324} to 1.7×10^{308}
decimal	128	28-29 decimal places	1.0×10^{-28} to 7.9×10^{28}

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The System.String type supports the following sequences.

Char	Meaning	Value
\'	Single quote	0x0027
\"	Double quote	0x0022
\\	Backslash	0x005C
\0	Null	0x0000
\a	Alert	0x0007
\b	Backspace	0x0008
\f	Form feed	0x000C
\n	New line	0x000A
\r	Carriage return	0x000D
\t	Horizontal tab	0x0009
\v	Vertical tab	0x000B

The escape character is while @ is the verbatim character. The following operators are supported by C#.

Category (by precedence)	Operator(s)	Associativity
Primary	x.y f(x) a[x] x++ x-- new typeof default checked unchecked delegate	left
Unary	+ - ! ~ ++x --x (T)x	right
Multiplicative	* / %	left
Additive	+ -	left
Shift	<< >>	left
Relational	< > <= >= is as	left
Equality	== !=	right
Logical AND	&	left
Logical XOR	^	left
Logical OR	|	left
Conditional AND	&&	left
Conditional OR	||	left
Null Coalescing	??	left
Ternary	?:	right
Assignment	= *= /= %= += -= <<= >>= &= ^= |= =>	right

Here is an example of using the Array type in C#.

```
Using System;
Class Array{
 Public static void Main(){
    Int[] myints = { 5, 10, 15 };
    Bool[][] mybools = new bool[2][];
    Mybools[0] = new bool[2];
    Mybools[1] = new bool[1];
    Double[,] mydoubles = new double[2, 2];
    String[] mystrings = new string[3];
    Console.writeline("myints[0]: {0}, myints[1]: {1}, myints[2]: {2}", myints[0],
myints[1], myints[2]);
    Mybools[0][0] = true;
    Mybools[0][1] = false;
    Mybools[1][0] = true;
    Console.writeline("mybools[0][0]: {0}, mybools[1][0]: {1}", mybools[0][0],
mybools[1][0]);
    Mydoubles[0, 0] = 3.147;
    Mydoubles[0, 1] = 7.157;
    Mydoubles[1, 1] = 2.117;
    Mydoubles[1, 0] = 56.00138917;
    Console.writeline("mydoubles[0, 0]: {0}, mydoubles[1, 0]: {1}", mydoubles[0, 0],
mydoubles[1, 0]);
    Mystrings[0] = "Joe";
    Mystrings[1] = "Matt";
    Mystrings[2] = "Robert";
    Console.writeline("mystrings[0]: {0}, mystrings[1]: {1}, mystrings[2]: {2}",
mystrings[0], mystrings[1], mystrings[2]);
 }
}
```

You can make jagged arrays or multi-dimensional arrays. Jagged is basically array of arrays. Array size is any integer type value. It uses a zero-based index.

Control statements like if and else work much like C++.

```
If (myint < 0 || myint == 0) {

Console.writeline("Your number {0} is less than or equal to zero.", myint);

}  else  if (myint > 0 && myint <= 10){

Console.writeline("Your number {0} is in the range from 1 to 10.", myint);

}  else  if (myint > 10 && myint <= 20){

Console.writeline("Your number {0} is in the range from 11 to 20.", myint);

} else if (myint > 20 && myint <= 30){

Console.writeline("Your number {0} is in the range from 21 to 30.", myint);

} else{

Console.writeline("Your number {0} is greater than 30.", myint); }
```

Switch statement can work for booleans, enums, integral types, and strings. You break the switch statement

Branching statement	Description
break	Leaves the switch block
continue	Leaves the switch block, skips remaining logic in enclosing loop, and goes back to loop condition to determine if loop should be executed again from the beginning. Works only if switch statement is in a loop as described in Lesson 04: Control Statements - Loops.
goto	Leaves the switch block and jumps directly to a label of the form "<labelname>:"
return	Leaves the current method. Methods are described in more detail in Lesson 05: Methods.
throw	Throws an exception, as discussed in Lesson 15: Introduction to Exception Handling.

with one of the following branching statements:
The following four kinds of loops are available.

While (<boolean expression>) { <statements> }
Do { <statements> } while (<boolean expression>);
For (<initializer list>; <boolean expression>; <iterator list>) { <statements> }
foreach (<type> <iteration variable> in <list>) { <statements> }

Here is an example of a while loop:

```
Using System;
Class whileloop
{
    Public static void Main()
    {
        Int myint = 0;

        While (myint < 10)
        {
            Console.Write("{0} ", myint);
            Myint++;
        }
        Console.writeline();
    }
}
```

Another example with a for loop.

```
Using System;
Class forloop
{
    Public static void Main()
    {
        For (int i=0; i < 20; i++)
        {
            If (i == 10)
                Break;

            If (i % 2 == 0)
```


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```
                Continue;
            Console.Write("{0} ", i);
        }
        Console.writeline();
    }

}
```

And one of foreach loop.

```
Using System;
Class foreachloop
{
    Public static void Main()
    {
        String[] names = {"Cheryl", "Joe", "Matt", "Robert"};

        Foreach (string person in names)
        {
            Console.writeline("{0} ", person);
        }
    }

}
```

Methods are of the form.

Attributes modifiers return-type method-name(parameters) { statements }

Elements are accessed using the dot operator. Object variables are references to the original object not the object themselves. Here is an example of a method.

```
Using System;
Class onemethod
{
    Public static void Main()
    {
        String mychoice;
        Onemethod om = new onemethod();
            Mychoice = om.getchoice();
    }
    String getchoice()
    {
        Return “example”;
    }

}
```

The “this” pointer in methods refers to the object on which the method is called. Parameters can be “ref” parameter which are passed by reference, “out” parameters which are used for return values or the “params” argument for variable arguments.

Here is an example of the last:

```
// show  addresses
Void viewaddresses(params string[] names)
{
 Foreach (string name in names){
 Console.writeline("Name: {0}", name);
 }
}
```


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Chapter 6

Lecture 6

Namespaces allow name reuse. For example, a Console class can reside in multiple libraries. Here is an example of namespace declaration:

```
// Namespace Declaration

Using System;

// The C\# Station Namespace

Namespace csharp_station
{
// Program start class
Class namespacecss  {
 // Main begins program execution.

Public static void Main()
 {

// Write to console

  Console.writeline("This is the new C\# Station Namespace.");

}

 }


}
```

Nested namespaces can be used like this:

Namespace testnamespace {

Namespace tutorial { OR

Namespace testnamespace.tutorial {

Calling this namespace from within testnamespace we use "tutorial.". The using directive can be used to rename a long namespace in the current file like:

Using theexample = testnamespace.tutorial.myexample

Class use the word class followed by name, curly braces and the class body. It has constructor (ctor) to create objects. Ctor does not return any values and it initializes class members.

```
// Namespace Declaration
Using System;
// helper class
Class outputclass
{
String mystring;

// Constructor
```


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```
Public outputclass(string inputstring)
    {

        Mystring = inputstring;
    }

// Instance Method

Public void printstring()
    {
        Console.writeline("{0}", mystring);

    }

// Destructor

~outputclass()
    {

        // Some resource cleanup routines

    }

}

// Program start class

Class exampleclass
{
    // Main begins program execution.

Public static void Main()
    {

    // Instance of outputclass
    Outputclass outcl = new
    Outputclass("This is printed by the output class.");
    // Call Output class' method
    Outcl.printstring();
    }

}
```

Multiple classes are conventionally stored in multiple files. To create a new class in IDE, use the new class wizard. Multiple .cs files can be compiled as csc a.cs b.cs c.cs. Default ctors are written with no arguments when no ctor is written. An initializer list can be used to use an alternate constructor.

```
Public outputclass() : this("Default Constructor String") { }
```

Multiple ctors can be written. Types of class members in C# are instance and static. For every new occurrence there are new instance vars.

Outputclass oc1 = new outputclass("outputclass1");

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Outputclass oc2 = new outputclass("outputclass2");

Now oc1.printstring or oc2.printstring will access different instance variables. On the other hand static members have only one copy.

```
Public static void staticprinter(){
 Console.writeline("There is only one of me.");

}
```

Then you could call that function from Main() like this: outputclass.staticprinter();

Static ctor exists to initialize class static members. Its called only once. It has no parameters. Destructors (dtors) are called by garbage collector. "public" methods accessed outside class. "public" classes accessed outside assembly. A class can contain Constructors, Destructors, Fields, Methods, Properties, Indexers, Delegates, Events, and Nested Classes.

Inheritance introduces the concept of base classes and derived classes. Declaration of inheritance is done as Derived : Base. Only single inheritance is supported. Multiple interface inheritance discussed later.

```
Using System;

Public class parentclass
{

  Public parentclass()
 {

    Console.writeline("Parent Constructor.");

  }

  Public void print() {

    Console.writeline("I'm a Parent Class.");

  }

}

Public class childclass : parentclass
{

  Public childclass()
 {

    Console.writeline("Child Constructor.");

  }

  Public static void Main()
 {

    Childclass child = new childclass();

    Child.print();
```


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```
  }

}
```

The output is:

Parent Constructor. Child Constructor. I'm a Parent Class

.

Derived class is exactly the same as base. Child class "is a" parent class except that derived is more special. Base are automatically initialized before derived. How can derived communicate with the base i.e. The specialized part communicate with the generic part. By using ": base()" at ctor time or "base.x()" later. "new" keyword used to override methods. Cast back to base type to call an overridden method of base.

Here is another example:

```
Using System;
Public class Parent {
  String parentstring;
  Public Parent() {
    Console.writeline("Parent Constructor.");
  }
  Public Parent(string mystring) {
    Parentstring = mystring;
    Console.writeline(parentstring);
  }

  Public void print() {
    Console.writeline("I'm a Parent Class.");
  }
}
  Public class Child : Parent {
    Public Child() : base("From Derived") {
      Console.writeline("Child Constructor.");
    }

    Public new void print() {
      Base.print(); Console.writeline("I'm a Child Class.");
    }

    Public static void Main() {
      Child child = new Child();
      Child.print(); ((Parent)child).print();
    }
  }
```

Output is:

From Derived
Child Constructor. I'm a Parent Class. I'm a Child Class. I'm a Parent Class.

Polymorphism means to invoke derived class methods through base class reference during run-time. Normally (as we saw) base class casting means base class method is called. Its handy when a group of objects is stored in array/list and we invoke some method on all of them. They dont have to be the same type but if related by inheritance, they can be cast.

```
Using System;
```


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```
Public class drawingobject
{
 Public virtual void Draw()
 {
     Console.writeline("I'm just a generic drawing object.");
   }

}

Public class Line : drawingobject
{
 Public override void Draw()
 {
     Console.writeline("I'm a Line.");
 }
}

Public class Circle : drawingobject
{
 Public override void Draw()
 {
     Console.writeline("I'm a Circle.");
  }
}

Public class Square : drawingobject
{
 Public override void Draw()
 {
    Console.writeline("I'm a Square.");
 }
}

Public class drawdemo
{
  Public static int Main( )
    {

    Drawingobject[] dobj = new drawingobject[4];
    Dobj[0] = new Line();
    Dobj[1] = new Circle();
    Dobj[2] = new Square();
    Dobj[3] = new drawingobject();
    Foreach (drawingobject drawobj in dobj)
    {
      Drawobj.Draw();
    }
    Return 0;
  }
}
```

Polymorphic methods are declared with “virtual” keyword and the “override” keyword is used to provide an implementation. Polymorphism needs the signatures to be the same. Because of inheritance, the three derived classes can be treated as the base class. Because of polymorphism, the methods from drived can still be called.

Output is:

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I'm a Line.
I'm a Circle. I'm a Square.
I'm just a generic drawing object.

Properties allow protected reads and writes to a field of a class. Other languages require custom getters and setters.

C# allows accessing properties like fields. Properties allows changing representation, public fields dont. Other languages use custom encapsulation to protect data.

```
Using System;

Public class Customer
{
    Private int m_id = -1;

    Public int getid()
    {
        Return m_id;
    }

    Public void setid(int id)
    {
        M_id = id;
    }

    Private string m_name = string.Empty;

    Public string getname()
    {
        Return m_name;
    }

    Public void setname(string name)
    {
        M_name = name;
    }
}

Public class customermanagerwithaccessormethods
{
    Public static void Main()
    {
        Customer cust = new Customer();

        Cust.setid(1);
        Cust.setname("Amelio Rosales");

        Console.writeline(
            "ID: {0}, Name: {1}",
            Cust.getid(),
            Cust.getname());

        Console.readkey();
    }
}
```


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Its such a common pattern that C# has a dedicated language feature for it.

```
Using System;
Public class Customer
{
    Private int m_id = -1;

    Public int ID
    {
        Get
        {
            Return m_id;
        }
        Set
        {
            M_id = value;
        }
    }

    Private string m_name = string.Empty;

    Public string Name
    {
        Get
        {
            Return m_name;
        }
        Set
        {
            M_name = value;
        }
    }
}

Public class customermanagerwithproperties
{
    Public static void Main()
    {
        Customer cust = new Customer();

        Cust.ID = 1;
        Cust.Name = "Amelio Rosales";

 Console.writeline(
            "ID: {0}, Name: {1}",
            Cust.ID,
            Cust.Name);

        Console.readkey();
    }
}
```

Getters “return”, setters use “value”. Next we discuss read-only peroperties. Here is an example that uses read-only properties.

```
Using System;

Public class Customer
{
```


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```
    Private int m_id = -1;
    Private string m_name = string.Empty;

    Public Customer(int id, string name)
    {
        M_id = id;
        M_name = name;
    }

    Public int ID
    {
        Get
        {
            Return m_id;
        }
    }

    Public string Name
    {
        Get
        {
            Return m_name;
        }
    }
}

Public class readonlycustomermanager
{
    Public static void Main()
    {
        Customer cust = new Customer(1, "Amelio Rosales");

        Console.writeline(
            "ID: {0}, Name: {1}",
            Cust.ID,
            Cust.Name);

        Console.readkey();
    }
}
```

This example has only get and no implementation of set. So what about write-only properties. Yes, they exist and have an implementation of set but not get.

```
  Using System;

Public class Customer
{
    Private int m_id = -1;

    Public int ID
    {
        Set
        {
            M_id = value;
        }
    }
```


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```
    Private string m_name = string.Empty;

    Public string Name
    {
        Set
        {
            M_name = value;
        }
    }

    Public void displaycustomerdata()
    {
        Console.writeline("ID: {0}, Name:
            {1}", m_id, m_name);
    }
}

Public class writeonlycustomermanager
{
    Public static void Main()
    {
        Customer cust = new Customer();

        Cust.ID = 1;
        Cust.Name = "Amelio Rosales";

        Cust.displaycustomerdata();

        Console.readkey();
    }
}
```


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Chapter 7

Lecture 7

Auto-implemented properties improve the common-case we saw. Here is the same example using auot-implemented properties. Properties have the same idea as getters and setters. They just allow simplified syntax. Same idea of backing store and a simplified syntax for the general case is enabled by auto-implemented properties.

```
  Using System;


Public class Customer
{
    Public int ID { get; set; }
    Public string Name { get; set; }
}

Public class autoimplementedcustomermanager
{
    Static void Main()
    {
        Customer cust = new Customer();

        Cust.ID = 1;
        Cust.Name = "Amelio Rosales";

        Console.writeline(
            "ID: {0}, Name: {1}",
            Cust.ID,
            Cust.Name);

        Console.readkey();
    }
}
```

An indexer enables your class to be treated like an array. However you have your internal data representation not an actual array. Its implemented using "this" keyword and square brackets syntax. It looks like a property implementation. Here is an example.

```
Class intindexer
{
    Private string[] mydata;

    Public intindexer(int size)
    {
        Mydata = new string[size];

        For (int i=0; i < size; i++)
        {
            Mydata[i] = "empty";
        }
```


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```
    }

    Public string this[int pos]
    {
        Get
        {
            Return mydata[pos];
        }
        Set
        {
            Mydata[pos] = value;
        }
    }

    Static void Main(string[] args)
    {
        Int size = 10;

        Intindexer myind = new intindexer(size);

        Myind[9] = "Some Value";
        Myind[3] = "Another Value";
        Myind[5] = "Any Value";

        Console.writeline("\nindexer Output\n");

        For (int i=0; i < size; i++)
        {
            Console.writeline("myind[{0}]: {1}", i, myind[i]);
        }
    }
}
```

The output of this program looks like this:

Indexer Output

Myind[0]: empty
Myind[1]: empty
Myind[2]: empty
Myind[3]: Another Value myind[4]: empty
Myind[5]: Any Value
Myind[6]: empty
Myind[7]: empty
Myind[8]: empty
Myind[9]: Some Value

Indexers can take any number of parameters. The parameters can be integers, strings, or enums. Additionally they can be overloaded with different type and number of parameters. The following example has overloaded indexers.

```
Using System;

/// <summary>
///     Implements overloaded indexers.
/// </summary>
Class ovrindexer
```


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```
{
    Private string[] mydata;
    Private int         arrsize;

    Public ovrindexer(int size)
    {
        Arrsize = size;
        Mydata = new string[size];

        For (int i=0; i < size; i++)
        {
            Mydata[i] = "empty";
        }
    }

    Public string this[int pos]
    {
        Get
       {
            Return mydata[pos];
        }
        Set
       {
            Mydata[pos] = value;
        }
    }

     Public string this[string data]
    {
        Get
       {
            Int count = 0;

            For (int i=0; i < arrsize; i++)
            {
                If (mydata[i] == data)
                    Count++;
            }
            Return count.tostring();
        }
        Set
       {
            For (int i=0; i < arrsize; i++)
            {
                If (mydata[i] == data)
                    Mydata[i] = value;
            }
        }
    }
    Static void Main(string[] args)
    {
        Int size = 10;
        Ovrindexer myind = new ovrindexer(size);

        Myind[9] = "Some Value";
        Myind[3] = "Another Value";
        Myind[5] = "Any Value";
```


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```
        Myind["empty"] = "no value";

        Console.writeline("\nindexer Output\n");

        For (int i=0; i < size; i++)
        {
            Console.writeline("myind[{0}]: {1}", i, myind[i]);
        }

        Console.writeline("\nnumber of \"no value\" entries: {0}", myind["no value"]);
    }
}
```

Indexers with multiple parameters look like this.

```
Public object this[int param1, ..., int paramn] {
Get {
 // process and return some class data
 }
Set {
 // process and assign some class data
 }

   }
```

Our next topic is a "struct". Its a value-type, whereas class is a reference-type. Value types hold their value in memory where they are declared, but reference types hold a reference to an object in memory. If you copy a class, C# creates a new copy of the reference to the object and assigns the copy of the reference to the separate class instance. Structs cant have destructors, cant have implementation inheritance (still can have interface inheritance). Many built-in types are structs like System.Int32 is a C# int. C# built-in types are aliases for .NET Framework types. Syntax of struct and class are very similar. Here is an example of using a struct.

```
/// <summary>
/// Custom struct type, representing
    A rectangular shape
/// </summary>
Struct Rectangle
{
    /// <summary>
    /// Backing Store for Width
    /// </summary>
    Private int m_width;

    /// <summary>
    /// Width of rectangle
    /// </summary>
    Public int Width
    {
        Get
        {
            Return m_width;
        }
        Set
        {
            M_width = value;
        }
    }
```


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```
    /// <summary>
    /// Backing store for Height
    /// </summary>
    Private int m_height;

    /// <summary>
    /// Height of rectangle
    /// </summary>
    Public int Height
    {
        Get
        {
            Return m_height;
        }
        Set
        {
            M_height = value;
        }
    }
}
```

Its instantiated and used just like a class.

```
Using System;

/// <summary>
/// Example of declaring and using
///    a struct
/// </summary>
Class structexample
{
    /// <summary>
    /// Entry point: execution starts
///        here
    /// </summary>
    Static void Main()
    {
        // instantiate a new Rectangle struct
        // where Width is set to 1 and Height is set to 3
       Rectangle rect1 = new Rectangle();
        Rect1.Width = 1;
        Rect1.Height = 3;

        // show the value of Width and Height for rect1
        Console.writeline("rect1: {0}:{1}", rect1.Width, rect1.Height);

        Console.readkey();
    }
}
```

The output of this program is: Output:

Rect1: 1:3

We can add constructors to a struct but cant overlaod the default ctor which initializes everything to default values. We can add methods like classes and they are called in exactly the same way.

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An object initializer is a syntax for initializing a struct without using a ctor. It looks like this:

```
// you can also use object initialization syntax
        Rectangle rect11 = new Rectangle
        {
            Width = 1,
            Height = 3
        };
```

Next we discuss "Interfaces". They are declared like a class but have no implementation. There is only declarations of events, indexers, methods and/or properties. They are inherited by classes which provide the real implementation. So, what are interfaces good for if they don't implement functionality? They are great for putting together plug-n-play like architectures where components can be interchanged at will. The interface forces each component to expose specific public members that will be used in a certain way.

Interfaces define a contract. For instance, if class foo implements the idisposable interface, it is making a statement that it guarantees it has the Dispose() method, which is the only member of the idisposable interface. Any code that wishes to use class foo may check to see if class foo implements idisposable. When the answer is true, then the code knows that it can call foo.Dispose()

To define an interface, we use the following syntax.

```
Interface imyinterface
{
Void methodtoimplement();
}
```

The "I" prefix is a convention. There is no implementation, only semi-colon. There are only method signatures.
Let's now see how its used.

```
Class  interfaceimplementer : imyinterface
{
Static void Main()
{
 Interfaceimplementer iimp = new interfaceimplementer();
 Iimp.methodtoimplement();
}
Public void methodtoimplement()
{
 Console.writeline("methodtoimplement() called.");
}
}
```

The syntax used is same as for inheritance. Any signature difference or not implementing some method in the interface is an error. Interfaces can also be inherited. Here is an example.

```
Using System;

Interface iparentinterface
{
    Void parentinterfacemethod();
}

Interface imyinterface : iparentinterface
{
    Void methodtoimplement();
}
```


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```
Class interfaceimplementer : imyinterface
{
    Static void Main()
    {
        Interfaceimplementer iimp = new interfaceimplementer();
        Iimp.methodtoimplement();
        Iimp.parentinterfacemethod();
    }

    Public void methodtoimplement()
    {
        Console.writeline("methodtoimplement() called.");
    }

    Public void parentinterfacemethod()
    {
        Console.writeline("parentinterfacemethod() called.");
    }
}
```

We can also do polymorphism using interfaces.

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Chapter 8

Lecture 8

A delegate is a reference to a method. Its like function pointers in some other languages. Methods are algorithms that operate on data. Sometimes the data needs a special operation e.g. A comparator in a sorting routine. We can use a bad solution which would an if then else for all types we want our sorting routine to work for. There are two good solutions. One is to implement a comparator interface in all types we want and then pass an instance of a type that provides that interface. The other solution is using delegates i.e. References to functions. Now we can pass a comparator delegate to the sorting algorithm.

```
Using System;

// this is the delegate declaration
Public delegate int Comparer(object obj1, object obj2);

Public class Name
{
    Public string firstname = null;
    Public string lastname = null;

    Public Name(string first, string last)
    {
        Firstname = first;
        Lastname = last;
    }

    // this is the delegate method handler
    Public static int comparefirstnames(object name1, object name2)
    {
        String n1 = ((Name)name1).firstname;
        String n2 = ((Name)name2).firstname;

        If (String.Compare(n1, n2) > 0)
        {
            Return 1;
        }
        Else if (String.Compare(n1, n2) < 0)
        {
            Return -1;
        }
        Else
        {
            Return 0;
        }
    }

    Public override string tostring()
    {
        Return firstname + " " + lastname;
    }
}
```


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```
Class simpledelegate
{
    Name[] names = new Name[5];

    Public simpledelegate()
    {
      Names[0] = new Name("Joe", "Mayo");
        Names[1] = new Name("John", "Hancock");
        Names[2] = new Name("Jane", "Doe");
        Names[3] = new Name("John", "Doe");
        Names[4] = new Name("Jack", "Smith");
    }
    Static void Main(string[] args)
    {
        Simpledelegate sd = new simpledelegate();

        // this is the delegate instantiation
        Comparer cmp = new Comparer(Name.comparefirstnames);
        Console.writeline("\nbefore Sort: \n");
        Sd.printnames();
        // observe the delegate argument
        Sd.Sort(cmp);
        Console.writeline("\nafter Sort: \n");
        Sd.printnames();
    }
    // observe  the delegate parameter
    Public void Sort(Comparer compare)
    {
        Object temp;

        For (int i=0; i < names.Length; i++)
        {
            For (int j=i; j < names.Length; j++)
            {
                // using delegate "compare" just like
                // a normal method
                If ( compare(names[i], names[j]) > 0 )
                {
                    Temp = names[i];
                    Names[i] = names[j];
                    Names[j] = (Name)temp;
                }
            }
        }
    }
    Public void printnames()
    {
        Console.writeline("Names: \n");

        Foreach (Name name in names)
        {
            Console.writeline(name.tostring());
        }
    }
}
```

A C# event is a class member that is activated whenever the event it was designed for occurs (fires).

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Anyone interested in the event can register and be notified as soon as the event fires. At the time an event fires, methods registered with the event will be invoked.

Events and delegates work hand in hand. Any class, including the same class that the event is declared in, may register one of its methods with the event. This occurs through a delegate, which specifies the signature of the method that is registered for the event. The delegate may be one of the pre-defined .NET delegates or one that you declare yourself. Then, you assign the delegate to the event, which effectively registers the method that will be called when the event fires.

Here is an example:

```
Using System;
Using System.Drawing;
Using System.Windows.Forms;

// custom delegate
Public delegate void Startdelegate();

Class Eventdemo : Form
{
    // custom event
    Public event Startdelegate startevent;
    Public Eventdemo()
    {
        Button clickme = new Button();
        Clickme.Parent = this;
        Clickme.Text = "Click Me";
        Clickme.Location = new Point(
            (clientsize.Width - clickme.Width) /2,
            (clientsize.Height - clickme.Height)/2);
        // an eventhandler delegate is assigned
        // to the button's Click event
        Clickme.Click += new eventhandler(onclickmeclicked);
        // our custom "Startdelegate" delegate is assigned
        // to our custom "startevent" event.
        Startevent += new Startdelegate(onstartevent);
        // fire our custom event
        Startevent();
    }
    // this method is called when the "clickme" button is pressed
    Public void onclickmeclicked(object sender, eventargs ea)
    {
        Messagebox.Show("You Clicked My Button!");
    }
    // this method is called when the "startevent" Event is fired
    Public void onstartevent()
    {
        Messagebox.Show("I Just Started!");
    }
    Static void Main(string[] args)
    {
        Application.Run(new Eventdemo());
    }
}
```

To run this example in Visual Studio you will either have to add a reference to System.Drawing.dll and

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Sys-tem.Windows.Forms.dll or you can make a new Windows Forms Project. This class inherits from Form, which essen-tially makes it a Windows Form. This automatically gives you all the functionality of a Windows Form, including Title.Bar, Minimize/Maximize/Close buttons, System Menu, and Borders. It is started by calling the Run() method of the static Application object with a reference to the form object as its parameter. The += syntax registers a delegate with the event. Its fired like a method call. The Click event is pre-defined, eventhandler delegate is also already in .NET. All you do is define your callback method (delegate handler method) that is invoked when someone presses the clickme button. The onclickmeclicked() method, conforms to the signature of the eventhandler delegate.

Next, we discuss exception handling. There can be unforeseen errors in the program. There are some normal errors and some exceptional errors e.g. File i/o error, system out of memory, null pointer exception. Exceptions are "thrown". They are derived from System.Exception class. They have a message property, contain a stacktrace, a tostring method. Identifying the exceptions you'll need to handle depends on the routine you're writing e.g. System.IO.File.openread() could throw securityexception, argumentexception, argumentnullexception, pathtoolongexception, directorynotfoundexception, unauthorizedaccessexception, filenotfoundexception, or notsupportedexception.

Here is an example of exception handling:

```
Using System;
Using System.IO;

Class trycatchdemo
{
    Static void Main(string[] args)
    {
        Try
        {
            File.openread("nonexistentfile");
        }
        Catch(Exception ex)
        {
            Console.writeline(ex.tostring());
        }
    }
}
```

A single exception can be handled differently as well:

```
1           catch (filenotfoundexception fnfex)
2           {
3               Console.writeline(fnfex.tostring());
4           }
5           catch (Exception ex)

6           {
7               Console.writeline(ex.tostring());
8           }
```

First handler that handles an exception catches it. Otherwise the exception bubbles up the caller stack until someone handles it.

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Chapter 9

Lecture 9

Exception can leave your program in an inconsistent state by not releasing resources or doing some other type of cleanup. Sometimes you need to perform clean up actions whether or not your program succeeds. These are good candidates for using a finally block e.g. A filestream must be closed.

```
Using System;
Using System.IO;
Class finallydemo
{
    Static void Main(string[] args)
    {
        Filestream outstream = null;
        Filestream instream = null;
        Try
        {
            Outstream = File.openwrite("destinationfile.txt");
            Instream = File.openread("bogusinputfile.txt");
        }
        Catch (Exception ex)
        {
            Console.writeline(ex.tostring());
        }
        Finally
        {
            If (outstream != null)
            {
                Outstream.Close();
                Console.writeline("outstream closed.");
            }
            If (instream != null)
            {
                Instream.Close();
                Console.writeline("instream closed.");
            }
        }
    }
}
```

The alternate would be to duplicate code after catch and in catch. But finally is a neat cleanup solution.

Next, we discuss attributes. Attributes add declarative information to your programs. They are used for various purposes during runtime. They can be used at design time by application development tools. For example, dllimpor-tattribute allows a program to communicate with Win32 libraries. Obsoleteattribute causes a compile-time warning to appear. Such things would be difficult to accomplish with normal code. Attributes add metadata to your programs.

When your C# program is compiled, it creates a file called an assembly, which is normally an executable or DLL library. Assemblies are self-describing because they have metadata. Via reflection, a program's

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attributes can be retrieved from its assembly metadata. Attributes are classes that can be written in C# and are used to decorate your code with declarative information. Its a powerful concept. It Let's you extend your language by creating customized declarative syntax with attributes.

Attributes are generally applied physically in front of type and type member declarations. They are declared with square brackets, "[" and "]" surrounding the attribute such as " [obsoleteattribute]". The "Attribute" part of the attribute name is optional i.e. "[Obsolete]" is correct as well. Parameter lists are also possible.

```
Using System;
Class basicattributedemo
{
    [Obsolete]
    Public void myfirstdeprecatedmethod()
    {
        Console.writeline("Called myfirstdeprecatedmethod().");
    }
    [obsoleteattribute]
    Public void myseconddeprecatedmethod()
    {
        Console.writeline("Called myseconddeprecatedmethod().");
    }
    [Obsolete("You shouldn't use this method anymore.")]
    Public void mythirddeprecatedmethod()
    {
        Console.writeline("Called mythirddeprecatedmethod().");
    }
    // make the program thread safe for COM
    [stathread]
    Static void Main(string[] args)
    {
        Basicattributedemo attrdemo = new basicattributedemo();
        Attrdemo.myfirstdeprecatedmethod();
        Attrdemo.myseconddeprecatedmethod();
        Attrdemo.mythirddeprecatedmethod();
    }
}
```

When you compile this program you'll see.

```
>csc basicattributedemo.cs
Microsoft (R) Visual C\# .NET Compiler version 7.10.2292.4 for Microsoft (R) .NET Framework version 1.1.4322
Copyright (C) Microsoft Corporation 2001-2002. All rights reserved.

Basicattributedemo.cs(29,3): warning CS0612:
'Basicattributedemo.myfirstdeprecatedmethod()'is obsolete basicattributedemo.cs(30,3): warning CS0612:
'Basicattributedemo.myseconddeprecatedmethod()' is obsolete basicattributedemo.cs(31,3):
warning CS0618:
'Basicattributedemo.mythirddeprecatedmethod()' is obsolete: 'You shouldn't use this method anymore.'
```

Stathread is a common attribute you will see later. It stands for Single Threaded Apartment model which is used for communicating with unmanaged COM.

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Attribute parameters can be either positional parameters or named parameters. Usually named parameters with optional stuff, but positional can be optional as well e.g. "[Obsolete("You shouldn't use this method anymore.", true)] public void mythirddeprecatedmethod()" will give an error instead of warning. The attribute dllimportattribute has

Both positional and named parameters "[dllimport("User32.dll", entrypoint="messagebox")]". Positional parameters come before named parameters. There is no order requirement on named parameters.

```
Using System;
[assembly: clscompliant(true)]
Public class attributetargetdemo
{
    Public void nonclscompliantmethod(uint nclsparam)
    {
        Console.writeline("Called nonclscompliantmethod().");
    }
    [stathread]
    Static void Main(string[] args)
    {
        Uint myuint = 0;
        Attributetargetdemo tgtdemo = new attributetargetdemo();
        Tgtdemo.nonclscompliantmethod(myuint);
    }
}
```

We rarely write new attributes but we use them extensively.

Enums (or enumerations) are strongly typed constants. They are unique types that allow you to assign symbolic names to integral values. Enum of one type may not be implicitly assigned to an enum of another type (even though the underlying value of their members is the same). All assignments between different enum types and integral types require an explicit cast. It allows you to work with integral values, but using a meaningful name. North, South, East, and West or the zet of integers 0, 1, 2, and 3. C# type, enum, inherits the Base Class Library (BCL) type, Enum.

```
Using System;
// declares the enum
Public enum Volume
{
    Low,_
    Medium,
    High
}
// demonstrates how to use the enum
Class enumswitch
{
    Static void Main()
    {
        // create and initialize
        // instance of enum type
        Volume myvolume = Volume.Medium;
        // make decision based
        // on enum value
        Switch (myvolume)
        {
            Case Volume.Low:
                Console.writeline("The volume has been turned Down.");
                Break;
```


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```
            Case Volume.Medium:
                Console.writeline("The volume is in the middle.");
                Break;
            Case Volume.High:
                Console.writeline("The volume has been turned up.");
                Break;
        }
        Console.readline();
    }
}
```

Default underlying type of an enum is "int". It can be changed by specifying a "base". Valid base types include byte, sbyte, short, ushort, int, uint, long, and ulong. Default value of first member is 0. You can assign any member any value. If it is unassigned it gets +1 the value of its predecessor.

```
Public enum Volume  : byte
{
 Low= 1,
 Medium,
 High
}
```

To convert user input to enum:

```
// get value user provided
String volstring  = Console.readline();
Int volint  = Int32.Parse(volstring);
// perform  explicit cast from
// int to Volume enum type

Volume myvolume  =(Volume)volint
```

Enum inherits from System.Enum in base class library.

```
// get a list of member names from Volume enum,
// figure out the numeric value, and display
Foreach  (string volume in Enum.getnames(typeof(Volume)))
{
Console.writeline("Volume Member:  {0}\n Value:  {1}",
Volume,  (byte)Enum.Parse(typeof(Volume), volume));
}
```

To enumerate values.

```
// get all values  (numeric values) from the Volume
// enum type, figure out member name, and display
Foreach  (byte val in Enum.getvalues(typeof(Volume)))
{
Console.writeline("Volume Value:  {0}\n Member:  {1}",
Val, Enum.getname(typeof(Volume), val));
}
```

Next topic is operator overloading. You can add operators to your own types e.g. A matrix type can have an add and a dot product operator etc.

```
Matrix result  = mat1.Add(mat2); // instance
Matrix result  = Matrix.Add(mat1, mat2);
Matrix result  = mat1.dotproduct(mat2).dotproduct(mat3);
Matrix result  = mat1  + mat2;
Matrix result  = mat1  * mat2; 6  Matrix result = mat1 * mat2 * mat3 * mat4;
```

It should not be used when its not natural to be used. Its implementation syntax is:

```
Public static Matrix operator *(Matrix mat1, Matrix mat2)
```


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```
{
 // dot product  implementation
}
```

Here is an example.

```
Using System;

Class Matrix
{
    Public const int dimsize = 3;
    Private double[,] m_matrix = new double[dimsize, dimsize];

    // allow callers to initialize
    Public double this[int x, int y]
    {
        Get { return m_matrix[x, y]; }
        Set { m_matrix[x, y] = value; }
    }

    // let user add matrices
    Public static Matrix operator +(Matrix mat1, Matrix mat2)
    {
        Matrix newmatrix = new Matrix();

        For (int x = 0; x < dimsize; x++)
            For (int y = 0; y < dimsize; y++)
                Newmatrix[x, y] = mat1[x, y] + mat2[x, y];

        Return newmatrix;
    }
}
```

Overloaded operators must be static. They must be declared in the class for which the operator is defined. It is required that matching operators are both defined, e.g. == and != so that the behavior is consistent. Access modifiers on types and assemblies provide encapsulation. They are: private, protected, internal, protected internal, and public.

Next, we see how generic collections are used: There is a very useful List collection. It is better than arraylist of objects in earlier .NET versions. Another useful collection is Dictionary¡tkey,tvalue¿ vs. A non-generic hashtable of objects. We write "using System.Collections.Generic" to include these collections.

```
List<int> myints  = new List<int>();

Myints.Add(1);
Myints.Add(2);
Myints.Add(3);

For(int i= 0; i < myints.Count; i++)
{
Console.writeline("myints:{0}", myints[i]);
}

Public class Customer
{
Public Customer(int id, string name)
{
ID= id;
Name = name;
```


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```
}
Public int ID
{
Get;
Set;
}
Public string Name
{
Get;
Set;
}
}
Dictionary<int, Customer> customers = new Dictionary<int, Customer>();

Customer cust1 = new Customer(1, "Cust 1");
Customer cust2 = new Customer(2, "Cust 2");
Customer cust3 = new Customer(3, "Cust 3");

Customers.Add(cust1.ID, cust1);
Customers.Add(cust2.ID, cust2);
Customers.Add(cust3.ID, cust3);

Foreach (keyvaluepair<int, Customer> custkeyval in customers)
{
Console.writeline(
"Customer ID:{0}, Name: {1}",
Custkeyval.Key,
Custkeyval.Value.Name);
}
```


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Chapter 10

Lecture 10

Let's discuss Anonymous methods. It is used with delegates and handlers and events. Anonymous methods result in much less code. Anonymous method is a method without a name. You don't declare anonymous methods, Instead they get hooked up directly to events. With delegates 1) you declare the delegate, 2) write a method with a signature defined by the delegate interface, 3) declare the event based on that delegate, and 4) write code to hook the handler method up to the delegate. To delcare an anonymous method, you just use keyword "delegate" followed by method body.

```
Public partial class Form1: Form
{
 Public Form1()
 {
  Button btnhello= new Button();
  Btnhello.Text= "Hello";
  Btnhello.Click +=
  Delegate
  {
   Messagebox.Show("Hello");
  };
  Controls.Add(btnhello);
 }
}
```

We can skip parameters when not using them. Here is an example with parameters.

```
Using System;
Using System.Windows.Forms;

Public partial class Form1 : Form
{
    Public Form1()
    {
        Button btnhello = new Button();
        Btnhello.Text = "Hello";

        Btnhello.Click +=
            Delegate
            {
                Messagebox.Show("Hello");
            };

        Button btngoodbye = new Button();
        Btngoodbye.Text = "Goodbye";
        Btngoodbye.Left = btnhello.Width + 5;
        Btngoodbye.Click +=
            Delegate(object sender, eventargs e)
            {
                String message = (sender as Button).Text;
                Messagebox.Show(message);
            };
```


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```
        Controls.Add(btnhello);
        Controls.Add(btngoodbye);
    }
}
```

We can create nullable value types by appending a question mark to a type name.

```
Int? Unitsinstock = 5;
Datetime? Startdate;
Startdate= datetime.Now;
Startdate= null;
Int availableunits;

If(unitsinstock== null)
{
Availableunits = 0;
}
Else
{
Availableunits = (int)unitsinstock;
}
// OR just
Int availableunits = unitsinstock ?? 0;
```

We will now discuss debugging in Visual Studio.By printing output frequently we can debug. We can use Console.Write to print. Breakpoints allow stopping the program during execution. Press F5 and execution will stop at breakpoint. You can hover over variabeles for their current values. Locals, watch, call stack, immediate window give you different information about the program. Your changes will have effect on the current execution. Conditional breakpoints have a hit count and when hit condition.

We now discuss XML. XML or extensible Markup Language is widely used for exchanging data. Its readable for both humans and machines. Its a stricter version of HTML. Its made of tags, attributes, and values.

```
<users>
<user name="John Doe" age="42" />
<user name="Jane Doe" age="39" />
</users >
```

And here is a larger XML file.

```
<gesmes:Envelope xmlns:gesmes="http://www.gesmes.org/xml/2002-08-01"
xmlns="http://www.ecb.int/vocabulary/2002-08-01/eurofxref">
 <gesmes:subject>Reference rates</gesmes:subject>
 <gesmes:Sender>
  <gesmes:name>European Central Bank</gesmes:name>
 </gesmes:Sender>
 <Cube>
  <Cube time="2012-12-18">
   <Cube currency="USD" rate="1.3178"/>
   <Cube currency="JPY" rate="110.53"/>
   <Cube currency="BGN" rate="1.9558"/>
   <Cube currency="CZK" rate="25.200"/>
   <Cube currency="DKK" rate="7.4603"/>
   <Cube currency="GBP" rate="0.81280"/>
   <Cube currency="HUF" rate="288.40"/>
   <Cube currency="LTL" rate="3.4528"/>
   <Cube currency="LVL" rate="0.6961"/>
```


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```
   <Cube currency="PLN" rate="4.0928"/>
   <Cube currency="RON" rate="4.4700"/>
   <Cube currency="SEK" rate="8.7378"/>
   <Cube currency="CHF" rate="1.2080"/>
   <Cube currency="NOK" rate="7.3850"/>
   <Cube currency="HRK" rate="7.5380"/>
   <Cube currency="RUB" rate="40.6850"/>
   <Cube currency="TRY" rate="2.3476"/>
   <Cube currency="AUD" rate="1.2512"/>
   <Cube currency="BRL" rate="2.7595"/>
   <Cube currency="CAD" rate="1.2972"/>
   <Cube currency="CNY" rate="8.2079"/>
   <Cube currency="HKD" rate="10.2131"/>
   <Cube currency="IDR" rate="12707.71"/>
   <Cube currency="ILS" rate="4.9615"/>
   <Cube currency="INR" rate="72.2880"/>
   <Cube currency="KRW" rate="1413.30"/>
   <Cube currency="MXN" rate="16.7795"/>
   <Cube currency="MYR" rate="4.0236"/>
   <Cube currency="NZD" rate="1.5660"/>
   <Cube currency="PHP" rate="54.068"/>
   <Cube currency="SGD" rate="1.6053"/>
   <Cube currency="THB" rate="40.285"/>
   <Cube currency="ZAR" rate="11.2733"/>
  </Cube>
 </Cube>
</gesmes:Envelope>
```

We will now learn to read XML with the xmlreader class. There are two methods to read XML document. Using xmldocument and xmlreader. Xmldocuments reads entire document in memory, Let's you go forward, backward, even apply xpath searches on it. Xmlreader is fast, uses less memory and provides one element at a time.

```
Using System;
Using System.Text;
Using System.Xml;

Namespace parsingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmlreader xmlreader =
xmlreader.Create("http://www.ecb.int/stats/eurofxref/eurofxref-daily.xml");
            While (xmlreader.Read())
            {
                If ((xmlreader.nodetype == xmlnodetype.Element) && (xmlreader.Name ==
"Cube"))
                {
                    If (xmlreader.hasattributes)
                        Console.writeline(xmlreader.getattribute("currency") + ": " +
xmlreader.getattribute("rate"));
                }
            }
            Console.readkey();
        }
```


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```
    }
}
```

Using xmldocument is easier. No order is required. Documentelement is the root element and childnodes is the set of children of any node.

```
Namespace  parsingxml
{
Class Program
{
Static void Main(string[] args)
{
Xmldocument  xmldoc = new xmldocument();
Xmldoc.Load("http://www.ecb.int/stats/eurofxref/eurofxref-daily.xml");
Foreach (xmlnode xmlnode in xmldoc.documentelement.childnodes[2].childnodes[0].Child
Console.writeline(xmlnode.Attributes["currency"].Value + ": "+ xmlnode.Attribut
Console.readkey();
}
}
}
```

An xmlnode is derived from xmlelement and contains Name, innertext, innerxml, outerxml, and Attributes. Xpath is a cross-platform Xml Query language. We will look at basic examples only. We will see xmldocument methods that take xpath queries and return xmlnode(s) in particular selectsinglenode and selectnodes.

We will use the RSS feed from CNN, located at http://rss.cnn.com/rss/edition world.rss. It has a hierarchy of rss containing channel containing item.

```
Using System;
Using System.Text;
Using System.Xml;

Namespace parsingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmldocument xmldoc = new xmldocument();
            Xmldoc.Load("http://rss.cnn.com/rss/edition_world.rss");
            Xmlnode titlenode = xmldoc.selectsinglenode("//rss/channel/title");
            If (titlenode != null)
                Console.writeline(titlenode.innertext);
            Console.readkey();
        }
    }
}
```

Another example with selectnodes instead of selectsinglenode.

```
Using System;
Using System.Text;
Using System.Xml;

Namespace parsingxml
{
    Class Program
    {
```


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```
        Static void Main(string[] args)
        {
            Xmldocument xmldoc = new xmldocument();
            Xmldoc.Load("http://rss.cnn.com/rss/edition_world.rss");
            Xmlnodelist itemnodes = xmldoc.selectnodes("//rss/channel/item");
            Foreach (xmlnode itemnode in itemnodes)
            {
                Xmlnode titlenode = itemnode.selectsinglenode("title");
                Xmlnode datenode = itemnode.selectsinglenode("pubdate");
                If ((titlenode != null) && (datenode != null))
                    Console.writeline(datenode.innertext + ": " + titlenode.innertext);
            }
            Console.readkey();
        }
    }

}
```

If we only need title:

```
Using System;
Using System.Text;
Using System.Xml; 4
Namespace  parsingxml
{
Class Program
{
Static void Main(string[] args)
{
Xmldocument  xmldoc = new xmldocument();
Xmldoc.Load("http://rss.cnn.com/rss/edition_world.rss");
Xmlnodelist  itemnodes= xmldoc.selectnodes("//rss/channel/item");
Foreach (xmlnode  itemnode in itemnodes)
{
Xmlnode  titlenode= itemnode.selectsinglenode("title");
Xmlnode  datenode = itemnode.selectsinglenode("pubdate");
If((titlenode != null) && (datenode != null))
Console.writeline(datenode.innertext + ": "+ titlenode.innertext);
}
Console.readkey();
}
}
}
```

Let's write some XML using xmlwriter.

```
Using System;
Using System.Text;
Using System.Xml;

Namespace writingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmlwriter xmlwriter = xmlwriter.Create("test.xml");

            Xmlwriter.writestartdocument();
```


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```
            Xmlwriter.writestartelement("users");

            Xmlwriter.writestartelement("user");
            Xmlwriter.writeattributestring("age", "42");
            Xmlwriter.writestring("John Doe");
            Xmlwriter.writeendelement();

            Xmlwriter.writestartelement("user");
            Xmlwriter.writeattributestring("age", "39");
            Xmlwriter.writestring("Jane Doe");

            Xmlwriter.writeenddocument();
            Xmlwriter.Close();
        }
    }
}
```

The above code will create the following XML:

```
<users>
  <user age="42">John Doe</user>
  <user age="39">Jane Doe</user>
</users>
```

Let's now see writing with xmldocument (especially for updates).

```
Using System;
Using System.Text;
Using System.Xml;
Using System.Xml.Serialization;

Namespace writingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmldocument xmldoc = new xmldocument();
            Xmlnode rootnode = xmldoc.createelement("users");
            Xmldoc.appendchild(rootnode);

            Xmlnode usernode = xmldoc.createelement("user");
            Xmlattribute attribute = xmldoc.createattribute("age");
            Attribute.Value = "42";
            Usernode.Attributes.Append(attribute);
            Usernode.innertext = "John Doe";
            Rootnode.appendchild(usernode);

            Usernode = xmldoc.createelement("user");
            Attribute = xmldoc.createattribute("age");
            Attribute.Value = "39";
            Usernode.Attributes.Append(attribute);
            Usernode.innertext = "Jane Doe";
            Rootnode.appendchild(usernode);

            Xmldoc.Save("test-doc.xml");
        }
    }
}
```


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Updating is hard with a combination of xmlreader and xmlwriter.

```
Using System;
Using System.Text;
Using System.Xml;
Using System.Xml.Serialization;

Namespace writingxml
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Xmldocument xmldoc = new xmldocument();
            Xmldoc.Load("test-doc.xml");
            Xmlnodelist usernodes = xmldoc.selectnodes("//users/user");
            Foreach (xmlnode usernode in usernodes)
            {
                Int age = int.Parse(usernode.Attributes["age"].Value);
                Usernode.Attributes["age"].Value = (age + 1).tostring();
            }
            Xmldoc.Save("test-doc.xml");
        }
    }
}
```


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Chapter 11

Lecture 11

Let's start discussing extension methods.

```
Public class myutils
{
Public static bool isnumeric(string s)
{
Float output;
Return float.tryparse(s, out output);
}
}
String test = "4";
If(myutils.isnumeric(test))
Console.writeline("Yes");
Else

Console.writeline("No");
```

We can simplify the above example with an extension method.

```
Public class myutils
{
Public static bool isnumeric(string s)
{
Float output;
Return float.tryparse(s, out output);
}
}

String test = "4";
If(myutils.isnumeric(test))
Console.writeline("Yes");
Else

Console.writeline("No");
```

Let's now see how we can handle files.

```
Using System;
Using System.IO;

Namespace filehandlingarticleapp
{
    Class Program
    {
        Static void Main(string[] args)
        {
            If(File.Exists("test.txt"))
            {
                String content = File.readalltext("test.txt");
                Console.writeline("Current content of file:");
                Console.writeline(content);
            }
            Console.writeline("Please enter new content for the file:");
```


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```
            String newcontent = Console.readline();
            File.writealltext("test.txt", newcontent);
        }
    }
}
```

And another example.

```
Using System;
Using System.IO;

Namespace filehandlingarticleapp
{
    Class Program
    {
        Static void Main(string[] args)
        {
            If(File.Exists("test.txt"))
            {
                String content = File.readalltext("test.txt");
                Console.writeline("Current content of file:");
                Console.writeline(content);
            }
            Console.writeline("Please enter new content for the file - type exit and
press enter to finish editing:");
            String newcontent = Console.readline();
            While (newcontent != "exit")
            {
                File.appendalltext("test.txt", newcontent + Environment.newline);
                Newcontent = Console.readline();
            }
        }
    }
}
```

If using keyword is not used, manually call the Close method of the stream.

```
Console.writeline("Please enter new content for the file - type exit and press enter to
finish”);
Using (streamwriter sw= new streamwriter("test.txt"))
{
String  newcontent = Console.readline();
While (newcontent != "exit")
{
Sw.Write(newcontent+ Environment.newline);
Newcontent = Console.readline();
}
}
```

We can delete a file like this.

```
Console.writeline("Please enter new content for the file  - type exit and press enter to
finish”);
Using (streamwriter sw= new streamwriter("test.txt"))
{
String  newcontent = Console.readline();
While (newcontent != "exit")
{
Sw.Write(newcontent+ Environment.newline);
Newcontent = Console.readline();
}
```


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```
}
```

We can delete a directory like this.

```
Console.writeline("Please enter new content for the file -type exit and press enter to
finish”);
Using (streamwriter sw= new streamwriter("test.txt"))
{
String  newcontent = Console.readline();
While (newcontent != "exit")
{
Sw.Write(newcontent+ Environment.newline);
Newcontent = Console.readline();
}
}
```

We can rename a file like this.

```
If (File.Exists("test.txt"))
{
Console.writeline("Please enter a new name for this file:");
String  newfilename= Console.readline();
If (newfilename != String.Empty)
{
File.Move("test.txt", newfilename);
If (File.Exists(newfilename))
{
Console.writeline("The file was renamed to " + newfilename);
Console.readkey();
}
}
}
```

We can rename a directory like this.

```
If (Directory.Exists("testdir"))
{
Console.writeline("Please enter a new name for this directory:");
String  newdirname = Console.readline();
If (newdirname != String.Empty)
{
Directory.Move("testdir", newdirname);
If (Directory.Exists(newdirname))
{
Console.writeline("The directory was renamed to "+ newdirname);
Console.readkey();
}
}
}
```

We can create a new directory like this.

```
Console.writeline("Please enter a name for the new directory:");
String  newdirname = Console.readline();
If (newdirname != String.Empty)
{
Directory.createdirectory(newdirname);
If (Directory.Exists(newdirname))
{
Console.writeline("The directory was created!");
```


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```
Console.readkey();
}
}
```

A fileinfo object gives you information about a file.

```
Static void Main(string[] args)
{
Fileinfo fi = new fileinfo(System.Reflection.Assembly.getexecutingassembly().Location);
If(fi!= null)
Console.writeline(String.Format("Information about file: {0}, {1} bytes, last modified o
Console.readkey();
}
```

Directoryinfo gives you information about a directory.

```
Directoryinfo di= new
directoryinfo(Path.getdirectoryname(System.Reflection.Assembly.getexecuti
If(di!= null)
{
Fileinfo[] subfiles= di.getfiles();
If(subfiles.Length> 0)
{
Console.writeline("Files:");
Foreach (fileinfo  subfile in subfiles)
{
Console.writeline("" + subfile.Name + " (" + subfile.Length + " bytes)");
}
}
Console.readkey();
}
```

Finding directories as well is as easy.

```
Directoryinfo[] subdirs = di.getdirectories();
If(subdirs.Length > 0)
{
Console.writeline("Directories:");
Foreach (directoryinfo  subdir in subdirs)
{
Console.writeline("" + subdir.Name);
}
}
```

We can use reflection to get information about types in our program.

```
Using System;
Using System.Collections.Generic;
Using System.Text;
Using System.Reflection;

Namespace  reflectiontest
{
Class Program
{
Static void Main(string[] args)
{
String test = "test";
Console.writeline(test.gettype().fullname);
Console.writeline(typeof(Int32).fullname);
Console.readkey();
}
}
```


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```
}
```

Here is another example.

```
Using System;
Using System.Collections.Generic;
Using System.Text;
Using System.Reflection;

Namespace reflectiontest
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Assembly assembly = Assembly.getexecutingassembly();
            Type[] assemblytypes = assembly.gettypes();
            Foreach(Type t in assemblytypes)
                Console.writeline(t.Name);
            Console.readkey();
        }
    }

    Class dummyclass
    {
        //Just here to make the output a tad less boring :)
    }
}
```

Let's see how methods are shown in reflection.

```
Using System;
Using System.Collections.Generic;
Using System.Text;
Using System.Reflection;

Namespace reflectiontest
{
    Class Program
    {
        Static void Main(string[] args)
        {
            Type testtype = typeof(testclass);
            Constructorinfo ctor = testtype.getconstructor(System.Type.emptytypes);
            If (ctor != null)
            {
                Object instance = ctor.Invoke(null);
                Methodinfo methodinfo = testtype.getmethod("testmethod");
                Console.writeline(methodinfo.Invoke(instance, new object[] { 10 }));
            }
            Console.readkey();
        }
    }

    Public class testclass
    {
        Private int testvalue = 42;

        Public int testmethod(int numbertoadd)
        {
```


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```
            Return this.testvalue + numbertoadd;
        }
    }
}
```

We now have a more detailed example. We will now build a complete profile save load system with reflection.

```
Using System;
Using System.IO;
Using System.Reflection;


Class Program
{
    Public class Person
    {
        Public void Load()
        {
            If (File.Exists("settings.dat"))
            {
                Type type = this.gettype();

                String propertyname, value;
                String[] temp;
                Char[] splitchars = new char[] { '|' };
                Propertyinfo propertyinfo;

                String[] settings = File.readalllines("settings.dat");
                Foreach (string s in settings)
                {
                    Temp = s.Split(splitchars);
                    If (temp.Length == 2)
                    {
                        Propertyname = temp[0];
                        Value = temp[1];
                        Propertyinfo = type.getproperty(propertyname);
                        If (propertyinfo != null)
                            This.setproperty(propertyinfo, value);
                    }
                }
            }
        }

        Public void Save()
        {
            Type type = this.gettype();
            Propertyinfo[] properties = type.getproperties();
            Textwriter tw = new streamwriter("settings.dat");
            Foreach (propertyinfo propertyinfo in properties)
                Tw.writeline(propertyinfo.Name + "|" + propertyinfo.getvalue(this,
null));
            Tw.Close();
        }

        Public void setproperty(propertyinfo propertyinfo, object value)
        {
```


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```
            Switch (propertyinfo.propertytype.Name)
            {
                Case "Int32":
                    Propertyinfo.setvalue(this, Convert.toint32(value), null);
                    Break;
                Case "String":
                    Propertyinfo.setvalue(this, value.tostring(), null);
                    Break;
            }
        }

        Public int Age { get; set; }
        Public string Name { get; set; }
    }

    Static void Main(string[] args)
    {
        Person person = new Person();
        Person.Load();
        If ((person.Age > 0) && (person.Name != String.Empty))
        {
            Console.writeline("Hi " + person.Name + " - you are " + person.Age + " years
old!");
        }
        Else
        {
            Console.writeline("I don't seem to know much about you. Please enter the
following information:");
            Type type = typeof(Person);
            Propertyinfo[] properties = type.getproperties();
            Foreach (propertyinfo propertyinfo in properties)
            {
                Console.writeline(propertyinfo.Name + ":");
                Person.setproperty(propertyinfo, Console.readline());
            }
            Person.Save();
            Console.writeline("Thank you! I have saved your information for next time.");
        }
        Console.readkey();
    }
}
```


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Chapter 12

Lecture 12

We will start discussing WPF (Windows Presentation Foundations) now. It was publicly announced in 2003 (codenamed Avalon). WPF 4 was released in April 2010. It has a steep learning curve. Code has to be written in many places. There are multiple ways to do a particular task.

WPF enables polished user interfaces which are getting a lot of attention. It enables rapid iterations and major interface changes throughout the development process. It allows to keep user interface description and implementation separate. Developers can create an "ugly" application which designers can re-theme. Win32 style of programming makes such re-theming difficult. The code to re-paint the user interface is mixed up with program logic. GDI was an earlier user interface library introduced in windows 1.0 in 1985. Opengl was a leap ahead introduced in the 90's with directx coming in 95 and directx 2 in 96. GDI+ is a newer user interface library based on directx. It is also used behind Xbox graphics. Next was Windows Forms which is the primary way of user interface design in C#. XNA comes with managed library for directx and is great for game development (.net/com interoperability not required). A simple example is drawing bitmaps on buttons which can be efficiently done using GDI.

The highlights of WPF are 1) broad integration (2D, 3D, video, speech libraries etc.) 2) Resolution Independence with WPF giving emphasis on vector graphics 3) Hardware accelration as it is based on Direct3D but it can work using software pipeline also if Direct3D hardware is not available. 4) Declrative progrmaming using extensible Application Markup Language (XAML; pronounced "Zammel"). Custom attribute and configuration files were always there but XAML is very rich. 5) Rich composition and customization e.g. You can create a combobox filled with animated

Buttons or a Menu filled with live video clips! And it is quite easy to skin applications.

In short, WPF aims to combine the best attributes of systems such as directx (3D and hardware acceleration), Windows Forms (developer productivity), Adobe Flash (powerful animation support) and HTML (declarative markup). The first release in November 2006 was WPF 3.0 because it shipped as part of the .NET Framework 3.0. WPF 3.5 came an year later. Next version as part of .NET 3.5 SP1 came in August 2008. WPF Toolkit released in Aug 2008 was experimental. The toolkit has quick releases. Regarding tool support, WPF extensions for Visual Studio 2005 came a few months after the first WPF release and a public release of expression blend. Now, Visual Studio 2012 is a first class WPF devevelopment environment. Its mostly re-written using WPF and expression blend is 100% WPF and is great for designing and prototyping WPF apps.

New things that came in WPF 3.5/3.5SP1 include Interactive3D with 2d elements in 3d scenes, first class interoperability with directx, Better data binding using XLINQ and better validation and debugging which reduces code, Better special effects, High performance custom drawing, Text improvements, enhancements to Partial-trust apps, improved deployment, and improved performance.

Things that came with WPF 4.0 include multi-touch support — compatible with surface API v2, win7 support like jump lists, new common dialogs etc., new controls like datagrid, calendar etc, easing animation functions (bounce, elastic), enhanced styling with visual state manager, improved layout on pixel boundaries, non-blurry text but some limitations so must opt-in, deployment improvements, and performance improvements.

Silverlight in comparison is a light-weight version for web. It chose to follow WPF appraoch. First released in 2007 and in April 2010 4th version was released near WPF 4. There is often confusion when to use one or

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the other. Both can run in and outside the web. Silverlight is ostly a subset of WPF but there are some incompatibilities. Should I use full .net or partial .net and should I have the ability to run on other devices e.g. Macs. Ideally common codebase should work for both but today best case is using #ifdefs to handle incompatibilities.

XAML is primarily used to describe interfaces in WPF and Silverlight. It is also used to express activities and configurations in workflow foundation (WF) and windows communication foundation (WCF). Its a common language for programmers and other experts e.g. UI design experts. Field specific development tools can be made. Field experts are graphic designers. They can use a design tool such as expression blend. Other than co-ordinating with designers, XAML is good for a concise way to represent UI or hierarchies of objects, encourages separation of front-end and back-end, tool support with copy and paste, used by all WPF tools.

XAML is xml with a set of rules. Its a declrative programming language for creating and initializing objects. Itsa way to use .net apis.

Comparisons with SVG, HTML etc are misguided. It has few keywords, not much elemetns. It doesn't make sense without .net, like C# doesnt make sense without .net. Microsoft formalized XAML vocabluries, e.g. WPF XAML vocabluary.

Online specifications for XAML object specification language, and WPF and Silverlight vocabluaries are available. XAML is used by other technologies as well although originally it was designed for WPF. Also using XAML in WPF projects is optional. Everything can be done in procedural code as well (althought its rare to find it done that way).

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Chapter 13

Lecture 13

XAML specification defines rules that map .NET namespaces, types, properties, and events into XML namespaces, elements, and attributes. Let's see XAML and equivalent C#.

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK"/>
```

And the corresponding C# code is

```
System.Windows.Controls.Button b= new System.Windows.Controls.Button();

B.Content= "OK";
```

Declaring an XML element in XAML (known as an object element) is equivalent to instantiating the corresponding .NET object via a default constructor. Setting an attribute on the object element is equivalent to setting a property of the same name (called a property attribute) or hooking up an event handler of the same name (called an event attribute).

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK"
Click="button_Click"/ and the corresponding C# code is
System.Windows.Controls.Button b=new System.Windows.Controls.Button();
B.Click += new System.Windows.routedeventhandler(button_Click);

   B.Content = "OK";
```

XAML can no longer run standalone in the browser becuase of the butto click method — event handlers are attached before properties are set.

Let's discuss namespaces. Mapping to the namespace used above and other WPF namespaces is hard-coded in-side the WPF assemblies. Its just an arbitrary string like any namespace. The root object element in XAML must specify at least one XML namespace that is used to qualify itself and any child lements. Additional XML namespaces (on the root or on children) must be given a distinct prefix to be used on any identifiers from that namespace. E.g, typically a second namespace with the prefix x (denoted by using xmlns:x instead of just xmlns): xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"

This is the XAML language namespace, which maps to types in the System.Windows.Markup namespace but also defines some special directives for the XAML compiler or parser.

All of the following are mapped with http://schemas.microsoft.com/winfx/2006/xaml/presentation.

```
System.Windows
System.Windows.Automation
System.Windows.Controls
System.Windows.Controls.Primitives.System.Windows.Data
System.Windows.Documents
System.Windows.Forms.Integration
System.Windows.Ink
System.Windows.Input
System.Windows.Media
System.Windows.Media.Animation
System.Windows.Media.Effects
System.Windows.Media.Imaging
```


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```
System.Windows.Media.Media3D
System.Windows.Media.textformatting.System.Windows.Navigation
System.Windows.Shapes
System.Windows.Shell
```

Property elements in XAML enable rich composition.

```
System.Windows.Controls.Button b= new System.Windows.Controls.Button();
System.Windows.Shapes.Rectangle r = new System.Windows.Shapes.Rectangle();
R.Width = 40;
R.Height = 40;
R.Fill= System.Windows.Media.Brushes.Black;
B.Content= r; // Make the square the content of the Button
```

And in XAML

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
<Button.Content>
<Rectangle  Height="40" Width="40" Fill="Black"/>
</Button.Content ></Button >
```

The period distinguishes property elements from object elements. It doesn't have attributes except x:uid for localization. The syntax can be used for simple properties as well. The following two are equivalent.

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK"
Backgroun
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
<Button.Content>OK</Button.Content >
<Button.Background>White</Button.Background >
</Button >
```

Let's discuss type converters.

```
System.Windows.Controls.Button b= new System.Windows.Controls.Button();
B.Content= "OK";
B.Background =System.Windows.Media.Brushes.White;
```

WPF provides type converters for many common data types: Brush, Color, fontweight, Point, and so on. Classes deriving from typeconverter (brushconverter, colorconverter, and so on) convert from string to the corresponding types. You can also write your own type converters for custom data types.

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK">
<Button.Background>
<solidcolorbrush  Color="White"/>
</Button.Background >
</Button>
```

Without the type converter for color

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Content="OK">
<Button.Background>
<solidcolorbrush>
<solidcolorbrush.Color>
<Color A="255" R="255" G="255" B="255"/>
</solidcolorbrush.Color>
</solidcolorbrush>
</Button.Background>
</Button>
```

Even that requires a type converter for Byte.

```
System.Windows.Controls. Button b = new System.Windows.Controls. Button ();
B.Content = "OK";
```


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```
B.Background = (Brush) System.componentmodel. Typedescriptor. Getconverter (
Typeof (Brush)). Converterfrominvariantstring ("White");
```

Constants in strings cause problems with run time exception not caught at compile time. Althought for xaml Visual Studio will check it at compile time. Let's see how to find the type converter for a type.

Let's discuss markup extensions now. Markup extensions, like type converters, enable you to extend the expres-siveness of XAML. Both can evaluate a string attribute value at runtime (except for a few built-in markup extensions for performance reasons) and produce an appropriate object based on the string. As with type converters, WPF ships with several markup extensions built in.

Unlike type converters, however, markup extensions are invoked from XAML with explicit and consistent syntax. For this reason, using markup extensions is a preferred approach for extending XAML.

For example, if you want to set a control's background to a fancy gradient brush with a simple string value, you can write a custom markup extension that supports it even though the built-in brushconverter does not. Whenever an attribute value is enclosed in curly braces ({}), the XAML compiler/parser treats it as a markup extension value rather than a literal string (or something that needs to be type-converted). Here are few examples:

```
<Button xmlns=''http://schemas.microsoft.com/winfx/2006/xaml/presentation''
Content=''{}{This is not a markup extension!}''/>
<Button xmlns=''http://schemas.microsoft.com/winfx/2006/xaml/presentation''>
<Button.Content>{This is not a markup extension!}</Button.Content> </Button>
```

Because markup extensions are just classes with default constructors, they can be used with property element syntax.

```
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
            Xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
            Background = "{x:Null}"
            Height = "{x:Static systemparameters.iconheight}"
            Content = "{Binding Path=Height, relativesource={relativesource
Mode=Self}}"/>
```

It works because positional parameters have corresponding property value. Markup extension has the real code to be executed.

An object element can have three types of children: a value for a content property, collection items, or a value that can be type-converted to the object element. Designated property is the content property.

```
<Button xmlns=''http://schemas.microsoft.com/winfx/2006/xaml/presentation''
Content=''OK''/>
```

Could be rewritten as follows:

```
<Button xmlns=''http://schemas.microsoft.com/winfx/2006/xaml/presentation''> OK
</Button>
```

Also

```
<Button xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”>
<Button.Content>
<Rectangle Height=”40” Width=”40” Fill=”Black”/> </Button.Content>
</Button>
```

Could be rewritten as follows:

```
<Button xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”> <Rectangle
Height=”40” Width=”40” Fill=”Black”/>
```


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```
</Button>
```

There is no requirement that the content property must actually be called Content; classes such as combobox, listbox, and tabcontrol (also in the System.Windows.Controls namespace) use their Items property as the content property. Its designated with a custom attribute.

XAML enables you to add items to the two main types of collections that support indexing: lists and dictionaries.

```
<listbox xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”>
<listbox.Items>
<listboxitem Content=”Item 1”/>
<listboxitem Content=”Item 2”/> </listbox.Items>
</listbox>
```

Its equivalent to:

```
System.Windows.Controls.listbox listbox= new System.Windows.Controls.listbox();
System.Windows.new System.Windows.Controls.listboxitem();
System.Windows.Controls.listboxitem item2= new System.Windows.Controls.listboxitem();
Item1.Content=Item 1 ;
Item2.Content= Item 2;
```

OR

```
<listbox xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”> <listboxitem
Content=”Item 1”/>
<listboxitem Content=”Item 2”/>
</listbox>
```

Let’s see how to use a dictionary.

```
<resourcedictionary xmlns=”http://schemas.microsoft.com/winfx/2006/xaml/presentation”
xmlns:x=”http://schemas.microsoft.com/winfx/2006/xaml”>
<Color x:Key=”1” A=”255” R=”255” G=”255” B=”255”/>
<Color x:Key=”2” A=”0” R=”0” G=”0” B=”0”/>
</resourcedictionary>
```

We used the XAML Key keyword. The equivalent code is:

```
System.Windows. Resourcedictionary d = new System.Windows. Resourcedictionary ();
System.Windows.Media. Color color1 = new System.Windows.Media. Color ();
System.Windows.Media. Color color2 = new System.Windows.Media. Color ();
Color1.A = 255; color1.R = 255; color1.G = 255; color1.B = 255;
Color2.A = 0; color1.R = 0; color1.G = 0; color1.B = 0;
D.Add ("1", color1);
D.Add ("2", color2);
```


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Chapter 14

Lecture 14

<solidcolorbrush>White</solidcolorbrush>

As discussed in the last lecture, this is equivalent to the following:

<solidcolorbrush Color="White"/>

It works because a typeconverter exists that converts string to solidcolorbrush, although no designated content property exists.
<Brush>White</Brush>

In fact, the above also works even although Brush is abstract. It works because of a type converter can convert a string to a solidcolorbrush.

XAML works with classes designated with marked attributes. Usually these classes have default ctors and useful properties. But what about other classes not designed for XAML. E.g. Hashtable.

```
System.Collections.Hashtable h
H.Add("key1",  7);
H.Add("key2",  23);
```

Heres how it can be represented in XAML:

```
<Collections: Hashtable
       Xmlns: collections = "clr-namespace: System.Collections; assembly = mscorlib"
       Xmlns: sys = "clr-namespace: System; assembly = mscorlib"
       Xmlns: x = "
<Sys: Int32 x: Key = "key1"> 7 </ sys: Int32>
<Sys: Int32 x: Key = "key2"> 23 </ sys: Int32>
</ Collections: Hashtable>
```

Let's study XAML child element rules. If the type implements ilist, call ilist.Add for each child. Otherwise, if the type implements idictionary, call idictionary.Add for each child, using the x:Key attribute value for the key and the element for the value. (Although XAML2009 checks idictionary before ilist and supports other collection interfaces, as described earlier.)

Otherwise, if the parent supports a content property (indicated by System.Windows.Markup.content property attribute) and the type of the child is compatible with that property, treat the child as its value. Otherwise, if the child is plain text and a type converter exists to transform the child into the parent type (and no properties are set on the parent element), treat the child as the input to the type converter and use the output as the parent object instance. Otherwise, treat it as unknown content and potentially raise an error

We can mix XAML and procedural code. Let's try this with xamlreader and xamlwriter.

```
Window window= null;
Using (filestream fs = new filestream("mywindow.xaml", filemode.Open, fileaccess.Read))
{
//Get the root element, which we know is a Window
Window= (Window)xamlreader.Load(fs);
}
```

Let's see how to grab the OK button by walking the children (with hard-coded knowledge!)

```
Stackpanel panel= (stackpanel)window.Content;
```


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```
Button okbutton = (Button)panel.Children[4];
```

For easier fetching, we can name XAML elements.

```
 <Button x:Name="okbutton">OK</Button>
```

Let's grab the OK button, knowing only its name

```
 Button okbutton=(Button)window.findname("okbutton");
```

Classes that already have a name property designate it as the special name using a custom attribute.

Let's see how to use named properties. Let's first use a Binding markup extension.

```
<stackpanel
  Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
  <Label Target="{Binding elementname=box}" Content="Enter _text:"/>
  <textbox Name="box"/>
</stackpanel>
```

It gives it focus when the Label's access key is pressed.) WPF 4 includes a new, simpler markup extension which works at parse time.

```
<stackpanel
  Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
  Xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"> <Label
  Target="{x:Reference box}" Content="Enter _text:"/> <textbox
  Name="box"/>
  </stackpanel>
```

If a relevant property is marked with the System.Windows.Markup.namereferenceconverter type converter:

```
  <stackpanel
  Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
  <Label Target="box" Content="Enter _text:"/> <textbox Name="box"/>
  </stackpanel>
```

Compiling xaml involves 3 steps — converting a XAML file into a special binaryformat, embedding the converted content asa binary resource in the assembly being built,and performing the plumbing that connects XAML with procedural code automatically. C# and VB have best support in typical case, the first step is specifying a subclass for the root element in a XAML file. Use the Class keyword for that.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
Xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
X:Class="mynamespace.mywindow"></Window >
Namespace  mynamespace
{
Partial class mywindow: Window
{
Public mywindow()
{
// Necessary to call in order to load XAML-defined content!
Initializecomponent();

}
Any other members can go here...
}
}
```

This is often refered as the code-behind file. We define event handlers here. The partial keyword is important. When we add new items to certain WPF projects, Visual Studio does all this automatically for us.

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Chapter 15

Lecture 15

BAML is binary application markup language. It just a compressed representation of XAML. There is even a BAML reader available. Earlier there was CAML which stands for compiled application markup language but its not used now.

Some glue code is generated when we use x:Class. Its kind of same as loading and parsing the XAML file. We must call initializecomponent and we can refer named elements like class members.

Procedural code can even be written inside the XAML file.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml" x:Class="mynamespace.mywindow">
    <Button Click="button_Click">OK</Button>
    <x:Code>
        <![CDATA[
            Void button_Click(object sender, routedeventargs e)
            {
                This.Close();
            }
        ]]>
    </x:Code>
</Window>
```

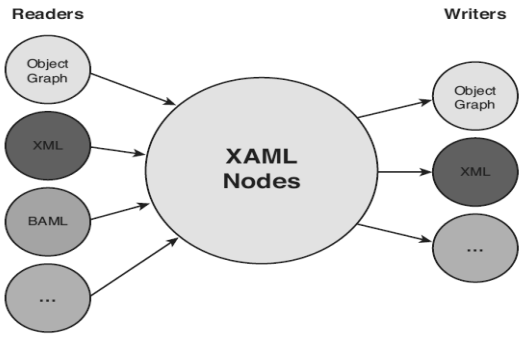
We must avoid]]. If we have to use them, we need &lt &amp etc. There is no good reason to embed code. Internally its build into .cs file by build system. BAML can be converted back to XAML.

```
System.Uri uri = new System.Uri("/wpfapplication1;component/mywindow.xaml",
System.urikind.Relative);
Window window= (Window)Application.loadcomponent(uri);
String xaml = xamlwriter.Save(window);
```

There are different loading mechanisms. We can load from resource, identified by original xaml file, or actually integrated baml loaded.

Key features of XAML 2009 include full generics support using typearguments, dictionary keys of any type, builtin system data types, creating objects with non-default ctors, getting instances via factory methods, event handlers can be markup extensions returning delegates, define additional members and properties.

System.Xaml.xamlxmlreader etc. Can be extended with readers and writers for a lot of formats. It abstracts differences in Xaml formats like accessing a content property, property element, or a property attribute.

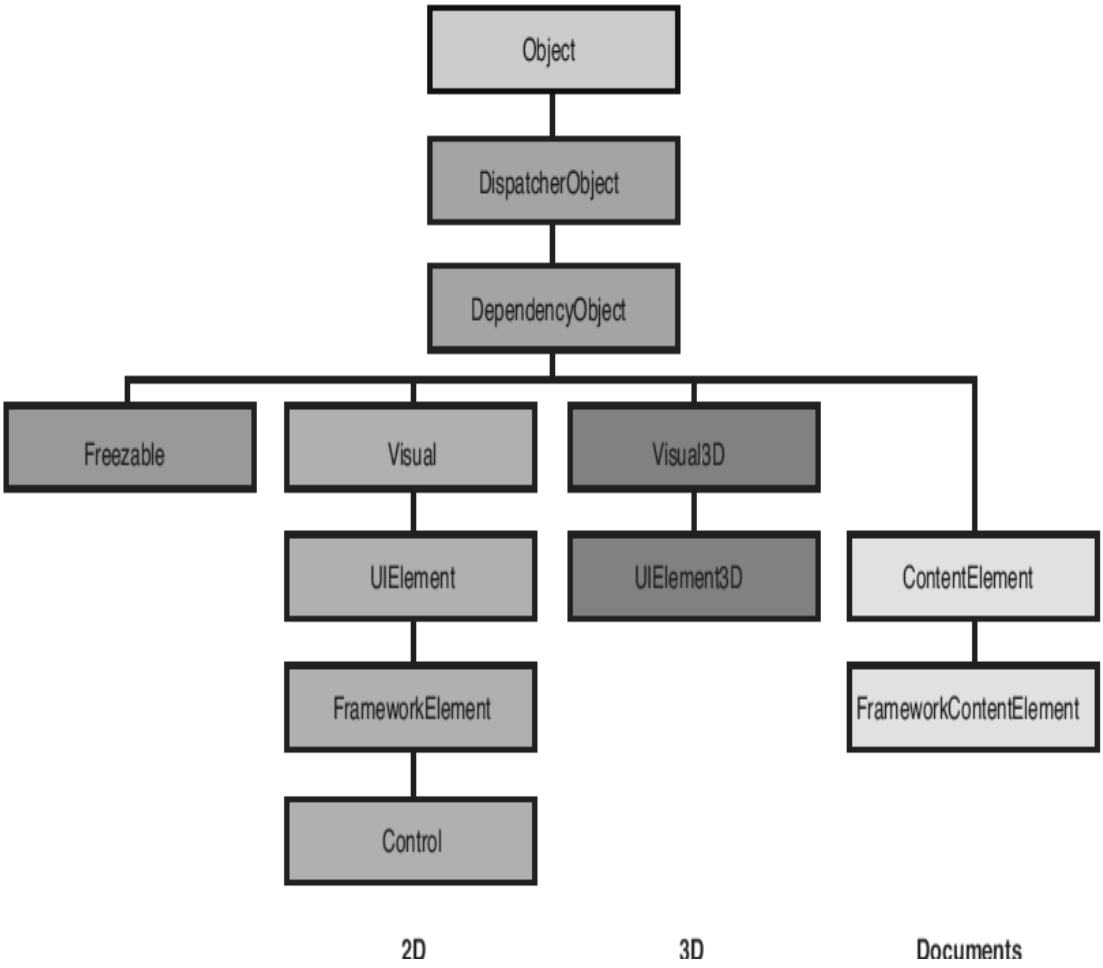


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XAML 2006 keywords include x:asyncrecords - Controls the size of asynchronous XAML-loading chunks, x:Class, x:classmodifier - visibility, public by default, x:Code, x:connectionid - not for public use, x:fieldmodifier - field visibility (internal def), x:Key, x:Name, x:Shared - =false means same resource instance not shared, x:subclass - only needed when partial not supported, x:Synchronous mode - xaml loaded in async mode, x:typearguments - used only with root with x:Class in xaml2006, x:Uid - represents system.uri, and x:xdata - data opaque for xaml parser.

Markup extensions often confused as keywords are x:Array - use with x:Type to define type of array, x:Null, x:Reference - to named element, x:Static - static property field, and x:Type - just like typeof operator. In summary, we can convert xaml into C#, often apis are optimized for xaml and can look clunky in C#, wpf applications have deep hierarchy, small building blocks. Two complaints are that xml too verbose to type and slow to parse (tools and baml fix it partially).

Let's discuss wpf fundamentals now. WPF concepts are above and beyond .net concepts. Its the cause of the steep learning curve of wpf. It has a deep inheritance hierarchy. There are a handful of fundamental classes.



Object is the base class for all .NET classes and the only class in the figure that isn't WPF specific.

Dispatcherobject is the base class meant for any object that wishes to be accessed only on the thread that created it. Most WPF classes derive from dispatcherobject and are therefore inherently thread-unsafe. The Dispatcher part of the name refers to wpfs version of a Win32-like message loop, discussed further in

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Chapter 7, “Structuring and Deploying an Application” of the book.

Dependencyobject is the base class for any object that can support dependency properties, one of the main topics in this chapter.

Freezable is the base class for objects that can be “frozen” into a read-only state for performance reasons. Freezables, once frozen, can be safely shared among multiple threads, unlike all other dispatcherobjects. Frozen objects can never be unfrozen, but you can clone them to create unfrozen copies. Most Freezables are graphics primitives such as brushes, pens, and geometries or animation classes.

Visual is the base class for all objects that have their own 2D visual representation. Visuals are discussed in depth in Chapter 15, “2D Graphics” of the book.

Uielement is the base class for all 2D visual objects with support for routed events, command binding, layout, and focus. These features are discussed in Chapter 5, “Layout with Panels,” and Chapter 6, “Input Events: Keyboard, Mouse, Stylus, and Multi-Touc.” Of the book.

Visual3D is the base class for all objects that have their own 3D visual representation. Visual3Ds are discussed in depth in Chapter 16, “3D Graphics” of the book.

Uielement3d is the base class for all 3D visual objects with support for routed events, command binding, and focus, also discussed in Chapter 16.

Contentelement is a base class similar to uielement but for document-related pieces of content that don’t have rendering behavior on their own. Instead, contentelements are hosted in a Visual-derived class to be rendered onthe screen. Each contentelement often requires multiple Visuals to render correctly (spanning lines, columns, and pages).

Frameworkelement is the base class that adds support for styles, data binding, resources, and a few common mechanisms for Windows-based controls, such as tooltips and context menus.

Frameworkcontentelement is the analog to frameworkelement for content. Chapter 11, “Images, Text, and Other Controls,” of the book examines the frameworkcontentelements in WPF.

Control is the base class for familiar controls such as Button, listbox, and statusbar. Control adds many properties to its frameworkelement base class, such as Foreground, Background, and fontsize, as well as the ability to be completely restyled. Part III, “Controls,” examines wpfs controls in depth.

Let’s discuss logical and visual trees.

Xaml good for UI because of hierarchical nature. Logical tree exists even if there is no xaml. Properties, events, resources are tied to logical trees. Properties propagated down and events can be routed up or down the tree. Its a simplification of whats actually going on when rendered. Visual tree can be thought of as an extension of the logical tree though some things can be dropped as well. Visual tree exposes visual implementation details e.g. A listbox is a border, two scrollbars and more. Only things from visual or visual3d appear in a visual tree. Avoid depending on visual tree in your code.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Title="About
WPF 4 Unleashed" sizetocontent="widthandheight" Background="orangered">

    <stackpanel>
        <Label fontweight="Bold" fontsize="20" Foreground="White">WPF 4 Unleashed
        </Label>
        <Label>© 2010 SAMS Publishing
        </Label>
```


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```
        <Label>Installed Chapters:
        </Label>
        <listbox>
            <listboxitem>Chapter 1</listboxitem>
            <listboxitem>Chapter 2</listboxitem>
        </listbox>
        <stackpanel Orientation="Horizontal" horizontalalignment="Center">
            <Button minwidth="75" Margin="10">Help</Button>
            <Button minwidth="75" Margin="10">OK</Button>
        </stackpanel>
        <statusbar>You have successfully registered this product.</statusbar>
    </stackpanel>
</Window>
```

We can traverse using System.Windows.logicaltreehelper and System.Windows.Media.visualtreehelper. Here is a code-behind file that will print these trees, assuming the gluing code is there i.e. X:Class and xmlns:x.

```
Using System;
Using System.Diagnostics;
Using System.Windows;
Using System.Windows.Media; 5

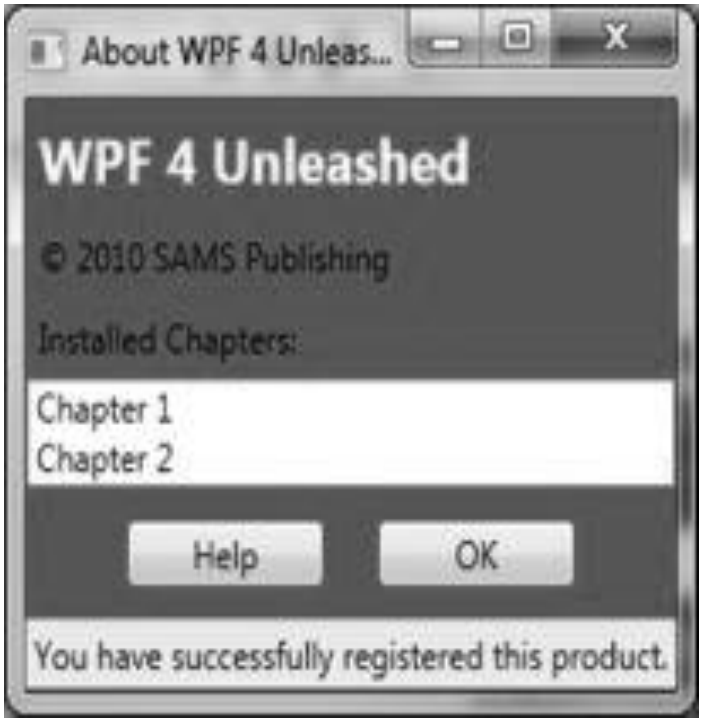
Public partial class aboutdialog  : Window
{
Public  aboutdialog()
{
Initializecomponent();
Printlogicaltree(0, this);
}
Protected  override void oncontentrendered(eventargs e)
{
Base.oncontentrendered(e);
Printvisualtree(0, this);
}
Void printlogicaltree(int depth, object obj) 19 {
// Print the object with preceding  spaces that represent its depth
Debug.writeline(new string(’ ’, depth)+ obj);
// Sometimes leaf nodes  aren  t  dependencyobjects (e.g. Strings)
If (!(obj is dependencyobject))
Return;

// Recursive call for each logical child
Foreach (object child in logicaltreehelper.getchildren( obj as dependencyobject))
Printlogicaltree(depth+ 1, child);
}
Void printvisualtree(int depth, dependencyobject obj)
{
// Print the object with preceding  spaces that represent its depth
Debug.writeline(new string(’ ’, depth)+ obj);
// Recursive call for each visual child
For(int i= 0; i < visualtreehelper.getchildrencount(obj); i++)
Printvisualtree(depth + 1, visualtreehelper.getchild(obj, i));

}

}
```

Visual tree is empty until the dialog box is rendered. Navigating either can be done in instance methods of the elements themselves e.g. Visual class has protected members visualparent, visualchildrencount, and getvisualchild. Frameworkelement and frameworkcontentelement have Parent property and logicalchildren property.



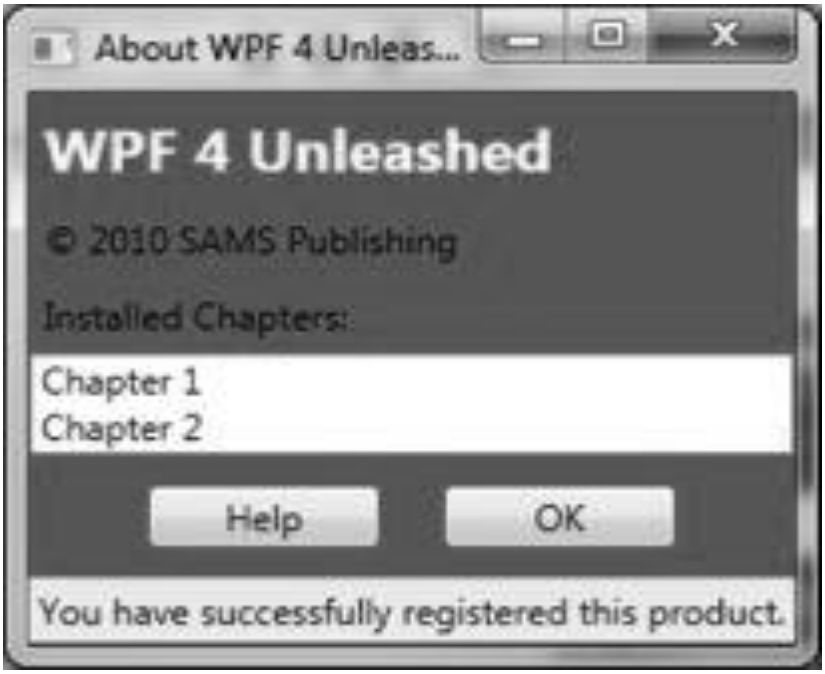
Chapter 16

Lecture 16

Let's discuss dependency properties. They complicates .net types but once you realize the problem it solves, you realize its importance. Dependency properties depends on multiple providers for determining its value at any point in time. These providers can be an animation continuously changing its values. A parent whose property propagates down. Arguably biggest feature is change notification. Motivation is to add rich functionality from declarative markup. Key to declarative-friendly design is heavy use of properties. Button e.g. Has 111 public properties (98 inherited). Properties can be set using xaml, directly or in a design tool, without procedural code. Without the extra work in dependency properties, it would be hard. We looked at implementation of a dependency property and then see its benefits on top of .net properties. Change notification, property value inheritance, and support for multiple providers are key features.

Understanding most nuances is important only for custom control authors. However what they are and how they work important for everyone. Can only style and animate dependency properties. After using wpf for a while, one wishes all properties were dependency properties. In practice, dependency properties are normal .net properties with extra wpf infrastructure. No .net language other than xaml has an intrinsic understanding of dependency properties.

Let's look at a standard dependency property implementation.




```
Public class Button: buttonbase
{
// The dependency  property
Public static readonly  dependencyproperty  isdefaultproperty;
Static Button()
{
```


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```
// Register the property
Button.isdefaultproperty = dependencyproperty.Register("isdefault", typeof(bool), typeof

}
// A .NET property  wrapper(optional)
Public bool isdefault
{
Get
{
Return(bool)getvalue(Button.isdefaultproperty);
}
Set
{
Setvalue(Button.isdefaultproperty, value);
}
}
// A property  changed  callback(optional)
Private static void onisdefaultchanged(_dependencyobject o,
dependencypropertychangedeventar
{
//...
}
//...
}
```

Dependency properties are represented by System.Windows.dependencyproperty. By convention, public static and Property suffix are used. Its required by several infrastructure pieces e.g. Localization tools, xaml loading. Optionally (via different overloads of Register), you can pass metadata that customizes how the property is treated by WPF, as well as callbacks for handling property value changes, coercing values, and validating values. .net prop wrapper optional. It helps setting from xaml and isnatural. Otherwise getvalue and setvalue are to be used. Getvalue and setvalue methods inherited from System.Windows.dependencyobject. Getvalue setvalue does not support generic. Dependency properties were introduced before generic support was added in C#.

Visual Studio has a snippet called propdp that automatically expands into a definition of a dependency property, which makes defining one much faster than doing all the typing yourself! .NET property wrappers are bypassed at runtime when setting dependency prop- erties in XAML! Although the XAML compiler depends on the property wrapper at compile time, WPF calls the underlying getvalue and setvalue methods directly at runtime! Therefore, to maintain parity between setting a property in XAML and procedural code, its crucial that property wrappers not contain any logic in addition to the getvalue/setvalue calls. If you want to add custom logic, thats what the registered callbacks are for. All of wpfs built-in property wrappers abide by this rule, so this warning is for anyone writing a custom class with its own dependency properties.

On the surface, it too much code but saves per-instance cost. Only static field and an efficient sparse storage system. If all were .net props with backing store, would consume lot more space e.g. 111 fields for button, 104 for label but 89 and 82 are dependency properties. Also code to check thread access, prompt containing element to be re-rendered.

Dependency properties support change notification. Its based on metadata at register time. Actions can be re-rendering the appropriate elements, updating the current layout, refreshing data bindings, and much property triggers - imagine you want color change on hovering.

```
<Button  mouseenter="Button_mouseenter" mouseleave="Button_mouseleave" minwidth="75"
Margin="10">
<Button  mouseenter="Button_mouseenter" mouseleave="Button_mouseleave" minwidth="75"
```


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```
Margin="10">

// Change the foreground to blue when the mouse enters the button
Void Button_mouseenter(object sender, mouseeventargs e)
{
Button b = sender as Button;
If(b != null)
B.Foreground = Brushes.Blue;
}
// Restore the foreground to black when the mouse exits the button
Void Button_mouseleave(object sender, mouseeventargs e)
{
Button b = sender as Button;
If(b != null)
B.Foreground = Brushes.Black;
}
```

And with Xaml

```
<Trigger  Property="ismouseover" Value="True">
<Setter Property="Foreground" Value="Blue"/>
</Trigger >
Eed to worry about reverting but cannot be applied directly to button elements.

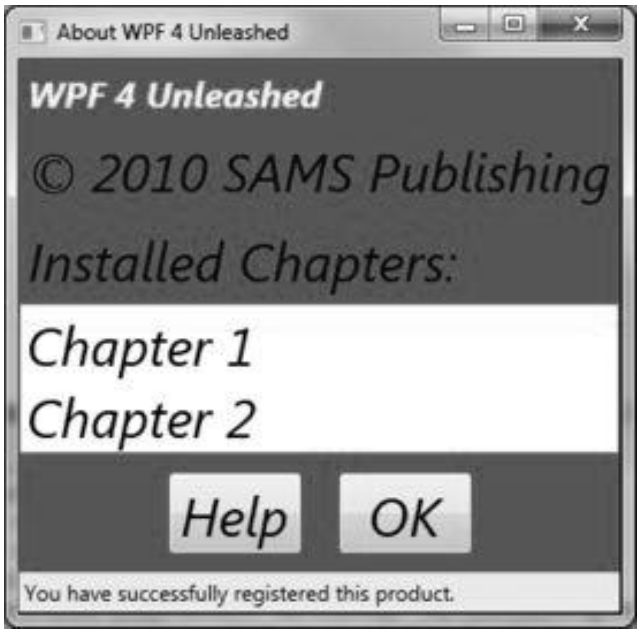
<Button minwidth="75" Margin="10">
<Button.Style>
<Style targettype="{x:Type Button}">
<Style.Triggers>
<Trigger  Property="ismouseover" Value="True">
<Setter Property="Foreground" Value="Blue"/>
</Trigger >
</Style.Triggers >
</Style >
</Button.Style >  OK
</Button >
```

Property value inheritance is flowing of prop values down an element tree.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Title="About WPF
 4 Unl
<stackpanel>
<Label fontweight="Bold" fontsize="20" Foreground="White">
WPF4 Unleashed
</Label >
<Label> 2010 SAMS Publishing</Label>
<Label>Installed  Chapters:</Label>
<listbox>
<listboxitem>Chapter 1</listboxitem>
<listboxitem>Chapter 2</listboxitem>
</listbox >
<stackpanel  Orientation="Horizontal" horizontalalignment="Center">
<Button minwidth="75" Margin="10">Help</Button>
<Button minwidth="75" Margin="10">OK</Button>
</stackpanel >
<statusbar>You have successfully  registered this product.</statusbar>
</stackpanel >
```


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```
</Window >
```



It does not effect status bar. Not every dependency property participates in inheritance (Actually they can opt-in). May be other higher priority sources setting property value. Some controls like status bar internally set their values to system defaults. Property value inheritance in other places like to triggers inside a definition.

WPF contains many powerful mechanisms that independently attempt to set the value of dependency properties. Dependency properties were designed to depend on these providers in a consistent and orderly manner. Its a 5 step process

Step 1: determine base value. Most providers factor into base value determination. Highest to lowest precedence. Ten providers that can set value for most dep. Props. 1. Local valuedependencyobject.setvalue, prop assignment in xaml or2. Parent template trigger3. Parent template4. Style triggers5. Template triggers6. Style setters7. Theme style triggers8. Theme style setters9. Property value inheritancealready seen10. Default valueinitial value registered with the property. Thats why status bar did not get font propagated. Use dependencypropertyhelper.getvaluesource to find which was the source used.

Step 2: evaluate. Expressions need evaluation. It is used in data binding.

Step 3: apply animation. Animations can alter value from step 2 or replace it.

Step 4: coerce. The almost final value passed to coercevaluecallback delegate if one is registered.

Step 5: validate. Passed to validatevaluecallback delegate. If one was registered. Returning false causes exception canceling the entire process. Wpf 4 adds new value in dependencyobject called setcurrentvalue. It updates current value without changing value source.

Let's discuss attached properties now. They are special dependency properties that can be attached to arbitrary objects. Sounds strange but there are many applications for it. There is new xaml syntax for attached props.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
```


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```
Title="About WPF4 Unleashed" sizetocontent="widthandheight"
Background="orangered">
<stackpanel>
<Label fontweight="Bold" fontsize="20" Foreground="White">
WPF4 Unleashed
</Label >
<Label> 2010 SAMS Publishing</Label>
<Label>Installed  Chapters:</Label>
<listbox>
<listboxitem>Chapter 1</listboxitem>
<listboxitem>Chapter 2</listboxitem>
</listbox >
<stackpanel  textelement.fontsize="30" textelement.fontstyle="Italic"
Orientation="Horizontal”>
<Button minwidth="75" Margin="10">Help</Button>
<Button minwidth="75" Margin="10">OK</Button>
</stackpanel >
<statusbar>You have successfully  registered this product.</statusbar>
</stackpanel >
</Window >
```


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Chapter 17

Lecture 17

We discussed this example in the last lecture.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
Title="About WPF4 Unleashed" sizetocontent="widthandheight"
Background="orangered">
<stackpanel>
<Label fontweight="Bold" fontsize="20" Foreground="White">
WPF4 Unleashed
</Label >
<Label> 2010 SAMS Publishing</Label>
<Label>Installed  Chapters:</Label>
<listbox>
<listboxitem>Chapter 1</listboxitem>
<listboxitem>Chapter 2</listboxitem>
</listbox >
<stackpanel  textelement.fontsize="30" textelement.fontstyle="Italic" Orientation="Horizo
<Button minwidth="75" Margin="10">Help</Button>
<Button minwidth="75" Margin="10">OK</Button>
</stackpanel >
<statusbar>You have successfully  registered this product.</statusbar>
</stackpanel >
</Window >
```

Xaml parser requires setfontsize, setfontstyle on textelement and therefore equivalent code in C# is.

```
Stackpanel panel= new stackpanel();
Textelement.setfontsize(panel, 30);
Textelement.setfontstyle(panel, fontstyles.Italic);
Panel.Orientation = Orientation.Horizontal;
Panel.horizontalalignment= horizontalalignment.Center;
Button  helpbutton= new Button();
Helpbutton.minwidth= 75;
Helpbutton.Margin = new Thickness(10);
Helpbutton.Content = "Help";
Button okbutton= new Button();
Okbutton.minwidth = 75;
Okbutton.Margin= new Thickness(10);
Okbutton.Content= "OK";
Panel.Children.Add(helpbutton);
Panel.Children.Add(okbutton);
```

Enumeration values such as fontstyles.Italic, Orientation.Horizontal, and horizontalalignment.Center were pre-viously specified in XAML simply as Italic, Horizontal, and Center, respectively. This is possible thanks to the enumconverter type converter C#. C# code shows no real magic, no .net property involved internally. There are just calls to dependencyobject.setvalue and getvalue like normal property wrappers, must not go anything else.

```
Public static void setfontsize(dependencyobject element, double value)
{
Element.setvalue(textelement.fontsizeproperty, value);
}

Public static double  getfontsize(dependencyobject  element)
{
```


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```
Return(double)element.getvalue(textelement.fontsizeproperty);
}
```

We are setting the fontsize property by unrelated class textelement. We could also have used textblock but textelement.fontsizeproperty is a separate dependencyproperty field from Control.fontsizeproperty.

```
Textelement.fontsizeproperty= dependencyproperty.registerattached(
 "fontsize", typeof(double), typeof(textelement), new frameworkpropertymetadata(
  Systemfonts.messagefontsize, frameworkpropertymetadataoptions.Inherits |
  Frameworkpropertymetadataoptions.affectsrender |
  Frameworkpropertymetadataoptions.affectsmeasure),
 New validatevaluecallback(textelement.isvalidfontsize));
```

Using registerattached us optimized for attached property metadata. Control on the other hand just calls addowner.

```
Control.fontsizeproperty = textelement.fontsizeproperty.addowner(
Typeof(Control), new frameworkpropertymetadata(systemfonts.messagefontsize,
Frameworkpropertymetadataoptions.Inherits));
```

These font related properties all controls inherit are from textelement in most cases, the class exposing the attached property is the same that defines the normal dependency propery. Many dependency properties have a Tag property for storing arbitrary custom data. A great mechanism for even extending "sealed" classes. In procedural code, you can actually use "any" property by calling setvalue.

```
// Attach an unrelated  property to a Button and set its value to true:
Okbutton.setvalue(itemscontrol.istextsearchenabledproperty, true);
```

We can use this property any way we want but its better way to use Tag.

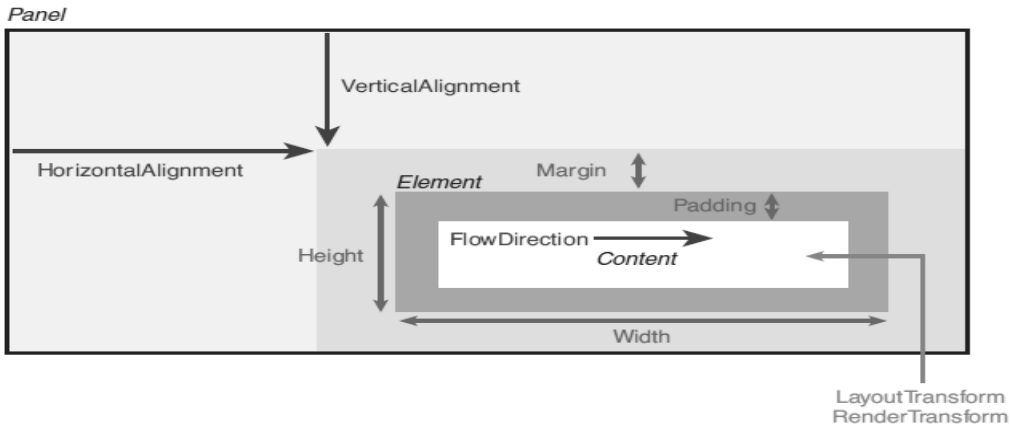
```
Geometrymodel3d  model= new geometrymodel3d();
Model.setvalue(frameworkelement.tagproperty, "my custom data");
```

Its most commonly used for layout of UI elements. Various panel derived classes define attached properties designed to be attached to their children. This way each panel can define custom behavior without burdening all possible children with their own set of properties. It allows extensibility.

In summary, wpf could have exposed its features by api's but instead they focused on adding features to the core. Multiple types of properties, multiple tree, multiple ways to achieve the same thing. Hopefully you can appreciate the value in coming lectures. These concepts will fade in the background as we accomplish different tasks in coming lectures.

Let's discuss sizing, positioning, and transforming elements. Sizing and positioning called "layout".

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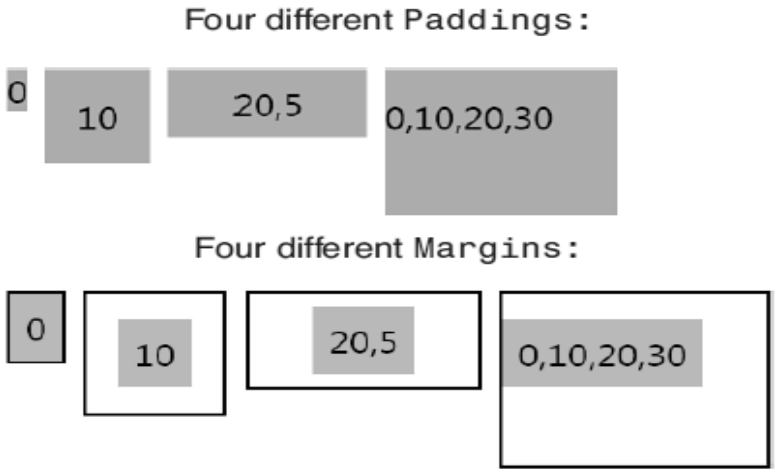


It boils down to parent child relationships. They work together to determine their final sizes and positions. Parents tell where to render and how much space they get. But more like collaborators they also ask children how much they really need. Panels are parents supporting layout of multiple children. They derive from the abstract System.Windows.Controls.Panel class. All the elements involved in the layout process derive from System.Windows.uielement. Several properties control various aspects of layout.

Size related properties are shown in blue, position related in red, green transforms can effect both. We will discuss various panels that arrange their children in specific ways.

Let's first discuss size. Children tell their size to parents. Wpf elements tend to size to content and no larger. Even the whole window when you set sizetocontent property. All frameworkelements have Height and Width properties of type double. Also have minheight, maxheight, minwidth, maxwidth. If explicit height width they takeprecedence when in range (avoid explicit). Min=0 and max=Infinity by default. Nan, Auto, Double.isnan mean size to content. Read only properties desiredsize, rendersize, actualheight, actualwidth. They are useful to act programatically on final size.

Next is margin and padding. All framework elements have margin, controls (and Border) have padding. Margin is extra space "outside" and padding is "inside" edges of the element. They are of type Thickness that can represent 1, 2, or 4 values. Negative margin is possible. Label's default padding=5. Xaml specification is helped by typeconverter.



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```
Mylabel.Margin = new   Thickness (10); //(10); // Same as Margin="10" in XAML
Mylabel.Margin = new Thickness (20,5,20,5); // Same as Margin="20,5" in XAML
Mylabel.Margin = new Thickness (0,10,20,30);// Same as Margin="0,10,20,30" in XAML
```

The lengthconverter type converter associated with the various length properties supports specifying explicit units of cm, pt, in, or px (the default). Default are device independent pixels. These logical pixels are 1/96 in regardless of screen DPI setting. They are always stored as "double". A typical display is 96 DPI. Important to note that all measurements are DPI independent.

Next is Visibility. It can have three values: Visible - The element is rendered and participates in layout, Collapsed The element is invisible and does not participate in layout, Hidden - The element is invisible yet still participates in layout.

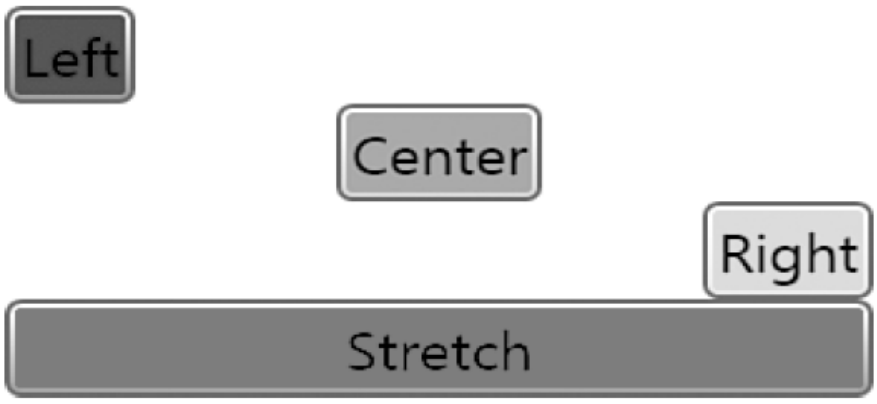
Compare the following two stackpanels.

```
<stackpanel  Height="100"  Background="Aqua">
<Button  Visibility="Collapsed">Collapsed Button</Button>
<Button>Below a Collapsed Button</Button>
</stackpanel >

<stackpanel  Height="100"  Background="Aqua">
<Button  Visibility="Hidden">Hidden Button</Button>
<Button>Below a Hidden Button</Button>
</stackpanel >
```

Next is Position and alignmnet. Position is not given as (x,y), instead its parent controlled. Rather alignment and flow direction is given vs. Explicit height. Horizontalalignment can taken values Left, Center, Right, and Stretch and verticalalignment can take values Top, Center, Bottom, and Stretch.

```
<stackpanel>
<Button  horizontalalignment="Left" Background="Red">Left</Button>
<Button  horizontalalignment="Center" Background="Orange">Center</Button>
<Button  horizontalalignment="Right" Background="Yellow">Right</Button>
<Button  horizontalalignment="Stretch" Background="Lime">Stretch</Button>
</stackpanel >
```

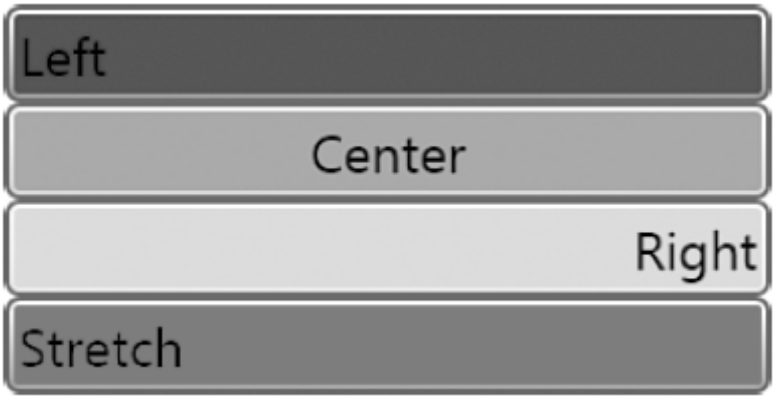


Its only useful when parent gives more space than needed.

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Control class also has horizontalcontentalignment and verticalcontentalignment. It sets how a control content fills the space inside, more like the relationship between margin n padding. Default is left and top except button which overrides it. Textblock doesnt strech like a control.

```
<stackpanel>
<Button  horizontalcontentalignment="Left" Background="Red">Left</Button>
<Button  horizontalcontentalignment="Center" Background="Orange">Center</Button>
<Button  horizontalcontentalignment="Right" Background="Yellow">Right</Button>
<Button  horizontalcontentalignment="Stretch" Background="Lime">Stretch</Button>
</stackpanel >
```



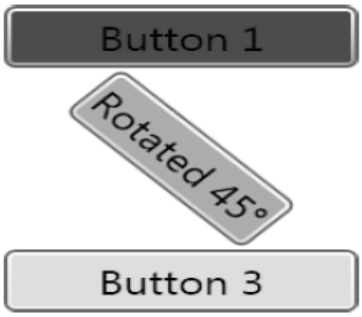
Frameworkelement.flowdirection can reverse the direction. It can be used for languages that are read right to left.

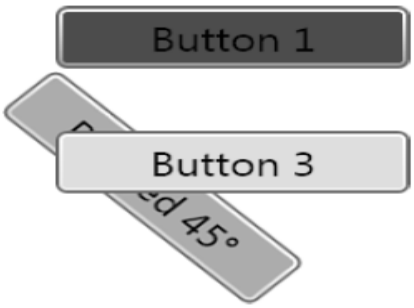
```
<stackpanel>
<Button  flowdirection="lefttoright" horizontalcontentalignment="Left"
verticalcontentalignme
<Button  flowdirection="righttoleft" horizontalcontentalignment="Left"
verticalcontentalignme
</stackpanel >
```

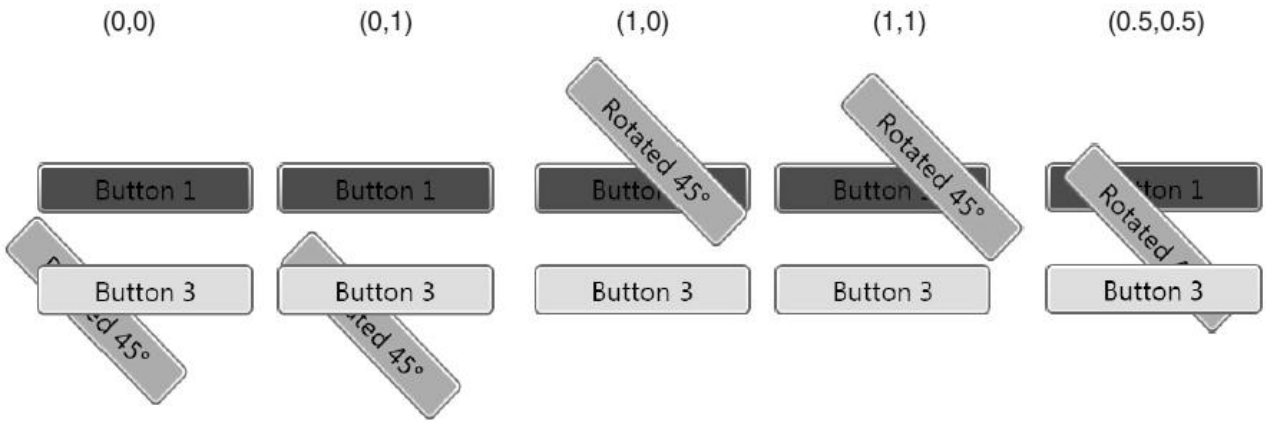


Transforms are from the base class System.Windows.Media.Transform. All framework elements have layouttransform and rendertransform. One applied before layout and one after. Uielements have rendertransformorigin.

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Button 1
Rotated 45°
Button 3


Button 1
Button 3
45°


(0,0)
(0,1)
(1,0)
(1,1)
(0.5,0.5)
Button 1
Button 3
Rotated 45°

Chapter 18

Lecture 18

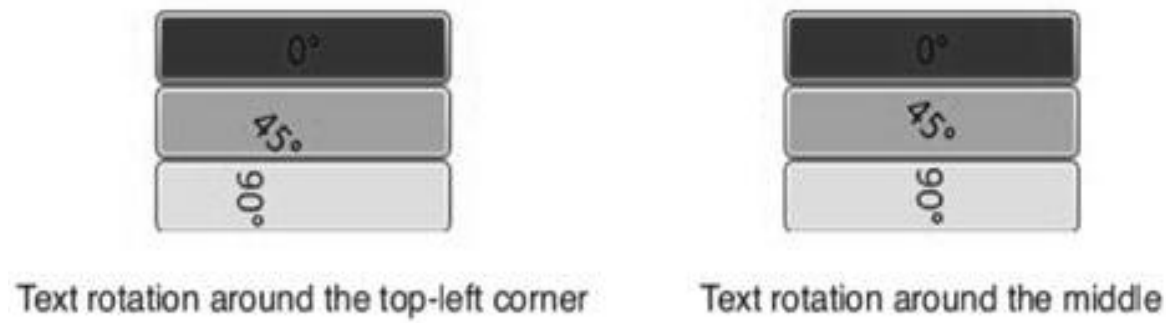
Let's revise size and position properties.
Layout transform vs render transform. One is applied before rendering and one after.

Pointconverter is used to specify the origin.

```
<Button  rendertransformorigin="0.5,0.5" Background="Orange">
<Button.rendertransform>
<rotatetransform  Angle="45"/>
</Button.rendertransform >
Rotated 45
</Button >
```

Builtin transforms are Rotate Transform, Scale Transform, Skew Transform, Translate Transform, and Matrix Transform. Rotatetransform has Angle, centerx, centery with defaults 0. Center useless for layout transform. Center is also useful when grouping rendertransforms.
Transform can be on inner element.

```
<Button  Background="Orange">
<textblock  rendertransformorigin="0.5,0.5">
<textblock.rendertransform>
<rotatetransform  Angle="45"/>
</textblock.rendertransform >
45
</textblock >
</Button >
```

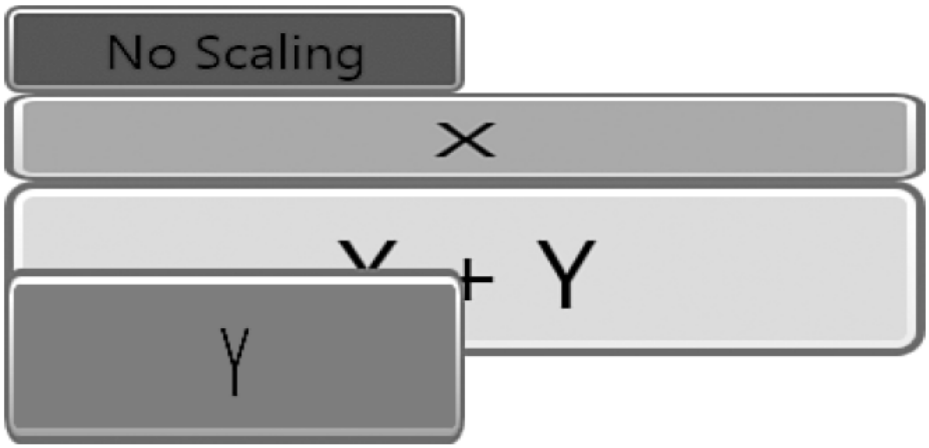


Scale Transform has scalex, scaley, centerx, centery.

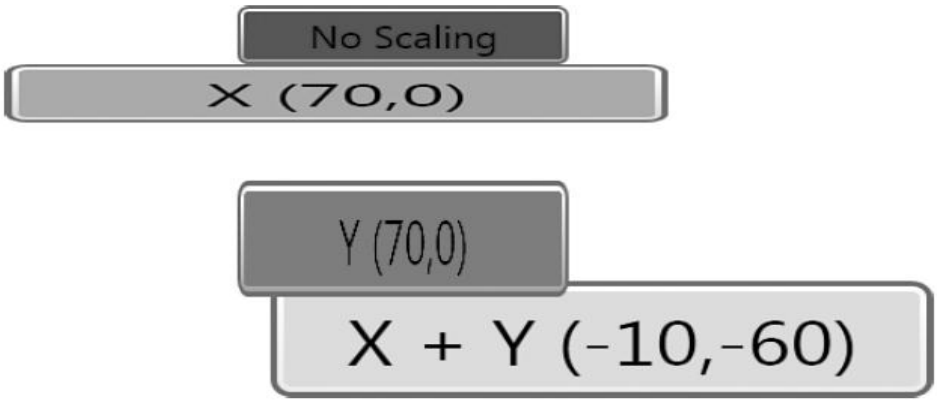
```
<stackpanel Width="100">
  <Button  Background="Red">No Scaling</Button>
  <Button  Background="Orange">
    <Button.rendertransform>
      <scaletransform  scalex="2"/>
    </Button.rendertransform >
    X
  </Button >
  <Button  Background="Yellow">
    <Button.rendertransform>
      <scaletransform  scalex="2" scaley="2"/>
    </Button.rendertransform >
```


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```
    X + Y
  </Button >
  <Button  Background="Lime">
    <Button.rendertransform>
      <scaletransform  scaley="2"/>
    </Button.rendertransform >
    Y
  </Button> 21
</stackpanel >
```

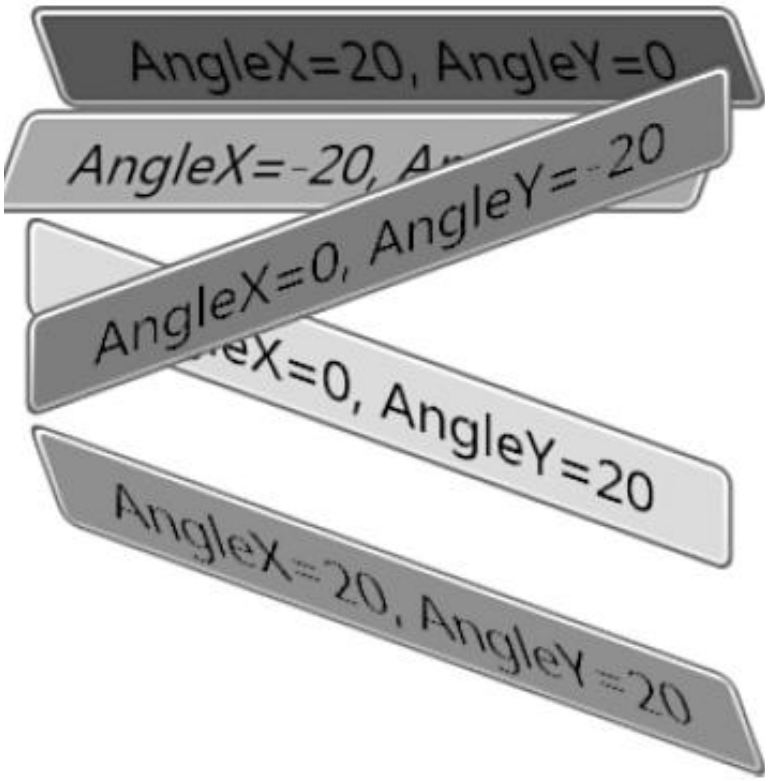


Stretch and scaletransform only effect if more than stretch. Padding is scaled but margin is not. Does not effect actualhieght actualwidth rendersize etc.
Skew Transform has anglex — Amount of horizontal skew (default value = 0), angley — Amount of vertical skew



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(default value = 0), centerx — Origin for horizontal skew (default value = 0), centery — Origin for vertical skew (default value = 0). It is applied as a render transform here.



Translate Transform has X — Amount to move horizontally (default value = 0), Y — Amount to move vertically (default value = 0). It has no effect as a layout transform.

Matrix Transform has a single Matrix property (of type System.Windows.Media.Matrix) representing a 3x3 affine transformation matrix.

```
<Button  rendertransform=1,0,0,1,10,20/>
```

M11	M12	0
M21	M22	0
OffsetX	OffsetY	1

Its the only transform with a type converter to convert a string.

Combining transforms can be done. We can apply both layout and render transforms. Can figure out a

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combined matrix and apply matrix transform or use transformgroup a Transform derived class and let it be calculated for you.

```
<Button>
  <Button.rendertransform>
    <transformgroup>
      <rotatetransform  Angle="45"/>
      <scaletransform  scalex="5" scaley="1"/>
      <skewtransform  anglex="30"/>
    </transformgroup >
  </Button.rendertransform >
  OK
</Button >
```



Not all elements can be transformed.

In Summary, we learned about layout of individual elements. The layout is finally decided by parent. We have used stackpanel. Let's see this and other panels for layout.

Static pixel based placement and sizes results in limited but different screen sizes. Builtin panels that make it easy.

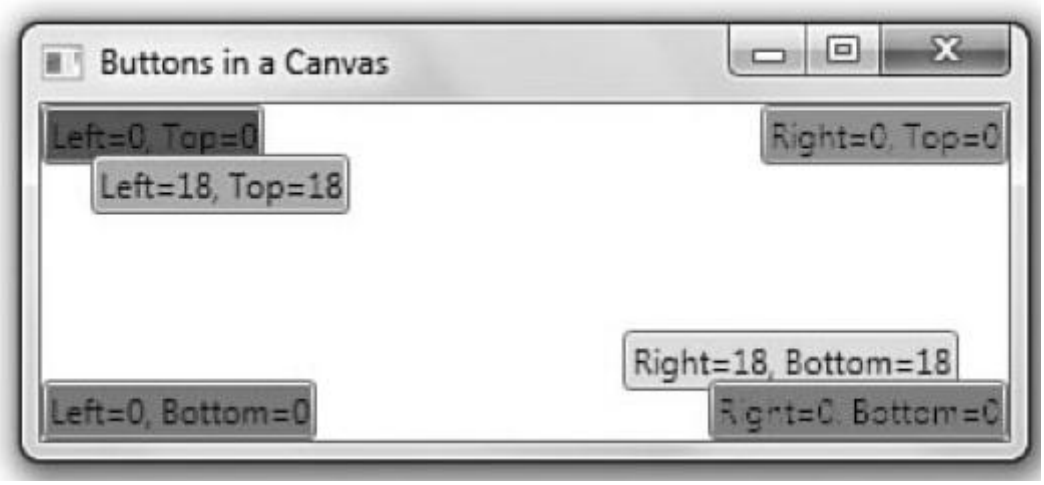
5 main builtin panels in System.Windows.Controls: Canvas, stackpanel, wrappanel, dockpanel, and Grid. Content overflow is when parents and children cant agree on the use of available space.

Canvas is the most basic. Probably never used. "classic" notion of explicit coordinates but device independent pixels and relative to any corner of the window. We can position elements in a Canvas by using its attached properties: Left, Top, Right, and Bottom. Default offsets are from top left.

```
<Window xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
Title="Buttons in a Canvas">
  <Canvas>
    <Button  Background="Red">Left=0, Top=0</Button>
    <Button Canvas.Left="18" Canvas.Top="18"
    Background="Orange"> Left=18, Top=18</Button>
    <Button Canvas.Right="18" Canvas.Bottom="18"
    Background="Yellow">Right=18, Bottom=18</Button>
```

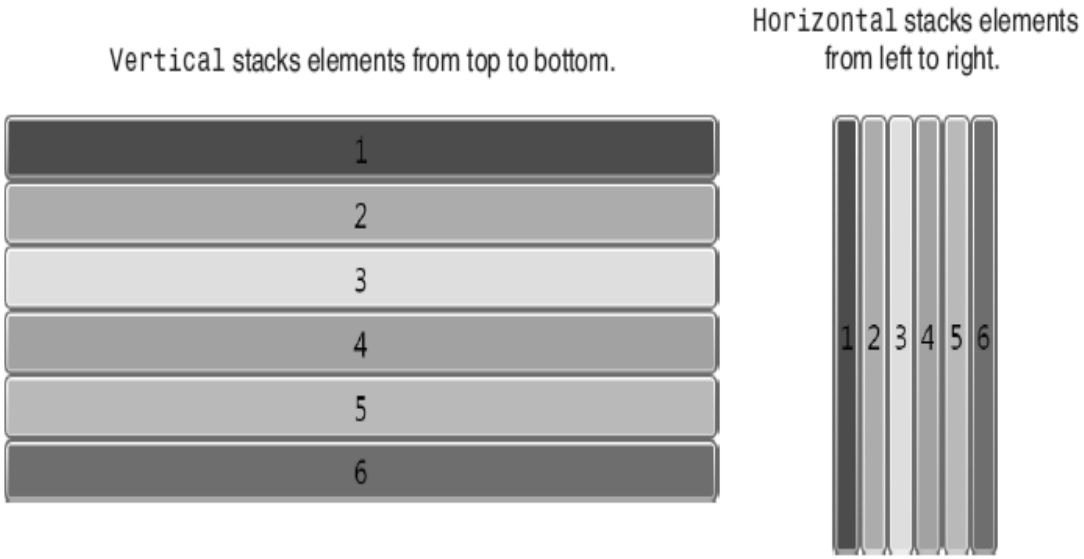

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```
    <Button Canvas.Right="0" Canvas.Bottom="0"
    Background="Lime">Right=0, Bottom=0</Button>
    <Button Canvas.Right="0" Canvas.Top="0"
    Background="Aqua">Right=0, Top=0</Button>
    <Button Canvas.Left="0" Canvas.Bottom="0"
    Background="Magenta">Left=0, Bottom=0</Button>
  </Canvas ></Window >
```



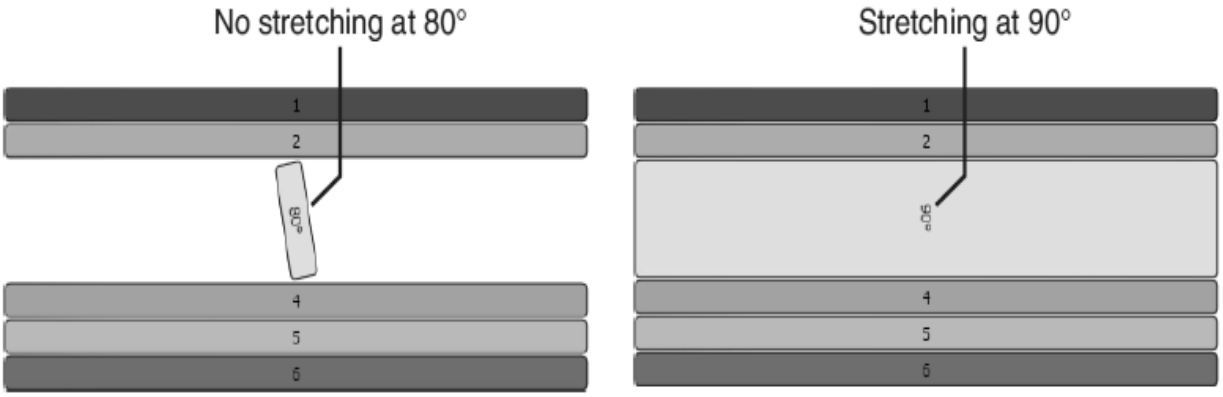
If left and right, right is ignored, top and bottom, bottom is ignored. Whats z order. Z-order decides which element to show in overlapping elements.

```
<Canvas>
  <Button Canvas.zindex="1" Background="Red">On Top!</Button>
  <Button  Background="Orange">On Bottom with a Default zindex=0</Button>
</Canvas >
```



Stack Panel is popular, simple, useful, stacks sequentially. It is one of the few panels that dont even define an attached property. Orientation can be horizontal or vertical (which is default). Default horizontal direction is basedon flowdirection.

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Virtualizing panels e.g. Virtualizingstackpanel save space for offscreen content when data binding e.g. Listbox uses it Wrap panel wraps to additional rows or columns when not enough space. It has no attached properties. Orientation is by default horizontal. Itemheight, itemwidth are uniform for all children and not set by default.



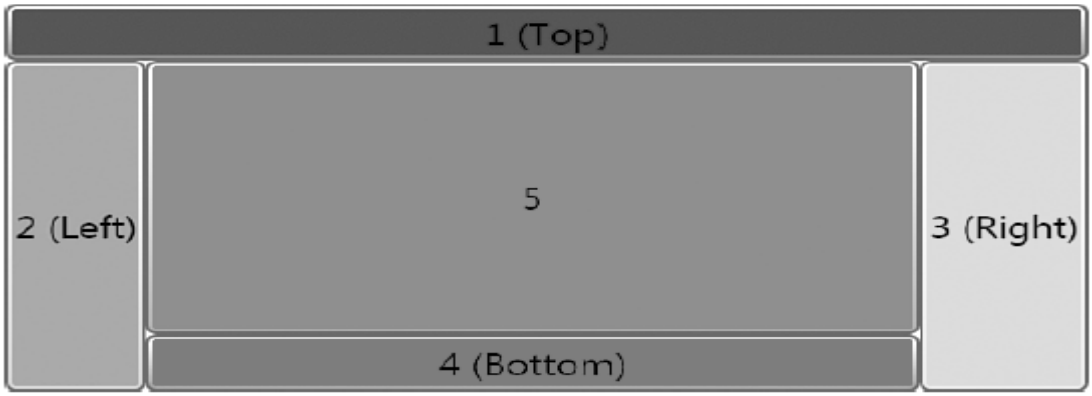
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Chapter 19

Lecture 19

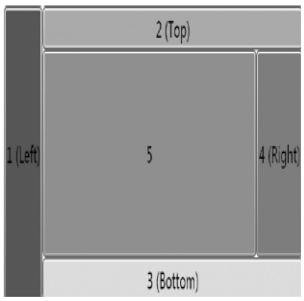
Dock panel allows easy docking of elements to an entire side. Dock attached property has left, right, top, bottom values. Last child fills space unless lastchildfill=false.

```
<dockpanel>
  <Button dockpanel.Dock="Top" Background="Red">1(Top)</Button>
  <Button dockpanel.Dock="Left" Background="Orange">2 (Left)</Button>
  <Button dockpanel.Dock="Right" Background="Yellow">3 (Right)</Button>
  <Button dockpanel.Dock="Bottom" Background="Lime">4 (Bottom)</Button>
  <Button  Background="Aqua">5</Button>
</dockpanel >
```



Why are elements streching. Let's change alignment and see.

```
<dockpanel>
  <Button dockpanel.Dock="Top" horizontalalignment=”Right” Background=”Red”>1 (Top,
Align=Right) </Button>
  <Button dockpanel.Dock=”Left” verticalalignment=”Bottom” Background=”Orange”>2 (Left,
Align=Bottom)</Button>
  <Button dockpanel.Dock=”Right” verticalalignment=”Bottom” Background=”Yellow”>3 (Right,
Align=Bottom)</Button>
  <Button dockpanel.Dock=”Bottom” horizontalalignment=”Right” Background=”Lime”>4
(Bottom, Align=Right)</Button>
  <Button Background=”Aqua”>5</Button></dockpanel>
```



Its useful in a toplevel interface. Order of adding matters for the corners.

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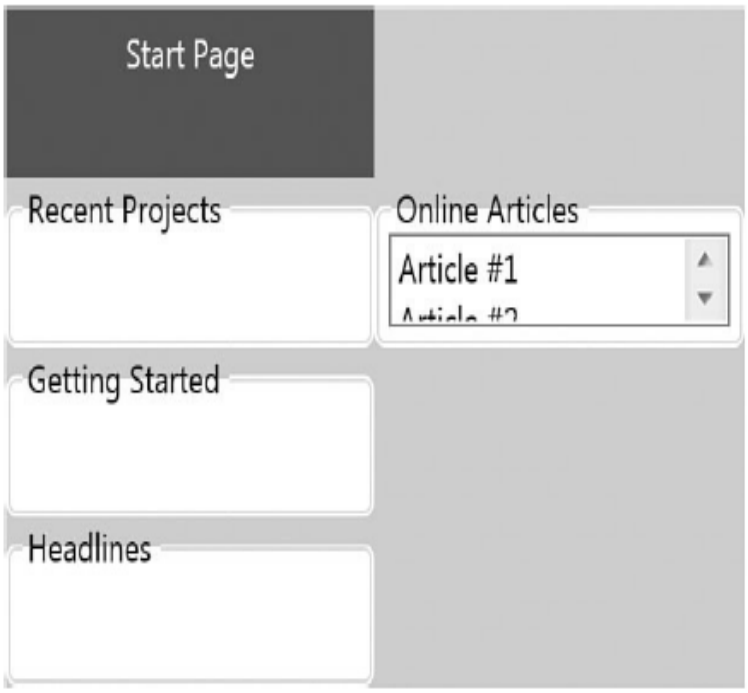


More elements can be added to any side. Its a superset of stackpanel.

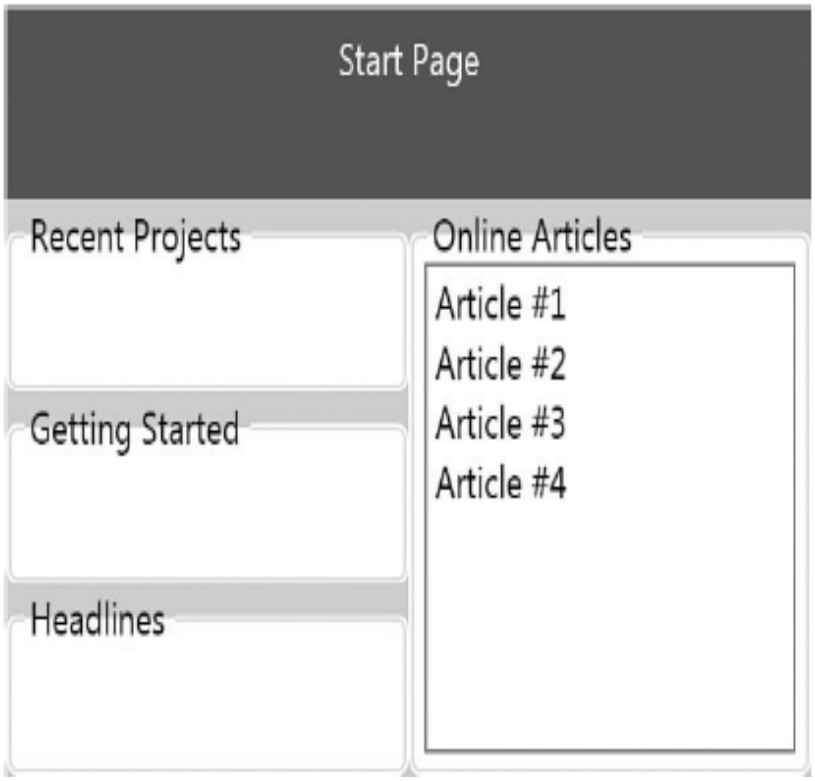
Grid is the most versatile. Its the default panel in VS and expression blend. You arrange things in multirow multicolumn, like a table in html. Also a Table class which is a frameworkcontentelement not a uielement.

```
<Grid Background="lightblue">
  <!--  Define  four  rows: -->
  <Grid.rowdefinitions>
    <rowdefinition/>
    <rowdefinition/>
    <rowdefinition/>
    <rowdefinition/>
  </Grid.rowdefinitions >
  <!--  Define  two  columns:-->
  <Grid.columndefinitions>
    <columndefinition/>
    <columndefinition/>
  </Grid.columndefinitions >
  <!--  Arrange  the  children: -->
  <Label   Grid.Row="0" Grid.Column="0" Background="Blue" Foreground="White"
horizontalcontenta=""
<groupbox Grid.Row="1" Grid.Column="0" Background="White" Header="Recent Projects">
 </groupbox>
  <groupbox Grid.Row="2" Grid.Column="0" Background="White" Header="Getting
Started"></groupbox>
  <groupbox Grid.Row="3" Grid.Column="0" Background="White"
Header="Headlines"></groupbox>
  <groupbox Grid.Row="1" Grid.Column="1" Background="White" Header="Online
Articles"></groupbox>
  20 <listbox>
    <listboxitem>Article #1</listboxitem>
    <listboxitem>Article #2</listboxitem>
    <listboxitem>Article #3</listboxitem>
    <listboxitem>Article #4</listboxitem>
  </listbox >
  </groupbox >
</Grid >
```

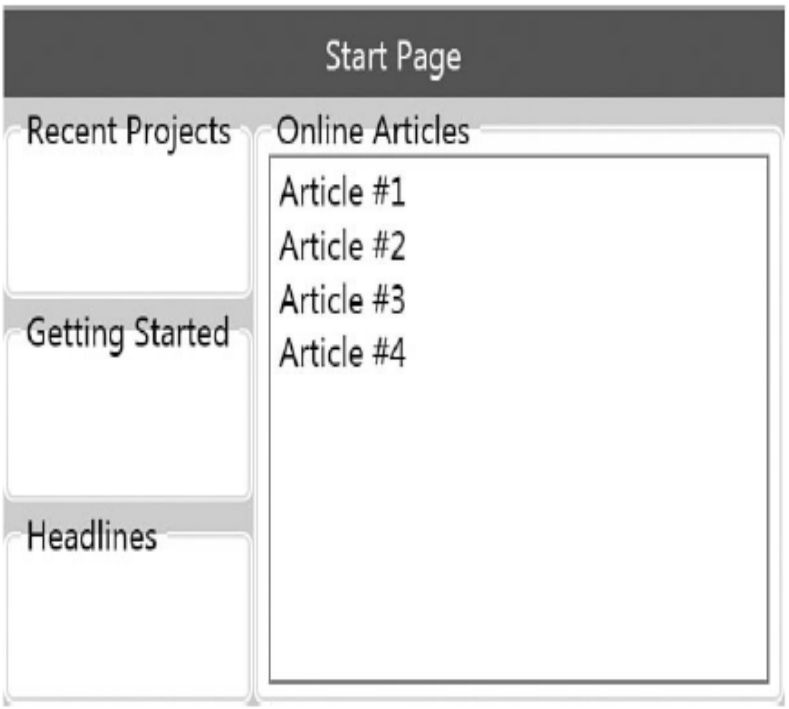

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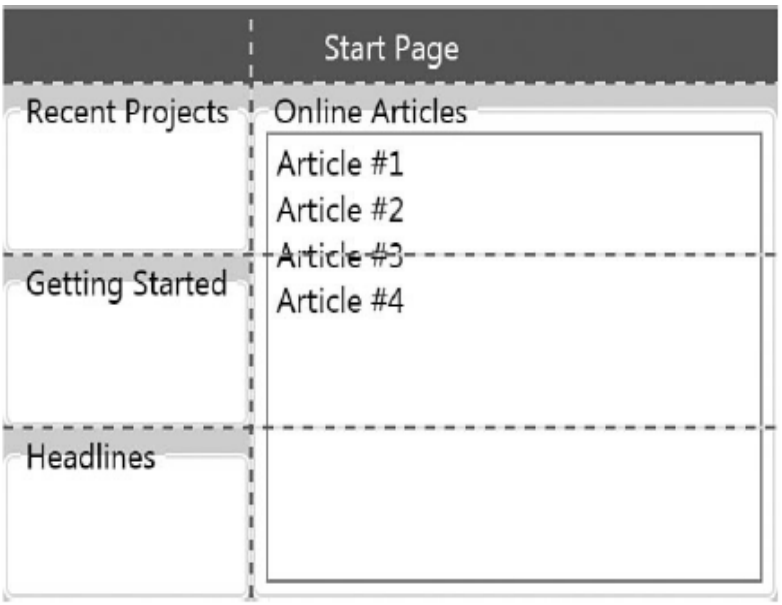


We specify individual rows and cols. Its verbose but useful to specify details. Default is 1 cell. We position elements using Row and Column attached properties (zero-based) default is 0,0. Multiple elements in same cell simply overlap by z-order. Cells can be empty.



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Online article list too small and and start page label not full width. We can make spanning rows and cols. Rowspan and columnspan 1 by default. By default height n width are same. Height and Width = "Auto" for sizing to content. Showgridlines = "True" to show grid lines.

For sizing rows and columns, use rowdefinition and Col. Height and Width not double but gridlength and not default to Auto or nan. Three types of sizing. Absoulte sizing: device independent pixels means no grow shrink, Autosizing: size to content, or Proportional or star sizing: grows or shrinks. When 1 row col * all remaining space is taken by it. When more row col * they divides remaining space. It can an be 2* or

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5.5*. 2* is twice as * (1*). 5.5* is twice as 2.75*. Remaining space is after absolute and autosized rows or cols. Default width and height are *.

Here is how gridlength is used in procedural code.




```
Gridlength  length = new gridlength(100);
Gridlength  length = new gridlength(0, gridunittype.Auto);
Gridlength  length = new gridlength(100, gridunittype.Pixel);
Gridlength  length = new gridlength(2, gridunittype.Star);
```

Grid splitter is used for interactive sizing. Any number of gridsplitters can be added to a Grid. Can be used to move entire row or column. Atleast one cell resizes and other cells behavior depends on weather absolute sizing or proportional is applied on them. It fits in one cell, but behavior affects entire row or col so better to use span. Which cells are affected depends on gridsplitters alignment values. Default horizontal alignment=right and vertical alignment=stretch. Reasonable use requires stretch in one dimension.

HorizontalAlignment

VerticalAlignment	Left	Right	Center	Stretch
Top	Current cell and cell to the left	Current cell and cell to the right	Cells to the left and right	Current cell and cell above
Bottom	Current cell and cell to the left	Current cell and cell to the right	Cells to the left and right	Current cell and cell below
Center	Current cell and cell to the left	Current cell and cell to the right	Cells to the left and right	Cells above and below
Stretch	Current cell and cell to the left	Current cell and cell to the right	Cells to the left and right	Cells to the left and right if `GridSplitter` is taller than it is wide, *or* cells to the top and bottom if `GridSplitter` is wider than it is tall

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When all proportionally sized, changes co-efficients accordingly. When absolute sizing, changes only top or left of the cells. Remaining cells pushed down or right. Also has resize direction (default Auto, or can be set to Rows or Cols) and resize behavior for explicit control. Resize direction effect only when stretching in both directions. Resize behavior defaults to based on alignment. It can be set to previous and current, current and next, or previous and next to control which 2 rows or columns should be directly affected by resizing. Best way is to place it in its own auto sized row or column so it doesn't overlap existing content.

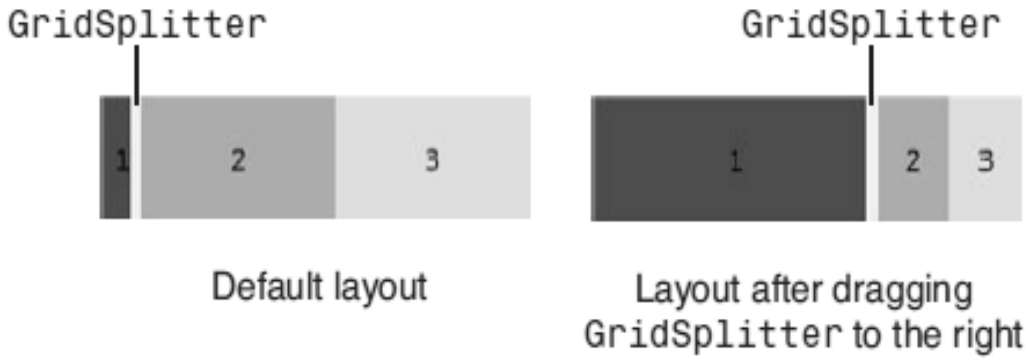
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Chapter 20

Lecture 20

Let's revise grid.

```
<Grid>
  <Grid.columndefinitions>
    <columndefinition  Width="Auto"/>
    <columndefinition/>
    <columndefinition/>
  </Grid.columndefinitions >
  <Label Grid.Column="0" Background="Red" horizontalcontentalignment="Center"
verticalcontenta=""
  </Label >
  <gridsplitter Grid.Column="0" Width="5"/>
  <Label Grid.Column="1" Background="Orange" horizontalcontentalignment="Center"
verticalconte=""
  </Label >
  <Label Grid.Column="2" Background="Yellow" horizontalcontentalignment="Center"
verticalconte=""
  </Label >
</Grid >
```

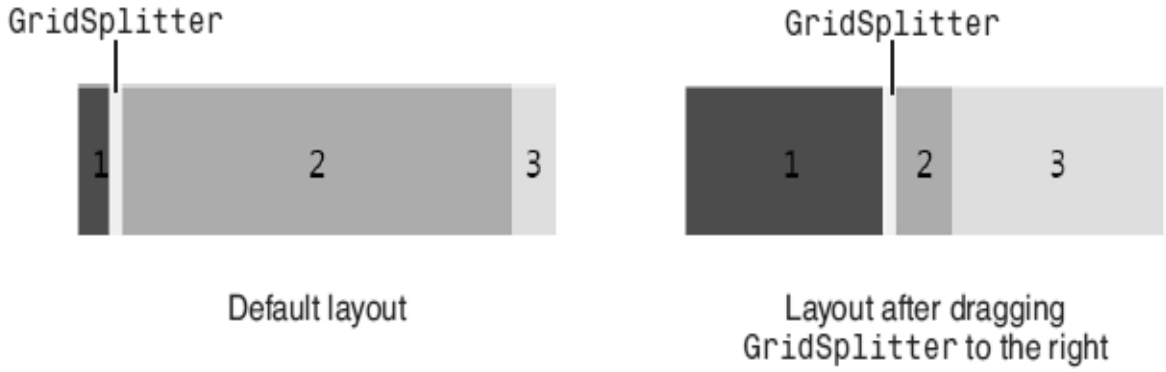


Sharedsizegroup enables multiple row cols to remain the same width height when length changed via gridsplitter.

```
<Grid issharedsizescope="True">
  <Grid.columndefinitions>
    <columndefinition  Width="Auto" sharedsizegroup="mygroup"/>
    <columndefinition/>
    <columndefinition  sharedsizegroup="mygroup"/>
  </Grid.columndefinitions >
  <Label Grid.Column="0" Background="Red" horizontalcontentalignment="Center"
verticalcontenta=""
  </Label >
  <gridsplitter Grid.Column="0" Width="5"/>
  <Label Grid.Column="1" Background="Orange" horizontalcontentalignment="Center"
verticalconte=""
  </Label >
  <Label Grid.Column="2" Background="Yellow" horizontalcontentalignment="Center"
verticalconte=""
```


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```
  </Label >
</Grid >
```



Issharedscope because sizegroups can be shared across multiple grids. All uses must be under a common parent with issharedscope set to true. It is an attached property of grid as well. Grid is usually the best choice. Except wraping, it can do what most panels do.

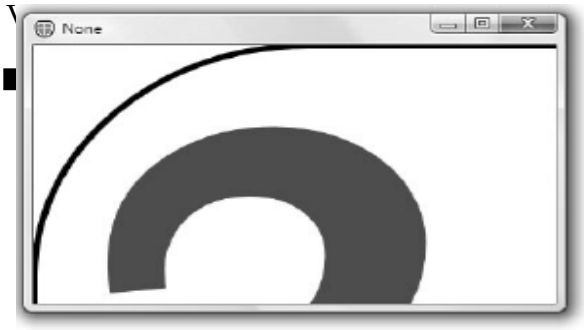


Content overflow can be dealt with Clipping, Scrolling, Scaling, Wrapping and Trimming. Wrapping already seen. Only way for non-text to wrap is using wrappanel. Trimming is intelligent form of clipping. Supported for text by textblock and accesstext. They have texttrimming=None (default), or characterellipsis, orwordellipsis.

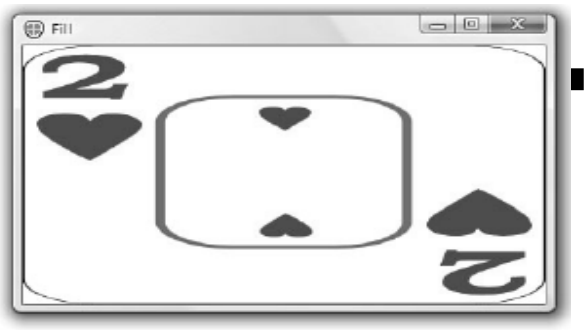
Clipping is default behavior. Edges of panel or cell area of dock area. All uielements have cliptobounds. Controls if it can draw outside bounds. Still cannot draw outside the window or page. Most panels clip regardless of this property. Canvas and uniformgrid dont unless this property is set. Button has this property false by default. Place canvas inside a grid cell to avert clipping. Clipping is done before rendertransform. Cant shrink back.



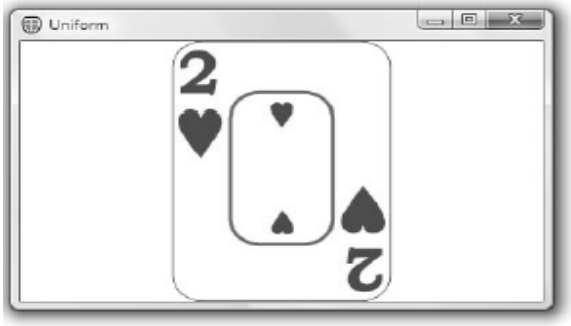
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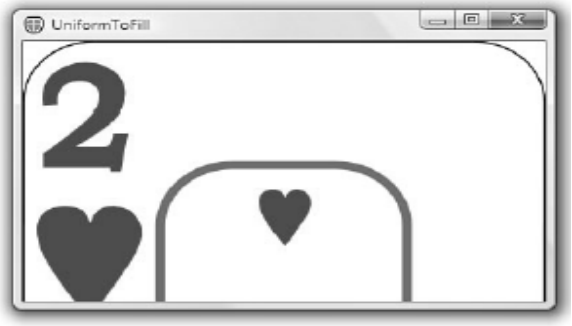
Stretch="None"



Stretch="Fill"



Stretch="Uniform"



Stretch="UniformToFill"

System.Windows.Controls.scrollviewer control can be used for scrolling. Its content property is set and verticalscrollbarvisibility and horizontalscrollbarvisibility properties determine the display.

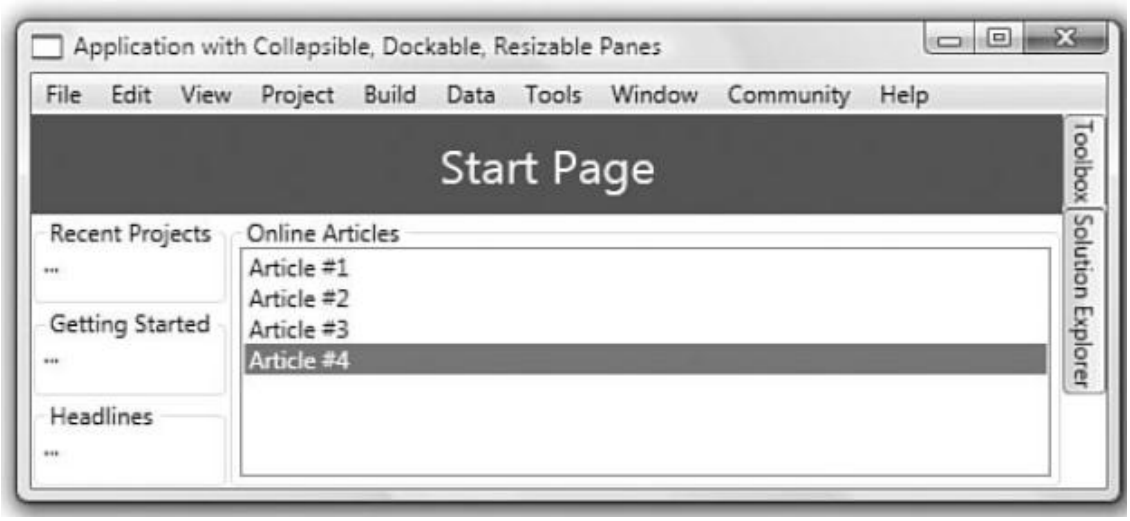
```
<Window Title="Using scrollviewer"
Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
    <scrollviewer>
        <stackpanel>
            …
        </stackpanel>
    </scrollviewer>
</Window>
```

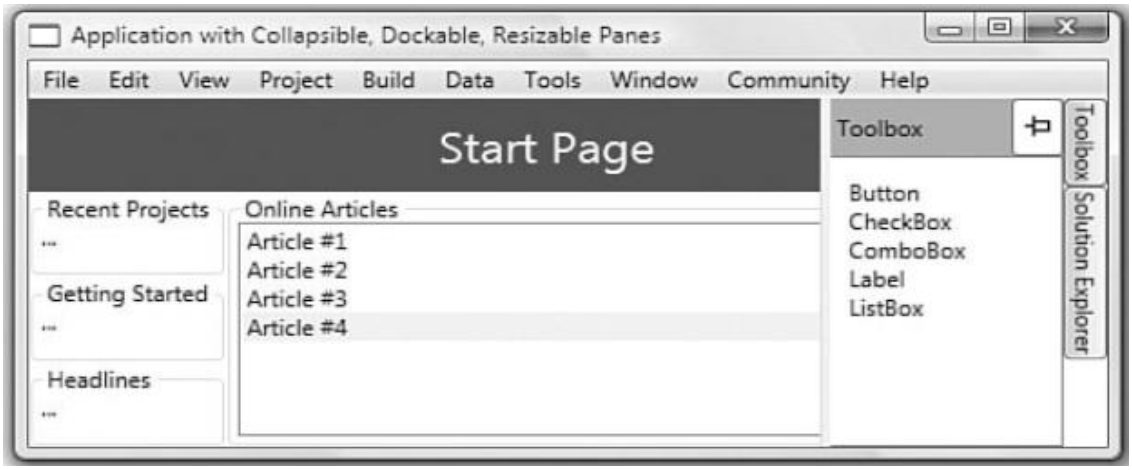

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Scrolling more popular but e.g. Card game needs scaling. Scaletransform works relative to their own size not the available size. System.Windows.Controls.Viewbox, a type of class called Decorator (also Border). A panel-like thing but has only one child. Stretches to fill available space by default but also Stretch=None (like not using it at all), Fill, Uniform (aspect ratio, default), uniformtofill (cropped) strechdirection=uponly, downonly, Both (default).

```
<Window Title="Using Viewbox"
Xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation">
    <Viewbox>
        <stackpanel>
            …
        </stackpanel>
    </Viewbox>
</Window>
```

Let's make a visual studio like interface.





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```
Using System.Windows;
Using System.Windows.Input;
Using System.Windows.Media;
Using System.Windows.Controls;

Public partial class aboutdialog : Window
{
    Public aboutdialog()
    {
        Initializecomponent();
    }
    Void aboutdialog_mouserightbuttondown(object sender, mousebuttoneventargs e)
    {
        // Display information about this event
        This.Title = "Source = " + e.Source.gettype().Name + ", originalsource = " +
            E.originalsource.gettype().Name + " @ " + e.Timestamp;
        // In this example, all possible sources derive from Control
        Control source = e.Source as Control;
        // Toggle the border on the source control
        If (source.borderthickness != new Thickness(5))
        {
            Source.borderthickness = new Thickness(5);
            Source.borderbrush = Brushes.Black;
        }
        Else
            Source.borderthickness = new Thickness(0);
    }
}
```


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CS411-Visual Programming
MID TERM OBJECTIVE
Prepared by: JUNAID MALIK

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LMS Activities Paid Task

Assignments 95% Results

Quizes 95% Results

GDB 95% Results

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Email: junaidfazal08@gmail.com

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1. Every ____ is represented by an event-object.

- Information
- Entity
- Object
- Event page10

2. Which statement is True about interfaces?

- An interface can contain only methods, variables, events, and indexers.
- An interface can contain only variables, properties, events, and indexers.
- An interface can contain only methods, properties, variables, and indexers.
- An interface can contain only methods, properties, events, and indexers. Page 40

3. Which one is the correct syntax for declaring attribute?

- [Obsolete] page 48
- {Obsolete}
- (Obsolete)
- *Obsolete

4. An "object" element can have____type/s of children.

- Four
- Three Page 73
- Two
- One

5_____allows to keep the user interface description and implementation separate.

- GDI

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- WPF (Page#61)
- Avalon
- Altia

6. There is/are _____type(s) of padding in WPF element.

- One
- Two
- Three page 91
- Four

7. Which of the following is not a feature of "C#"?

- Operator overloading is not allowed. Page 17
- Multiple Inheritances is not supported but interfaces are supported.
- Enumeration members are scoped.
- Global variables or functions are not allowed.

8. Stylus can behave like a_____ but has_____resolution.

- Keyboard, lower
- Mouse, lower
- Mouse, higher page 122
- Keyboard, higher

9. Attributes are declared_____the class/function.

- Within
- After
- Above
- Outside

10. If we want that "Stretch" property of child element takes the available "height or

width of parent" and shape of child element doesn't change, then we should write

- "Stretch =_____".

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- Fill page 109
- UniformToFill
- Uniform

11.The code given below is equivalent to ___.

- <SolidColorBrush>White</SolidColorBrush>
- <WhiteBrush >MyWhiteBrush </ WhiteBrush>
- <SolidColorBrush Color="White"/> (Page#65)
- <Brush Style= “SolidWhite”>

12.Visual Studio has a snippet called _____that automatically expands into a definition of a

dependency property, which makes defining one much faster than doing all the typing yourself!

- Propdp (Page#77)
- Property extension
- Dpprop
- Dependency property

13.Multiple_____are conventionally stored in multiple files.

- Functions
- Values
- Variables
- Classes (Page#22)

14. "Situation" is an event occurrence that requires a (n)_____.

- Reaction page 10
- Class
- Object
- Action

15. We have two buttons "b1" and "b2". If we want to place "b1" on "b2" then value of

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- "ZIndex" of "b1" should be___the value of "ZIndex" of "b2".
- Less than
- Greater than
- Equal to
- No need to give value of ZIndex of b1 page99

16. Events can be_____.

- Specialized
- Composed
- Generalized
- All of the given options page 15

17. Wpf 3.5 released in

- 2005
- 2006
- 2008 Page No 61 In AUGUST
- 2009

18.The corresponding type converter class for "Brush" class will be ___.

- BrushConverter
- CastBrush
- BrushCast
- None of the given options

19.Extension methods_____access the private data members of the class in which they are extended.

- may
- **cannot**
- may not
- can

20. Which "transform" property can help us to flip the element from its center?

- ScaleY

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- None of given options
- ScaleX
- RenderTransformOrigin page 126

21. CLI stands for_____.

- Compact Language Infrastructure
- Common Language Infrastructure page18
- Console Language Infrastructure
- Control Language Infrastructure

22. The elements in a/an___ are always stored in a contiguous block of memory.

- Tree
- Array
- Variable
- Struct

23. The _____statement iterates over each element in an "enumerable" object.

- Foreach page 191
- Dowhile
- While
- For

24. We are checking that whether an element is eligible for focus or not, which property

will help us in this testing?

- Focus
- IsFocus
- IsFocusable
- Focusable page 121

25. Directory is the class of_____.

- System.Collections.Generic

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- System.Reflection
- System.Text
- System.IO MPOV and GOOGLE

25. WCF stands for
- Windows Communication Foundation
- Windows Class Foundation
- Windows Classic Foundation
- None

26. C# is a ___and .NET is a____.
- Platform, Language
- Package, Language
- Language, Package
- Language, Platform

27. The ___,____ and ___of applications that use events, either directly or indirectly is called event-based programming.
- Model, Code, Operation
- Analyze, Design, Operation
- Code, Operation, Maintain
- Design, Coding, Operation

28. Suppose we want to restrict the user to don't decrease the height of a WPF element from a specific amount, then which property will be used?
- MinHeight
- None
- HeightMinimum
- MinimumHeight

29. Types not defined in any namespace are said to reside in the ___ namespace.
- Static
- Virtual
- Local
- Global

30. Overloaded operators must be ____
- Referenced
- Static

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- Objective
- Delegates

31. If we want to scale an object in the 2-D x-y coordinate system then which transform class will be used?

- Rotate Transform
- Matrix Transform
- Translate Transform
- Skew Transform

32. Polymorphism works on the basis that child class should have _____ feature/s of its parent class.

- Only one
- No
- Few
- All

33. The root object eement in XAML must specify at least_____XML namespace that is usedto qualify itself and any child elements.

- **one(Page#63)**
- two
- three
- five

34. In event-driven architecture (EDA),_____sends event to _____.

- Event Consumer, Event Producer
- **Event Producer, Event Consumer(Page#10)**
- Event Handler, Event Producer
- Event Generator, Event Producer

35. Auto-implemented properties do not allow simplified syntax.

- True
- **False(Page#30)**

36. Clean room design is useful as a defense against_____.

- Copyright
- Trade secret infringement
- **Copyright and trade secret infringement(Page#13)**
- None of the given options

37._____is an occurrence within a particular system or domain.

- Function
- **Event (Page#7)**
- Information
- Transaction

38.BAML stands for:

- binary application map language
- binary application markup linkage
- **binary application markup language(Page#69)**
- binary application map linkage

39. If we want to place a component at top in "dockpanel" then which one is correct syntax?

- PanelDock="Top"
- Dock="Top"
- **DockPanel.Dock="Top"(Page#93)**
- DockPanel="Top

40. Dock panel allows easy docking of elements to an entire side.

- **True(Page#91)**
- False

41. _____processing operations include reading, creating, transforming, and deleting events.

- Observed Events
- **Common event(Page#8)**
- Deducted Events
- Probabilistic Events

42. The _____in the namespace indicate a hierarchy of nested namespaces.

- Arrows
- Dots GOOGLE
- Semicolon
- **Brackets HO (Page#22)**

43. The interface forces each component to expose specific _____ members that will be used ina certain way.

- **Public(Page#35)**
- Private
- Both of above

- None of the given

44. Static ctor exists to initialize class static members and called only____.
- **Once(Page#23)**
- Twice
- Anytime
- All of the given

45. Interfaces define a contract.
- **True(Page#35)**
- False

46. Additional XML namespaces (on the root or on children) must be _____ to be used on anyidentifiers from that namespace.
- Given a common prefix
- Qualified with "Pre" keyword
- **Given a distinct prefix(Page#63)**
- Qualified with "Ext" keyword

47. We want the width of button should cover the whole width of StackPanel in WPF then whatshould be value of HorizontalAlignment property of button?
- Parent
- Whole
- Fill
- **Stretch(Page#97)**

48. _____decouple producers and consumers.
- Requests
- **Events(Page#10)**
- Functions
- All of these

49. To declare an anonymous method, you just use keyword "__".
- handler
- event
- **delegate (Page#47)**

- none of the given

50. A (n)_____is a subscription mechanism for events.

- Raw Event
- Event Stream
- **Event Channel (Page#10)**
- Event Producer

51. "Situation" is an event occurrence that requires a (n)_____.

- Class
- Object
- **Reaction**
- Action

52. XML is a stricter version of_____.

- CSS
- C#
- Java
- **HTML**

53. _____contacts the service provider or the server.

- Event-driven
- **Service Requester (Page#10)**
- Both of the above
- None of the given options

54. Any class, including the same class that the event is declared in, may register one of its methods with the_____.

- **Event (Page 39)**
- Delegate
- Function
- Class

55. There are_____methods to read XML document.

- **Two (1.xmldocument 2.xmlreader) Page#47**

- Three
- Four
- Five

56. releasing resources or doing some other type of cleanup.

- Functions
- **Exception (Page#41)**
- Abstraction
- All of these

57. _____are called by garbage collector.

- Constructors
- **Destructors (Page#23)**
- Both of above
- None of the given

58. Attributes are generally applied_____in front of type and type member declarations.

- Logically
- **Physically (Page#42)**
- Both of above
- None of given

59. Multiple_____are conventionally stored in multiple files.

- **Classes (Page#22)**
- Values
- Variables
- Functions

60. The_____,_____and_____of applications that use events, either directly
or indirectly is called event-based programming.

- Model, Code, Operation

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- Analyze, Design, Operation
- **Design, Coding, Operation (Page#8)**
- Code, Operation, Maintain

61. C# is a _____ and .NET is a _____.

- Platform, Language
- **Language, platform (Page#13)**
- Package, Language
- Language, Package

62. Windows Presentation Foundations (WPF) was publicly announced in _____.

- 2000
- 2001
- 2002
- **2003 (Page#61)**

63. WPF stands for:

- Windows Powerful Foundations
- Windows Presentation Formations
- Windows Presentation Functions
- **Windows Presentation Foundations (Page#61)**

64. Value-types hold their value in_____where they are declared.

- Variable
- **Memory (Page#33)**
- Object
- Reference

65. Indexers can take any number of _____.

- Arrays
- Variables
- **Parameters (Page#31)**

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- All of these

66. Attributes add declarative_____to your programs.

- Clearance
- **information (Page#41)**
- Meta Data
- Ambiguity

67. Cast back to base type to call a/an _____method of base.

- Instance
- Base Class
- **Overridden (Page#24)**
- Derived Class

68. _____is said to be stateless if the way it processes one event does not influence the way it processes any subsequent events.

- Event Producer
- Raw Event
- **Event Processing (Page#11)**
- Event Stream

69. _____is built from request-response and it moves away from monolithic applications.

- Event-driven architecture (EDA)
- **Service-oriented architecture (SOA)(Page#10)**
- Both of above
- None of the given option

70. Object variables are references to the_____object not the object themselves.

- **Original (Page#20)**
- Alternative
- Primary

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- None of the given

71. An indexer enables your class to be treated like a/an_____.

- Class
- Function
- **Array (Page#30)**
- Pointer

72. Polymorphism needs the signatures to be the _____.

- Different
- **Same (Page#25)**
- Alternative
- None of the given

73. Types of class members in C# are _____and_____.

- static, instance
- **instance, static (Page#22)**
- public, private
- static, private

74. Events may live _____events like in event logs.

- Inside
- **Outside (Page#10)**
- Both of above
- None of the given

75. Additional XML namespaces (on the root or on children) must be _____to be used on any identifiers from that namespace.

- Given a common prefix
- Qualified with "Pre" keyword
- **Given a distinct prefix (Page#63)**
- Qualified with "Ext" keyword

76. **The interface forces each component to expose specific _____ members that will be used in a certain way.**

- **Public (Page#35)**
- Private
- Both of above
- None of the given

77. _____allows to keep the user interface description and implementation separate.

- GDI
- **WPF (Page#61)**
- Avalon
- Altia

79. Dependency properties are represented by_____.

- **System.Windows.DependencyProperty (Page#75)**
- System.Windows.Dependency.Property
- System.Windows.Dependency
- System.Windows.Property.Dependency

80. Suppose we want to restrict the user to don‟t decrease the height of a WPF element from a specific amount then which property will be used?

- HeightMinimum
- MinimumHeight
- **MinHeight**
- None of the given

81. If we want to scale an object in the 2-D x-y coordinate system then which transform class will be used?

- Rotate Transform
- Matrix Transform
- **Translate Transform**
- Skew Transform

82. Which of the following is/are not derived from “Control” class?

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- ListBox
- StatusBar
- Button
- **None of the given**

83. Wait for a single event is_____operation.

- Waiting
- Waste
- **Blocking (Page#8)**
- Idle

84. _____method reads the whole document in memory.

- XPath
- **XmlDocument (Page#49)**
- XmlReader
- XmlLine

85. Which of the following events may or may not relate to an actual occurrence?

- **Probabilistic Events (Page#7)**
- Actual Events
- Real Events
- Expected Events

86. The System.Console is a_____ which enables us to do Console Input and Output.

- Structure
- **Class**
- Namespace
- Library

88. Clean room design is useful as a defense against _____.

- Copyright
- Trade secret infringement
- **Copyright and trade secret infringement (Page#13)**

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- None of the given options

89. Clean room design is also called_____.

- Clear Room Design
- Wall Technique Design
- **Chinese Wall Technique (Page#13)**
- French Wall Technique

90. Which of the following is an occurrence within a particular system or domain?

- Object
- **Event (Page#7)**
- Result
- Message

91. During the program execution, program can be stopped with the help of _____.

- Attributes
- Compiler
- **Breakpoints (Page#48)**
- Exception Handling

92. Which of the following does NOT include in event processing operations?

- Deleting Events
- Reading Events
- **Terminating Events (Page#8)**
- Transforming Events

93. The result of Exception can bring_____in the program.

- Reliability
- Stability
- **Inconsistency**
- Consistency

95. Graphical user interfaces and the event-driven model is not applied on which of the following?

- Web Application

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- **Console Application (Page#4)**
- Mobile Application
- Desktop Application

96. Events are based on the principle of_____.

- Request
- Response
- Coupling
- **Decoupling (Page#10)**

97. _____methods do not have names.

- Attribute
- Non-Anonymous
- **Anonymous (Page#47)**
- None of give

98. Which of the following is an entity that introduces event into the system?

- Event Channel
- **Event Producer (Page#11)**
- Event Stream
- Event Consumer

99. Anonymous methods_____the code size.

- Double
- Expand
- **Reduce (Page#47)**
- Extend

100. _____events may or may not relate to an actual occurrence

- Layer
- Properties
- Event properties
- **Probabilistic (Page#7)**

101. Task Parallel_____exploit multicore for real parallel tasks.

- Structure
- **Library (Page#175)**

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- Class
- Namespace

102. The Main method returns a non-zero value which indicates the_.

- Source of Program
- **Error in Program**
- Termination of Program
- Exception in Program

103. WPF4 was released in_____.

- 2012
- **2010 (Page#61)**
- 2005
- 2003

104. Extension methods can only be defined in_____class.

- **Static class**
- Dynamic class
- Normal class
- String class

105. BAML is compressed representation of_____.

- CAML:NS
- **XAML (Page#69)**
- XAML:NS
- CAML

106. _____processing operations include reading, creating, transforming,

- **and deleting events.**
- Observed Events
- **Common event (Page#8)**
- Deducted Events
- Probabilistic Events

107. Static ctor exists to initialize class static members and called only _____.

- **Once (Page#23)**
- Twice
- Anytime
- All of the given

108. _____Interact with web page layout.

- Java script
- **HTML**
- XML
- DOM

109. _____supports the creation of applications that run directly in a web browser.

- C#
- **WPF**
- HTML
- C/C++

110. Which one of the following panels is most powerful, versatile and customizable?

- Doc panel
- Stack panel
- **Grid Panel (Page#92)**
- Canvas panel

111. With the help of WPF, an ugly looking application can be _____ by the designers.

- Rearranged
- Re-created
- Rejected

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- **Re-themed (Page#61)**

112. Graphical user interfaces and the event-driven model are not applied on which of the following?

- Web Applications
- Mobile applications
- Desktop Applications
- **Console Applications (Page#4)**

113. C# is most often used with some implementation of the _____.

- Compact Language Interface
- **Common Language Infrastructure (Page#14)**
- Control Language Infrastructure
- Control Language Interface

114. In C#, comments are written using_____.

- # and */ /*
- **// and /* */**
- # and /*
- // and */ /*

115. XML document is readable by_____.

- Human only
- Machine only
- No one
- **Machine and Human both (Page#48)**

116. "Console" is a/an_____class in the "System" namespace.

- Input
- Dynamic
- **Static (Page#15)**
- Virtual

117. The only _____.Net language has an intrinsic understanding of dependency Properties.

- Vb.net

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- **Xaml (Page#74)**
- C#
- J#

118. Which of the following grabs the "OK" button by only knowing its name?

- Button okButton = (Button)window.GetName("okButton")
- Button okButton = (Button)window.getElementByName("okButton")
- Button okButton = (Button)window.FindName("okButton")
- **Button okButton = (Button)window.GetButton("okButton")**

120. Unlike type converters, however, markup extensions are invoked from XAML with _____.

- Bridge assemblies
- **Explicit and consistent syntax (Page#65)**
- Wrapper code
- None of the given options

121. Which statement is True about interfaces?

- An interface can contain only methods, variables, events, and indexers.
- An interface can contain only variables, properties, events, and indexers.
- An interface can contain only methods, properties, variables, and indexers.
- **An interface can contain only methods, properties, events, indexers.**

122. Events can be_____.

- Specialized
- Composed
- Generalized
- **All of the given options**

123. g++ is a command of_____.

- Dev C++

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- **Unix based system**
- Windows based system
- None of the given options

124. _____reads entire document in memory.

- **Xmldocuments (Page#49)**
- Xmlreader
- Both of above
- None of the given

125. _____may contain events.

- Requests
- Service
- **Messages (Page#10)**
- Applications

126. A customer order can be represented as_____.

- Event
- Request
- **Event or Request (Page#10)**
- All of the given

127. Markup extensions are invoked from_____with explicit and consistent syntax.

- C#
- HTML
- **XAML (Page#65)**
- None of the given

128. All the members of interfaces are implicitly_____.

- Private
- Static
- **Public**
- Abstract

129. A compiled C# file is called_____.

- Complied File

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- Source File
- **Assembly File**
- EXE File

130. _____add declarative information to your programs.

- **Attributes (Page#41)**
- Functions
- Exception
- Objects

131. Anonymous method is a method without a name.

- **True (Page#47)**
- False

132. "out" parameters which are not used for return values or the "params" argument for variable arguments.

- True
- **False (Page#20)**

133. _____types have the notion of referential identity.

- Value
- **Reference (Page#14)**
- Both of above
- None of the given

134. _____let any classes behave like an array.

- Properties
- Fields
- **Indexers**
- Finalizers

135. _____are algorithms that operate on data.

- Variables
- Functions
- **Methods (Page#37)**
- Classes

136. _____is an event that is introduced into an event processing system by an event producer.

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- Event Producer
- Event Consumer
- **Raw Event (Page#11)**
- Event Stream

137. Events and_____work hand in hand.

- Parameters
- **Delegates (Page#39)**
- Functions
- Data Members

138. C# event is a class member that is _____whenever the event it was designed for occurs (fires).

- **Activated (Page#38)**
- Deactivated
- Transformed
- Associated

139. _____is a static class in the System namespace.

- **Console (Page#15)**
- Pointer
- Variable
- Function

140. Default value of first member is _____.

- **(Page#44)**
- 1
- 2
- 3

141. _____are inherited by classes which provide the real implementation.

- FunctionsVariables
- **Interfaces (Page#35)**

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- Objects

142. _____is basically array of arrays.

- Stack
- Pointer
- **Jagged (Page#18)**
- Function

143. The initial name for C# was_____.

- COAL
- COOP
- COAP
- **COOL (Page#13)**

144. BAML is compressed representation of_____.

- CAML:NS
- **XAML (Page#69)**
- XAML:NS
- CAML

145. Extension methods_____access the private data members of the class in which they are extended.

- May
- **Cannot**
- May not
- Can

146. Which of the following is/are not derived from "Control" class?

- ListBox
- StatusBar
- Both of above
- **None of the given**

147. _____events may or may not relate to an actual occurrence.

- Observed Events
- Deducted Events
- **Probabilistic Events (Page#7)**
- None of the above

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148. _____is an entity that receives events from the system.

- Event Producer
- **Event Consumer (Page#11)**
- Event Channel
- Event Generator

149. The easiest way to declare an event is to put the event keyword in front of a_____member.

- Interface
- **Delegate**
- Class
- Struct

150. keyword is used to declare a base class method while performing overriding of baseclass method.

- Overload
- **Virtual**
- Extend
- Override

151. Choose the correct option among the following indexers which correctly allows to indexin sameway as an array?

- **A Class**
- A function
- An interface
- A property

152. A class can inherit from_____class/classes in C#.

- Three
- **Single**
- Two
- Multiple

153. Which one of the following is not the feature of C#?

- **Multiple inheritance**
- Interface supported
- Locals can not shadow global variables
- Operator Overloading is allowed

154. Which keyword is used to declare the indexers?

- super
- base
- **this**

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- extract

155. The common type system of C# has value types and_____types.

- Class
- Object
- **Reference**
- Enum

156. Select the correct statement about the structure in C#?

- **Its syntax is similar to class**
- It can implement destructors
- It is a reference type
- It can implement inheritance

157. Reaction of events are either_or_____.

- Asynchronous, Blocking
- Synchronous, Blocking
- Blocking, Synchronous
- **Synchronous, Asynchronous**

158. is an example of request and response based application

- **Web browser**
- E –Commerce Site
- Pressing a button
- Swiping of credit card

159. Which of the following is built from request-response?

- Event-based Architecture (EBA)
- Service-driven Architecture (SDA)
- **Service-oriented Architecture (SOA)**
- Event-processing Architecture (EDA)

160. and delegates work hand in hand. (Page-38)

- Static member
- **Event**
- Post-processor
- Pre-processor

161. In C#, when a value type is converted to object type, it is called_.

- Polymorphism
- Abstraction
- **Boxing**
- Data Conversion

162. Which keyword is used for correct implementation of an interface in C#?

- Intf
- **Interface**
- Int Interface

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- Interfaces

163. The main method return a non-zero value which indicates the

- Exception in program
- Termination of program
- **Error in program**
- Success of program

164. is an occurrence within a particular system or domain.

- Program
- Class
- **Event**
- Object

165. Select wrong statement about destructor inC#?

- **Destructor can have modifiers or parameter**
- Destructor cannot be inherited or overloaded
- Destructor work opposite to constructor
- Class can have one destructor only

166. The main method returns zero value which indicates the

- **Successful execution of program**
- Source of program
- Termination of program
- Exception in program

167. In C#, which of the following keyword is used for including the namespaces in theprogram?

- **using**
- extend
- import
- #

168. When your C# program is compiled, it creates a file called

- Class
- EXE
- Object
- Assembly According to the Instructor

169. Which of the following is an insertion operator in C++?

- <<
- >
- -
- &

170. C#_is a class member that is activated whenever the event . it was designed for.Occurs

- Attributes
- Arguments
- Method

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- **Event**

171. A fraud detection event on banking transaction is an example of.

- Linear event
- Sequential event
  - Predictive event
- **Probabilistic event**

172. In C++, which of the following library is include for 'cin' and 'cout' function?

- #include<fstream.h>
- #include<fstream>
- #include<conio.h>
- **#include<iostream>**

173. Which of the following doesn't have Graphical User Interface and the Event-DrivenModel?

- **Console-Application**
- Web Application
- Mobile Application
- Desktop Application

174. A stream in which the events may be of different types is referred to as a **heterogeneous** eventstream.

- Continuous
- Homogenous
- **Heterogeneous**
- Mutual

175. Defining a method in subclass having same name and type signature as a method in itssuperclass is known as

- Method hiding
- Method Overloading
- Method Binding
- **Method Overriding**

176. A single try block must be followed by which of these?

- Finally
- Catch
- **Both Finally and catch**
- Throw

177. By default the data member of a class are

- Protected, public
- Public
- **Private**
- Private, Public

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178. Exception are derived from___class. (Page-39)

- Windows error
- Exception error
- **System exception**
- Error Exception

179. Which of the following does NOT include in event processing operations?

- **Terminating Event**
- Deleting Event
- Transforming Event
- Reading Event

180. What is the return type of destruction?

- Float
- **No return type**
- Void
- Int

181. At the time an event is fired, the_method (with that event) will be invoked.

- **Public**
- Protected
- Registered
- Private

182. C# is a____and .NET is a_.

- Platform, Language
- Package, Language
- Language, Package
- **Language, Platform**

183. Which of the following is an entity that introduces event into the system?

- **Event producer**
- Event Stream
- Event Channel
- Event Consumer

184. Which of the following is an entity that receive event from the system?

- Event producer
- **Event Consumer**
- Raw event
- Event stream

185. Which of the following is the object-oriented way to handle run time errors?

- OnError () Method
- **Exception**
- Interface
- Error Code

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186. Why do we use cin get() in C++ program?

- ❖ It is used to print all values of array
- ❖ It is used to get numeric data
- ❖ **It is used to access character array**
- ❖ It is used to take input from user

187. Types not defined in any namespace are said to reside in the _namespace.

- Static
- Virtual
- C. Global

❖ D. Local

188. Arrays in C# are_object.

- ❖ Logical
- ❖ Arithmetic
- ❖ **Reference**
- ❖ Value

189. The initial name for C# was_.

- ❖ COAL
- ❖ COOP
- ❖ COAP
- ❖ **COOL**

190. Which of the following is declared like a class but have no implementation?

- ❖ Encapsulation
- ❖ Structure
- ❖ **Interface**
- ❖ Inheritance

191. Which of the following keyword is used to change data and behavior of a base class byreplacinga member of the base class with the new derived member.

- ❖ Base
- ❖ Overrides
- ❖ Overloads
- ❖ **New**

192. A stream in which all the events must be of same types is called **Homogenous** eventstream

- ❖ Continuous
- ❖ **Homogenous**
- ❖ Heterogeneous
- ❖ Mutual

193. Select the correct statement about an exception>

- ❖ It occurs during Just-in-time compilation
- ❖ **It occurs at runtime**
- ❖ It occurs during loading of program

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- It occurs during data binding

194. is not a defense against patents and clean room design?

- **Independent Invention**
- Event Invention
- Dependent Invention
- Clear room design

195. COOL stands for_____.

- Class Oriented Object Language
- Consumer Oriented Object Language
- **C-like Object Oriented Language**
  - C++-like Oriented Language

196. operator is used to signifies the destructor in C#?

  - #

- B. =
- C. &
- D. ~

197. What is the return type of destruction?

- Int
- Float
- Void
- **No return types**

198. An Event that is generated as a result of event processing that take place inside an eventprocessing system is called . _

- Event Stream
- **Derived Event**
- Raw Event
- Event Consumer

199. Choose the incorrect statement about delegates in C#?

- Delegates are value type
- Delegates are like function pointers in other language
- Delegates are object oriented
- **Delegates are type safe**

200. is the method of copying a design by reverse engineering and then recreating it without infringing any of the copyrights and trade secrets associated with the originaldesign.(Page-12)

- Clear Room Design
- Event Room design
- **Clean Room Design**
- Wall Room Design

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201. Which of the following statement is TRUE about raw event?

- It is a subscription mechanism for event
- It is introduced into an event processing system by an event consumer
- It is generate as a result of event processing
- **It is introduced into an event processing system by an event producer**

202. Which symbol is used to declare the preprocessor directives?

- *
- **#**
- $
- ^

203. Which of these keyword is used to manually throw an exception in C#?

- Try
- Finally
- Catch
- **Throw**

204. On successful compilation of a java source code_____is generated.

- Output
- **Bytecode**
- Hexadecimal code
- Octal code

205. _____ is the necessary condition to implement delegates.

- **Class Declaration**
- Function overloading
- Inheritance
- Runtime Polymorphism

206. Select correct statement about properties of read and write in C#

- **A read only property will only have get accessor**
- A write only property will only have get accessor
- A property cannot be either read only or write only
- A property can simultaneously be read or write only

207. <p> What will be the output of the following C# code?Static void Main{string[] args}

```
{
  int i;   for  (i = 0;  ;)
  {
    Console.WriteLine(&amp;quot;Virtual University&amp;quot;);
  }
  Console.ReadLine();
}
```

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- || will print "Virtual University" in continuous loop
- ** Syntax error **
- Virtual University
  - D. =p= No output =/p=

208. Choose the wrong statement about structures in C#.Net.

- **Structures can have destructors**
- Structures can implement an interface but they cannot inherit from another structure
- Structure members cannot be declared as protected
- A structure cannot be empty

209. Which of the following keyword is used to monitor exceptions?

- Catch
- Finally
- **Try**
- Throw

210. _____ library is used for dictionary in C#.

- System.Windows.Dictionary
- System.Windows.DictionaryResource
- System.Windows.ResourceDictionary
- System.Windows.Resource

211. In WPF,_____class define an object which has read-only(forzen)state.

- Read-Only
- Freeze
- Frozen
- Freezable

212. Which one of the following is the default panel in Visual Studio and expression blend?

- Stack Panel
- Wrap Panel
- Grid
- Canvas

213. Which one of the following is the first step in compiling XAML?

Performing the plumbing that connects XAML with procedural code automatically

- Converting XAML into.exe file
- Converting XAML into a special binaryformat
- Converting XAML into bytecode

214. Which of the following property will be used to restrict the user to not increase the width of a WPF element from a specific value.

- MinimumWidth
- MinWidth

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- WidthMinumum
- DecreaseWidth

215. ____transform requires an angle in transformation.

- Button transform
- Rotate transform Not SUre
- Element transform
- Scale transform

216. "Viewbox" element uses different types of stretching with the help of stretch property. There is/are _____possible value(s) of stretch property.

- 4
- 3
- 1
- 2

217. Markup extensions are surrounded by____

- Curly braces
- Square Brackets
- Parentheses
- Angle Brackets

218. ____is a timer based change of value.

- Themes
- Template
- Animation
- Styles Not Sure

219. _____Element is used in XAML to embed the C# code in XAML file.

- X:Code
- X:Content
- X:Name
- X:Button

220. The reason behind why XAML is good for user Interface is because of its_____nature.

- Class
- Hierarchical OR UI
- Base

221. _____ is used to declare a class into multiple files and all these files are combined into a single class file when the application is compiled.

- Property Element
- Abstract Class
- Inheritance
- Partial Class

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222. Margin defines an extra space_____while considering edges of the element.

- Outside
- Inside
- Between
- Anywhere

223. Which provider has the highest priority while providing a value to a dependency property?

- Style Setters
- Theme Style Setters
- Default Value
- Local Value

224. 'FrameworkElement' is the base class that adds support for _____

- Polymorphism
- Inheritance
- Abstract Class
- Data binding

225. Type Converter exists for_____transform.

- Render
- Matrix
- Scale
- Translate

226. _____classes are used to restrict the inheritance in C#.

- Sealed
- Property
- Object

227. _____class provides type conversion to convert a string name into an object reference to the object with that name.

- NameReferenceConverter
- NameConverter
- NameObjectConverter

228. Dockpanel.Dock is the___ in which child element describe its dock position in dock panel.

- Button Property
- Grid Property
- Attach Property

229. Type converters in WPF are similar to_____

- Multimedia Integration
- Markup extensions Not Sure
- Namespaces

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- Rich compositions

230. Which of the following device is sensitive to pressure?

- Mouse
- Printer
- Stylus
- Keyboard

231. Which will take precedence in sizing if explicit given width and height value of WPF application are not in range?

- Width=-1,height =-1
- Min,max(default values)
- Width=0,height=0

232. Which one is considered to be a keyword with reference to XAML 2006?

- X Name
- X Static
- X Null
- X Array

233. In WPF Grids have _____base indexing for Rows& Columns.

- One
- -1
- Zero
- Two

234. By default there will be ____ number of columns if we do not mention any column definition in grids.

- 2
- 3
- 1
- None

235. Translate Transform work on____ axes.

- X,Y
- X,Y,Z
- Y

236. In WPF, if Trigger is _____then its Setter's value will be Set.

- Hidden
- Inactive
- Active

237. Default value if Grid Row is_____

- 0
- 2
- **1**

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- 3

238. Superset of Canvas is ____

- Stack Panel
- Dock Panel
- Wrap Panel
- Grid

239. _____function will be register a Dependency property in WPF.

- DependencyProperty.Registration()
- DependencyProperty.Registered()
- DependencyProperty.Reg()
- DependencyProperty.Register()

240. Equivalent norms on a vector space 'X'defines _____topology for X.

- Coarser
- None of these
- Same Not Sure
- Finer

241. Dock panel is a ___of stack panel.

- Intersection
- Superset
- Union
- Subset

242. All classes of Windows Presentation foundation are derived form___class.

- Object
- Element
- XAML
- Abstract

243. Dock Panel is used in____interface.

- Right level
- Bottom level
- Top level
- Left level

244. In XAML namespace other than the default one is denoted by_____

- Xmlns(x)
- Xlmns/x
- Xlmns.x
- Xlmns:x

245. In WPF, if there is any name property than that property is also be taken as a substitute of_____

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- X:static
- X:reference
- X:binding
- X:name

246. In XAML 2006,X:Classmodifier by default access level is_____

- Public
- Internal
- Private
- Protected

247. _____sequence is not allowed within a CDATA section.

- {}
-]]
- <>
- ()

248. If we want to place a component at top in "dockpanel" then which one is correct syntax?

- DockPanel.Dock="top"
- Dock="Top"
- PanelDock="Top"
- DockPanel="Top"

249. In WPF, the default value of Horizontal/Alignment properties of child Element is_____

- Right
- Center
- Stretch
- Left

250. Which property is used to change the Z order of elements in WPF?

- ZPanel
- Zvalue
- ZIndex
- Z-coordinates

251. 1/96 pixel are equal to ___inch(es) in DPI display setting.

- 3
- 2
- 4
- 1

252. In WPF,_____control will allow to you visually group a set of control together.

- GroupBox
- Group

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- Box_Group
- Boxed_group

253. In WPF,_____function registers an attached property with the property system.

- Register_Attached(String,Type,Type,PropertyMetadata,ValidateValueCallback)
- RegisterAttached(String,Type,Type,PropertyMetadata,ValidateValueCallback)
- Reg_Attached(String,Type,Type,PropertyMetadata,ValidateValueCallback)
- None

254. In grid, elements are arranged in multicolumn and multirow like a____in HTML.

- Column
- Paragraph
- List
- Table

255. In XAML, tag declaration is equivalent to C#'s_____

- Object declaration
- Object Validation
- Variable declaration
- Object instantiation

256. Mapping to the WPF namespaces is _____inside the WPF assemblies.

- Dynamically specified
- Hard-coded
- Statically declared
- Specified at compile time

257. _____convert from a string in XAML markup into particular objects/data types in an object graph.

- Rich composition
- Resolution Independence
- Type converters
- Data binding

258. In WPF, if we try to add listbox item in listbox object which is already null, it will give us____

- Runtime exception
- Syntax Error
- Compile time error
- Logical error

259. In WPF, Subtags don't add as a(n)_____

- List
- Array
- Designated Attribute

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- Dictionary

260. Tag of namespace in XAML is just a_____

- Float
- Property
- Integer
- String

261. Is SharedSizeScope share size information in _____grid(s).

- Multiple
- Single
- Both 1&2
- Double

262. How can you specify null property in markup extension?

- :null:
- {x:null}
- (null)
- [x:null]

263. _____property value is used to set width and height of element in WPF according to its content.

- Tree
- Nan
- X:class
- Name

264. In WPF,____library is used for scrolling.

- Systems.Windows.Controls.ScrollViewer
- Systems.Windows.Controls.Scroll
- Systems.Windows.Controls.Viewer
- Systems.Media.ScrollViewer

265. When element in XAML overlaps in that scenario which technique is used to decide which element should come at the top of other element?

- Scaling
- Alignment
- Rotate
- Z Order

266. Which one of the following is the built-in transform function in WPF?

- Left Move
- Query transform
- Top transform
- Matrix transform

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267. In WPF, _____use(s) the method to add additional data with the sealed class object.

- Class
- Tree Structure
- Xmlns
- Dependency Properties

268. Grid act as a stack panel if it has____

- 1 row and autosize column
- 1 row and autosize rows
- Multiple rows and single column
- Single row & multiple column

269. In context of XML child element rules, if the child is plain text and a type converter existsto transform the child into the parent type (and no properties are set on the parent element), treat the child as input to the type converter and use the output as the _____.

- Child Object instance
- **Parent object instance**
- Interface object instance
- None of these

270. The root object in XAML must specify a least___XML namespace(s) that is/are used toqualify itself and any child elements (Page-62)

- **One**
- Two
- Three
- None

271. _____library is used to import the dependency poperties in WPF project (Page-74)

- System.Windows.Property.Dependency
- System.Windows.Dependancy
- System.Windows.Dependancy.Property
- **System.Windows.DependancyProperty**

272. A_transforms elements before they are laid out by the parent panel.

- Skew transform
- Render transform
- **Layout transform**
- Matrix transform

273. Defining any type of controls as a content of another control is __

- Vector based rendering
- **Rich Composition**
- Data Bining
- Resolution dependency

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274. The goal of WPF and XAML is used to keep separate the user interface design and ___

- Tools
- Content
- Element tree
- **Implementation**

275. Wgen elements in XAML overlaps, in that scenario which technique is used to decidewhich element should come at the top of other element?

- **Z order**
- Rotate
- Scaling
- Alignment

276. Visual Studio has a snippet called _ that automatically expands into a definition of a dependencyproperty, which makes defining one much faster than doing all the typing yourself.(Page-74)

- Property Extension
- **Propdp**
- Dependency property
- Dpprop

277. CAML stands for_.

- **Compiled Application Markup language**
- Composite Application Markup language
- Compiled Application Matrix language
- Centralized Application Markup language

278. _____is the callback which can be registered with property that can change the value ofproperty.

- Bubbling
- Element Property
- Tunneling
- **Coerce**

279. _library is used for Transforms in C#? (Page-82)

- System.Control.Transform
- System.Windows.Transform
- **System.Windows.Media.Transform**
- System.Windows.Controls.Transform

280. In WPF, Translate transform has_default value for X and Y? (Page-84)

- **0**
- 3
- 2
- 1

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281. DependancyProperty.Register is the__type of method of dependency property.

- Abstract
- Element
- **Static**
- Class

282. In XML, CDATA Section looks like

- <[CDATA[…]]>
- <![CDATA[…]]
- **<![CDATA[…]]>**
- ![CDATA[…]]>

283. Default Value of Grid.Row is

- 1
- 3
- 2
- **0**

284. ____is a base class similar to UIElement but for document-related pieces ofcontent that don't have rendering behavior on their own.

- VisualElement
- SystemElement
- **ContentElement**
- XAMLElement

285. In WPF ,_____function adds another type as an owner of a dependency property that hasalready been registered to a type.

- Owner(Type.propertyMetadata)
- Add_Owner (Type.Propert Metadata)
- **AddOwner(Type.PropertyMetadata)**
- Owner_Add(Type.Property Metadata)

286. xmlns namespace must be declare once in a XAML file at_element. (Page-67)

- Form
- Middle
- **Root**
- Child

287. Which type of panel is used when we have to customize the builtin controls in wpf?

- Wrap
- Grid
- Primitive
- **Stack**

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288. If we want to rotate an object clockwise about a specified point in a 2-D x-y coordinate systemthen which transform class will be used? (Page 87)

- **Rotate Transform**
- Translate Transform
- Skew Transform
- Scale Transform

289. A___enables to extend XAML and set any property that can be set in XAML using attributessyntax.

- **Markup extension**
- Reduction independence
- Data binding
- Rich composition

290. In WPF, Subtags don't add as a(n)___

- **Designated attribute**
- List
- Dictionary
- Array

291. ____key will provide a label access in WPF

- **Alt**
- CTRL
- Ctrl+alt
- Shift

292. Which one of the following is the first step in compiling XAML?

- Converting XAML to byte code
- Performing the plumbing that connects XAML with procedural code automatically
- **Converting a XAML file into a special binary format**
- Converting XAML into .exefile

293. "Viewbox" element uses different types of stretching with the help of stretch property .There is/are _____possible value(s) of stretch property

- 2
- 1
- 3
- **4**

294. Singleton method has__instance

- 4
- 3
- 2
- **1**

295. Suppose the property of x coordinate at the button timely changes in such a way thatafter every 100 second value of x coordinate increments by 1, then this case refers to

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- Transformation
- Property element
- Tunneling
- **Animation**

296. __refers to the space between an element and the content inside it.

- Tunneling
- Margin
- Pading
- **Style**

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Visual Programming CS411

Grand Quiz 2020 (COVID-19)

Solved by Sonu Ilyas Mughal

Sonu Mughal

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CS411:Midterm Quiz

Question # 1 of 30 (Start time: 04:28:38 PM, 11 July 2020)

In WPF, If we try to add listbox item in listbox object which is already null, it will give us

Select the correct option

- ○ Runtime exception
- ● Compile time error
- ○ Syntax error
- ○ Logical error

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Question # 2 of 30 (Start time: 04:29:57 PM, 11 July 2020)

XML is a stricter version of _____.

Select the correct option

- ○ Java
- ○ CSS
- ● HTML
- ○ C#

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Question # 3 of 30 (Start time: 04:30:26 PM, 11 July 2020)

Which of these methods used to read strings from the console?

Select the correct option

- Get()
- Getline()
- Read()
- ReadLine()

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CS411:Midterm Quiz

Question # 4 of 30 (Start time: 04:31:22 PM, 11 July 2020)

Which of the following statement is correct about WPF?

Select the correct option

- ○ WPF can use Python language for designing
- ○ WFP can developed mobile apps
- ● WPF is resolution independence
- ○ WPF not support declarative programming

Original

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Question # 5 of 30 (Start time: 04:31:47 PM, 11 July 2020)

The code given below is equivalent to _____ .
<SolidColorBrush>White</SolidColorBrush>

Select the correct option

- <WhiteBrush >MyWhiteBrush </ WhiteBrush>
- None of the given options
- <SolidColorBrush Color="White"/>
- <Brush Style= “SolidWhite” >

Mughal

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Question # 6 of 30 (Start time: 04:32:13 PM, 11 July 2020)

WPF4 was released in _____.

Select the correct option

- 2012
- 2010
- 2005
- 2003

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Question # 7 of 30 (Start time: 04:32:45 PM, 11 July 2020)

Which of the following events may or may not relate to an actual occurrence?

Select the correct option

- Probabilistic Events
- Actual Events
- Real Events
- Expected Events

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Question # 8 of 30 (Start time: 04:33:30 PM, 11 July 2020)

All elements of WPF have following property/properties.

Select the correct option

- ○ Width
- ○ None of given options
- ● Height and Width
- ○ Height

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Question # 9 of 30 (Start time: 04:34:13 PM, 11 July 2020)

_____ is used to declare a class into multiple files and all these files are combined into a single class file when the application is compiled.

Select the correct option

- (●) Inheritance
- ( ) Partial Class
- ( ) Abstract Class
- ( ) Property Element

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Question # 10 of 30 (Start time: 04:35:18 PM, 11 July 2020)

Which of the following is an example of tunnel key event?

Select the correct option

- Key up
- Preview key entered
- Preview key down
- Key down

ughal

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Question # 11 of 30 (Start time: 04:35:54 PM, 11 July 2020)

IsSharedSizeScope shares size information in _____ grid(s).

Select the correct option

- (●) Multiple
- ( ) Single
- ( ) Both 1 & 2
- ( ) Double

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MC180400061: SIDRA FATIMA

CS411:Midterm Quiz

Question # 12 of 30 (Start time: 04:37:24 PM, 11 July 2020)

Wait for a single event is _____ operation.

Select the correct option

- Waiting
- Waste
- Idle
- Blocking

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Question # 13 of 30 (Start time: 04:37:58 PM, 11 July 2020)

Which of the following components generates an input Event?

Select the correct option

- ○ Printer
- ● Multi Touch
- ○ Plotter
- ○ Speaker

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Question # 14 of 30 (Start time: 04:38:52 PM, 11 July 2020)

Arrays in C# are _____ objects.

Select the correct option

- Logical.
- Arithmetic.
- Reference.
- Value.

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A language supported by MS.Net platform.

Select the correct option

- C
- C++
- Java
- C#

Original

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Question # 16 of 30 (Start time: 04:40:46 PM, 11 July 2020)

Extra space inside the element in WPF is known as _____

Select the correct option

- Margin
- Style
- Padding
- Tunneling

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Question # 17 of 30 (Start time: 04:41:16 PM, 11 July 2020)

Which of the following statements is TRUE about raw event?

Select the correct option

- (●) It is introduced into an event processing system by an event producer.
- ( ) It is generated as a result of event processing.
- ( ) It is introduced into an event processing system by an event consumer.
- ( ) It is a subscription mechanism for events.

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Question # 18 of 30 (Start time: 04:42:10 PM, 11 July 2020)

DependencyObject.ClearValue method clears the _____ value in a dependency property.

Select the correct option

- Default
- Local
- Setter
- Getter

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Question # 19 of 30 (Start time: 04:43:23 PM, 11 July 2020)

All the members of interfaces are implicitly _____.

Select the correct option

- Private
- Static
- Public
- Abstract

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Question # 20 of 30 (Start time: 04:44:05 PM, 11 July 2020)

Dock panel is used in _____ interface.

Select the correct option

- (●) Top level
- ( ) Bottom level
- ( ) Left level
- ( ) Right level

The data members of a class by default are?

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Question # 21 of 30 (Start time: 04:45:23 PM, 11 July 2020)

The data members of a class by default are?

Select the correct option

- Protected, Public
- Private, Public
- Public
- Private

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Question # 22 of 30 (Start time: 04:46:28 PM, 11 July 2020)

The Main method returns zero value which indicates the_____.

Select the correct option

- Source of Program
- Successful execution of Program
- Termination of Program
- Exception in Program

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Question # 23 of 30 (Start time: 04:47:38 PM, 11 July 2020)

The _____, _____ and _____of applications that use events, either directly or indirectly is called event-based programming.

Select the correct option

- Model, Code, Operation
- Analyze, Design, Operation
- Code, Oeration, Maintain
- Design, Coding , Operation

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Question # 24 of 30 (Start time: 04:48:18 PM, 11 July 2020)

Silverlight is the subset of _____

Select the correct option

- (●) .NET
- ( ) Python
- ( ) Window Forms
- ( ) C++

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Question # 25 of 30 (Start time: 04:49:16 PM, 11 July 2020)

If we insert some objects in a single cell in Grid panel, then these objects are placed on _____ .

Select the correct option

- (●) One on Top of the other
- ( ) One after the other
- ( ) One on Bottom of the other
- ( ) One on Side of the other

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Question # 26 of 30 (Start time: 04:50:17 PM, 11 July 2020)

A class can inherit from _____class/classes.

Select the correct option

- ○ Two
- ● Single
- ○ Three
- ○ Multiple

ughal

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Question # 27 of 30 (Start time: 04:50:59 PM, 11 July 2020)

If we exclude the mechanism of calling the objects through name then:

Select the correct option

- Code will not compile
- Code will be difficult to debug
- Code will have errors
- Code will become design dependent

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Margin defines an extra space_____ while considering edges of the element

Select the correct option

- Between
- Inside
- Outside
- Anywhere

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Question # 29 of 30 (Start time: 04:52:33 PM, 11 July 2020)

Suppose we want to restrict the user to don't decrease the height of a WPF element from a specific amount then which property will be used?

Select the correct option

- (●) MinHeight
- ( ) None of given
- ( ) HeightMinimum
- ( ) MinimumHeight

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Attribute parameters can be either positional parameters or _____ parameters.

Select the correct option

- (●) Named
- ( ) Static
- ( ) Global
- ( ) None of the given options

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Question # 1 of 30 (Start time: 04:56:51 PM, 11 July 2020)

Suppose we have two elements in grid one is button and other is rectangle,which property will be helpful to set position of button on top of the rectangle?

Select the correct option

- ○ Z index
- ● Sizing
- ○ Transform
- ○ Stack Panel

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Question # 2 of 30 (Start time: 04:58:03 PM, 11 July 2020)

_____ graphics work with shapes.

Select the correct option

- Scalar
- Vector
- Both 1 & 2
- None of the above

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Question # 3 of 30 (Start time: 04:59:20 PM, 11 July 2020)

UI element focusable value is _____ by default.

Select the correct option

- (●) True
- ( ) False
- ( ) Nil
- ( ) Zero

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Question # 4 of 30 (Start time: 05:00:23 PM, 11 July 2020)

Tag of namespace in XML is just a_____

Select the correct option

- String
- Integer
- Float
- Property

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Question # 3 of 30 (Start time: 05:01:44 PM, 11 July 2020)

In which type does the enumerators are stored by the compiler?

Select the correct option

- string
- integer
- float
- none of these

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Question # 6 of 30 (Start time: 05:03:00 PM, 11 July 2020) **Total Marks**

Visual Studio has a snippet called _____ that automatically expands into a definition of a dependency property, which makes defining one much faster than doing all the typing yourself!

Select the correct option

- (●) Propdp
- ( ) Property extension
- ( ) Dpprop
- ( ) Dependency property

Saving

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Question # 7 of 30 (Start time: 05:03:30 PM, 11 July 2020)

Suppose we want to restrict the user to don't increase the height of a WPF element from a specific amount then which property will be used?

Select the correct option

- MaximumHeight
- MaxHeight
- None of given
- HeightMaximum

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Which of the following events may or may not relate to an actual occurrence?

Select the correct option

- (●) Probabilistic Events
- ( ) Actual Events
- ( ) Real Events
- ( ) Expected Events

SOL

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Question # 9 of 30 (Start time: 05:05:18 PM, 11 July 2020)

Attribute parameters can be either positional parameters or _____ parameters.

Select the correct option

- (●) Named
- ( ) Static
- ( ) Global
- ( ) None of the given options

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Question # 10 of 30 (Start time: 05:06:01 PM, 11 July 2020)

Which of the following keyword is used to monitor exceptions?

Select the correct option

- finally
- try
- throw
- catch

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Question # 11 of 30 (Start time: 05:07:04 PM, 11 July 2020)

Which keyword is used to declare the indexers?

Select the correct option

- base
- this
- extract
- super

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We want the width of button should cover the whole width of StackPanel in WPF then what should be value of HorizontalAlignment property of button?

Select the correct option

- Parent
- Whole
- Fill
- Stretch

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Question # 13 of 30 (Start time: 05:09:15 PM, 11 July 2020)

Which of these methods used to read strings from the console?

Select the correct option

- Get()
- Getline()
- Read()
- ReadLine()

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Question # 14 of 30 (Start time: 05:09:38 PM, 11 July 2020)

Which symbol is used to declare the preprocessor directives?

Select the correct option

- $
- *
- #
- ^

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Question # 15 of 30 (Start time: 05:10:16 PM, 11 July 2020)

Which method is used to write single byte to the file?

Select the correct option

- Write( )
- WriteLine( )
- WriteByte ( )
- WriteKey( )

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Question # 10 of 50 (Start time: 08:11:44 PM, 11 July 2020)

Graphical user interfaces and the event-driven model is not applied on which of the following?

Select the correct option

- Web Applications
- Console Applications
- Mobile applications
- Desktop Applications

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Question # 18 of 30 (Start time: 05:13:37 PM, 11 July 2020)

_____ Function will register a dependency property in WPF.

Select the correct option

- DependencyProperty.Reg()
- DependencyProperty.Registeration()
- DependencyProperty.Registered()
- DependencyProperty.Register()

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Question # 19 of 30 (Start time: 05:14:43 PM, 11 July 2020)

Derived Event is:

Select the correct option

○	Introduced into an event processing system by an event consumer.
●	Generated as a result of event that takes place inside an event processing system.
○	Introduced into an event processing system by an event producer.
○	A set of associated events.

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_____ key can be represented as _text in XAML.

Select the correct option

- Protected
- Focus
- Public
- None of the above

Solved

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Question # 20 of 50 (Start time: 05:15:09 PM, 11 July 2020)

_____ key can be represented as _text in XAML.

Select the correct option

- Protected
- Focus
- Public
- None of the above

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Question # 21 of 50 (Start time: 08:10:07 PM, 11 July 2020)

Silverlight is the subset of _____

Select the correct option

- [x] .NET
- [ ] Python
- [ ] Window Forms
- [ ] C++

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Question # 22 of 30 (Start time: 05:17:31 PM, 11 July 2020)

Which type of sizing is used in Wpf?

Select the correct option

- ○ Line
- ○ Percentage
- ● Proportional
- ○ Element

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Question # 23 of 30 (Start time: 05:18:34 PM, 11 July 2020)

By convention _____ event names are prefixed with Preview.

Select the correct option

- (●) Tunneling
- ( ) Bubbling
- ( ) Direct
- ( ) None of the given

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Question # 24 of 30 (Start time: 05:19:31 PM, 11 July 2020)

We use "UIElement" property ClipToBounds= “false” when _____.

Select the correct option

- We want area child element don’t cross the boundary of parent
- None of given
- We want area of child element is allowed to cross the boundary of parent
- Parent area also increases on increasing of area of child element

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Question # 25 of 30 (Start time: 05:20:44 PM, 11 July 2020)

The _____ in the namespace indicate a hierarchy of nested namespaces.

Select the correct option

- Arrows
- Dots
- Semicolon
- Brackets

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Question # 26 of 30 (Start time: 05:21:39 PM, 11 July 2020)

If we want to place a component at top in "dockpanel" then which one is correct syntax?

Select the correct option

○	PanelDock="Top"
○	Dock="Top"
●	DockPanel.Dock="Top"
○	DockPanel="Top"

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Question # 27 of 30 (Start time: 05:22:45 PM, 11 July 2020)

If we insert some objects in a single cell in Grid panel, then these objects are placed on _____ .

Select the correct option

- (●) One on Top of the other
- ( ) One after the other
- ( ) One on Bottom of the other
- ( ) One on Side of the other

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Question # 28 of 30 (Start time: 05:23:51 PM, 11 July 2020)

Which of the following is the object-oriented way to handle run time errors?

Select the correct option

- OnError
- Error codes
- Interface
- Exceptions

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Question # 29 of 30 (Start time: 05:24:54 PM, 11 July 2020)

The code given below is equivalent to _____ .
<SolidColorBrush>White</SolidColorBrush>

Select the correct option

- <WhiteBrush >MyWhiteBrush </ WhiteBrush>
- None of the given options
- <SolidColorBrush Color="White"/>
- <Brush Style= “SolidWhite” >

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_____ is an entity that receives events from the system.

Select the correct option

- ○ Event Producer
- ■ Event Consumer
- ○ Event Channel
- ○ Event Generator

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Question # 1 of 30 (Start time: 05:33:42 PM, 11 July 2020)

Wait for a single event is _____ operation.

Select the correct option

- Waiting
- Waste
- Idle
- Blocking

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Question # 2 of 50 (Start time: 08:04:07 PM, 11 July 2020)

_____ is used to declare a class into multiple files and all these files are combined into a single class file when the application is compiled.

Select the correct option

- Inheritance
- Partial Class
- Abstract Class
- Property Element

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Question # 3 of 30 (Start time: 05:35:05 PM, 11 July 2020)

<Button.Content> tag in XAML is the syntax for _____

Select the correct option

- Namespace
- Property Element
- Class
- Form

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Question # 4 of 30 (Start time: 05:36:21 PM, 11 July 2020)

COOL stands for _____.

Select the correct option

- Class Oriented Object Language
- Consumer Oriented Object Language
- C-like Object Oriented Language
- C++-like Object Oriented Language

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Against _____ keyword a partial class is generated which we write in C# code.

Select the correct option

- (●) X:class
- ( ) Xmlns
- ( ) <button>
- ( ) X:Element

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Question # 8 of 30 (Start time: 05:39:08 PM, 11 July 2020)

Suppose we have declared a button in XAML as shown in given code.Choose the correct C# code which get this button in C#.
<Button x:Name="okbutton">OK</Button>

Select the correct option

○	Button okbutton=(Button)window.("okbutton");
○	Button okbutton=(Button)window.searchname("okbutton");
○	Button okbutton=(Button)window.getbutton("okbutton");
●	Button okbutton=(Button)window.findname("okbutton");

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Question # 9 of 50 (Start time: 08:40:10 PM, 11 July 2020)

Additional XML namespaces (on the root or on children) must be _____ to be used on any identifiers from that namespace.

Select the correct option

- Given a common prefix
- Qualified with “Pre” keyword
- Given a distinct prefix
- Qualified with “Ext” keyword

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Question # 10 of 30 (Start time: 05:41:23 PM, 11 July 2020)

Arrays in C# are _____ objects.

Select the correct option

- Logical.
- Arithmetic.
- Reference.
- Value.

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Question # 11 of 30 (Start time: 05:41:51 PM, 11 July 2020)

When your C# program is compiled, it creates a file called _____

Select the correct option

- (●) Assembly
- ( ) Exe
- ( ) Object
- ( ) Class

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Question # 12 of 30 (Start time: 05:42:50 PM, 11 July 2020)

_____ property value is used to set width and height of element in WPF according to its content.

Select the correct option

- Name
- X:class
- Nan
- Tree

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Question # 13 of 30 (Start time: 05:43:34 PM, 11 July 2020)

What is the return type of destructors?

Select the correct option

- int
- float
- void
- no return types

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Question # 14 of 30 (Start time: 05:44:34 PM, 11 July 2020)

In context of XAML child element rules, if the parent supports a content property and the type of the child is compatible with that property, treat the child as its _____

Select the correct option

- Super class
- Sub class
- Value
- Interface

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Question # 20 of 30 (Start time: 05:49:44 PM, 11 July 2020)

Event-based programming is also known as:

Select the correct option

- Service-oriented architecture (SOA)
- Event-driven Architecture (EDA)
- Service-driven Architecture (SDA)
- Event-based Architecture (EBA)

“FrameworkElement” is the base class that adds support for _

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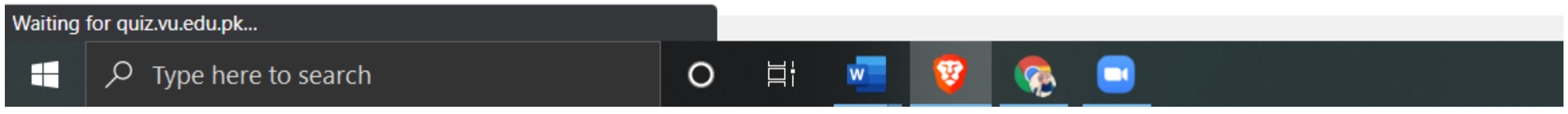
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Question # 21 of 30 (Start time: 05:51:13 PM, 11 July 2020)

When a user select element from listbox, after selecting compiler marked the selected element as____

Select the correct option

- Fix
- Selected
- Handled
- Bubble



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Design of Silverlight mostly follow the design of _____

Select the correct option

- C++
- WPF
- Console
- C#

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Question # 23 of 30 (Start time: 05:54:00 PM, 11 July 2020)

Which of the following is/are not derived from “Control” class?

Select the correct option

- ListBox
- StatusBar
- None of the given
- Button

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Question # 24 of 30 (Start time: 05:55:08 PM, 11 July 2020)

To design a complex Interface, which of the following technique is best to achieve the required results?

Select the correct option

- ○ Panels composed in three panels
- ● Panels composed within panels
- ○ Panels composed in two panels
- ○ Panels composed in single panel

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Question # 25 of 30 (Start time: 05:56:09 PM, 11 July 2020)

In tunneling, event move from _____.

Select the correct option

- ○ Left to Right
- ○ Right to Left
- ● Top to Down
- ○ Diagonally

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Question # 26 of 30 (Start time: 05:57:17 PM, 11 July 2020)

Attributes of a program can be retrieved from its assembly metadata through _____.

Select the correct option

- Static member
- Reflection
- Garbage collector
- Constructor

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Question # 27 of 30 (Start time: 05:58:20 PM, 11 July 2020)

Clean room design is also called _____.

Select the correct option

- Clear Room Design
- Wall Technique Design
- Chinese Wall Technique
- French Wall Technique

Sohail Mughal

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Question # 28 of 30 (Start time: 05:58:46 PM, 11 July 2020)

The code given below is equivalent to _____ .
<SolidColorBrush>White</SolidColorBrush>

Select the correct option

- <WhiteBrush >MyWhiteBrush </ WhiteBrush>
- None of the given options
- <SolidColorBrush Color="White"/>
- <Brush Style= “SolidWhite” >

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CS411:Midterm Quiz

Question # 29 of 30 (Start time: 05:59:27 PM, 11 July 2020)

If we want to scale an object in the 2-D x-y coordinate system then which transform class will be used?

Select the correct option

- ○ Rotate Transform
- ○ Matrix Transform
- ● Translate Transform
- ○ Skew Transform

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Question # 30 of 30 (Start time: 06:00:43 PM, 11 July 2020)

Clean room design is useful as a defense against _____.

Select the correct option

○	Copyright
○	Trade secret infringement
●	Copyright and trade secret infringement
○	None of the given options

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Question # 1 of 30 (Start time: 06:44:55 PM, 11 July 2020)

_____ Function will register a dependency property in WPF.

Select the correct option

- DependencyProperty.Reg()
- DependencyProperty.Registeration()
- DependencyProperty.Registered()
- DependencyProperty.Register()

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Question # 1 of 30 (Start time: 06:44:55 PM, 11 July 2020)

_____ Function will register a dependency property in WPF.

Select the correct option

- DependencyProperty.Reg()
- DependencyProperty.Registeration()
- DependencyProperty.Registered()
- DependencyProperty.Register()

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Question # 2 of 30 (Start time: 06:46:10 PM, 11 July 2020)

The root object element in XAML must specify at least _____ XML namespace(s) that is/are used to qualify itself and any child elements.

Select the correct option

- Two
- None of the given options
- Four
- One

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CS411:Midterm Quiz

Question # 3 of 30 (Start time: 06:47:19 PM, 11 July 2020)

Anonymous methods _____ the code size.

Select the correct option

- Double
- Expand
- Reduce
- Extend

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Question # 4 of 30 (Start time: 06:48:15 PM, 11 July 2020)

_____ works with Graphic Cards.

Select the correct option

- OpenGL
- GDI
- DirectX
- XNA

Sughal

03063548328 For Solving Quiz



Question # 5 of 30 (Start time: 06:49:37 PM, 11 July 2020)

System.Windows.Controls.Button b = new System.Windows.Controls.Button();
b.Content = "OK";

For the above code (written in C#) the corresponding XAML code will be:

Select the correct option

●	<Button xmlns=http://schemas.microsoft.com/winfx/2006/xaml/presentation Content="OK"/>
○	<Button xmlns= “OK” Content="http://schemas.microsoft.com/winfx/2006/xaml/presentation "/>
○	<Button xmlns=http://schemas.microsoft.com/winfx/2006/xaml/presentation Event="OK"/>
○	<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Call="OK"/>

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Which of these methods used to read strings from the console?

Select the correct option

- Get()
- Getline()
- Read()
- ReadLine()

Sumaira Mughal

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Question # 7 of 30 (Start time: 06:50:57 PM, 11 July 2020)

<Button.Content> tag in XAML is the syntax for _____

Select the correct option

- Namespace
- Property Element
- Class
- Form

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Question # 8 of 30 (Start time: 06:51:50 PM, 11 July 2020)

We use "UIElement" property ClipToBounds= “false” when _____.

Select the correct option

- We want area child element don’t cross the boundary of parent
- None of given
- We want area of child element is allowed to cross the boundary of parent
- Parent area also increases on increasing of area of child element

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Question # 9 of 30 (Start time: 06:53:14 PM, 11 July 2020)

To check that the user either pressed left Alt key or right Alt key, the _____ is used.

Select the correct option

- IsKeyDown
- IsDown
- keyboardDevice.IsKeyDown
- KeyStates.IsKeyDown

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Question # 10 of 30 (Start time: 06:54:02 PM, 11 July 2020)

Wait for a single event is _____ operation.

Select the correct option

- Waiting
- Waste
- Idle
- Blocking

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UI element focusable value is _____ by default.

Select the correct option

●	True
○	False
○	Nil
	Zero

Sonia Mughal

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Question # 12 of 30 (Start time: 06:54:59 PM, 11 July 2020)

XAML specification defines rules that map ".NET" namespaces, types, properties, and events into _____.

I) XML namespaces
II) XML elements
III) XML Attributes

Select the correct option

○	(I) Only
○	(II) Only
●	(I), (II) and (III) only
○	(III) Only

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Question # 13 of 30 (Start time: 06:55:17 PM, 11 July 2020)

Which provider has the highest priority while providing a value to a dependency property?

Select the correct option

- Style Setters
- Default Value
- Local Value
- Theme Style Setters

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Question # 14 of 30 (Start time: 06:56:53 PM, 11 July 2020)

Which of the following statements is TRUE about raw event?

Select the correct option

- [x] It is introduced into an event processing system by an event producer.
- [ ] It is generated as a result of event processing.
- [ ] It is introduced into an event processing system by an event consumer.
- [ ] It is a subscription mechanism for events.

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CS411:Midterm Quiz

Question # 15 of 30 (Start time: 06:57:40 PM, 11 July 2020)

WPF Margin is defined by _____ type.

Select the correct option

- Thin
- Thinnest
- Thick
- Thickness

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Question # 16 of 30 (Start time: 06:59:08 PM, 11 July 2020)

Which property is used to change the Z Order of elements in WPF?

Select the correct option

- Z-Coordinate
- ZPanel
- ZIndex
- ZValue

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Question # 18 of 30 (Start time: 07:01:45 PM, 11 July 2020)

If we run the program from within Visual Studio, the console window disappears quickly so we can use _____ to keep screen from going away.

Select the correct option

- (●) Console.ReadLine()
- ( ) Console.WriteLine()
- ( ) Console.Title
- ( ) Console.Clear

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Question # 19 of 30 (Start time: 07:02:41 PM, 11 July 2020)

When parent element space is not enough for the content of child element,this type of situation arises due to _____

Select the correct option

- Short memory
- Content Overflow
- Indexing
- Grid

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Question # 20 of 30 (Start time: 07:04:06 PM, 11 July 2020)

_____ is a base class similar to UIElement but for document-related pieces of content that don't have rendering behavior on their own.

Select the correct option

- ● ContentElement
- ○ VisualElement
- ○ SystemElement
- ○ XAMLElement

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Question # 21 of 30 (Start time: 07:04:41 PM, 11 July 2020)

Which of these methods of the class String is used to obtain length of string object?

Select the correct option

- get()
- Sizeof()
- length()
- lengthof()

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Question # 22 of 30 (Start time: 07:05:37 PM, 11 July 2020)

Overloaded operators must be _____.

Select the correct option

- Referenced
- Static
- Objective
- Delegates

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Question # 23 of 30 (Start time: 07:06:01 PM, 11 July 2020)

Which symbol is used to declare the preprocessor directives?

Select the correct option

- $
- *
- #
- ^

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Question # 24 of 30 (Start time: 07:06:39 PM, 11 July 2020)

Suppose we want to restrict the user to don't increase the height of a WPF element from a specific amount then which property will be used?

Select the correct option

- MaximumHeight
- MaxHeight
- None of given
- HeightMaximum

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Question # 25 of 30 (Start time: 07:07:00 PM, 11 July 2020)

A(n) _____ is a reference to a method.

Select the correct option

- (●) Delegate
- ( ) XAML
- ( ) Function
- ( ) Constructor

ughal
SO

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Question # 26 of 30 (Start time: 07:07:47 PM, 11 July 2020)

Default value of Grid.Row is _____

Select the correct option

- (●) 0
- ( ) 1
- ( ) 2
- ( ) 3

Vughal

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Question # 28 of 30 (Start time: 07:09:53 PM, 11 July 2020)

In XAML 2006, x:AsyncRecords works on _____ thread(s).

Select the correct option

- Single
- Multiple
- Both 1 & 2
- None of the given

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Question # 29 of 30 (Start time: 07:10:51 PM, 11 July 2020)

How can you specify null property in markup extension?

Select the correct option

- (null)
- :null:
- {x:null}
- [x:null]

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Question # 30 of 30 (Start time: 07:11:25 PM, 11 July 2020)

DependencyObject.ClearValue method clears the _____ value in a dependency property.

Select the correct option

- Default
- Local
- Setter
- Getter

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Question No: 1 (Marks: 1) - Please choose one
Event-based programming is also known as:

Service-oriented architecture (SOA)
Event-driven Architecture (EDA) (Page 11)
Service-driven Architecture (SDA)
Event-based Architecture (EBA)

Question No: 2 (Marks: 1) - Please choose one
Polymorphism works on the basis that child class should have_____feature/s of its parent class.

1. Only one
2. No
3. Few
4. All

Question No: 3 (Marks: 1) - Please choose one
The base class will have the _____ functionality of all derived classes

1. Common
2. Specialized
3. Unique
4. Different

Question No: 4 (Marks: 1) - Please choose one
Clean room design is also called_____

1. Clear Room Design
2. Wall Technique Design (Page 13)
3. Chinese Wall Technique
4. French Wall Technique

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Question No: 5 (Marks: 1) - Please choose one
Indexers are similar to properties, but are accessed via a/an_____rather than a property name.

1. Index Argument Click here 4 detail
2. Reference
3. Pointer
4. Integer Argument

Question No: 6 (Marks: 1) - Please choose one
Types not defined in any namespace are said reside in the_____namespace.

1. Local
2. Static
3. Virtual ✓
4. Global Click here 4 detail ✗

Question No: 7 (Marks: 1) - Please choose one
A compiled C# file is called

1. Complied File
2. Source File
3. Assembly File (Page 41)
4. EXE File

Question No: 8 (Marks: 1) - Please choose one
Cool stands for_____.

1. Class Oriented Object Language
2. Consumer Oriented Object Language
3. C-like Object Oriented Language ✓
4. C++-like Object Oriented Language (Page 13) ✗

Question No: 9 (Marks: 1) - Please choose one
The initial name for C# was_____.

1. COAL
2. COOP
3. COAP
4. COOL (Page 13)

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Question No: 10 (Marks: 1) - Please choose one
The Main method returns a non-zero value which indicates the_____.

1. Source of Program
2. Error in Program Click here 4 more detail
3. Termination of Program
4. Exception in Program

Question No: 11 (Marks: 1) - Please choose one
Attributes add _____to year program.

1. Clearance
2. Errors
3. Meta Data (Page 40)
4. Ambiguity

Question No: 12 (Marks: 1) - Please choose one
Wait for a single event is _____operation.

1. Waiting
2. Waste
3. Idle
4. Blocking (Page 8)

Question No: 13 (Marks: 1) - Please choose one
To use kbhit() command, you need to include which header file in your program?

1. fstream.h
2. stdio.h
3. iostream.h
4. conio.h Click here 4 more detail

Question No: 14 (Marks: 1) - Please choose one
Which of the following is built from request-response?

1. Event-processing Architecture (EDA)
2. Service-driven Architecture (SDA)
3. Service-oriented Architecture (SOA) (Page 11)
4. Event-bas Architecture (EBA)

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Question No: 15 (Marks: 1) - Please choose one
_____let any classes behave like an array.

1. Properties
2. Fields
3. Indexers (Page 29)
4. Finalizers

Question No: 16 (Marks: 1) - Please choose one
All the members of interfaces are implicitly_____.

1. Private
2. Static
3. Public (Page 34)
4. Abstract

Question No: 17 (Marks: 1) - Please choose one
_____methods do not have names.

1. Attribute
2. Anonymous (Page 46)
3. None of given
4. Non-Anonymous

Question No: 18 (Marks: 1) - Please choose one
Derived Event is:

Introduced into an event processing system by an event consumer.
Generated as a result of event that takes place inside an event processing system. (Page 11)
Introduced into an event processing system by an event producer.
A set of associated events.

Question No: 19 (Marks: 1) - Please choose one
Which of the following is an occurrence within a particular system or domain?

Object
Event (Page 7)
Result
Message

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Question No: 20 (Marks: 1) - Please choose one
_____ is an entity that receives events from the system.

Event Producer
Event Consumer (Page 11)
Event Channel
Event Generator

Question No: 21 (Marks: 1) - Please choose one
Writing higher-level functions that call upon lower-level functions _____ a program.

Decode
Harder
Refactor
Simplify

Question No: 22 (Marks: 1) - Please choose one
Which statement is True?

A "catch" block always executes whether or not an exception is thrown and whether or not the "try" block runs to completion.
A "finally" block executes only, when an exception is thrown and when the "try" block runs to completion.
A "finally" block may or may not executes whether or not an exception is thrown and whether or not the "try" block runs to completion.
A "finally" block always executes whether or not an exception is thrown and whether or not the "try" block runs to completion. Click here 4 detail

Question No: 23 (Marks: 1) - Please choose one
Attributes of a program can be queried at run time through _____.

Reflection (Page 41, for more detail**)**
Value
Reference
Extension

Question No: 24 (Marks: 1) - Please choose one
During the program execution, program can be stopped with the help of _____.

Attributes
Compiler
Exception Handling
Breakpoints (Page 47)

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Question No: 25 (Marks: 1) - Please choose one
The _____, _____ and _____of applications that use events, either directly or indirectly is called event-based programming.

Model, Code, Operation
Analyze, Design, Operation
Code, Operation, Maintain
Design, Coding, Operation (Page 8)

Question No: 26 (Marks: 1) - Please choose one
Clean room design is useful as a defense against _____.

Copyright
Trade secret infringement
Copyright and trade secret infringement (Page 13)
None of the given options

Question No: 27 (Marks: 1) - Please choose one
The result of Exception can bring _____ in the program.

Reliability
Stability
Inconsistency (Page 40)
Consistency

Question No: 28 (Marks: 1) - Please choose one
_____ method reads the whole document in memory.

XmlDocument (Page 48)
XmlLine
XPath
XmlReader

Question No: 29 (Marks: 1) - Please choose one
The _____ in the namespace indicate a hierarchy of nested namespaces.

Arrows
Dots
Semicolon
Brackets (Page 22)

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Question No: 30 (Marks: 1) - Please choose one
If we run the program from within Visual Studio, the console window disappears quickly so we can use _____ to keep screen from going away.

Console.ReadLine() (Page 15)
Console.WriteLine()
Console.Title
Console.Clear

Question No: 31 (Marks: 1) - Please choose one
Which of the following is an entity that introduces event into the system?

Event Channel
Event Stream
Event Consumer
Event Producer (Page 11)

Question No: 32 (Marks: 1) - Please choose one
At the time an event is fired, the _____ methods will be invoked.

Registered Click here 4 detail
Public
Static
Unregistered

Question No: 33 (Marks: 1) - Please choose one
The easiest way to declare an event is to put the event keyword in front of a _____ member.

Interface
Delegate Click here 4 detail
Class
Struct

Question No: 34 (Marks: 1) - Please choose one
Which of the following does NOT include in event processing operations?

Deleting Events
Reading Events
Terminating Events (Page 8)
Transforming Events

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Question No: 35 (Marks: 1) - Please choose one
An event is a list of_____

constants
variable
pointers
delgates

Question No: 36 (Marks: 1) - Please choose one
C# is a _____ and .NET is a _____.

Platform, Language
Package, Language
Language, Package
Language platform (Page 13)

Question No: 37 (Marks: 1) - Please choose one
Anonymous methods _____ the code size.

Double
Expand
Reduce (Page 46)
Extend

Question No: 38 (Marks: 1) - Please choose one
g++ is a command of _____.

Dev C++
Unix based system
Windows based system
None of the given options

Question No: 39 (Marks: 1) - Please choose one
A class can inherit from _____class/classes.

Two
Single Click here 4 detail
Three
Multiple

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Question No: 40 (Marks: 1) - Please choose one
Which of the following events may or may not relate to an actual occurrence?

Probabilistic Events (Page 7)
Actual Events
Real Events
Expected Events

Question No: 41 (Marks: 1) - Please choose one
Which of the following statements is TRUE about raw event?

It is introduced into an event processing system by an event producer. (Page 11)
It is generated as a result of event processing.
It is introduced into an event processing system by an event consumer.
It is a subscription mechanism for events.

Question No: 42 (Marks: 1) - Please choose one
A method performs an action in a series of statements, called a _____.

Code Block
Method Block
Routine Block
Statement Block Click here for detail

Question No: 43 (Marks: 1) - Please choose one
Graphical user interfaces and the event-driven model are not applied on which of the following?

Web Applications
Console Applications
Mobile applications
Desktop Applications

Question No: 44 (Marks: 1) - Please choose one
The System.Console is a _____ which enables us to do Console Input and Output.

Structure
Class (Page 14)
Namespace
Library

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Question No: 45 (Marks: 1) - Please choose one
Events are based on the principle of _____.

Request
Response
Coupling
Decoupling (Page 10)

Question No: 46 (Marks: 1) - Please choose one
In event-driven architecture (EDA), _____ sends event to _____.

Event Consumer, Event Producer
Event Producer, Event Consumer (Page 10)
Event Handler, Event Producer
Event Generator, Event Producer

Question No: 47 (Marks: 1) - Please choose one
XML is a stricter version of _____.

Java
CSS
HTML (Page 47)
C#

Question No: 48 (Marks: 1) - Please choose one
Extension methods _____ access the private data members of the class in which they are extended.

May
Cannot click here 4 more detail
May not
Can

Question No: 49 (Marks: 1) - Please choose one
BAML is compressed representation of _____.

CAML:NS
XAML (Page 71)
XAML:NS
CAML

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Question No: 50 (Marks: 1) - Please choose one
In context of XAML child element rules, if the child is plain text and a type converter exists to transform the child into the parent type (and no properties are set on the parent element), treat the child as the input to the type converter and use the output as the _____.

Parent object instance (Page 68)
Interface object instance
none
Child object instance

Question No: 51 (Marks: 1) - Please choose one
IsNumeric(string s) is a function of _____ class.

Math class
Program class
String class
MyUtills class (Page 53)

Question No: 52 (Marks: 1) - Please choose one
XAML specification defines rules that map ".NET" namespaces, types, properties, and events into _____.
I) XML namespaces II) XML elements III) XML Attributes

(I) Only
(II) Only
(I), (II) and (III) only (Page 63)
(III) Only

Question No: 53 (Marks: 1) - Please choose one
Visual Studio has a snippet called _____ that automatically expands into a definition of a dependency property, which makes defining one much faster than doing all the typing yourself!

Propdp (Page 77)
Property extension
Dpprop
Dependency property

Question No: 54 (Marks: 1) - Please choose one
Suppose we want to restrict the user to don't decrease the height of a WPF element from a specific amount then which property will be used?

MinHeight (Page 83)
none
HeightMinimum
MinimumHeight

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Question No: 55 (Marks: 1) - Please choose one
WPF element in "DockPanel" has property "dock". There is/are _____ possible value(s) of "dock" property.

two
three
four (Page 93)
one

Question No: 56 (Marks: 1) - Please choose one
WPF4 was released in

2012
2010 Click here for detail
2005
2003

Question No: 57 (Marks: 1) - Please choose one
_____ allows to keep the user interface description and implementation separate.

GDI
WPF (Page 61)
Avalon
Altia

Question No: 58 (Marks: 1) - Please choose one
When the data is passed from one place to another in chunks, this data is called _____.

Bits
Line
Byte
Stream

Question No: 59 (Marks: 1) - Please choose one
We want the width of button should cover the whole width of StackPanel in WPF then what should be value of HorizontalAlignment property of button?

Parent
Whole
Fill
Stretch (Page 97)

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Question No: 60 (Marks: 1) - Please choose one
Which of the following is/are not derived from “Control” class?

ListBox
StatusBar
None of the given (Page 73)

Question No: 61 (Marks: 1) - Please choose one
If we want to move an object in the 2-D x-y coordinate system then which transform class will be used?

Translate Transform (Page 89)
Skew Transform
Skew Transform
Rotate Transform

Question No: 62 (Marks: 1) - Please choose one
Dependency properties are represented by _____ .

System.Windows.DependencyProperty (Page 77)
System.Windows.Dependency.Property
System.Windows.Dependency
System.Windows.Property.Dependency

Question No: 63 (Marks: 1) - Please choose one
System.Windows.Controls.Button b = new System.Windows.Controls.Button(); b.Content = "OK"; For the above code (written in C#) the corresponding XAML code will be:

<Button xmlns=http://schemas.microsoft.com/winfx/2006/xaml/presentation Content="OK"/> (Page 63)
<Button xmlns= “OK” Content="http://schemas.microsoft.com/winfx/2006/xaml/presentation "/>
<Button xmlns=http://schemas.microsoft.com/winfx/2006/xaml/presentation Event="OK"/>
<Button xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" Call="OK"/>

Question No: 64 (Marks: 1) - Please choose one
_____ is a base class similar to UIElement but for document-related pieces of content that don’t have rendering behavior on their own.

ContentElement (Page 73)
VisualElement
SystemElement
XAMLElement

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Question No: 65 (Marks: 1) - Please choose one
In context of XAML child element rules, if the parent supports a content property and the type of the child is compatible with that property, treat the child as its _____.

Super class
Sub class
Value (Page 68)
Interface

Question No: 66 (Marks: 1) - Please choose one
Mapping to the WPF namespaces is _____ in-side the WPF assemblies.

Specified at compile time
Hard-coded (Page 63)
None of the given
Dynamically specified

Question No: 67 (Marks: 1) - Please choose one
The _____ user interfaces, enabled by WPF, is getting a lot of attention.

Loutish
Polished (Page 61)
Coarse
Distracted

Question No: 68 (Marks: 1) - Please choose one
The _____ method is used to read the whole file at once.

ReadAllText("test.txt"); Click here for detail
ReadAll();
Read("test.txt");
ReadLine();

Question No: 69 (Marks: 1) - Please choose one
With the help of WPF, an ugly looking application can be _____ by the designers.

Rearranged
Re-created
Re-themed(Page 61)
Rejected

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Question No: 70 (Marks: 1) - Please choose one
Additional XML namespaces (on the root or on children) must be _____ to be used on any identifiers from that namespace.

Given a common prefix
Qualified with “Pre” keyword
Given a distinct prefix (Page 63)
Qualified with “Ext” keyword

Question No: 71 (Marks: 1) - Please choose one
Extension methods can only be defined in _____ class.

Dynamic class
Normal class
String class
Static class Click here for detail

Question No: 72 (Marks: 1) - Please choose one
If we want to rotate an object clockwise about a specified point in a 2-D x-y coordinate system then which transform class will be used?

None of given (Page 87)
Translate Transform
Skew Transform
Scale Transform

Question No: 73 (Marks: 1) - Please choose one
The code given below is equivalent to _____ . <SolidColorBrush>White</SolidColorBrush>

<WhiteBrush >MyWhiteBrush </ WhiteBrush>
None of the given options
<SolidColorBrush Color="White"/> (Page 65)
<Brush Style= “SolidWhite” >

Question No: 74 (Marks: 1) - Please choose one
“FrameworkElement” is the base class that adds support for _____.

Some common mechanisms for Windows-based controls
Data binding
Styles
All of the given options (Page 73)

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Question No: 75 (Marks: 1) - Please choose one
All elements of WPF have following property/properties.

Width
None of given options
Height and Width
Height

Question No: 76 (Marks: 1) - Please choose one
If we exclude the mechanism of calling the objects through name then:

Code will not compile
Code will be difficult to debug
Code will have errors
Code will become design dependent

Question No: 77 (Marks: 1) - Please choose one
If we want to place a component at top in "dockpanel" then which one is correct syntax?

PanelDock="Top"
Dock="Top"
DockPanel.Dock="Top" (Page 93)
DockPanel="Top"

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